



Verified Carbon Standard

KA'A PYAHU

Project title	KA'A PYAHU
Project ID	5087
Monitoring period	01/05/2021-31/12/2023
Crediting period	01/05/2021-30/04/2041
Original date of issue	01-may-2024
Most recent date of issue	02-september-2024
Version	1.3
VCS Standard Version	4.7
Prepared by	BIOFIX CONSULTORIA LTDA.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

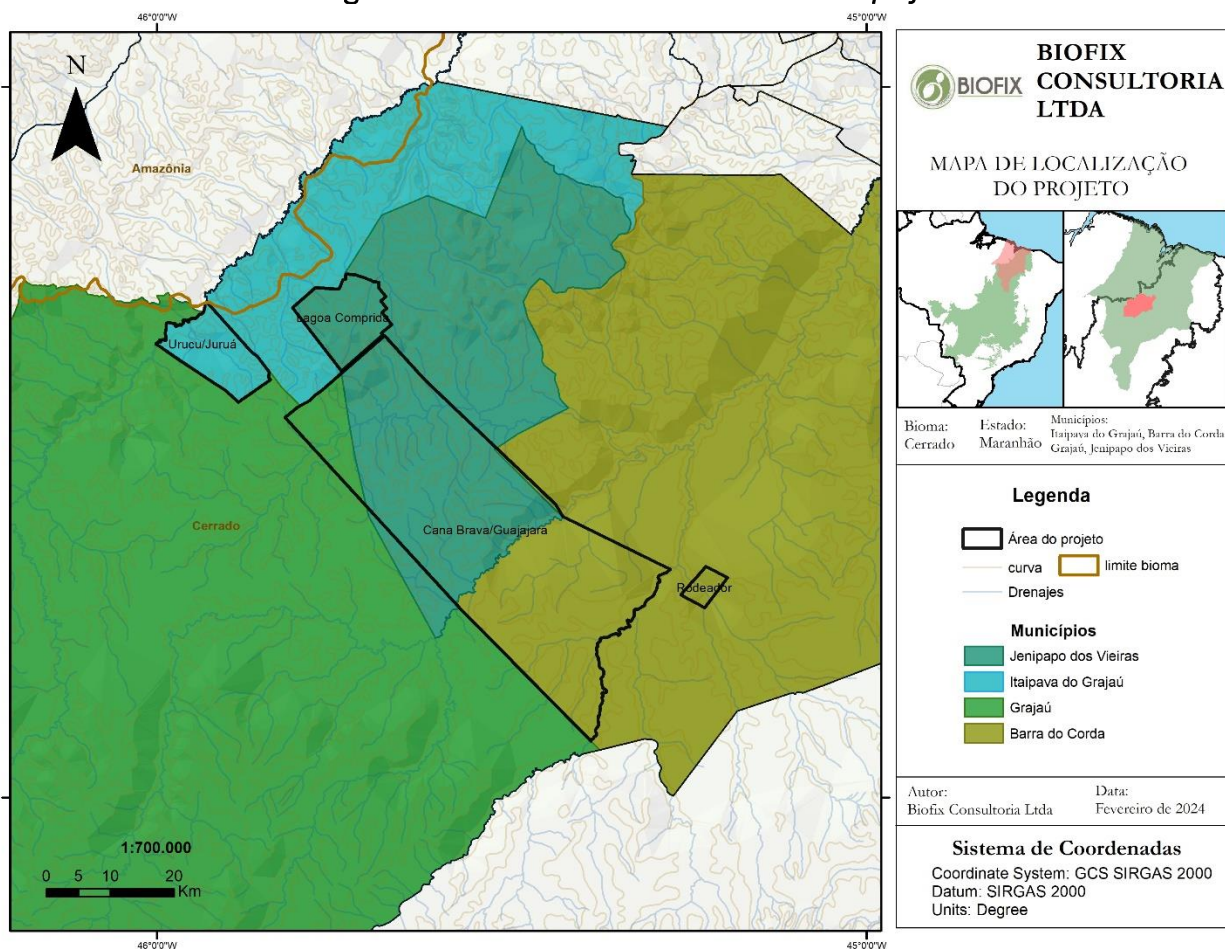
The KA'A PYAHU REDD+ project is located in the state of Maranhão, in the northeast region of Brazil, under the jurisdiction of the Legal Amazon. The proponents of the project are the Guajajara people, inhabitants and holders of three indigenous lands (TI) and one indigenous domain area (DI), namely: Cana Brava Indigenous Land, Urucu-Juruá Indigenous Land, Rodeador Indigenous Dominion Area and Lagoa Comprida Indigenous Land. The Indigenous Land is the traditionally used space by the indigenous peoples, of form constant, for its productive activities and for to preserve the environmental resources necessary for their well-being and physical and cultural reproduction. Domain Lands are properties of indigenous communities, acquired according to foreseen in civil legislation. The Canabrava Indigenous Land and the Area Domain Indigenous Rodeador find in Barra do Corda municipality, the Earth Indigenous Lagoa Comprida is in Jenipapo dos Vieira municipality, and the Earth Urucu/Juruá indigenous group covers the municipality of Grajaú, all in Maranhão, northeastern Brazil.

These territories are historically marked by risk factors such as deforestation, illegal logging that threatens the integrity of the territory and the people who live there, and socio-environmental impacts due to the installation of power lines within the territories. These aspects cause harm to the way of life of these people, who need greater attention to health, education, protection and territorial monitoring. The Ka'a Pyahu project aims to reinforce the aspects covered in the REDD+ investment lines to bring benefits to communities. According to what was discussed in the workshops, the project will recover degraded areas, monitor illegal invasions and control fires. In addition, it will support the production of articles made from tucum straw, an abundant species in the region in the green business line, and basic sanitation to alleviate problems such as the lack of drinking water, among others that are extremely important and directly impact people's lives.

In this way, BIOFIX and the Guajajara of the four territories under the governance of the indigenous association Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara aim to promote the improvement and well-being of the people, in accordance with the priorities listed in the workshops by Guajajara leaders. The project, which is in agreement with the leaders and the association, respects Guajajara cultural principles and is premised on good environmental and social management of the territories. In the assembly, lines of action were established that aim, on the one hand, to meet the most urgent needs of the communities and, on the other, to offer tools for preservation.

The project began in March 2019 when the indigenous communities living in the four proposed territories organized themselves internally to found the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara. It is from this starting point in the governance of the Guajajara indigenous peoples that it is possible to identify the beginnings of mobilization for their well-being and wellbeing, attracting other initiatives that share the common ideal of sustainable and social development.

Figure 1. Location of the KA'A PYAHU REDD+ project



Source: KA'A PYAHU REDD+ Project

With implementation of the activities for unplanned deforestation activities - AUDD, the KA'A PYAHU REDD+ project expects to reduce emissions by 200,000 tCO₂e per year.

1.2 Audit History

The project has not yet undergone any validation or verification process. However, the first validation period is expected to occur with the following dates:

Table 1. Audit history of the KA'A PYAHU REDD+ project

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	01-05-2021-30-04-2041	Verified Carbon Standard VCS		20

Source: KA'A PYAHU REDD+ Project

1.3 Sectoral Scope and Project Type

Table 2. Sectoral scope and type of the KA'A PYAHU REDD+ Project

Sectoral scope	Agriculture, Forestry, and Other Land Use
AFOLU project category¹	Reduction of emissions from deforestation and degradation (REDD+)
Project activity type	Avoiding unplanned Deforestation (AUDD)

Source: KA'A PYAHU REDD+ Project

1.4 Project Eligibility

1.4.1 General eligibility

The KA'A PYAHU REDD+ project is eligible under the scope of the VCS program for responding to REDD+ activities, under an independent project scheme for the validation and verification of GHG emission reductions in accordance with ISO 14064-2:2019 and ISO 14064-3: 2019. According with section 2.1 of the VCS Standard (V 4.7), the activities of the project are not excluded.

1.4.2 AFOLU project eligibility

According to the VCS Standard V 4.7, eligible REDD activities are those that reduce net GHG emissions by reducing deforestation and/or degradation of forests, which is the case of the KA'A PYAHU REDD+ project. Deforestation is the direct, human-induced conversion of forestland to non-forest land.

The analysis of land use categories in the project is based mainly on Mapbiomas² which utilizes a satellite imagery dataset used composed by Landsat 5 (Thematic Mapper), Landsat 7 (Enhanced Thematic Mapper Plus) and Landsat 8 (Operational Land Imager), Landsat sensors. The MapBiomas classification scheme is a hierarchical system with a combination of LULC classes compatible with the Food and Agriculture Organization (FAO) and IBGE classification systems (Souza, et al., 2020). In this regard, the project meets the internationally accepted definition of forest, and was qualified as forest for a minimum of 10 years before the project start date. The definition of forest may include mature forests, secondary forests, and degraded forests.

According with the VCS standard, activities covered under the REDD project category are those that are designed to stop planned (designated and sanctioned) deforestation or unplanned (unsanctioned) deforestation and/or degradation. In this context, as described in section 1.9, the actions carried out in the KA'A PYAHU REDD+ project, with the creation of the association, are oriented towards the defense, preservation and conservation of the environment. Specifically, by taking action to combat the environmental and social impacts caused by the construction of the power line. This project activity stops unplanned deforestation and reduces GHG emissions by stopping deforestation of forests.

1.4.3 Transfer project eligibility

The project has not been registered under another GHG program.

1.5 Project Design

² <https://brasil.mapbiomas.org/atbd---entenda-cada-etapa/>

- ☐ Single location or installation
- ☒ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

Grouped Project Design

Not applicable as it is not a grouped project.

1.6 Project Proponent

The project proponents are the **indigenous association Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara**, which is responsible for the governance of the Guajajara people in the four territories and indigenous domains: Cana Brava Indigenous Land, Urucu-Juruá Indigenous Land, Rodeador Indigenous Domainial and Lagoa Comprida Indigenous Land, on which emission reduction actions are carried out. The Guajajara people have rights over land use and carbon rights. **Biofix Consultoria Ltda**, as an allied company that carries out the structuring of the project and manages the commercialization of VCU.

Table 3. Contact information

Organization name	Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara
Contact person	José Luciano Clemente Guajajara
Title	President
Address	Coquinho Village Area, S/N Countryside JENIPAPO DOS VIEIRAS - MA
Telephone	+ 55 99 98524-5222
Email	conselhosupremodecaciques@gmail.com

Organization name	BIOFIX CONSULTORIA LTDA.
Contact person	Cristiane Vancine Leme
Title	Director of operations Biofix Brasil
Address	Rua Cerqueira Cesar, Avenida Paulista 1636 CJ.4 PAV.15 SAO PAULO-SP
Telephone	+55 11 99522-0749
Email	crisleme@biofix.com.br

Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, under its authority, appoints the company **BIOFIX CONSULTORIA LTDA** to prepare the project document, through a temporary association; with which it exclusively grants the rights to issue and sell the emission reduction certificates it generates, without prejudice to the fact that carbon rights will always belong to the indigenous council.

1.7 Other Entities Involved in the Project

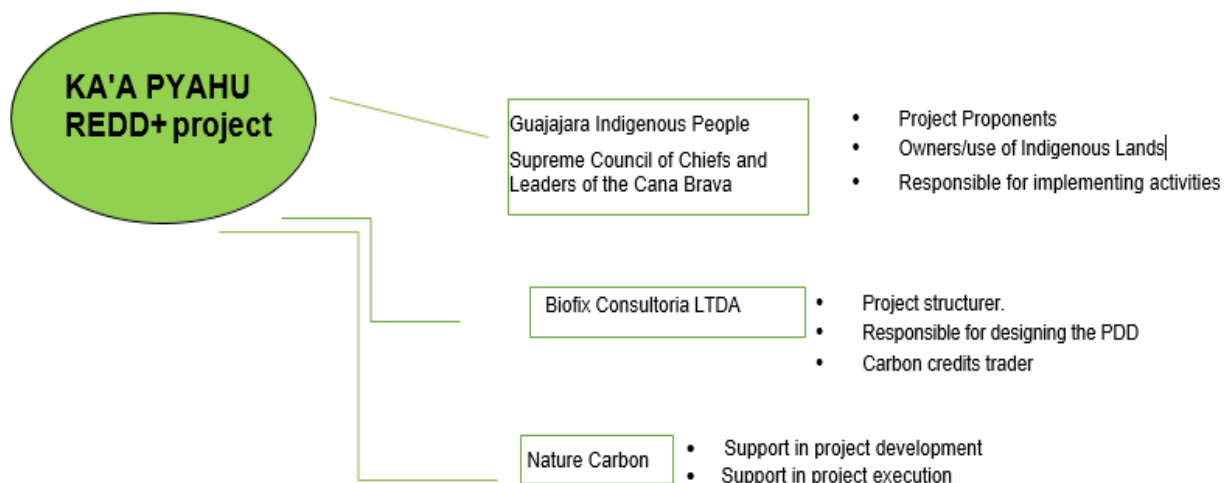
The KA'A PYAHU REDD+ Project has an external participant, the company Nature Carbon, a partner of the Guajajara indigenous association, responsible for collaborating with the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara and with Biofix in the execution and development of the project.

Table 4. Other internal entities involved in the project

Organization name	Nature Carbon LTDA
Role in the project	Collaborate with the Association and Biofix to execute and develop the Project. Collect the requested documentation. Dialogue, interact and maintain a good relationship with ethnic authorities and administrative communities in the region. Witness and monitor socialization activities, socio-environmental workshops and Project actions in Indigenous communities. Inform about situations that impact the Project. When necessary, send the Minutes of recognition or possession of traditional authorities.
Contact person	Daniel Cavalcante
Title	Legal Representative
Address	Rua Urbano Santos, 155, room 402, Empress –MA.
Telephone	+55 99 8444-8181
Email	danielcavalcant.adv@gmail.com

Source: KA'A PYAHU REDD+ Project

Figure 2. Project organization chart



Source: KA'A PYAHU REDD+ Project

1.8 Ownership

The usufruct of the territories where the KA'A PYAHU Project takes place belongs to the communities belonging to the Guajajara ethnic group, which are legally represented by the indigenous association Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara (Annex 011). The Cana Brava Indigenous Land, approved by DECREE No. 246, OF OCTOBER 29, 1991 (Annex 001); Urucu-Juruá Indigenous Land, approved by DECREE No. 382, OF DECEMBER 24, 1991 (Annexes 002), there was a rectification of the geographic coordinates of Decree No. 382, of December 24, 1991, which approved the administrative demarcation of the Indigenous Land Urucu/Juruá on FEBRUARY 10, 2003 (Annex 003); Lagoa Comprida Indigenous Land, DECREE OF FEBRUARY 10, 2003 (Annex 004) and Rodeador Indigenous Domain Area approved by DECREE N° 88,813, OF OCTOBER 4, 1983 (Annex 005), are under the possession of the people proposing the project, therefore owners of the carbon credits and of tradable carbon credits.

The Guajajara indigenous people who are part of the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, making use not only of their rights of autonomy and self-determination, but also of the legal capacity that has been legally granted to them, which allows them to enter into contracts under the terms of civil and commercial law, through the signing of the Partnership and Cooperation Agreement (Annex 012), granted to BIOFIX CONSULTORIA LTDA, as the project's structuring entity, the express power to generate the issuance and commercialization of carbon credits (certified reductions in emissions GHG emissions) originating from the territories in which the Community Council exercises its jurisdiction.

The representation of the Conselho Supremo dos Caciques e Lideranças da Terra Indígena Cana Brava covers was built through the autonomy of the Guajajara people representing the four territories, as mentioned in the association's statute. In this sense, the association is able to represent the regions according to its basic representation capacity (Annexes 006a, 006b, 006c, 006d, 006e, 006f, 007, 008, 009, 026, 027, 030, 031, 032, 033, 034, 035, 036 and 037). Based on all the documents that support the Conselho Supremo dos Caciques e Lideranças da Terra Indígena Cana Brava and the consent and

signature of the 280 villages below, the project achieved consent through the centrality of this association.

The communities proposing the KA'A PYAHU REDD+ project are legally organized in a representative manner before the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara:

Table 5. Villages in the indigenous land of the project area

Nº	Villages	Indigenous Land
1.	Piçarreira	Cana Brava – BR 226
2.	Cajazeira BR	Cana Brava – BR 226
3.	Maria Joana BR	Cana Brava – BR 226
4.	Betânia BR	Cana Brava – BR 226
5.	Jardim BR	Cana Brava – BR 226
6.	Taboca BR	Cana Brava – BR 226
7.	Chuveiro BR	Cana Brava – BR 226
8.	Cia Norte	Cana Brava – BR 226
9.	Cabeça da Onça	Cana Brava – BR 226
10.	Nova Tamarinda BR	Cana Brava – BR 226
11.	Pentecoste	Cana Brava – BR 226
12.	Nova Angico	Cana Brava – BR 226
13.	Nova Maria	Cana Brava – BR 226
14.	Orizonte	Cana Brava – BR 226
15.	Capim BR	Cana Brava – BR 226
16.	Maranata BR	Cana Brava – BR 226
17.	Cajueira BR	Cana Brava – BR 226
18.	Coquinho	Cana Brava – BR 226
19.	Ywaté	Cana Brava – BR 226
20.	Sítio BR	Cana Brava – BR 226
21.	Jenipapinho BR	Cana Brava – BR 226
22.	Nova Cafarnaum BR	Cana Brava – BR 226
23.	Nova Moisés BR	Cana Brava – BR 226
24.	Canaã	Cana Brava – BR 226
25.	Myramá BR	Cana Brava – BR 226
26.	Mussum BR	Cana Brava – BR 226
27.	Bitá Moreira BR	Cana Brava – BR 226
28.	Jatobar	Cana Brava – BR 226
29.	Bueira	Cana Brava – BR 226
30.	Sumauma II BR	Cana Brava – BR 226
31.	Nova Carnaúma	Cana Brava – BR 226
32.	Nova Belém BR	Cana Brava – BR 226
33.	Zahytata	Cana Brava – BR 226

34.	Nova Barreirinha BR	Cana Brava – BR 226
35.	Nova Missão	Cana Brava – BR 226
36.	Monte Moriá	Cana Brava – BR 226
37.	Eliza BR	Cana Brava – BR 226
38.	Nova Jerusalém BR	Cana Brava – BR 226
39.	Eliza BR	Cana Brava – BR 226
40.	Sobradinho BR	Cana Brava – BR 226
41.	Michael	Cana Brava – BR 226
42.	Jerusalém	Cana Brava – BR 226
43.	Vila Nova III BR	Cana Brava – BR 226
44.	Albanio BR	Cana Brava – BR 226
45.	Sapucaí	Cana Brava – BR 226
46.	Jurema II BR	Cana Brava – BR 226
47.	Araçar BR	Cana Brava – BR 226
48.	EL-Betel BR	Cana Brava – BR 226
49.	Mykaza	Cana Brava – BR 226
50.	Silvino BR	Cana Brava – BR 226
51.	Igarana	Cana Brava – BR 226
52.	Nova Egito BR	Cana Brava – BR 226
53.	Nova Esperança BR	Cana Brava – BR 226
54.	Boa Vista BR	Cana Brava – BR 226
55.	Jericó BR	Cana Brava – BR 226
56.	Fazendinha BR	Cana Brava – BR 226
57.	Zaryr Hu	Cana Brava – BR 226
58.	Orizonte BR	Cana Brava – BR 226
59.	Ywyzar	Cana Brava – BR 226
60.	Kitara BR	Cana Brava – BR 226
61.	Barreirinha BR	Cana Brava – BR 226
62.	Horizonte	Cana Brava – BR 226
63.	Mangueirinha BR	Cana Brava – BR 226
64.	Kurumiwahy	Cana Brava – BR 226
65.	Naiva	Cana Brava – BR 226
66.	Macaúba	Cana Brava – BR 226
67.	Maraca Ypú	Cana Brava – BR 226
68.	Ilha Branca	Cana Brava – BR 226
69.	Imbaúba	Cana Brava – BR 226
70.	Pantanal BR	Cana Brava – BR 226
71.	Canoeiro	Cana Brava – BR 226
72.	Nova Belém II	Cana Brava – BR 226
73.	Cajazeiro II	Cana Brava- Braço da Cachoeira Comprida Mearim
74.	Cachoeira Comprida	Cana Brava- Braço da Cachoeira Comprida Mearim

75.	Alfredo Neto	Cana Brava- Braço da Cachoeira Comprida Mearim
76.	Ywyting	Cana Brava- Braço da Cachoeira Comprida Mearim
77.	Ywyrapong	Cana Brava- Braço da Cachoeira Comprida Mearim
78.	Bananal	Cana Brava- Braço da Cachoeira Comprida Mearim
79.	Ventura	Cana Brava- Braço da Cachoeira Comprida Mearim
80.	Coroatá	Cana Brava- Braço da Cachoeira Comprida Mearim
81.	Elomar Ribeiro	Cana Brava- Braço da Cachoeira Comprida Mearim
82.	Boa Esperança	Cana Brava- Braço da Cachoeira Comprida Mearim
83.	Portu Velho	Cana Brava- Braço da Cachoeira Comprida Mearim
84.	Guegara	Cana Brava- Braço da Cachoeira Comprida Mearim
85.	Wyrapog	Cana Brava- Braço da Cachoeira Comprida Mearim
86.	Celina	Cana Brava- Braço da Cachoeira Comprida Mearim
87.	Galileia	Cana Brava- Braço da Cachoeira Comprida Mearim
88.	Nececir	Cana Brava- Braço da Cachoeira Comprida Mearim
89.	Igarapé do Mearim	Cana Brava- Braço da Cachoeira Comprida Mearim
90.	Maravilha Mearim	Cana Brava- Braço da Cachoeira Comprida Mearim
91.	Tukura'i	Cana Brava- Braço da Cachoeira Comprida Mearim
92.	Beira Rio	Cana Brava – Rio Mearim
93.	Beira Rio II	Cana Brava – Rio Mearim
94.	Arroz	Cana Brava – Rio Mearim
95.	Valdemar Viana	Cana Brava – Rio Mearim
96.	Remanso	Cana Brava – Rio Mearim
97.	Canoeira	Cana Brava – Rio Mearim
98.	Chupé	Cana Brava – Rio Mearim
99.	Tawewa	Cana Brava – Rio Mearim
100.	Miliana	Cana Brava – Rio Mearim
101.	Aréia	Cana Brava – Rio Mearim
102.	Belém II	Cana Brava – Rio Mearim
103.	Haká Peahu	Cana Brava – Rio Mearim
104.	Colonia	Cana Brava – Rio Mearim
105.	Nova Copaíba	Cana Brava – Rio Mearim
106.	Copaíba	Cana Brava – Rio Mearim
107.	Aliança	Cana Brava – Rio Mearim
108.	Nova Aliança	Cana Brava – Rio Mearim
109.	Jatobá	Cana Brava – Rio Mearim
110.	Brejinho	Cana Brava – Rio Mearim
111.	Ourives	Cana Brava – Rio Mearim
112.	Cajazeira	Cana Brava – Rio Mearim
113.	Pé de Galinha	Cana Brava – Rio Mearim
114.	Santa Rosa	Cana Brava – Rio Mearim
115.	Sapucaio	Cana Brava – Rio Mearim

116.	Cajueira	Cana Brava – Rio Mearim
117.	Jatobazinho	Cana Brava – Rio Mearim
118.	São Mateus	Cana Brava – Rio Mearim
119.	Santa Nova	Cana Brava – Rio Mearim
120.	Altamira I	Cana Brava – Rio Mearim
121.	Lizete	Cana Brava – Rio Mearim
122.	João Pereira	Cana Brava – Rio Mearim
123.	Altamira II	Cana Brava – Rio Mearim
124.	Zahy I	Cana Brava – Rio Mearim
125.	Miliã	Cana Brava – Rio Mearim
126.	Haka Pyahu	Cana Brava – Rio Mearim
127.	Celino	Cana Brava – Rio Mearim
128.	Tukuman	Cana Brava – Rio Mearim
129.	Turyweteahaw	Cana Brava – Rio Mearim
130.	Sítio Novo	Cana Brava – Rio Mearim
131.	Terra Boa	Cana Brava – Rio Mearim
132.	Zé Luiz	Cana Brava – Rio Mearim
133.	Montanha	Cana Brava – Ilha de São Pedro e Cacimba
134.	Raimundão	Cana Brava – Ilha de São Pedro e Cacimba
135.	Moacir	Cana Brava – Ilha de São Pedro e Cacimba
136.	Canafístula	Cana Brava – Ilha de São Pedro e Cacimba
137.	Josiel	Cana Brava – Ilha de São Pedro e Cacimba
138.	Crioli II	Cana Brava – Ilha de São Pedro e Cacimba
139.	Capim Duro	Cana Brava – Ilha de São Pedro e Cacimba
140.	Maria Elena	Cana Brava – Ilha de São Pedro e Cacimba
141.	Norinda	Cana Brava – Ilha de São Pedro e Cacimba
142.	Francisco Artur	Cana Brava – Ilha de São Pedro e Cacimba
143.	Mangueiro	Cana Brava – Ilha de São Pedro e Cacimba
144.	Ilha de São Pedro	Cana Brava – Ilha de São Pedro e Cacimba
145.	Jorge L.	Cana Brava – Ilha de São Pedro e Cacimba
146.	Cacimba Velho	Cana Brava – Ilha de São Pedro e Cacimba
147.	Crioli I	Cana Brava – Ilha de São Pedro e Cacimba
148.	Egito	Cana Brava – Ilha de São Pedro e Cacimba
149.	Alexandre Tomaz	Cana Brava – Ilha de São Pedro e Cacimba
150.	Cocalin Pyahu	Cana Brava – MA 328
151.	Uchoa	Cana Brava – MA 328
152.	Nativa	Cana Brava – MA 328
153.	Firmino	Cana Brava – MA 328
154.	Ribeiro Guajajara	Cana Brava – MA 328
155.	Cana Brava I	Cana Brava – MA 328
156.	Raimundo Filho	Cana Brava – MA 328

157.	Progresso	Cana Brava – MA 328
158.	Karu	Cana Brava – MA 328
159.	São João	Cana Brava – MA 328
160.	Taboqui Martins	Cana Brava – MA 328
161.	Alvorada	Cana Brava – MA 328
162.	Taquari	Cana Brava – MA 328
163.	Santa Fé	Cana Brava – MA 328
164.	Yrykaw	Cana Brava – MA 328
165.	Tamburi	Cana Brava – MA 328
166.	Parari	Cana Brava – MA 328
167.	Cajueiro Real	Cana Brava – MA 328
168.	Colina	Cana Brava – MA 328
169.	Mardônio	Cana Brava – MA 328
170.	Joa	Cana Brava – MA 328
171.	Manoel Braga	Cana Brava – MA 328
172.	Daniel	Cana Brava – MA 328
173.	Porto Alegre	Cana Brava – MA 328
174.	Vila Nova	Cana Brava – MA 328
175.	Ana Rosa	Cana Brava – MA 328
176.	Criulzinho	Cana Brava – MA 328
177.	Tabokim	Cana Brava – MA 328
178.	Porco	Cana Brava – MA 328
179.	Dalto e Dantas	Cana Brava – MA 328
180.	Capinzal	Cana Brava – MA 328
181.	Da Paz	Cana Brava – MA 328
182.	Cocalinho	Cana Brava – MA 328
183.	Planalto	Cana Brava – Rio Corda
184.	Micaele	Cana Brava – Rio Corda
185.	Travessia I	Cana Brava – Rio Corda
186.	Estrela	Cana Brava – Rio Corda
187.	Travessia II	Cana Brava – Rio Corda
188.	Pantanal	Cana Brava – Rio Corda
189.	Yaweté	Cana Brava – Rio Corda
190.	Capim	Cana Brava – Rio Corda
191.	Tamburi II	Cana Brava – Rio Corda
192.	Sardinha Nova	Cana Brava – Rio Corda
193.	Fazendinha	Cana Brava – Rio Corda
194.	Cumarú	Cana Brava – Rio Corda
195.	Abrão	Cana Brava – Rio Corda
196.	Planície	Cana Brava – Rio Corda
197.	Tamburi I	Cana Brava – Rio Corda

198.	Fazendinha II	Cana Brava – Rio Corda
199.	Jitiarana	Cana Brava – Rio Corda
200.	Geremias	Cana Brava – Rio Corda
201.	Baixão do Peixe	Cana Brava – Rio Corda
202.	Carlos Augusto	Cana Brava – Rio Corda
203.	Cachoeira do Meio	Cana Brava – Rio Corda
204.	Itahu	Cana Brava – Rio Corda
205.	Vila Nova II	Cana Brava – Rio Corda
206.	Cruzeiro	Cana Brava – Rio Corda
207.	Peetro	Cana Brava – Rio Corda
208.	Cajueiro do Rio Corda	Cana Brava – Rio Corda
209.	Sardinha	Cana Brava – Rio Corda
210.	Jenipapo	Cana Brava – Rio Corda
211.	Kariné	Cana Brava – Rio Corda
212.	Anastaci	Cana Brava – Rio Corda
213.	Inazá	Cana Brava – Rio Corda
214.	Sapuão II	Cana Brava – Rio Corda
215.	Tawmirim	Cana Brava – Rio Corda
216.	Tajara	Cana Brava – Rio Corda
217.	Mirion	Cana Brava – Rio Corda
218.	Eucalipto	Cana Brava – Rio Corda
219.	Kawan	Cana Brava – Rio Corda
220.	Augusto	Cana Brava – Rio Corda
221.	Talhada	Cana Brava – Rio Corda
222.	Macaúba	Cana Brava – Rio Corda
223.	Mundial	Cana Brava – Rio Corda
224.	Caxias	Cana Brava – Rio Corda
225.	Mangueira	Cana Brava – Rio Corda
226.	Mayra	Cana Brava – Rio Corda
227.	Maranato	Cana Brava – Rio Corda
228.	Aroeira	Cana Brava – Rio Corda
229.	Tabajara	Cana Brava – Rio Corda
230.	Yonar	Cana Brava – Rio Corda
231.	Vituriano	Cana Brava – Rio Corda
232.	Taiwa	Cana Brava – Rio Corda
233.	Nova Paraíso	Cana Brava – Rio Corda
234.	Formosa	Cana Brava – Rio Corda
235.	Bacuriu	Cana Brava – Rio Corda
236.	Bom Jesus	Cana Brava – Rio Corda
237.	Irailda	Cana Brava – Rio Corda
238.	Mukazá	Cana Brava – Rio Corda

239.	Zahy	Cana Brava – Rio Corda
240.	Nuzan	Cana Brava – Rio Corda
241.	Patizau	Rodeador
242.	Canafistula I	Rodeador
243.	Rodeador I	Rodeador
244.	Canafistula II	Rodeador
245.	Chapadinha	Rodeador
246.	Taboca	Rodeador
247.	Ourive	Rodeador
248.	Curió	Rodeador
249.	Rodeador II	Rodeador
250.	Boa Sorte	Rodeador
251.	Flor	Rodeador
252.	Genivaldo	Lagoa Comprida
253.	Lagoa do Leite	Lagoa Comprida
254.	Bom Jesus	Lagoa Comprida
255.	Maria Rodrigues	Lagoa Comprida
256.	Antônio Alfredo	Lagoa Comprida
257.	Bernardino	Lagoa Comprida
258.	Nonato	Lagoa Comprida
259.	Tarquínio	Lagoa Comprida
260.	Chico Naru	Lagoa Comprida
261.	Cocal	Lagoa Comprida
262.	Tamarindo	Lagoa Comprida
263.	Descendência	Lagoa Comprida
264.	Matias	Lagoa Comprida
265.	Mussun	Lagoa Comprida
266.	Jaborandi	Lagoa Comprida
267.	Wyraity	Lagoa Comprida
268.	Lagoa Comprida I	Lagoa Comprida
269.	Lagoa Comprida	Lagoa Comprida
270.	Artuzinho	Lagoa Comprida
271.	Babaçu	Urucu Juruá
272.	Araruna	Urucu Juruá
273.	PÁRÁÑÁ I	Urucu Juruá
274.	Pr'nhza	Urucu Juruá
275.	Divisão	Urucu Juruá
276.	Urucu'í II	Urucu Juruá
277.	Márákáná	Urucu Juruá
278.	Urucu'i	Urucu Juruá
279.	Milagres	Urucu Juruá

280.	Viliano	Urucu Juruá
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Source: KA'A PYAHU REDD+ Project

According to updated data from March 2024 provided by SESAI (Special Secretariat for Indigenous Health), the Cana Brava Indigenous Land has an indigenous population of 11.528 people, has a health network of a DSEI (Special Indigenous Health District) and two base centers of health. The Urucu Juruá Indigenous Land in total covers a population of 994 people, with a DSEI and a Health Hub in its network. The Lagoa Comprida Indigenous Land in total covers a population of 727 people, with a DSEI and a Base Health Center, And Dominial Rodeador, 752 people, a base hub and a DSEI. The KA'A PYAHU REDD+ project benefits a total of 14.001 Guajajara indigenous people.

In turn, the indigenous association expressly stated that it did not hire another natural or legal person before signing the contract and committed itself not to hire another natural or legal person for the same purposes. Consequently, BIOFIX CONSULTORIA LTDA, as another project proponent, will be responsible for carrying out the necessary activities to achieve validation, verification, certification and commercialization of certified emission reduction units before the competent entities.

1.9 Project Start Date

The KA'A PYAHU REDD+ project begins on March 20, 2019, with the opening and founding of the indigenous association the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara (Annex 011). This activity is aligned with the governance line and is responsible for the legal representation of the Guajajara peoples of the four territories to access public policies, compensation, mitigations and the fight for the people's rights through legal means.

- Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara.

The association called Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara was founded on March 20, 2019 in Aldeia Coquinho, Cana Brava Indigenous Land, in the municipality of Jenipapo dos Vieiras in the state of Maranhão, northeast of Brazil. It is a non-profit civil association of indefinite duration that was founded to work with the Guajajara communities mentioned above and legally represent them in accordance with Brazilian legislation.

The objective of the association is, according to its statute, to develop actions to defend and contribute to improving the quality of life of the indigenous communities that are represented by the council, through: promotion of social assistance; promotion of culture and defense of historical and artistic communities. Defense, preservation and conservation of the environment, and promotion of sustainable development. Promotion of economic and social development and combating poverty in communities. Promotion of ethics and peace, respecting citizenship and human rights, in addition to democracy and others values present in the communities belonging to the Council.

To be able to carry out such actions, the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara is organized for its deliberations as follows: general assembly, deliberative council, board of directors, fiscal council, president and secretary. The form of internal organization occurs as follows, according to the statute: president, vice-president, secretary and treasurer (Annex 011). The deliberative council is made up of the president, treasurer, secretary and deputy secretary. The association has its own forms of application and intervention in accordance with collectively agreed principles.

The Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, which represents the 267 villages that make up the KA'A PYAHU REDD+ project, has as one of its significant achievements, which is fundamental to this project, a prior consultation meeting with the communities of the Cana Brava, Lagoa Comprida, Urucu Juruá and Rodeador indigenous communities on February 11, 2022 at their headquarters in the Coquinho village, in the municipality of Jenipapo dos Vieiras, with the

purpose of formalizing the communities' interest in the structure, certify and commercialization of carbon credits project through an intermediation contract for these purposes (Annex 013 and 017).

The Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, in accordance with the representation objectives mentioned above, promotes actions with the Brazilian state and competent bodies in order to ensure the right to health (Annexes 014 and 016) and the practice of citizenship and access to the registration documents necessary to access rights (Annex 015).

Furthermore, the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara has been fighting since its creation against the impacts in the environmental and social areas suffered by the construction of the energy line along the BR 226 highway in the Cana Brava Indigenous Land resulting from the Tucuruí Hydroelectric Plant, in the state of Pará, This is the first large-scale plant in the Amazon region, inaugurated at the end of 1984, and the transmission line towers for the interconnection of the North/Northeast System that runs through the Guajajara villages and causes social and environmental harm in the region (Ribeiro Filho & de Campos, 2023). (Annexes 018, 018a, 018b, 018c, 018d, 018e, 018f, 018g, 018h, 018i and 018j).

Therefore, the activities carried out by the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara are aligned with the objective of the REDD+ Project, because: 1. It is carried out in the project area, 2. With people from the four Territories and Indigenous Domain, 3. Under the leadership of the Guajajara people in order to directly defend and ensure their rights by exercising indigenous citizenship under Brazilian legislation.

Complying with the start date requirements set out in the VCS standard for AFOLU projects, about the pipeline listing process (3.8.2 section), the start date of the project is set as follows:

Project start date

01-may-2021

Source: KA'A PYAHU REDD+ Project

1.10 Project Crediting Period

Table 6. KA'A PYAHUU Project Crediting Period

Crediting period	20 years. According to the guidelines of the VCS Standard V4.7 the credit period of the project is 20 years.
Start and end date of first or fixed crediting period	01-May-2021 to 30-April-2041

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Table 7. Estimated emissions reductions of the KA'A PYAHUU Project

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-May-2021 to 31-December-2021	199,786

01-January-2022 to 31-December-2022	297,640
01-January-2023 to 31-December-2023	297,640
01-January-2024 to 31-December-2024	297,640
01-January-2025 to 31-December-2025	297,640
01-January-2026 to 31-December-2026	297,640
01-January-2027 to 31-December-2027	297,640
01-January-2028 to 31-December-2028	297,640
01-January-2029 to 31-December-2029	297,640
01-January-2030 to 31-December-2030	297,640
01-January-2031 to 31-December-2031	297,640
01-January-2032 to 31-December-2032	297,640
01-January-2033 to 31-December-2033	297,640
01-January-2034 to 31-December-2034	297,640
01-January-2035 to 31-December-2035	297,640
01-January-2036 to 31-December-2036	297,640
01-January-2037 to 31-December-2037	297,640
01-January-2038 to 31-December-2038	297,640
01-January-2039 to 31-December-2039	297,640
01-January-2040 to 31-December-2040	297,640
01-January-2041 to 31-April-2041	97,854
Total estimated ERRs during the first or fixed crediting period	5,952,805
Total number of years	20 years
Average annual ERRs	297,640

Source: KA'A PYAHU REDD+ Project

1.12 Description of the Project Activity

The forests in the project area are important areas for conservation. Although they are strategic ecosystems in the provision of environmental services, they are at risk of deforestation and invasion due to the exploitation of wood from local species, in addition to the risk of burning.

To promote the preservation of the four territories and domains and consequently support local cultures, actions must be generated that inject resources and means to protect the diversity of local fauna and

flora in addition to the sustainable use of natural resources together with the strategic management of ecosystems.

It is essential that the risks of forest fires caused by human action are observed throughout the development of the project through strong monitoring in partnership with local fire fighters, in this case, Prevfogo.

REDD+ principles require that anthropogenic activities of deforestation and forest degradation are monitored and combatted strategically with the Guajajara people. In this way, actions were designed and coordinated with communities to reduce the aforementioned threats to forests in the project area and mitigate the impacts associated with GHG emissions.

The KA'A PYAHU REDD+ Project aims to formulate and promote local solutions for the development of a sustainable, participatory project to ensure environmental sustainability and mitigate the threats posed by the main agents and drivers of deforestation and forest degradation.

Through the implementation of actions in the four REDD+ investment lines: 1. Governance, 2. Health/Education/Infrastructure, 3. Monitoring and Reforestation and 4. Green Business, the aim is to avoid threats to local species through the implementation of a project REDD+ Reduction of Emissions due to deforestation and Forest Degradation, conservation of carbon reserves, sustainable management of forests and improvement of forest reserves in developing countries), providing payments for compensation for ecosystem services provided by the Guajajara people in the three territories and dominion.

Thus, the collective territories linked to the project seek, through their territorial autonomy, to carry out conservation processes and consequent valorization of their cultural identities and to promote the protection of biodiversity and the environmental wealth of their territories. Characteristics directly linked to the purpose of REDD+ projects, the desired sustainable local development, as well as the purpose of reducing emissions generated by deforestation.

The Project Proponent is distributing the finances, providing support for social development activities and implementing the monitoring program for the activities that will oversee BIOFIX BRASIL. The project activities, with which it is intended to comply with the resource plan and the development of the proposed objective, can be seen in the following table.

Table 8. KA'A PYAHU REDD+ lines of action.

Activity	Objective	Sub-activities	Indicator	Measuring Period	Related Stakeholders
CROSS LINES OF ACTION					
Institutional governance strengthening and	Apply training, provision, and authority strengthening actions as well as improve their culture and traditions.	Training workshops for leaders and communities in REDD+.	Indicators Number of people trained in REDD+.	Annual	State and Federal Universities Community Council Municipal Authorities Communities
		Territorial Monitoring – Internal Management Plan for territories.	Monitoring Plan produced by the community.		
		Acquisition of equipment to strengthen associations.	Purchase of Equipment.		
		Construction of cultural houses.	Number of houses built.		
		Support the participation of regional and national indigenous mobilizations (cultural exchanges).	Number of people who attended the event and minutes of participation.		
		Support for the movement of leaders for internal coordination in the territories.			

Participatory restoration, recovery, and/or rehabilitation of deforested and/or degraded ecosystems.	Conduct reforestation and native vegetation recovery in the riparian zone and degraded soils for conflicts of use with participation of the local communities.	Irrigation structure-distribution of water for the growth of crops.	Indicators	Annual	Community Council Municipal Authorities IBAMA ICMBIO FUNAI
		Planting of native species: buriti, tucumã, native seeds that support local crafts.	Number of structures built in communities. Number of native species planted.		
		Construction of nurseries for native species.	Number of nurseries of native species planted. Number of trained agents.		
		Training of local agents and forest guardians.			
Monitoring and control plan of deforestation and forest degradation.	Monitor the changes due to deforestation in influence through satellite image processing and/or the ones taken from the project.	Territorial inspection.	Indicators	Monthly	BIOFIX will oversee the monitoring program through the joint work between expert trainers and keepers of the community.
		Preventing fires during critical periods.	Number of alerts reported per year.		
		Monitoring against harassment and threats of theft of internal natural resources.	Parameters for monitoring. Supports Satellite images Images captured by drones.		

			Images and videos captured in the field. Periodical monitoring reports		
Strengthening and encouragement of education	Encourage the enrollment in quality education and the completion of the study period by the students, guaranteeing access to higher education.	Construction, improvement and adaptation of the communities' school structure (Aldeia Flor reference). Construction of a computer laboratory Construction of sports courts for physical education and strengthening cultural sports activities (archery), Strengthening the mother tongue, training and internal training of education professionals in the mother tongue (support) Strengthening local culture in schools (crafts, traditional practices, indigenous	Indicators Observation of built and improved school structures. Number of students benefiting. Acquisition of specific equipment. Number of trained professionals. Number of workshops offered and the audience benefited.	Annual	Municipal Authorities Community Council Universities MEC

		school education – indigenous education) through workshops on traditional cultural activities with teachers and masters from the community), Support for indigenous students in the city to continue their education, Strengthening traditional festivals, such as: the girl's festival, the honey festival, the corn festival, the boys' festival, the cassava, potato, yam, pumpkin festival, and “Imitation”, moments of cultural transmission between generations through festivals and rituals.	Number of students benefiting. Number of festivals and rituals benefited through the REDD project.		
Prevention and health care program	Encourage health prevention and care of the community council settlers, guaranteeing a good quality of life.	Construction of hospital structure. Strengthening traditional medicine. Collection of waste and garbage that remains in the territory and generates disease	Indicators Evidence from the construction of health structures. Number of collections.	Annual	Municipal Authorities Community Council SESAI communes FUNAI

		Disease prevention through the planting of native species that are used in traditional medicine.	Number of medicinal species planted.		
		Support in the fight against pollution (collective effort).	Number of joint efforts carried out with funding from the project.		
		Training in traditional medicine, support – conversation circles (studies in the area of pharmaceuticals, biology, partnerships in the manufacture of traditional medicines.	Number of trained agents.		
		Purchase of vehicles for emergency/health trips.	Material evidence of benefits in this type of mobility.		
Green Business	Encouragement and support of income generation through sustainable activities.	Bee farming – beekeeping.	Number of trained creators and creations.	Yearly	Communities Community Council and leaders ICMBIO IBAMA
		Native seed crafts.	Number of sales of crafts through activities supported by the project.		
		Fish farming.			

		Native seedling bed and creation of native fruits, for example: mango, cashew nuts	Number of tanks and capabilities. Number of captives of native species.		
Infrastructure	Works and structures that facilitate and support local ways of life.	Drinking water treatment.	Number of beneficiaries and diseases avoided.	Yearly	Community Council Leadership FUNAI
		Installation of artesian well.	Number of wells installed in communities.		
		Creation of cooperatives to strengthen internal productive activities	Formalization and number of initiatives created.		
		Construction of a flour house.			
		Construction of houses/housing/housing structure.	Number of houses built.		
		Road construction.	Structures built and improvements generated.		

Source: KA'A PYAHU REDD+ Project

1.13 Project Location

The KA'A PYAHU REDD+ project is located in the state of Maranhão, in the northeast region of Brazil, under the jurisdiction of the Legal Amazon. Specifically, the project is located on indigenous lands of the Guajajara peoples: Cana Brava Indigenous Land, Urucu-Juruá Indigenous Land, Rodeador Indigenous Dominion Area and Lagoa Comprida Indigenous Land (figure 1). The total area of the indigenous land (declared area) is 165,544.30 hectares, (166,228.99 area calculated using GIS software based on the coordinate system WGS_1984_UTM Zone 22S).

Table 9. Geometric centroid coordinates of the Project Area

Collective Territory	Extension (Ha) ³	Coordinates	
		Latitude	Longitude
Cana Brava/Guajajara	137.329,54	5° 37'6.02900"S	45° 32'43.01100 W
Lagoa Comprida	13.198,27	5° 19'28.17800"S	45° 44'5.278000 W
Rodeador	2.319,45	5° 42'12.19300"S	45° 13'42.49000 W
Urucu/Juruá	12.697,04	5° 22'30.76700"S	45° 55'0.432000 W

Source: SII FUNAI⁴

The indigenous territories are located in the state of Maranhão, in the municipalities of Barra do Corda, Grajaú, Itaipava do Grajaú and Jenipapo dos Vieiras. Of the indigenous territories in the project area, approximately 50% are located in the Jenipapo dos Vieiras municipality. Secondly, 36.5% of the territories are located in the municipality of Barra do Corda. Finally, the remaining percentage is distributed in the municipalities of Grajaú and Itaipava do Grajaú.

Table 10. Municipalities in the project area

Municipality	Area (km ²) ⁵	Percentage of Municipal Area in the Indigenous Lands (%)
Barra do corda	5,187.67	36.48
Grajaú	8,861.72	6.73
Itaipava do Grajaú	1,244.10	7.60
Jenipapo dos Vieiras	1,962.36	49.19

³ Declared area

⁴ http://sii.funai.gov.br/funai_sii/informacoes_indigenas/visao/visao_terras_indigenas.wsp

⁵ <https://cidades.ibge.gov.br/>

1.14 Conditions Prior to Project Initiation

1.14.1 Social parameters

1.14.1.1 Historic Contact-Guajajara

The KA'A PYAHU project takes place in four territories and one indigenous domain, all with the Guajajara indigenous population in the northeast region of Brazil, state of Maranhão. The Guajajara began contact history around 1615 with French explorers. The self-styled Tenetehára people who were found along the Pindaré River went through periods of subjugation through Jesuit missions with a slavery bias around the 17th and 18th centuries. After the expulsion of the Jesuit colonies, the Guajajara freed themselves from the slavery system, but from the 19th century onwards, forced integration occurred through regional patronage systems (Schröder, 2021).

In the current Cana Brava Indigenous Land, which is part of the KA'A PYAHU REDD+ project, there was the biggest revolt ever known by the Guajajara at the end of the 19th century, where the chief Cauré Imana united a large number of villages and expelled the non-indigenous people in the current regions of Barra do Corda and Grajaú, the municipalities where the territories that are part of the project are located (Consiglio, 1997). This revolt was called the “Alto Alegre” revolt and is a historical milestone for the Guajajara, it ended with the invasion of the army and military policy that persecuted the Guajajara for years.

Since the last decades of the 20th century until then, the Guajajara people have suffered due to territorial invasions in which squatters threaten the security of the people within their lands. The threat of large projects such as the Grande Carajás Program and groups such as loggers also threaten their lands and the Ka'a Pyahu project through monitoring, reforestation and infrastructure activities is fundamental to supporting the security of the Guajajara people (Soares, 1983).

In addition to their history of contact with non-indigenous people, the Guajajara also had moments of hostility, such as with the Canela, Urubu-ka'apor and Guajá peoples.

The Guajajara, project proponents, have the right to possess and enjoy the territories that are registered in the KA'A PYAHU REDD+ project. They are: Cana Brava Indigenous Land, approved and registered in accordance with Decree N° 246, OF OCTOBER 29, 1991, the Lagoa Comprida Indigenous Land, approved and registered in accordance with Decree N° 313, OF OCTOBER 29, 1991, Dominial Rodeador, approved and registered in accordance with Decree N° 88,813, OF OCTOBER 4, 1983, and the Urucu Juruá Indigenous Land, approved and registered in accordance with Decree N° 382, OF DECEMBER 24, 1991. The four territories of the Guajajara Indigenous peoples are represented by the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara. The Guajajara people have usufruct and possession, the right to use local land and biodiversity, and the right to the carbon contained in these territories.

1.14.1.2 Population

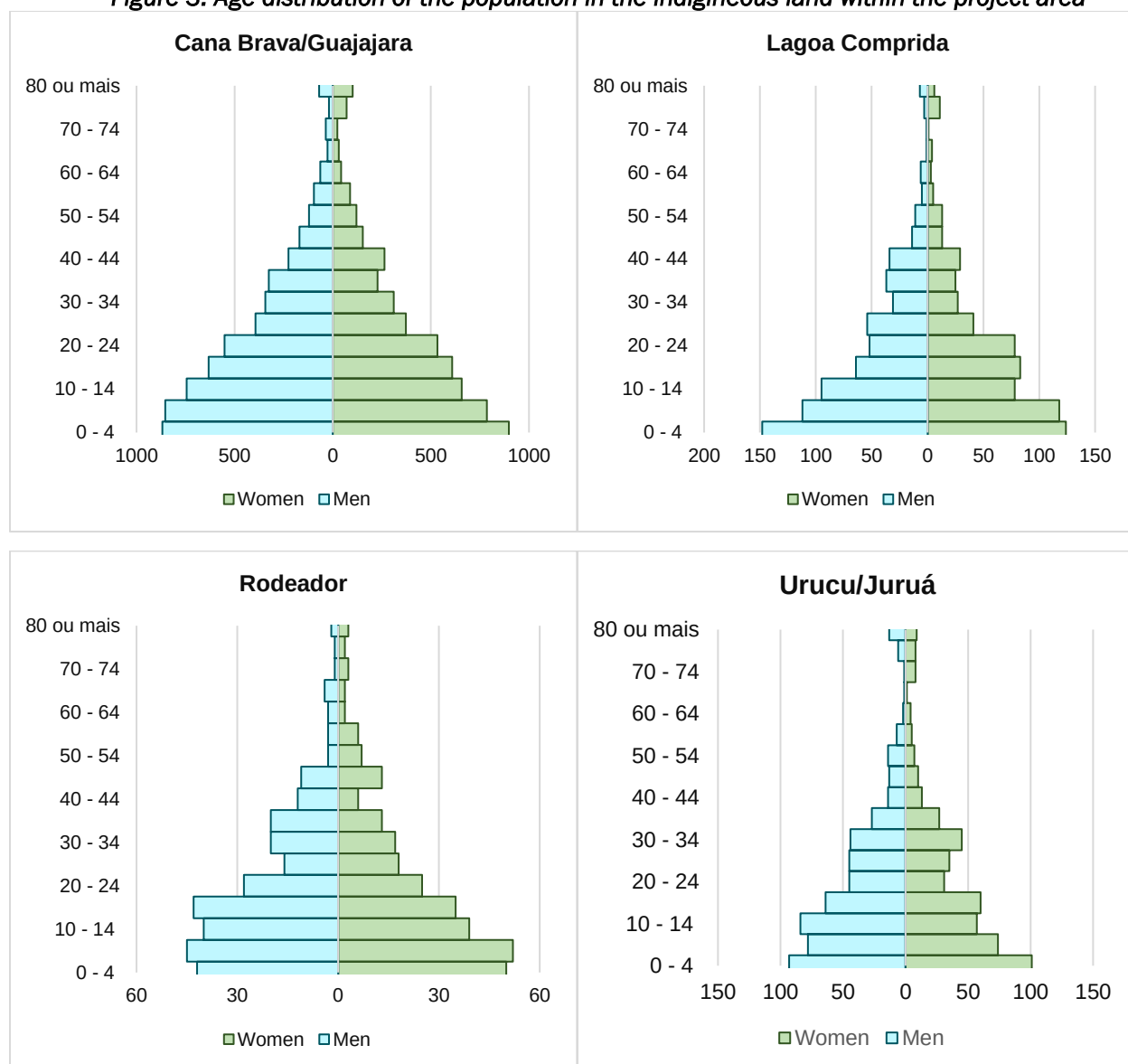
According to the last demographic census carried out in 2022 (IBGE, 2022), the total indigenous population residing in the indigenous territories that are part of the KA'A PYAHU Project are 13,791 people, 7,057 men and 6,734 women. The indigenous territory with the highest number of residents is the Cana Brava/Guajajara territory, with 10,824. For its part, the Rodeador indigenous territory is the territory with the fewest inhabitants, with a total of 587 residents in 2022. Regarding the sex ratio, for the 4 indigenous territories, 51.1% correspond to men and 48.9% to women respectively.

Table 11. People residing in the indigenous territories of the 2022 project area

Indigenous land	Population Total	Men	Women
Cana Brava/ Guajajara	10.824	5.537	5.287
Lagoa Comprida	1.334	675	659
Rodeador	587	294	293
Urucu/Juruá	1.046	551	495
Total	13.791	7.057	6.734

Source: 2022 IBGE Demographic Census, adapted.

Figure 3. Age distribution of the population in the indigineous land within the project area



According to data from the 2020 demographic census by IBGE, the four indigenous land in the project area have a population structure with progressive growth. In the Urucu/Juruá, Rodeador and Cana Brava/Guajajara indigenous land, half of the population is concentrated in the age group of 0-19 years (58.41, 59.84 and 55.86, respectively), and in the Lagoa Comprida indigenous lands, in the age group of 0-14 years (50.60%). Similarly, in the four indigenous land, the smallest population is concentrated in the age group from the 65 years.

Regarding the municipalities, the indigenous territories where the Guajajara people are located, represented by the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, are mainly in three municipalities in the state of Maranhão: Barra do Corda, Grajaú and Jenipapo do Vieiras.

The Cana Brava indigenous territory is located in the Municipality of Barra do Corda and is 309 km from Imperatriz, capital of the State of Maranhão, with the geographic coordinates: Latitude: 5°33' 09.1"S, Longitude: 45°39' 13.0"W, and an altitude of 200 meters. The Lagoa Comprida Indigenous territory, located in the Municipality of Jenipapo dos Vieira, is 289 km from Imperatriz of the State of Maranhão, with the geographic coordinates: Latitude: 5°03' 28.0"S, Longitude 46°33' 26.7"W, and an altitude of 100 meters. The Urucu/Juruá territory, located in the Municipality of Grajaú, is 190 km from Imperatriz, State of Maranhão, with geographic coordinates: Latitude: 5°21'32.3"S Longitude, 45°55'10.9"W, and at an altitude of 100 meters. Finally, the Rodeador Territory, located in the Municipality of Barra do Corda, is 337 km from Imperatriz, State of Maranhão, with the geographic coordinates: Latitude: 5°43' 18.8"S, Longitude: 45°13' 28.8"W, and the altitude of 200 meters ([figure 1](#)).

Barra do Corda

In the municipality of Barra do Corda are the Cana Brava and Rodeador indigenous territories. In this municipality, data from the last 2022 census, carried out by IBGE, the Brazilian Institute of Geography and Statistics⁶, indicates that a total of 5,335 people were identified who declared themselves indigenous, 53,270 brown people, 58 yellow people, 6,214 black people and 19,649 white people.

Table 12. Population of the municipality Barra do Corda.

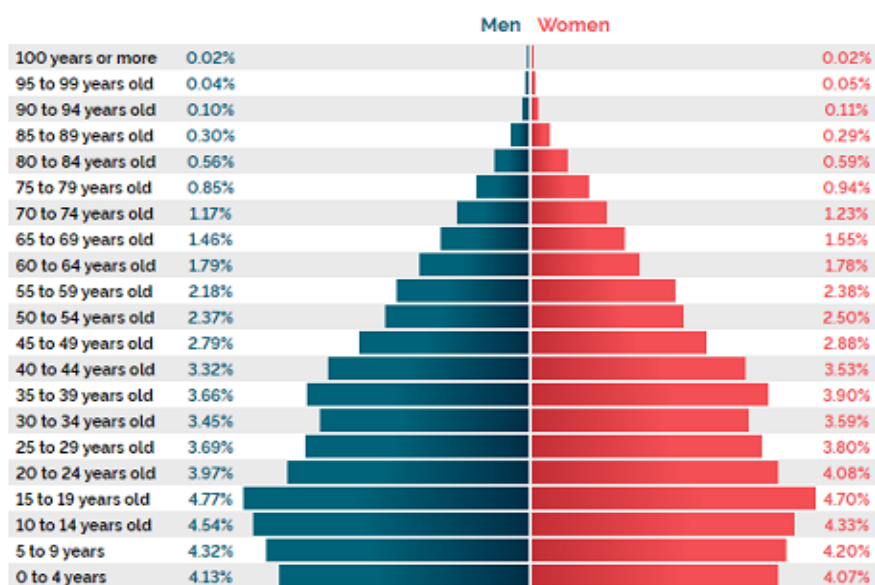
Population (2010)	Population (2022)	Color/Race	Municipality
20,325	19,649	White	Barra do Corda
4,577	6,214	Black	Barra do Corda
543	58	Yellow	Barra do Corda
53,953	53,270	Brown	Barra do Corda
3,432	5,335	Indigenous	Barra do Corda

Source: 2022 IBGE Demographic Census, adapted.

Of the total of 84.532 people living in the municipality of Barra do Corda, 5.552 are indigenous, representing 6.57% of the population. Of these, in indigenous territories 99% are indigenous and 1% non-indigenous. According to 2022 IBGE data, there are 4,500 people (81.05%) residing on Indigenous Land and 1,052 people (18.95%) outside indigenous territories. In the figure below is the age pyramid with the distribution of gender and age with the estimated lifespan of people in the municipality of Barra do Corda.

⁶ <https://cidades.ibge.gov.br/brasil>

Figure 4. Age distribution of the population in the Barra do Corda municipality.



Source: 2022 IBGE Demographic Census

Grajaú

The Urucu-Juruá Indigenous Land is located in the municipality of Grajaú, in the state of Maranhão. According to the most recent data from 2022, provided by IBGE, there are 7,646 indigenous people, which corresponds to 10.73% of the total population of the municipality. Since 2010 the growth in the rate of indigenous people was 3,510 people.

Table 13. Population of the municipality Grajaú.

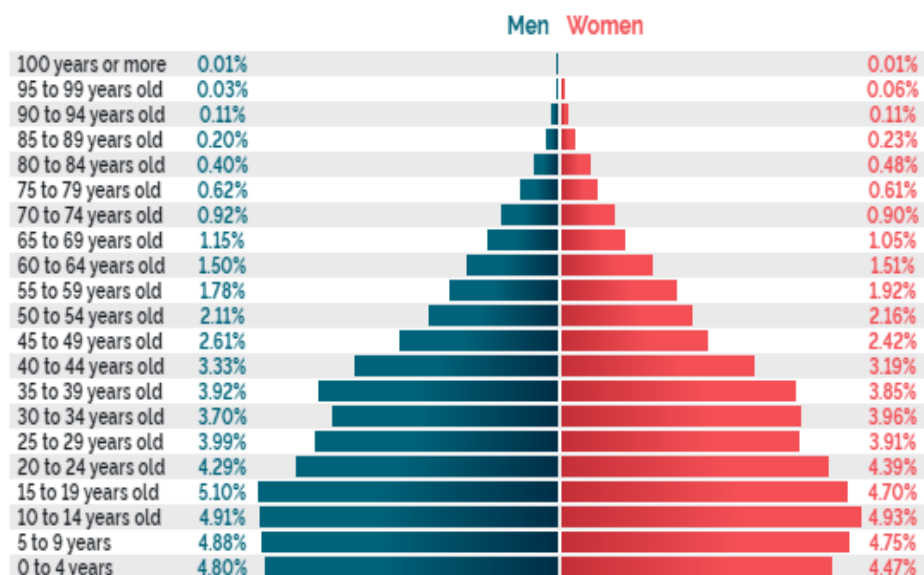
Population (2010)	Population (2022)	Color/Race	Municipality
15,369	15,459	White	Grajaú
4,483	6,336	Black	Grajaú
676	66	Yellow	Grajaú
37,430	44,365	Brown	Grajaú
4,135	7,646	Indigenous	Grajaú

Source: 2022 IBGE Demographic Census, adapted.

Of the region's total indigenous population, according to IBGE and 2022 data, 5,992 (75.59%) live within indigenous territory and 1,935 (24.41%) live outside indigenous territory. Within indigenous lands, 5,992 people (91.69%) are indigenous and 543 (8.31%) are non-indigenous.

As the age pyramid below indicates, the municipality's male population totals is 37,226 people (50.39%) and 36,646 females (49.61%).

Figure 5. Age distribution of the population in the Grajaú municipality.



Source: 2022 IBGE Demographic Census

Jenipapo das Vieiras

The Lagoa Comprida Indigenous Land, also a proponent of the project, is in the municipality of Jenipapo das Vieiras. Of the total 17,076 population residing in the municipality, 7,880 are indigenous, corresponding to 46.15% of the total population, with a growth of 44%.

Table 14. Population of the municipality Jenipapo das Vieiras.

Population (2010)	Population (2022)	Color/Race	Municipality
15,369	15,459	White	Grajaú
4,483	6,336	Black	Grajaú
676	66	Yellow	Grajaú
37,430	44,365	Brown	Grajaú
4,135	7,646	Indigenous	Grajaú

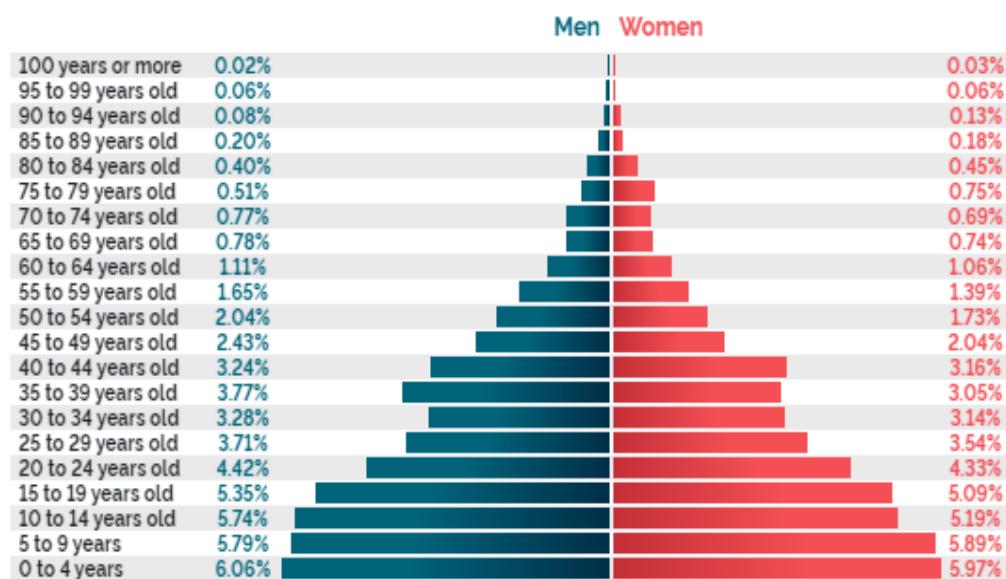
Source: 2022 IBGE Demographic census, adapted.

The indigenous populations residing in the demarcated territory correspond to 97.91%, with 7,701 people and residing outside the territory, 164 people, 2.09%. Within the indigenous territory, 97.3% are indigenous and 2.27% are non-indigenous.

The age pyramid of the municipality of Jenipapo das Vieiras, is made up of 8777 (51.40%) men and 8,299 women, which are equivalent to 48.60% of the total population, with an annual growth rate of 0.63 %.

In the three municipalities where the project proponents are located, there are no official records of the quilombola population.

Figure 6. Age distribution of the population in the Jenipapo das vieiras municipality.



Source: 2022 IBGE Demographic Census, adapted.

1.14.1.3 Education

Barra do Corda

In the municipality of Barra do Corda, where the Cana Brava and Rodeador indigenous territories are located, it is observed in relation to the educational profile, according to data from the last IBGE census in 2010 (IBGE, 2010), education chapter, the schooling rate for 6 to 14 year olds was of 95.8%. In comparison with other municipalities in the state, it was in position 152 out of 217. In comparison with municipalities across the country, it was in position 4,692 out of 5570. In relation to Basic Education Development Index (IDEB), in 2021, IDEB for the initial years of education fundamental level in the public school system was 4.6 and for the final years, 3.9. In comparison with other municipalities in the state, it was in positions 90 and 120 out of 217. In comparison with municipalities across the country, it was in positions 4,347 and 4,435 out of 5570.

Grajaú

In 2010, the school enrollment rate for 6 to 14 year olds was 96.3%. In comparison with other municipalities in the state, it was in position 132 out of 217. In comparison with municipalities across the country, it was in position 4,359 out of 5,570. In relation to IDEB, in 2021, IDEB for the initial years of education fundamental level in the public school system was 4.3 and for the final years, 3.8. In comparison with other municipalities in the state, it was in positions 143 and 137 out of 217. In comparison with municipalities across the country, it was in positions 4716 and 4524 out of 5570

Jenipapo dos Vieiras

In 2010, the schooling rate for 6 to 14 year olds was 80.7%. In comparison with other municipalities in the state, it was in position 217 out of 217. In comparison with municipalities across the country, it was in position 5547 out of 5570. In relation to IDEB, in 2021, IDEB for the initial years of education fundamental level in the public school system was 3.9 and for the final years, it was (no data). In comparison with other municipalities in the state, it was in positions 195 and (there is no data) 217. In comparison with municipalities across the country, it was in positions 4973 and (there is no data) 5570.

1.14.1.4 Health care

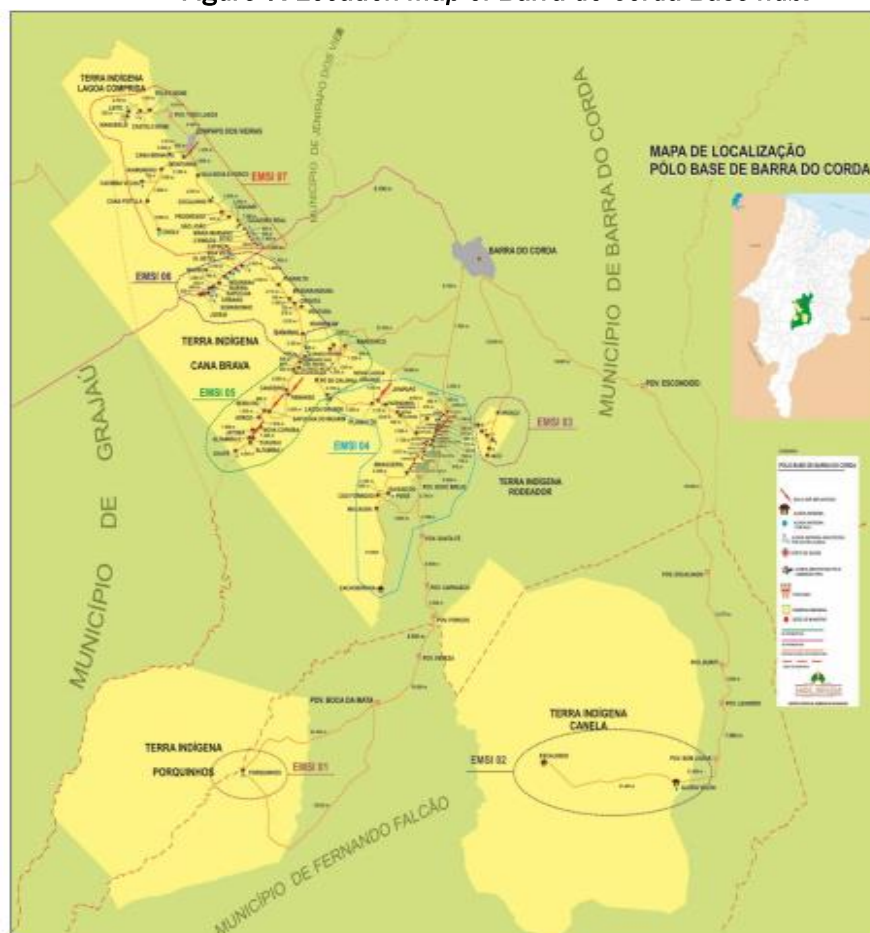
Table 15. Health indicators for the municipalities in the project area

Health indicator	Barra do Corda	Grajaú	Jenipapo dos Vieiras
Infant Mortality (2021). Deaths per thousand live births	16.68	20.88	33.99
Hospitalizations for diarrhea (2022) per 100 thousand inhabitants	2.4	60.9	146.4
SUS Health Establishments (2009). Establishments	42	29	7

Source: 2022 IBGE Demographic Census, adapted.

Barra do Corda

Figure 7. Location map of Barra do Corda Base hub.



Source: Secretariat of Indigenous Health / Special Indigenous Health District

The average infant mortality rate in the city is 16.68 per 1,000 live births. Hospitalizations due to diarrhea are 2.4 per 1,000 inhabitants. Compared to all municipalities in the state, it is ranked 84 out of 217 and 147 out of 217, respectively. When compared to cities across Brazil, these positions are 1608 out of 5570 and 2388 out of 5570, respectively

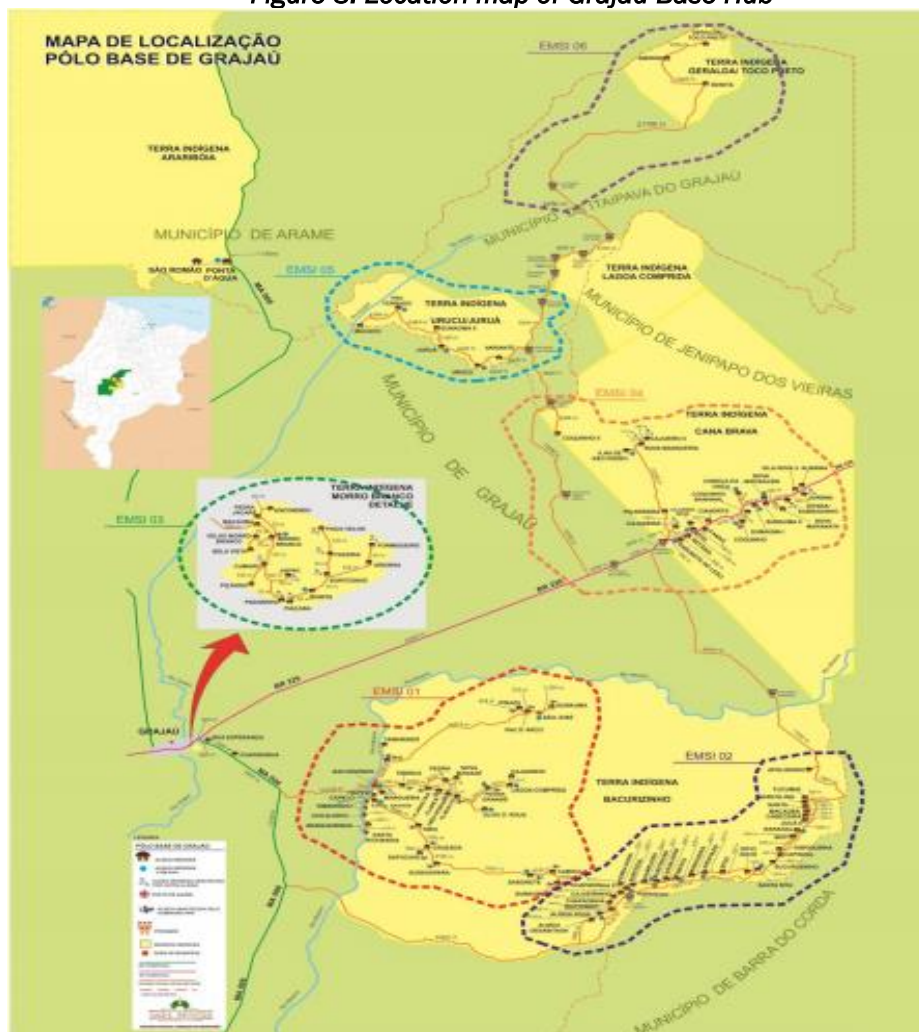
Barra do Corda Base hub

The municipality of Barra do Corda has a Barra do Corda Base Center with specific health care for the indigenous populations of the region, it has seven Multidisciplinary Indigenous Health Teams-EMSI, distributed and operating in seven contiguous territories, as shown in the map.

This indigenous health hub serves the Lagoa Comprida, Rodeador and Cana Brava Indigenous Lands, proponents of the KA'A PYAHU REDD+ project. It covers the three municipalities registered in the project and covers a total of seven indigenous territories, 273 villages and a total population of 15,712 people according to data from the Special Secretariat for Indigenous Health (SESAI) from 2023.

Grajaú

Figure 8. Location map of Grajaú Base Hub



Source: Secretariat of Indigenous Health / Special Indigenous Health District

The average infant mortality rate in the city is 20.88 per 1,000 live births. Hospitalizations due to diarrhea are 60.9 per 1,000 inhabitants. Compared to all municipalities in the state, it is ranked 48th out of 217

and 103rd out of 217, respectively. When compared to cities across Brazil, these positions are 1,072 out of 5,570 and 909 out of 5,570, respectively

Grajaú Base Hub:

The Grajaú Base Hub based in the municipality of Grajaú has six Multidisciplinary Indigenous Health Teams-EMSI, distributed and operating in six contiguous territories, as shown in the map.

The DSEI Grajaú covers six Indigenous Lands, among which three Indigenous Lands from the KA'A PYAHU REDD+ project are covered: Cana Brava, Lagoa Comprida and Grajaú. It covers a total population of 8,571 people and four municipalities

Jenipapo das vieiras

The average infant mortality rate in the city is 33.99 per 1,000 live births. Hospitalizations due to diarrhea are 146.4 per 1,000 inhabitants. Compared to all municipalities in the state, it ranks 9th out of 217 and 86th out of 217, respectively. When compared to cities across Brazil, these positions are 352 out of 5570 and 459 out of 5570, respectively.

1.14.1.5 Unmet Basic Needs

Barra do Corda

It presents 13.5% of households with adequate sanitation, 74% of urban households on public roads with trees and 3.4% of urban households on public roads with adequate urbanization (presence of drains, sidewalks, paving and curbs). When compared to other municipalities in the state, it is ranked 72nd out of 217, 77th out of 217 and 54th out of 217, respectively. When compared to other cities in Brazil, its position is 4,090 out of 5570, 2,844 out of 5,570 and 3,771 out of 5570, respectively (IBGE, 2022).

Grajaú

It presents 4.8% of households with adequate sewage, 89.5% of urban households on public roads with trees and 4.5% of urban households on public roads with adequate urbanization (presence of drains, sidewalks, paving and curbs). When compared to other municipalities in the state, it is ranked 136 out of 217, 24 out of 217 and 46 out of 217, respectively. When compared to other cities in Brazil, its position is 4823 out of 5570, 1544 out of 5570 and 3588 out of 5570, respectively (IBGE, 2022)

Jenipapo dos Vieiras

It presents 2.6% of households with adequate sewage, 95.4% of urban households on public roads with trees and 4.6% of urban households on public roads with adequate urbanization (presence of drains, sidewalks, paving and curbs). When compared to other municipalities in the state, it is ranked 173 out of 217, 11 out of 217 and 45 out of 217, respectively. When compared to other cities in Brazil, its position is 5108 out of 5570, 850 out of 5570 and 3563 out of 5570, respectively (IBGE, 2022).

1.14.1.6 Economic activities

The main subsistence activity of the Guajajara people, proponents of the KA'A PYAHU REDD+ project, is farming, especially the planting of cassava, corn, rice, pumpkin, watermelon, beans, broad beans, yams, yams, sesame, peanuts. In the dry season, from May to November, drilling, felling, burning, digging and cleaning are carried out, while from November to February planting and meadows are carried out.

According to (Schröder, 2021), the planted areas per residential unit vary between 1.25 ha and 3.55 ha per domestic unit, or between 0.25 ha and 0.71 ha per individual, respectively. This variation depends mainly on the involvement of communities and individuals in the marketing of agricultural products.

Fishing is mostly practiced in riverside villages. The Guajajara usually fish around 36 different species, with the yam, cascudo, lamprey, mandi, pacu, piau and traíra being the most common. In recent years, however, in several community projects, small dams have been built near some villages that are far from rivers. For the residents of these villages, the dams allow both subsistence and commercial fishing.

The Guajajara traditionally hunt more than fifty-six species, the most common being peccary, agouti, jacamim, jacu, peccary and several species of monkeys and armadillos. In a part of the Guajajara lands, hunting became more productive again during the 1990s after more efficient controls over land limits were initiated by the Indians themselves.

Collecting is still practiced by almost all Guajajara. Collection activities, however, are increasingly being replaced by fruit growing in villages and farms. Currently the Guajajara plant around thirty types of fruit trees and palm trees. The only forest product still collected in larger quantities for commercial purposes is honey.

Economic relations with non-indigenous people are based on both material and monetary exchanges. The most common sources of income are the sale of agricultural products, the sale of handicrafts and temporary work (for the settlers) or permanent work (for Funai). A very serious problem is the predatory commercialization of the areas' natural resources through concessions to loggers and hunters, in order to obtain small profits in the short term to, for example, buy medicines not provided by deficient government services.

1.14.1.7 Social and political organization

According to (Schröder, 2021) Guajajara villages are long, round or quadrangular. They are preferably located on the banks of rivers or, in the absence of watercourses, near lakes in the forest. The proximity to a road can be another attractive factor, for selling handicrafts, for example.

The villages, formerly very small and of temporary existence, are now permanent and rarely transferred. They can be made up of a single family, but in some cases they can have up to four hundred or more residents. The houses, built in the regional peasant style, are generally inhabited by nuclear families. Villages usually maintain their independence and rarely form regional coalitions, but there are diverse kinship, marriage and ritual relationships between communities.

In kinship, the most important unit is the extended family, which is made up of a number of matrilineal nuclear families linked together by kinship ties. It is, in essence, a group of related women under the leadership of a man. There are no moieties, clans or lineages, nor any right or obligation that is transmitted through a specific line of descent.

Residence after marriage is with the woman's parents, the man's in-laws, (uxorilocality). Many heads of extended families try to keep as many women as possible, even adopting the daughters of deceased men they used to call "brothers". They try to arrange marriages for these girls to obtain sons-in-law, who must live at least one or two years with their in-laws, providing various types of services. If the head of the family has enough prestige, he manages to get his sons-in-law to permanently settle with him, thus increasing the number of collaborators and attracting co-partisans to form a faction in the village.

The main traditional criteria for assuming leadership involving kinship politics (individual qualities and a base of co-partisans through consanguinity and affinity) have become less important, compared to the demands of knowing how to deal with the surrounding society. This concerns, firstly, the ability to relate to government bodies and have a political stance in relation to the local community, and individual qualities (knowledge of Portuguese and diplomatic talent, among others).

Each village has its own chief, but there are villages with more than one because of rivalries between several extended families. Some chiefs try to extend their influence to neighboring villages, but their authority is very unstable and can be challenged at any time by competitors from their own village. In this

game for power, the indigenous body usually intervenes to promote its own protégés, who may be weak characters, with no true base in the villages.

1.14.1.8 Cosmology

Guajajara cosmology is typical of the Tupi-Guarani peoples, distinguishing four categories of supernatural beings, which receive the generic designation of karowara: (1) the creators or cultural heroes, responsible for the creation and transformation of the world, being Maíra and the twins Maíra-ira and Mucura-ira the most important and Zurupari, the creator of pests and insects, venomous snakes and spiders, a much feared cultural hero; (2) the "owners" of the forests (Ka'a'zar), the waters (Y'zar), the hunts (Miar'i'zar) and the trees (Wira'zar), who are hostile and greatly feared by its evil power; (3) the azang, wandering spirits of the dead, also greatly feared; and (4) the piwara, animal spirits. Many Guajajara no longer believe in these beings, because of missionary activities (Wagley, 1973).

The mythology is a mix of Tupi, European and African motifs. There are three main categories of myths: (1) cultural hero myths; (2) myths that point to a moral; and (3) animal myths. In all the myths recorded so far, the role of Maíra stands out. A very important myth to explain the world from the Guajajara point of view is that of the twins Maíra-ira and Mucura-ira.

The major traditional rituals include the important Honey Festival, held in September or October, during the dry season, and which requires several months to prepare. It played a very important role in good relations between villages. The Corn Festival (awashire-wehuhau), also called the "shaman festival", takes place every year during the rainy season, during the growing season of this vegetable. Its purpose was to ensure a good harvest and protect the corn against the actions of the Azang. Therefore, its main characteristic was the pajelança. The Moqueado rite, held on the same occasion as part of the Corn Festival, marks the end of puberty for the participating teenagers. Moqueado is still practiced at irregular intervals, but it has become somewhat profane, often leaving only the culinary part of the rite to accompany political meetings.

A person's life cycle is still usually accompanied by a series of rituals. Among these, initiation rituals, particularly those of girls, are the most eye-catching and rich in meaning. Additionally, there are a series of rituals to ask permission from Maíra to plant, from Miar'i'zar to hunt and from Y'zar to fish.

Shamanism

In ancient times, most Guajajara men tried, at any cost, to be shamans, but few were successful and gained fame. The power and reputation of the shamans depended on the number of supernatural beings they knew how to "call". Highly recognized shamans could also become powerful leaders. Pajelança is an almost exclusively male activity. The main function of the shamans is still to heal and celebrate the festivals of Maíra and the "allowance", a ritual of offerings in favor of sick people. Shamanism is often seen as ambiguous, because shamans' powers can be used for both positive and negative goals.

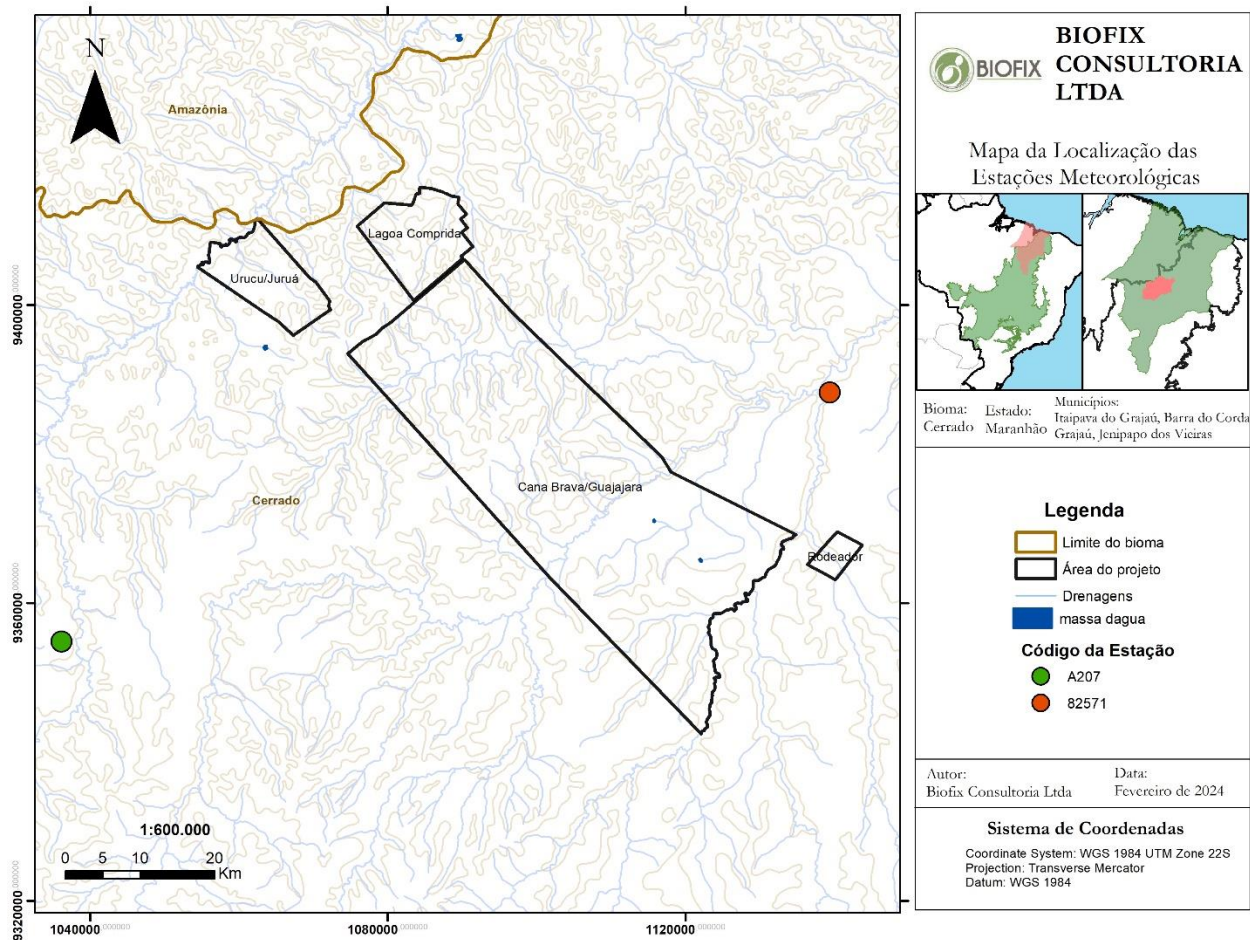
1.14.2 Biophysical Parameters

1.14.2.1 Climatology

To describe the climatology of the project area, particularly the variables of precipitation, temperature, and relative humidity, data from two meteorological stations in the National Institute of Meteorology's⁷ database were consulted. The two stations were selected because they are the closest stations to the project area, however, station Barra do Corda is the closest and less than 20 kilometers away. For the analysis of the two stations, data were consulted from the year 2000 (82571 station) and 2008 (A207 station), from which time they are available

⁷ <https://portal.inmet.gov.br/>

Figure 9. Location of the meteorological stations.



Source: KA'A PYAHU REDD+ Project.

Table 16. Data from meteorological stations

Station	Name	Type	Municipality	Latitude	Longitude	Altitude
82571	Barra do Corda	Conventional	Barra do Corda	-5.506666	-45.236944	154.19
A207	Grajaú	Automatic	Grajaú	-5.816111	-46.162222	232.29

Source: National Meteorology Institute INMET.

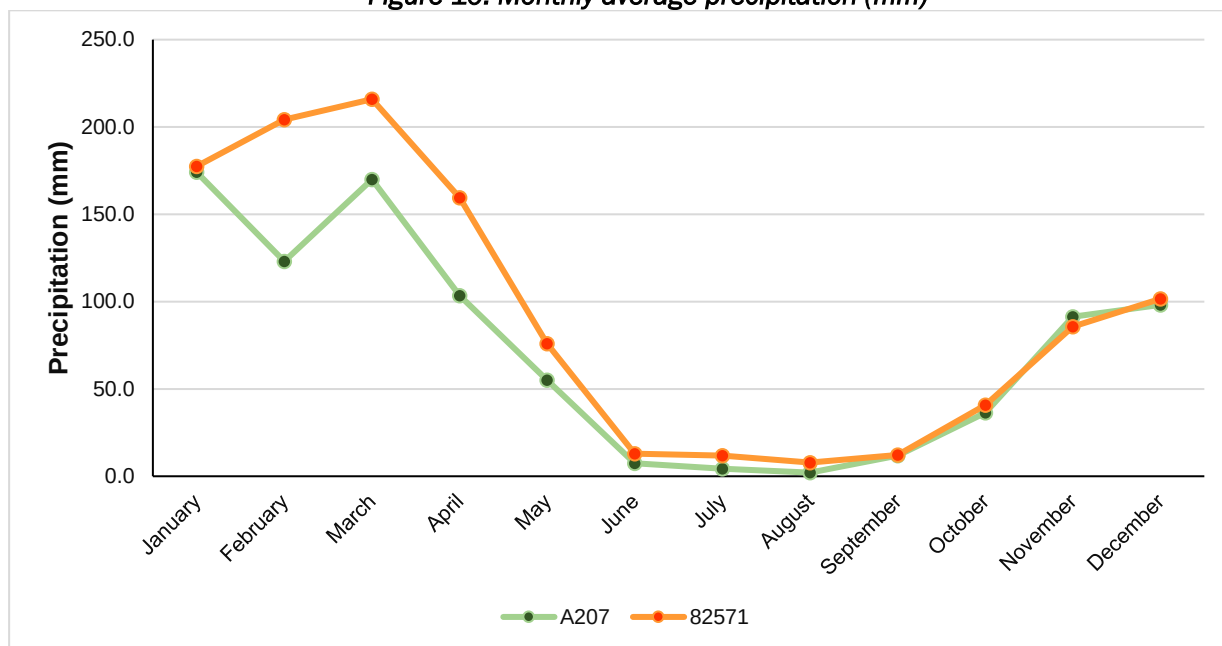
Precipitation

Rainfall in the project area is characterized by a long period of rain, starting in October and extending through May, with January and March being the months with the highest rainfall, with maximum values that can reach up to 215 millimeters, and an average of 17 days of rain.

Following this, between June and September there is a dry period, with monthly average rainfall of less than 10 mm and no rainy days.

According to the data from the stations, the total annual precipitation varies between 878 mm (A207 Station) and 1105 mm (82571 Station), with 991 mm being the average between the two stations. The total number of annual days varies between 101 and 133, respectively.

Figure 10. Monthly average precipitation (mm)

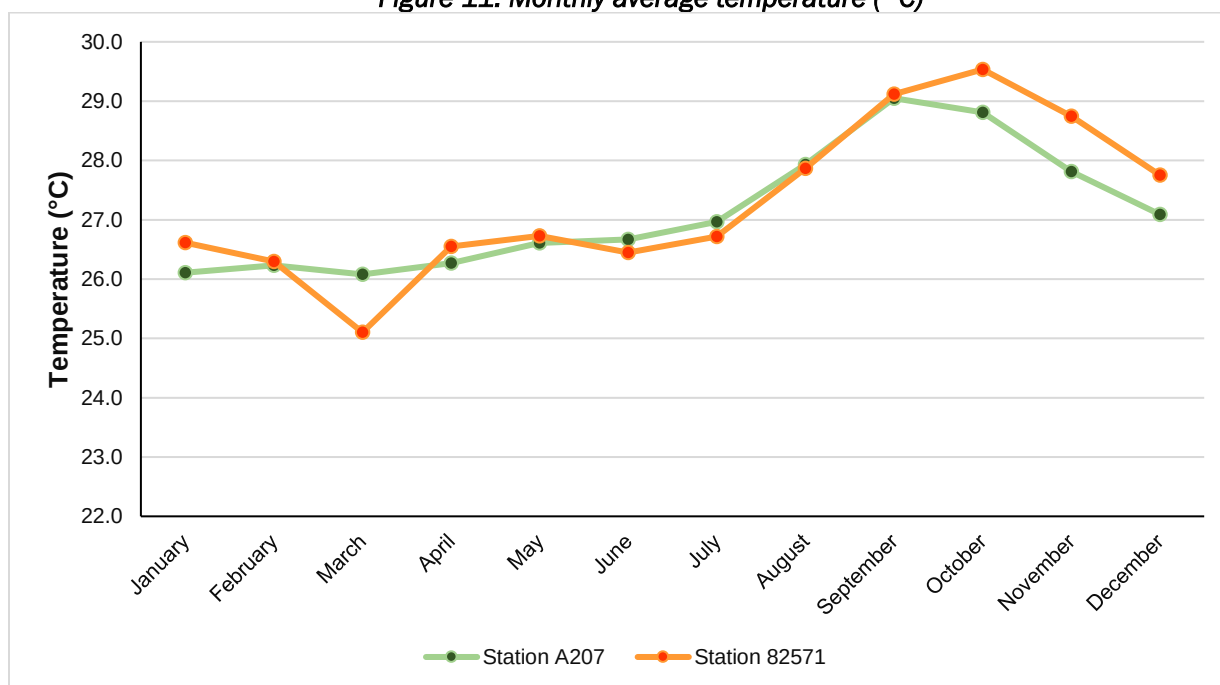


Source: Data from National Meteorology Institute INMET, adapted.

Temperature

According to data from the two automatic stations, the average annual temperature is 27.2 °C, with values varying between 25.6 °C and 29.3 °C.

Figure 11. Monthly average temperature (°C)



Source: Data from National Meteorology Institute INMET, adapted.

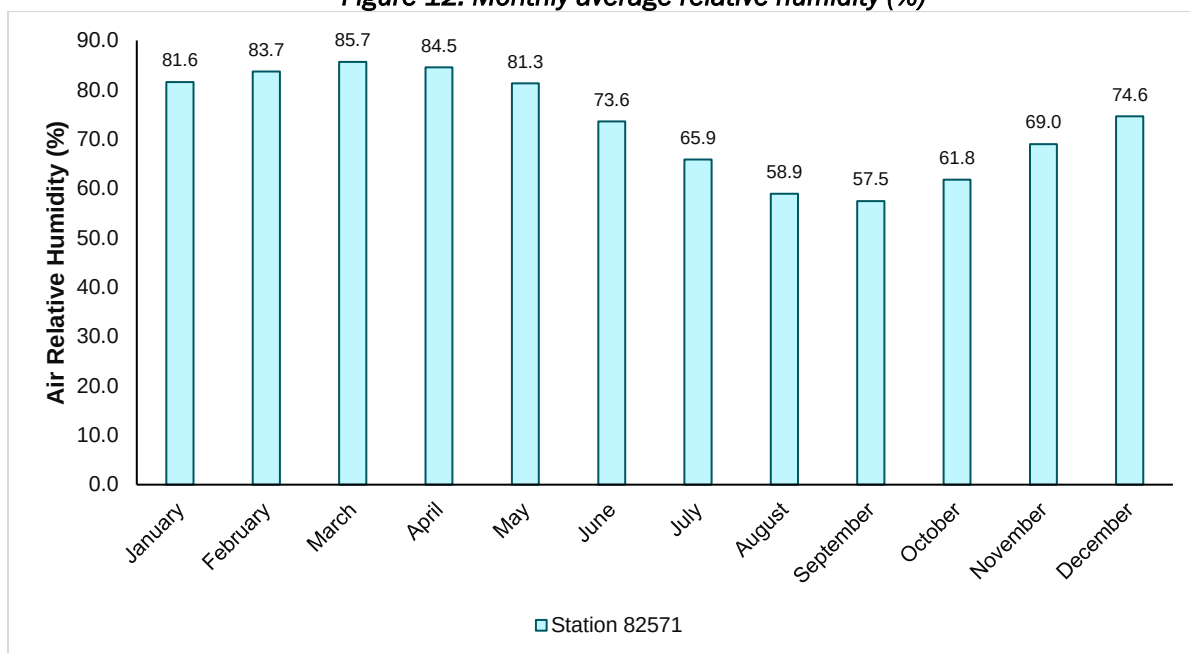
Generally, the temperature remains constant throughout the year. However, there is a difference of approximately 3.2°C between the hottest months (September and October) and the coldest months (February and March).

Air Relative Humidity

For the air relative humidity variable, only the conventional station was taken into consideration, as it was the only that presented data for this variable.

The annual average relative humidity in the project area is approximately 73.2%. According to the station data, there is a correlation between precipitation and relative humidity. That is, the months with the highest precipitation (January through April) correspond to the highest humidity values, reaching values of up to 85% relative humidity. Conversely, the driest months (July to September) have the lowest relative humidity values, with values of up to 65% relative humidity. The difference between the months with the highest and lowest humidity is approximately 30%.

Figure 12. Monthly average relative humidity (%)



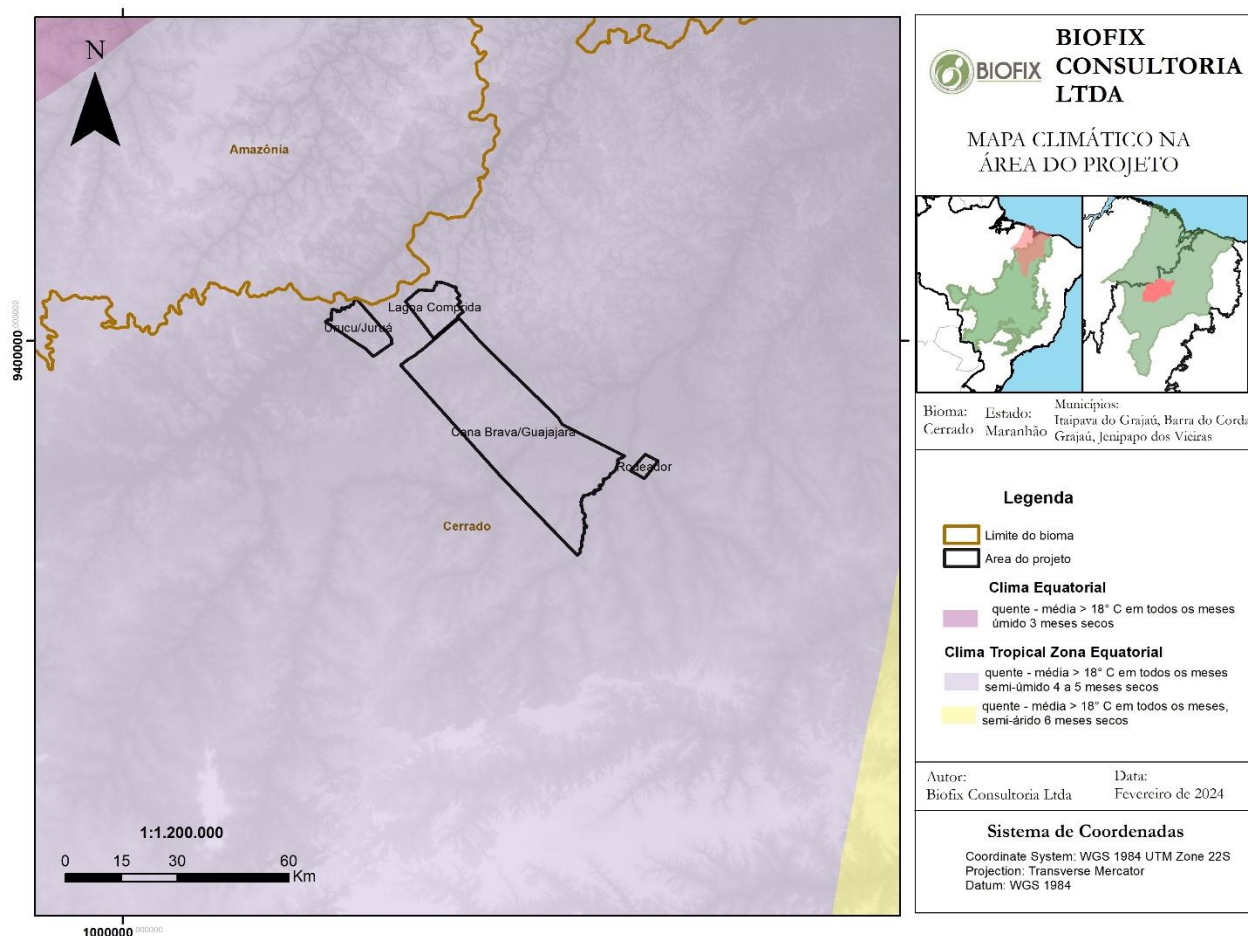
Source: Data from National Meteorology Institute INMET, adapted.

Climate Classification

According to the climate map of Brazil at a scale of 1:500.000 (Instituto Brasileiro de Geografia e Estatística IBGE, 2002), the project area is characterized by a humid equatorial climate, marked by the occurrence of 3 dry months and an average temperature above 18°C during all months of the year.

On the other hand, according to the Koppen Climate classification for Brazil, the project area has an **Aw** climate (Tropical with dry winter). The letter A indicates the climatic zone, defined by the temperature and precipitation, which in this case is *tropical*, with an annual average temperature above 18°C. The letter w considers the distribution of precipitation, in this case, it means with a dry winter, where the precipitation of the driest month is less than 25 mm (Alvares, Stape, Sentelhas, Gonçalves, & Sparovek, 2013).

Figure 13. Climate classification of the project area



Source: IBGE (2002), adapted.

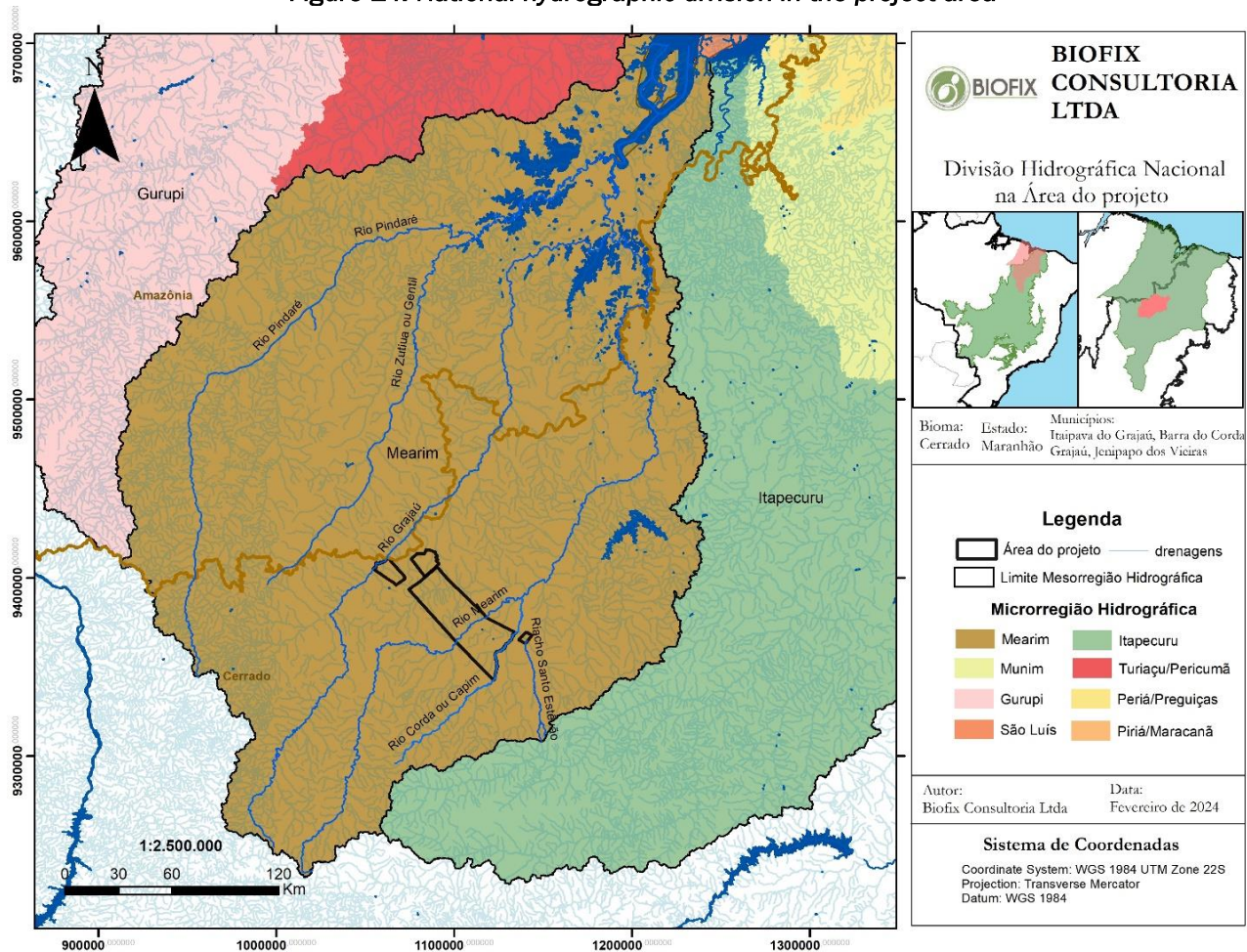
1.14.2.2 Hydrography

National Hydrographic Division DHN

The National Hydrographic Division of Brazil at a scale 1:250.000 was updated in 2021 in a joint effort by the Brazilian Institute of Geography And Statistics (IBGE) and the National Water Agency ANA (IBGE. Coordenação de Recursos Naturais e Estudos Ambientais, 2021). This division comprises three levels: The first level is the Hydrographic Macroregion (MacroRH), which divides Brazil into 12 hydrographic regions established in Resolution No. 32 of 15.10.2003 (MINISTÉRIO DO MEIO AMBIENTE. CONSELHO NACIONAL DE RECURSOS HÍDRICOS, 2003). The project area is located in the Atlântico Nordeste Ocidental macroregion, which has an extension of 275.000 km² and it is a region of great economic importance for the Northeast economy, since it has significant water potential for the development of agricultural activities.

The second level divides the country into 54 hydrographic mesoregions (MesoRH), based on hydrographic units that consider land use and resource management factor. The Nordeste Ocidental macroregion comprises 3 mesoregions, Gurupi, Itapecuru, and Mearim, being located the project area is located in the Mearim mesoregion, which has an extension of 101.998 km² and whose main rivers are Mearim river, Grajaú river, Pindaré river, and Zutiua river (Algarves, Pereira, Fernandes, Alves, & Machado, 2016).

Figure 14. National hydrographic division in the project area



Source: adapted from (IBGE. Coordenação de Recursos Naturais e Estudos Ambientais, 2021).

The third level comprises 302 hydrographic microregions micro RH, based on water planning units. The Mearim mesoregion comprises 8 microregions, being the project located in the Mearim microregion. This means that the project area is located in the same meso- and microregion. In general terms, the National Hydrographic Division (DHN) considers economic, land use, and resource management factors.

Hydrographic basins in Brazil.

In contrast to the National Hydrographic Division, the Brazilian river basin division is based on the concept of a river basin being "an area of land surface bounded by watersheds that collects and conveys, through slopes, rivers and streams, water from precipitation to an outfall, a single point of exit, located at a lower point in the relief".

The different basins according to the levels of the project area are shown below:

Table 17. Hydrographic basins in the project area

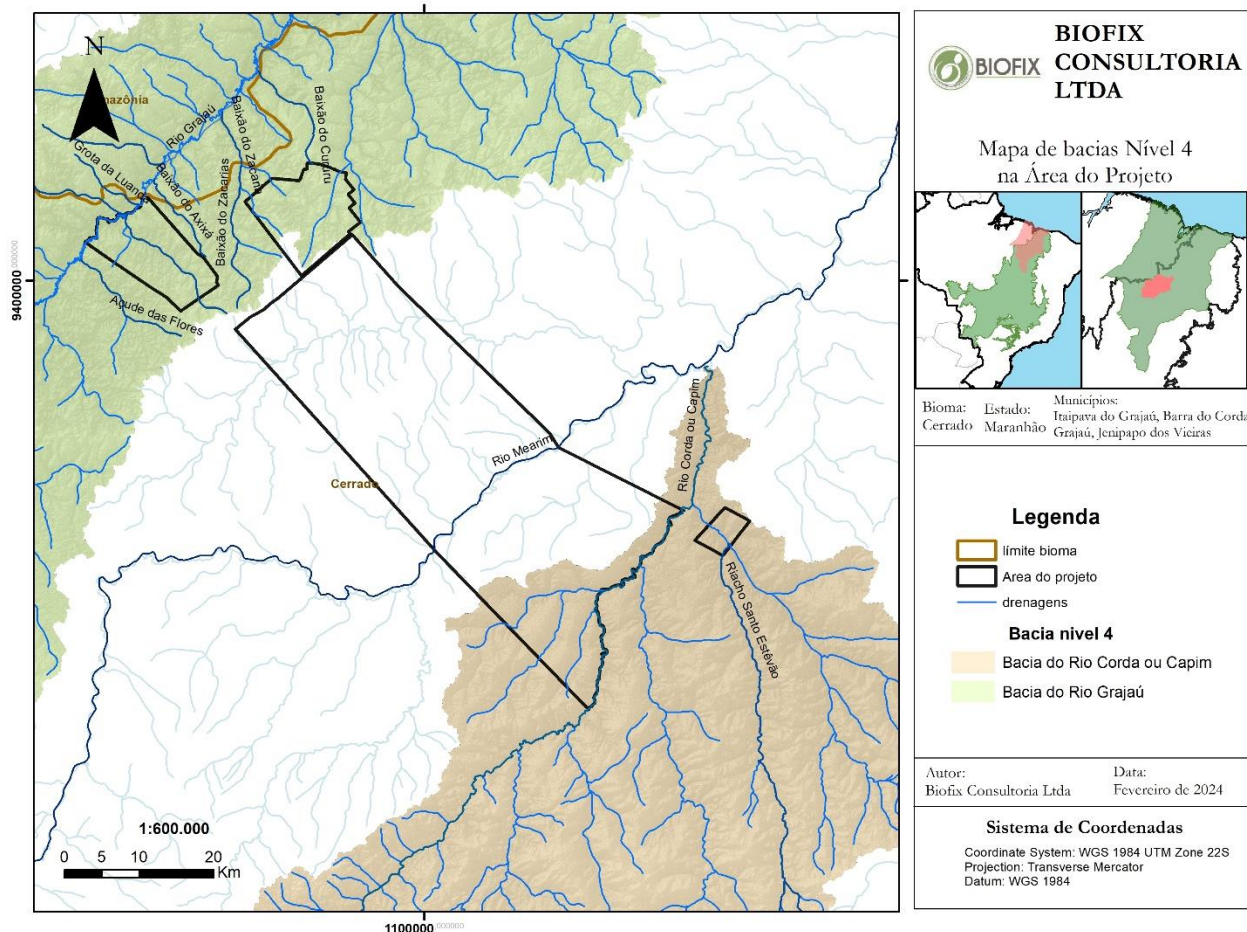
Level	Basin Name	Main River	Main tributaries
3	Rio Mearim	Mearim river	Pindaré, Grajaú, Zutiua, Santana, Piratininga, Ipixuna, Enjeitado, das flores, Corda.
4	Rio Corda	Corda river	Pau Grosso, Riacho Fundo, Ourives , Baixão do Sabonete, Baixão Pau Ferrado, Baixão do Zacarias, Baixão do Cururu.
	Rio Grajaú	Grajaú river	Santana, Riacho Santo Estêvão, Bacia do Baixão do Limão, Igarapé do Salgado.
5	Lagoa Comprida	Baixão da Lagoa Comprida	
	Capim Duro	Baixão do Capim Duro	Baixão do Fundo, Baixão do Francisco, Baixão do Crioli
	Baixão do Sabonete	Baixão do Sabonete	
	Rio Ourives	Ourives river	Brejo do Curicaca, Riacho Santo Estêvão.
6	Baixão do Crioli	Baixão do Crioli	
	Baixão do Zacaria	Baixão do Zacaria	
	Barro Branco-Fogo	Baixão do Barro Branco e Baixão do Fogo	
	Santo Estêvão	Riacho Santo Estêvão	

Data from (IBGE. Coordenação de Recursos Naturais e Estudos Ambientais, 2021).

The KA'A Y PYAHU project as mentioned in the Mearim river basin of the third level, as a mentioned before, the main rivers are Pindaré, Grajaú, Zutiua and Corda. The Mearim river originates on the northern slopes of Serra da Menina, in the extreme south of the Municipality of Grajaú, flowing into the Atlantic Ocean, 24 km after the confluence with the Pindaré river (Cunha, 2003).

There are two level 4 basins in the project area: The Grajaú river basin, which borders the Urucuru indigenous territory to the north. Of the main tributaries of the Grajaú River, the ones that pass through the project area are Baixão do Zacarias, Baixão do cururu, and Baixão do Axixá. The basin of the Capim river, which has as its main river the Capim or Corda river, which is the limit on the Cana Brava indigenous territory. One of the main tributaries of the Capim River is the Santo Estêvão stream, which runs through the Rodeador indigenous territory.

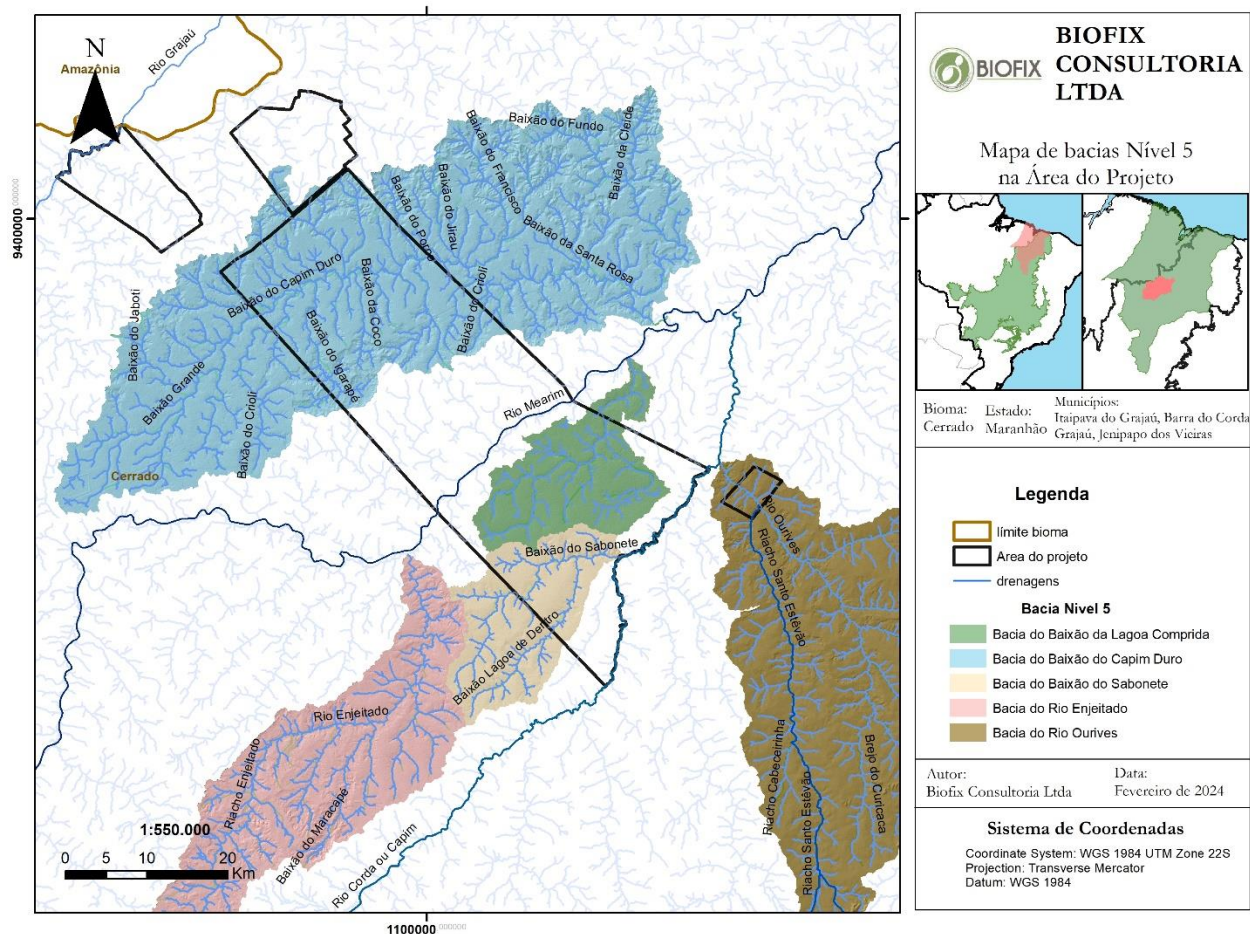
Figure 15. Level 4 basins in the project area



Source: adapted from (IBGE. Coordenação de Recursos Naturais e Estudos Ambientais, 2021).

Regarding type 5 level basins, the project area partially includes 4 basins: the Ourives river basin and the Sabonete basin, which both flow into the Capim river. The others two, the Lagao comprida basin and Capim duro basin flow into the Mearim river. With the exception of the Sabonete river basin, the other three basins cross the indigenous territory of Cana Brava. The Capim duro basins has as its main tributaries the Baixão do Francisco, Baixão do Crioli, and the Baixão do Coco, this which originates within the limits of the project area.

Similarly the Sabonete basin has the sabonete river as its main watercourse, and its originates within the limits of the project area, especially in the indigenous territory of Cana Brava. This demonstrates the importance of the project from a hydrographic point of view, since there are basins that feed important rivers such as Mearim river and Capim river.



BIOFÍX CONSULTORIA LTDA

Mapa de bacias Nível 6 na Área do Projeto

Legenda

- limite bioma
- Área do projeto
- drenagens

Bacia nível 6

- Bacia Barro Branco-Fogo
- Bacia de nome não identificado
- Bacia do Baixo do Curi
- Bacia do Baixo do Zacaia
- Bacia do Riacho Santo Estevão

Sistema de Coordenadas

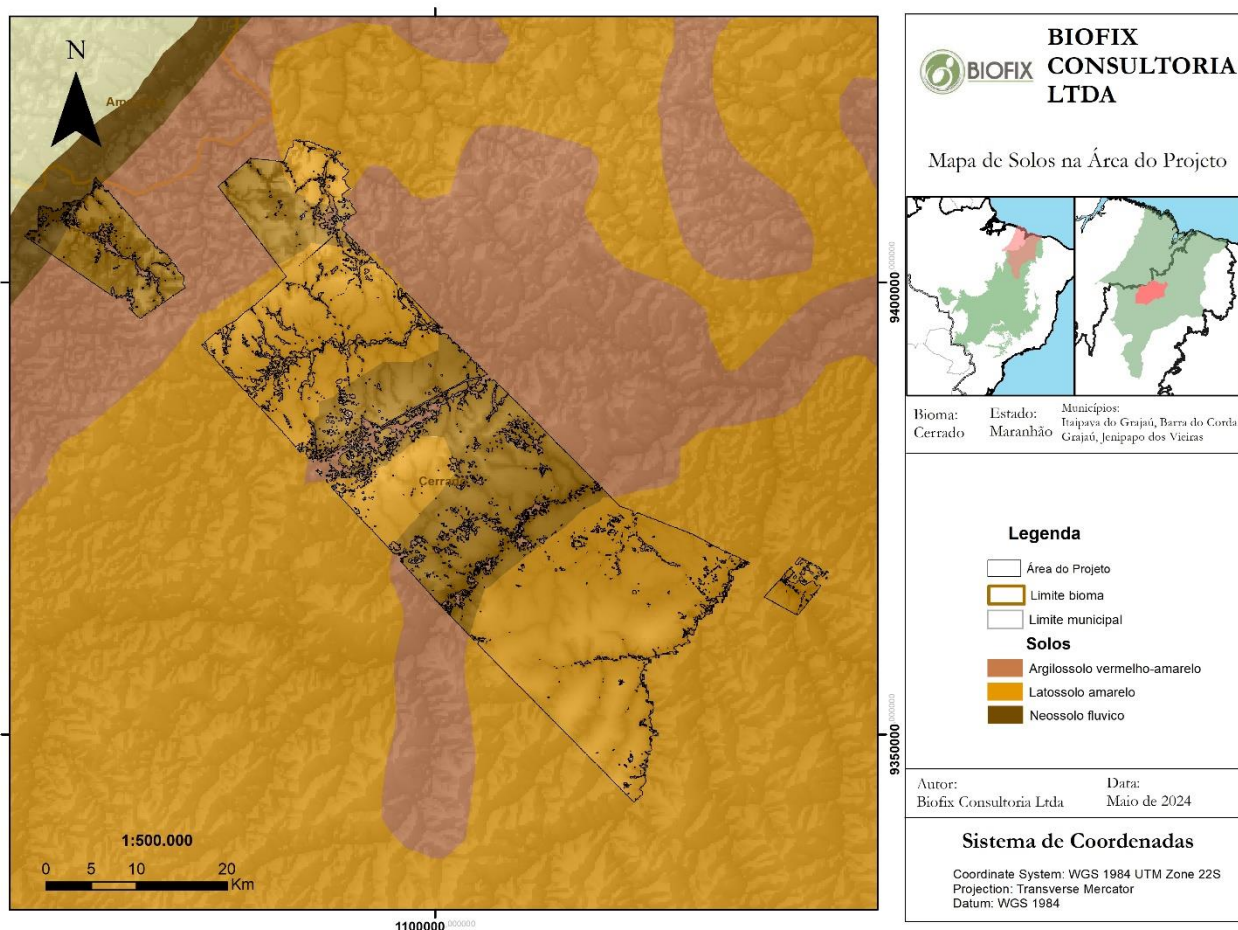
Coordinate System: WGS 1984 UTM Zone 22S
Projection: Transverse Mercator
Datum: WGS 1984

1.14.2.3 Geomorphology, edaphology and geology

According to the 2001 Brazilian Soil Map (EMBRAPA, 2001), 59.37% of the project area is located in Latosol order, yellow- latosol sub-order, which are soils made up of mineral material, clayey or very clayey, soils with more yellow color and with base saturation < 50% in most of the first 100 cm of the B horizon.

52

Figure 18. Map of soils in the project area



Source: (EMBRAPA, 2001), adapted.

Table 18. Soil order in the project area

Soil Orden	Sub-Order	Big group	Area (ha)	Area (%)
Argissolo	Red-yellow Argisols	Eutrophic Red-yellow Argisols	59,698.61	39.59
Latossolo	Yellow Latosol	Dystrophic yellow latosol	89,524.31	59.37
Neossolo	Fluvic Neosols	Eutrophic Fluvic Neosols	1577.38	1.05
Total			150,800.31	100.00

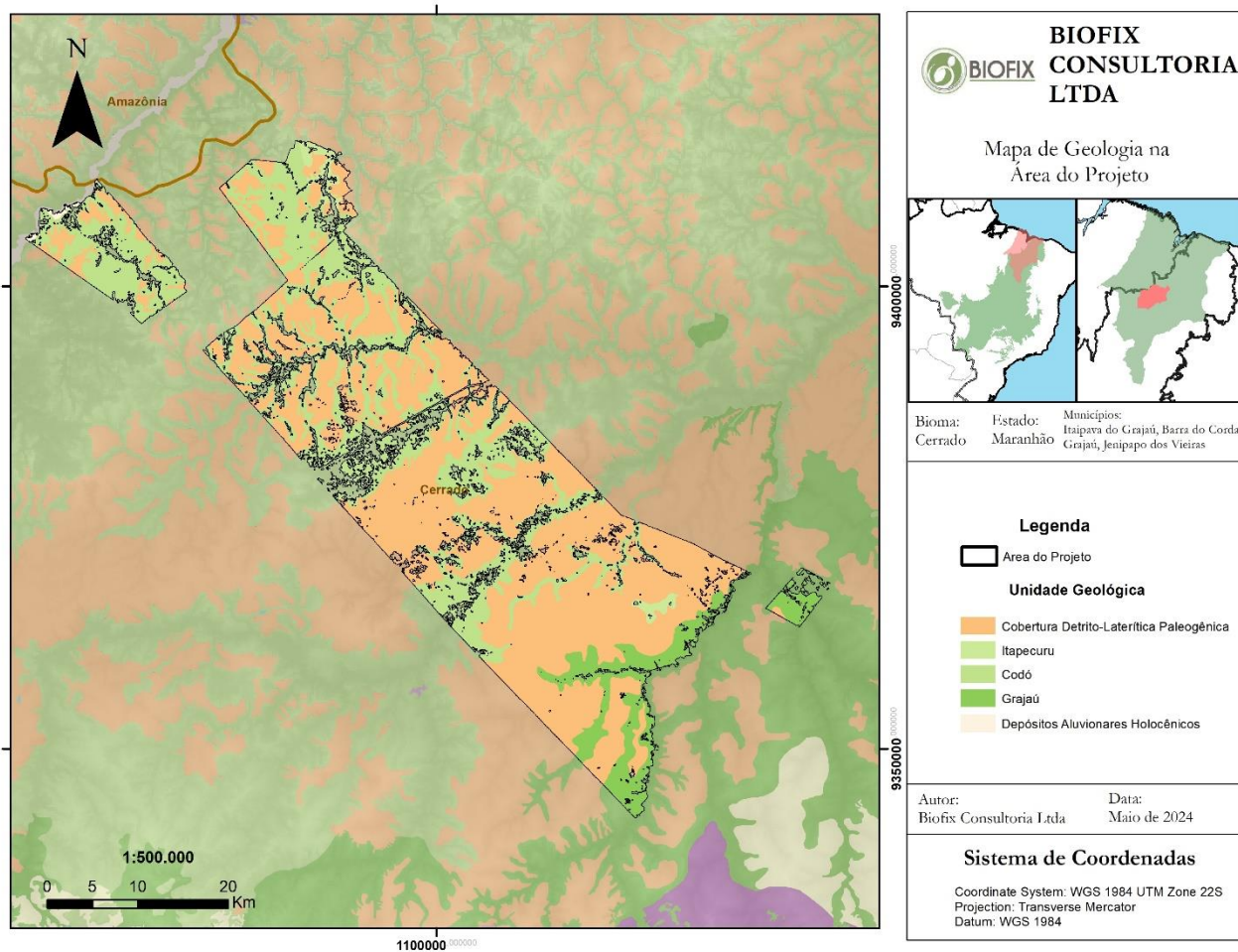
Source: (EMBRAPA, 2001), adapted.

Geology

In terms of geology, according to IBGE Natural Resources Mapping (IBGE, 2018), 5 geological units are present in the project area. The Palaeogene Detrital-Laterite Cover unit and the Holocenic Alluvial Deposits unit, which belong to geological time geological Cover and occupy 57.18% of the project area. These geological units are characterized by kaolinitic clay zone, bauxitic zone with concretions and glossitis lenses, ferruginous concretionary zone, nodular pisolitic zone, and clay capping on top (IBGE, 2011).

The geological units The Codó, Grajaú, and Itapecuru units occupy the remaining 42.82%, and which belong to the geological time Parnaíba. The formation Codó is characterized by cinza-escuras folhelhos, with intercalacoes of arenites, calcareous and glosite. On the other hand, the Grajaú formation is represented by yellowish firavels sandstones. Finally, the Itapecuru formation is made up of asphalted sandstones.

Figure 19. Map of geology in the project area



Source: (IBGE, 2018), adapted.

Table 19. Geology of the project area

Name of Geological Unit	Geological time	Name of structural province	Area (ha)	Area (%)
Palaeogene Detrital-Laterite Cover	Phanerozoic Cenozoic Paleogene Oligocene	Cenozoic cover	86,234.89	57.18
Codó	Phanerozoic Mesozoic Lower Cretaceous Aptian	Parnaíba	51,298.12	34.02
Grajaú	Phanerozoic Mesozoic Lower Cretaceous	Parnaíba	11,171.84	7.41
Itapecuru	Phanerozoic Mesozoic Lower Cretaceous Albian	Parnaíba	1,832.86	1.22
Holocene Alluvial Deposits	Phanerozoic Cenozoic Quaternary Holocene	Cenozoic cover	262.58	0.17
Total			150,800.31	100.00

Source: (IBGE, 2018), adapted.

Geomorphology

According to IBGE Natural Resources Mapping (IBGE, 2018), the geomorphology of the project area is characterized by 4 types of geomorphological units. The Chapada de Barra do Corda unit, occupies the 81.85% of the project area, which is characterized by a tray relief (tabuleiro), with a pediplane and convex top shape.

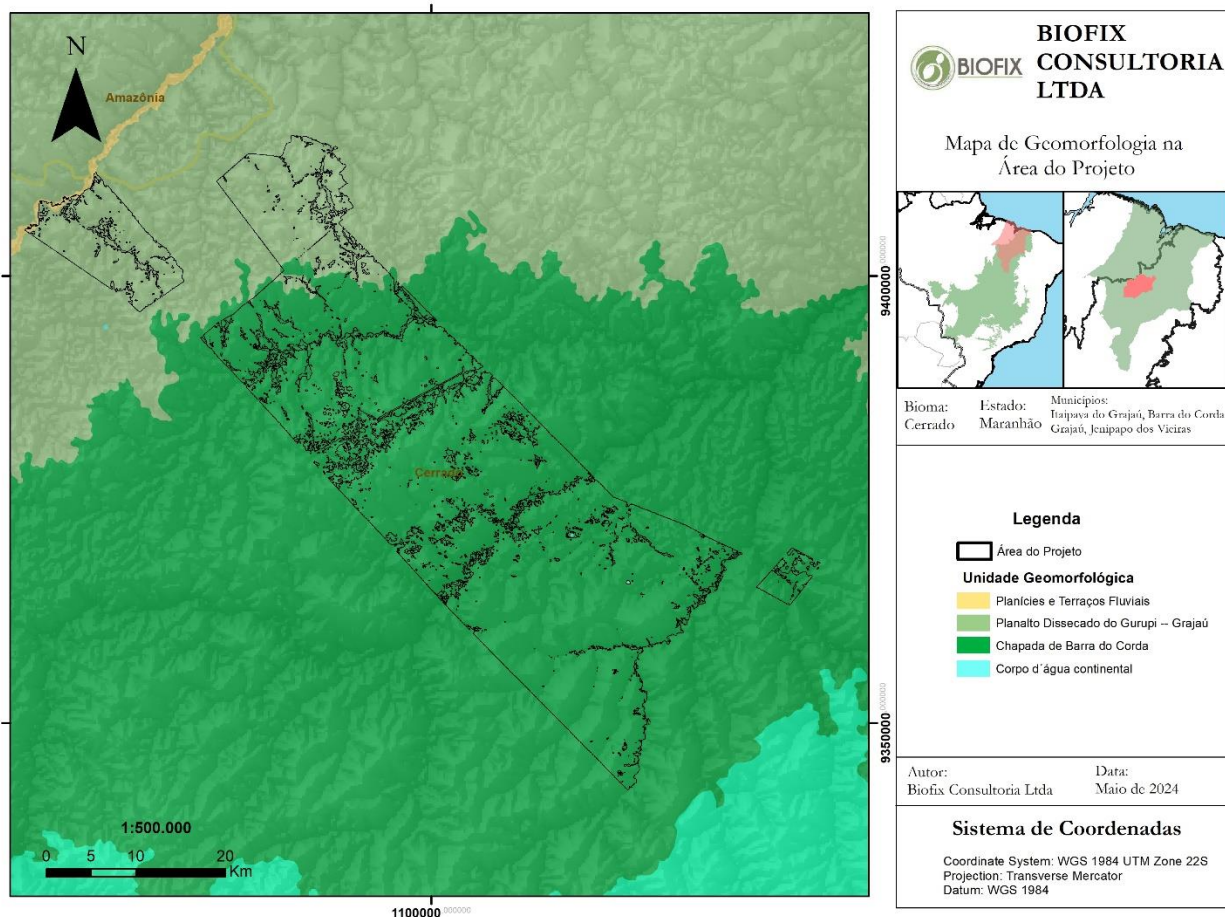
The Planalto Dissecado do Gurupi – Grajaú unit occupies 17.97% of the project area, which is characterized by patamares relief, with also a pediplane and convex top shape. The Chapada de Barra do Corda unit and the Planalto Dissecado do Gurupi – Grajaú unit, belong to the Phanerozoic Basins and Sedimentary Covers Morphostructural Domain.

Table 20. Geomorphology of the project area

Name of Geomorphical Unit	Relief compartment	Name of Morphostructural Domain	Area (ha)	Area (%)
Chapada de Barra do Corda	Tabuleiros	Phanerozoic Basins and Sedimentary Covers	123,431.75	81.85
Planalto Dissecado do Gurupi – Grajaú	Patamares	Phanerozoic Basins and Sedimentary Covers	27,100.17	17.97
Planícies e Terraços Fluviais	Planícies (plains)	Quaternary sedimentary deposits	266.34	0.18
Corpo d'água continental	Continental water body	Continental water body	2.06	0.00
Total			150,800.31	100.00

Source: (IBGE, 2018), adapted.

Figure 20. Map of Geomorphology in the project area



Source: (IBGE, 2018), adapted.

1.14.2.4 Ecosystems and vegetation

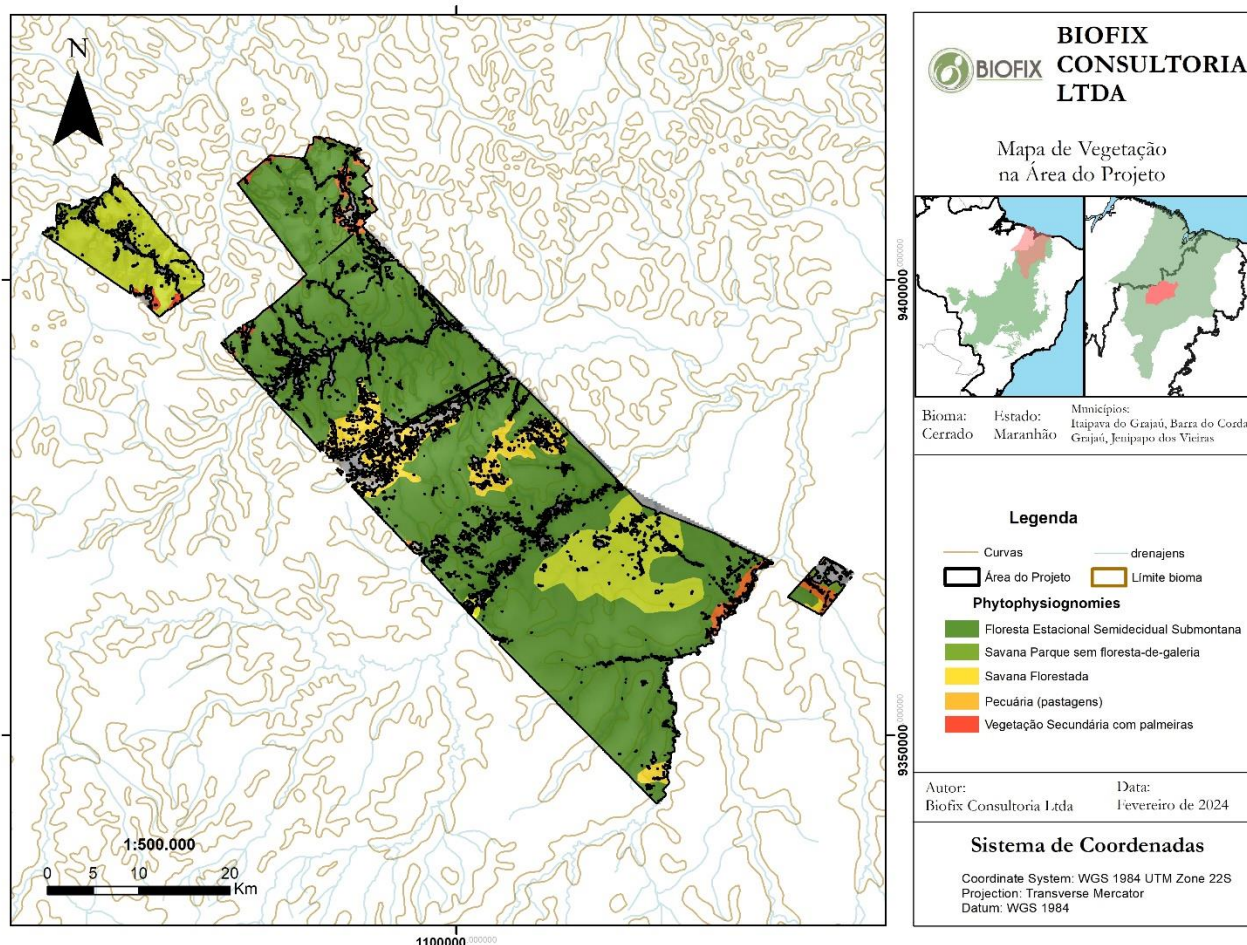
The Cerrado biome is the second largest biome in Brazil, occupying approximately 23.3% (1,983,017 km²) of the national territory and extending to the countries of Bolivia and Paraguay. In Brazil, the Cerrado covers 12 states of the country (IBGE, 2019).

Considered one of the world's biodiversity hotspots, the Cerrado has one of the highest biodiversities in the world with a high abundance of endemic species, harboring approximately 12,829 species of native plants and over 3,167 species of fauna (BFG The Brazil Flora Group, 2021).

The Cerrado is a mosaic of varied vegetation and gradients of coverage and height, ranging from forests to grasslands, which is why the biome is also the second largest in terms of number of phytophysiognomies, with a total of 28 (Eiten, 1972).

According to (RIBEIRO & WALTER, 2008), the main phytophysiognomies of the biome can be grouped into three main formations: **forest formations** (riparian forest, gallery forest, dry forest, and cerradão), **savanna formations** (dense cerrado, typical cerrado, sparse cerrado, rocky cerrado, vereda, cerrado park, and palm grove), and **grassland formations** (dirty field, clean field, and rocky field). Approximately 35% of the biome is represented by the park savanna and woody-grass savanna phytophysiognomies, savanna formations with predominantly herbaceous vegetation.

Figure 21. Vegetation and phytophysognomy map in the project area.



According to the vegetation map of Brazil (IBGE, 2018), 90.97% of the vegetation area is composed of Submontane semi Deciduous Seasonal Forest and 6.51% is savanna.

Table 21. Phytophysognomies in the project area

Abv.	Phytophysiognomy	Area (ha)	Area (%)
Fs	Submontane Semi Deciduous Seasonal Forest	137,182.64	90.97
Sd	Forested Savanna	9,815.07	6.51
Ap	Grasslands	2,774.82	1.84
Vsp	Secondary vegetation	1,027.78	0.68

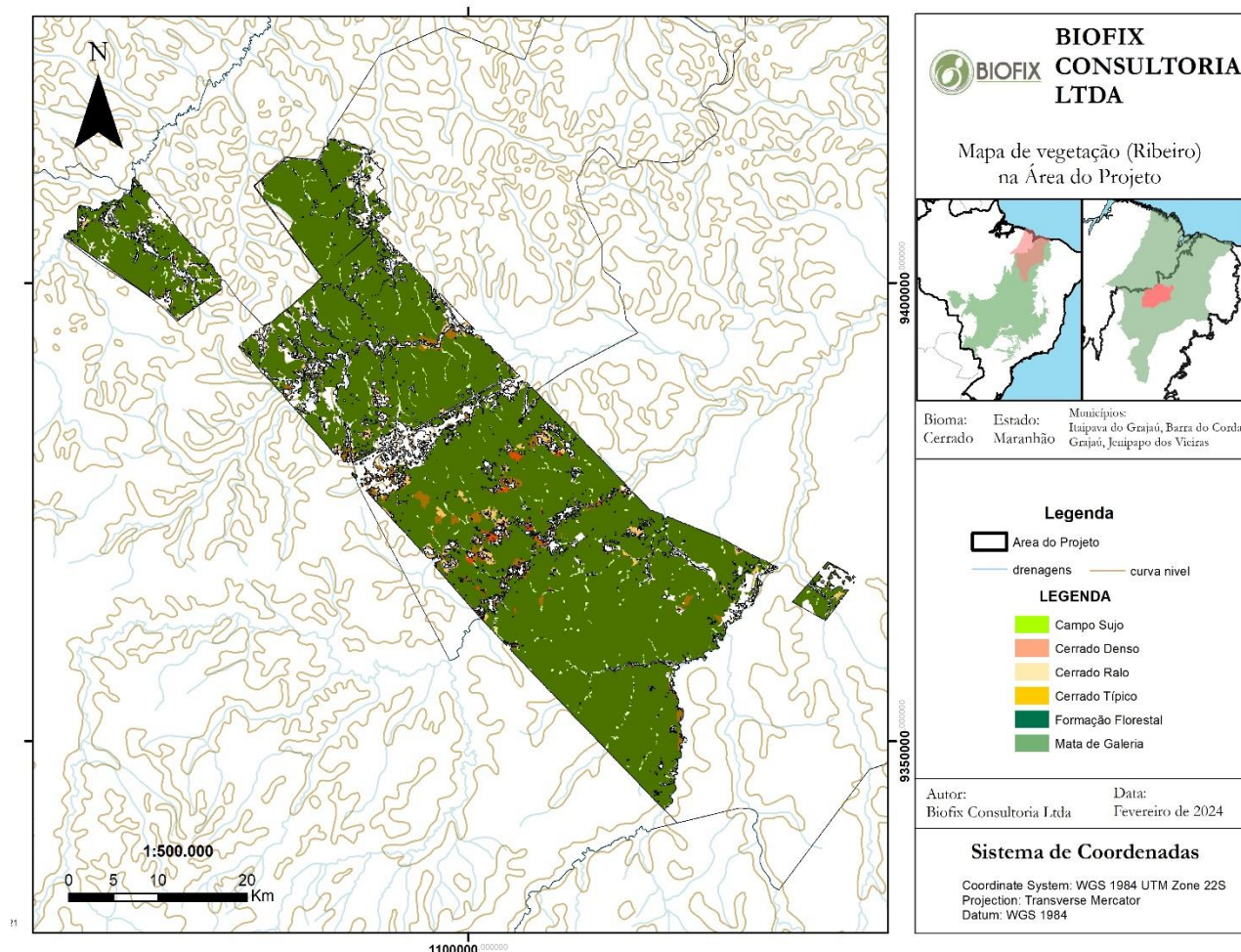
	Total	150,800.31	100.00
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Source: (IBGE, 2018), adapted.

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Figure 22. Vegetation map in the project area according to Ribeiro's phytophysognomias.



Source: (INPE, 2021), adapted.

According to the vegetation map of the Cerrado biome, which takes into account the phytophysiological classes of the Cerrado proposed by Ribeiro and Walter, 95% of the project area is made up of forest formations, including: Ciliary Forest, Gallery forest, Dry Forest and Cerradão⁸.

Ciliary forest: the forest vegetation that follows the medium and large size rivers on Cerrado region, where the trees vegetation do not form galleries. The forest does not occupy a long strip, usually 100 meters to the edges of water bodies. The Ciliary Forest occur normally on the rugged terrain, it may have a transition not always evident to other forestall physiognomies such as Dry Forest and Cerradão. The ciliary forest differs from the gallery forest by the floristic composition and by the deciduous condition, having the ciliary forest, different degrees of deciduousness in the dry season, unlike the gallery forests are perennial.

Gallery Forest: The gallery Forest can be present in two types: Non-Floodable Gallery Forest (where the water table is not close or on the surface of the soil most of the year) and Floodable Gallery Forest (where the water table is close or on the surface of the soil most of the year, even in the dry season).

In both types of forest, the average height of the vertical profile is 20-30 m and the tree coverage is over 90%. Some distinguished species include: breu (*Protium heptaphyllum*), erva-de-gralha (*Psychotria carthagenensis*), morototó (*Schefflera morototoni*), cuia-do-brejo (*Styrax camporum*), pombeiro (*Tapirira guianensis*), and Virola (*Virola sebifera*).

Dry Forest: Dry forest can present in three subtypes depending on the type of soil, floristic composition, and leaf fall in the dry period: Evergreen Dry Forest, Semideciduous Dry Forest, and Deciduous Dry Forest, the first being the most present in the project area. In all three types of dry forest, leaf fall increases organic matter in the soil.

Evergreen Dry Forest presents an average height of the tree layer between 15 and 25 m and an average tree coverage of 70-95%. Some tree species are: monjoliero (*Acacia polyphylla*), Imburana (*Amburana cearensis*), garapa (*Apuleia leiocarpa*), canafístula-preta (*Cassia ferruginea*), Cedro (*Cedro fissilis*), araribá (*Centrolobium tomentosum*), mu-tamba (*Guazuma ulmifolia*), carobá (*Jacaranda brasiliana*, *Jacaranda caroba*), jaca-randá-do-mato (*Machaerium villosum*), cegama-chado (*Physocallimma scaberrimum*), pau-pereira (*Platycamus regnellii*), ipês (*Tabebuia spp.*), capitão (*Terminalia spp.*).

Cerradão: is a forest formation with sclerophyllous characteristics (great occurrence of hard organs in plants, mainly leaves) and xeromorphic (with characteristics such as reduced leaves or succulence, to conserve water to support dry conditions). The average height of the Cerradão canopy varies between 8 and 15 m, with a continuous canopy and a tree cover of 50 to 90%. In the Cerradão, light conditions favor the formation of well-differentiated shrub and herbaceous layers.

The most frequent tree species are: jacaré-de-folha-grande (*Callisthene fasciculata*), jacarandá-do-cerrado (*Dalbergia miscolobium*), jacarandá-muchia (*Machaerium opacum*), maria-pobre (*Dilodendron bippinatum*), mutamba (*Guazuma ulmifolia*), cavalo (*Magonia pubescens*), pau-terra-grande (*Qualea grandiflora*).

As shrubs, the species are marmelada-de-cachorro (*Alibertia edulis*), mama-cadela (*Brosimum gaudichaudii*), unha-de-vaca (*Bauhinia brevipes*), café-bravo (*Casearia sylvestris*), folha-branca (*Miconia albicans*), pau d'olinho (*Copaifera oblongifolia*), and Capim-gordura (*Melinis minutiflora*). In the herbaceous strata, there are grass species of the genus *Aristida*, *Axonopus*, *Paspalum*, and *Trachypogon*.

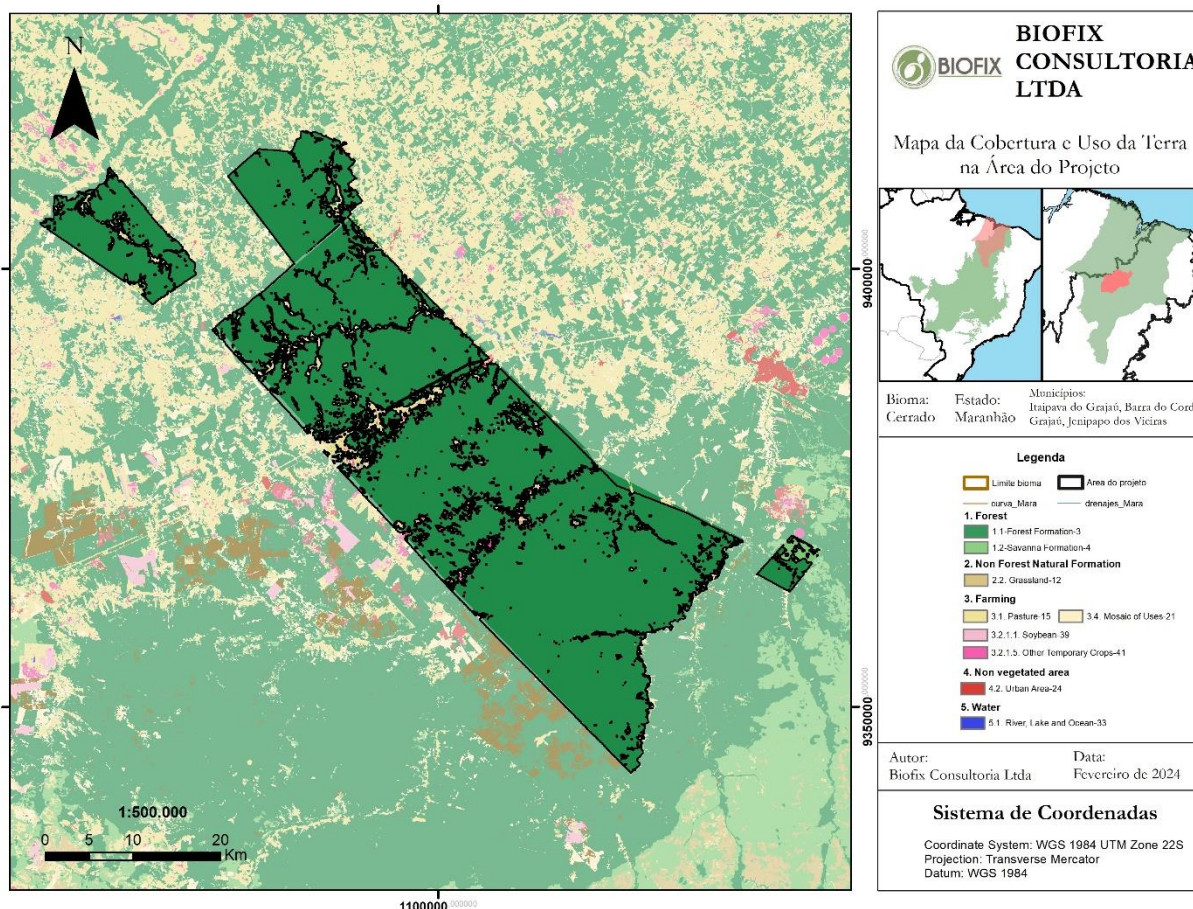
On the other hand less than 5% of the project area correspond to savanna formations, including:

The dense Cerrado is a predominantly arboreal vegetation subtype, with a coverage of 50 to 70% and an average height of 5 to 8 m, being the densest and tallest form of the strict sense Cerrado.

⁸ Translations for: Mata Ciliar, Mata de Galeria, Mata Seca and Cerradão

Coverage and land use

Figure 23. Land cover and land use map of the project area



Source: Mapbiomas⁹

In terms of land cover, according with the data of Mapbiomas, approximately 90% of the project area is covered by forest formations. The remaining percentage is covered by savanna formations, pastures and grasslands.

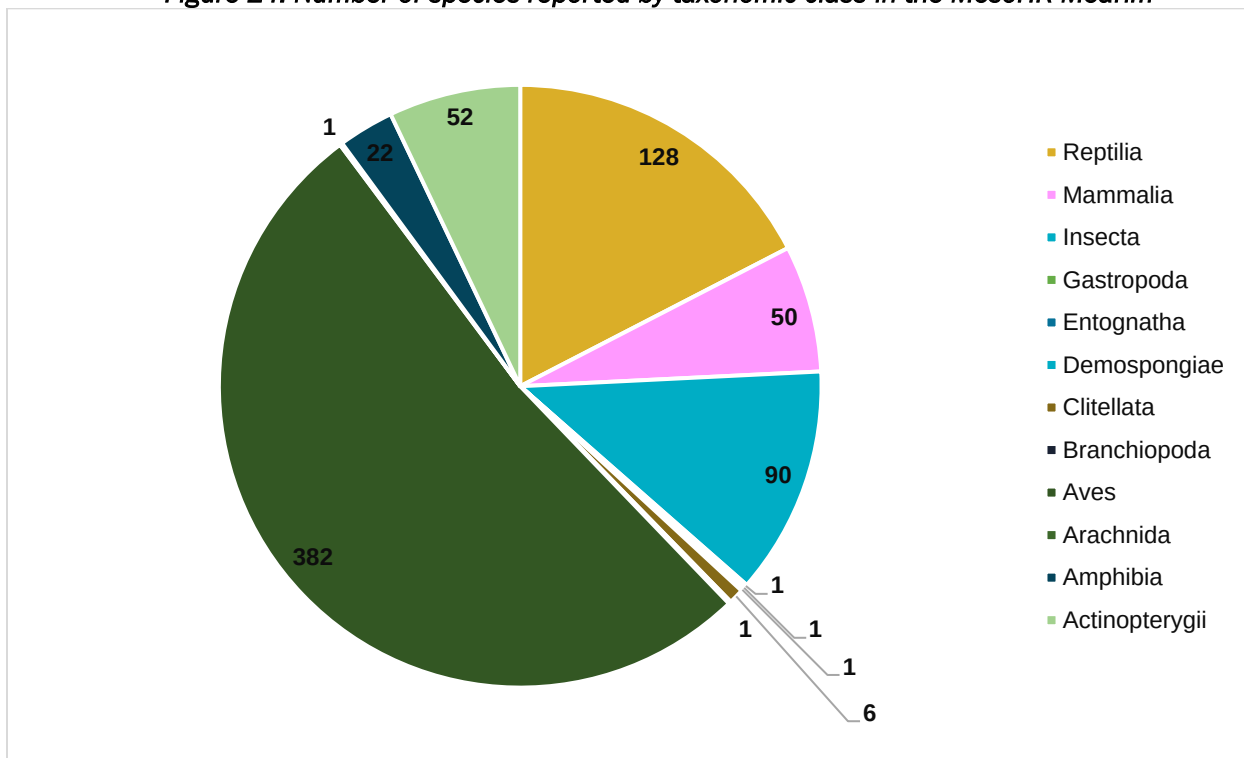
1.14.2.5 Fauna

As showed in 1.14.2.2 section, the project area is located in the Atlântico Nordeste Ocidental hydrographical macroregion, which comprises 3 mesoregions, Gurupi, Itapecuru, and Mearim, being located the project area is located in the Mearim mesoregion, whose main rivers are Mearim river, Grajaú river, Pindaré river, and Zutiua river.

⁹ <https://brasil.mapbiomas.org/colecoes-mapbiomas/>

According to the ICMBio data, in the Mearim mesoregion there is a record of 735 species, of which 52% correspond to the class of birds, 17.4% to the class of reptilia, 12.5% to the class of insects, 7.0% to the class Actinopterygii and 68% to the class mammalia (ICMBio, 2024).

Figure 24. Number of species reported by taxonomic class in the MesoHR Mearim



Source: adapted from ICMBio

Of the 735 species, it is worth highlighting that 15% are endemic, with birds and Actinopterygii being the classes that present the greatest number of endemic species or those with limited distribution in Brazil.

Table 22. Endemic and non-endemic species of Brazil in the MesoHR Mearim

Class	Non endemic	Endemic	Unknown
Actinopterygii	25	27	
Amphibia	20	2	
Arachnida	1		
Aves	342	40	
Branchiopoda	1		
Clitellata		6	
Demospongiae	1		
Entognatha	1		
Gastropoda	1		

Insecta	78	11	1
Mammalia	39	11	
Reptilia	112	16	
Total	621	113	

Source: adapted from ICMBio

Likewise, 91% of the species are classified as being of least concern. However, there are 5 species classified as Critically Endangered **CR**, corresponding to 3 birds: urutau-de-asa-branca (*Nyctibius leucopterus*), jacamim-de-costas-escuras (*Psophia obscura*), mutum-pinima (*Crax fasciolata pinima*); and 2 mammals: Cuxiú-preto/macaco-preto (*Chiropotes satanas*) and caiarara ka'apor/macaco-cara-branca (*Cebus kaapori*).

Likewise, there are 4 species classified as Endangered **EN**, corresponding to 3 birds: maçarico-de-costas-brancas (*Limnodromus griseus*), cabeça-de-prata (*Lepidothrix iris*), trinta-réis-real (*Thalasseus maximus*) and 1 mammals: Gato-do-mato (*Leopardus tigrinus*).

Table 23. Species listed as threatened in the region MesoHR Mearim

Class	Critically Endangered CR	Data Deficient DD	Endangered EN	Least Concern LC	Near Threatened NT	Vulnerable VU
Actinopterygii				52		
Amphibia				22		
Arachnida				1		
Aves	3		3	347	5	20
Branchiopoda				1		
Clitellata		3		2	1	
Demospongiae				1		
Entognatha						1
Gastropoda				1		
Insecta				90		
Mammalia	2	2	1	32	2	11
Reptilia		2		125	1	

Total	5	7	4	674	9	32
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Source: adapted from ICMBio

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Legal and Documentary Aspects of the KAA PYAHU REDD+ Project

The KAA PYAHU REDD+ Project is developed in compliance with local and national laws, statutes and regulations relating to environmental projects, REDD+ and climate change. Importantly, the formulation of the project is aligned with compliance with national and international legislation. This approach ensures strict adherence to regulations, promoting the effectiveness and legitimacy of the project in the legal and regulatory context.

Legislation:

International

UN Declaration on the Rights of Indigenous Peoples and ILO Convention 169

Federal Constitution

Articles 231 and 232 of the Federal Constitution provide for the recognition of the social organization, customs, languages, beliefs and traditions of indigenous communities, as well as the right to the lands they traditionally occupy and the usufruct of their wealth.

Federal legislation

The National Policy for Territorial and Environmental Management of Indigenous Lands - PNGATI, established by Decree 7.747/2012, has the following objectives: "to promote the participation of indigenous peoples, communities and organizations in environmental and territorial protection actions on indigenous lands", as well as "to promote actions aimed at recovering and restoring degraded areas on indigenous lands".

LAW No. 6.001/73 - Indian Statute

LAW No. 5.371/67 - establishing FUNAI

State of Maranhão

LAW No. 9.279 OF OCTOBER 20, 2010 (Establishes the State Environmental Education Policy and the State Environmental Education System of Maranhão)

LAW No. 9.370, OF MAY 13, 2011 (Adds item IV to art. 1 of Law No. 4.734/86, which prohibits the felling of babassu and makes other provisions)

LAW No. 9413, OF JULY 13, 2011 (Regulates art. 241 of the Constitution of the State of Maranhão, Chapter III, Section VII of State Law No. 5405, of April 8, 1992, Chapter II, Section VIII of State Decree No. 13494, of November 12, 1993, and establishes the State System of Nature Conservation Units of Maranhão and makes other provisions)

LAW No. 10.169 OF DECEMBER 5, 2014 (Provides for the protection of all animals at the state level)

LAW No. 10.595, OF MAY 24, 2017 (Establishes the Green Maranhão Program, aimed at fostering and developing projects to support environmental conservation and recovery)

LAW No. 11.365, OF OCTOBER 19, 2020 (Creates and organizes the Environmental School of the State of Maranhão, and makes other provisions.

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LAW Nº 11578 OF 01/11/2021 - Establishes the Policy for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation, the Conservation of Forest Carbon Stocks, Sustainable Forest Management and Increased Forest Carbon Stocks (REDD+), Environmental Asset Management and Payment for Environmental Services (PES) in the State of Maranhão, called the Jurisdictional REDD+ and PES System, and amends State Law No. 11. 000, of April 2, 2019, to expand the scope of action of Maranhão Parcerias - MAPA.

"IX - Private Projects Incentive and Harmonization Program;"

DECREE No. 37969 OF 10/26/2022 - Regulates State Law No. 11. 578, of November 1, 2021, which instituted the Policy for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Forest Management and Increasing Forest Carbon Stocks (REDD+), Management of Environmental Assets and Payment for Environmental Services (PES) in the State of Maranhão, called the Jurisdictional REDD+ and PES System.

"IX - Private Project Incentive and Harmonization Program: aims to encourage and promote private sector investment in projects that could potentially integrate, align and harmonize with the state REDD+ system, facilitating the formulation and implementation of REDD+ promotion, financing, investment and methodology rules by the private sector, contributing to the objectives of this Decree, with the conditions and rules to be applied respecting national legislation, Brazil's international commitments and state rules;"

Legal Requirements (Compliance with Legal Aspects)

The company BIOFIX Consultoria Ltda., as the structuring entity, will conduct an annual assessment of compliance with legal requirements, addressing:

- a. Identification of legal and environmental requirements.
- b. Adjustment of the reference level in accordance with the latest updates to the Forestry Emissions Reference Level (FREL).
- c. Verification of compliance with the new Land Use Schemes in the municipalities of Jenipapo dos Vieiras, Barra do Corda, Grajaú and Itaipava do Grajaú.
- d. Identifying social and environmental safeguards and ensuring their compliance.

Safeguards

Articulation with the Cancún Social and Environmental Safeguards

Context of Compliance with Social and Environmental Safeguards

Social and environmental safeguards originate from the mitigation actions of the United Nations Framework Convention on Climate Change (UNFCCC), defining measures to reduce emissions associated with deforestation and forest degradation, conservation and increasing forest carbon stocks (REDD+). The Cancun Safeguards were established to guide countries implementing REDD+.

The UNFCCC REDD+ Safeguards outline a global framework of social, environmental and governance principles. By complying with this global framework, countries can minimize risks and increase benefits,

both carbon and non-carbon. They apply to all REDD+ initiatives, adapted to local, regional and national scales.

The UNFCCC establishes that the implementation of measures to reduce emissions from deforestation and forest degradation must follow the "Cancun Safeguards". These safeguards include complementary actions with national forest program objectives, transparency in forest governance structures, respect for the rights of indigenous peoples, effective stakeholder participation, and consistency with the conservation of natural forests and biological diversity, actions to avoid risks of reversals of REDD+ results, and actions to reduce the displacement of carbon emissions.

Management of Legal Requirements (Safeguards A, B) - Property Rights and Land Tenure

The 4 territories have an area demarcated by Decree, as follows:

- Cana Brava - Homologated by Decree 246 - 30/10/1991.
- Lagoa Comprida - Homologated by Decree 313 - 30/10/1991.
- Urucu-Juruá - Approved by Decree 382 - 26/12/1991.
- Rodeador - Homologated by Decree 88.813 - 06/10/1983

Socialization and CPLI (Safeguards B, C, D) - Legal and Jurisprudential Framework

The KAA PYAHU project went through a process of socialization and prior, free and informed consultation (CPLI) with the indigenous communities, ensuring that their rights were respected. The consultation was carried out in accordance with international standards and Brazilian legislation. The communities' acceptance was formalized in General Assemblies.

The KAA PYAHU REDD+ Project complies with the Cancun Safeguards established by the UNFCCC. The effective management of legal requirements and the implementation of socio-environmental safeguards guarantee the mitigation of adverse impacts, promoting environmental, social and economic benefits for the communities involved and contributing positively to the fight against climate change.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

The lines of action were designed to improve the general quality of life of the community belonging to the indigenous territories. With the implementation of the REDD+ project, it is hoped that there will be significant contributions to sustainable development, according to each territory.

Institutional and governance strengthening: REDD+ training workshops for leaders and communities, territorial monitoring - Internal Management Plan for the territories, equipment to strengthen the associations, construction of cultural houses, support for participation in regional and national indigenous mobilizations.

Strengthening Education: Building and improving the school structure in the communities, strengthening the mother tongue. Strengthening local culture in schools (handicrafts, traditional crafts, indigenous school education - indigenous education), workshops on traditional cultural activities with teachers from the community), support for indigenous pupils in the municipality to continue their school education, strengthening of traditional festivals.

Participatory reforestation of degraded ecosystems: Irrigation structure, distribution of water for the growth of plantations, planting of native species and training of local agents and forest guardians.

Monitoring and control of deforestation and forest degradation: Territorial surveillance, prevention of fires during critical periods, monitoring of harassment and threats of theft of internal natural resources.

Green businesses: Beekeeping, native seed handicrafts, fish farming, native seedling nurseries, native fruit growing.

Infrastructure: Treatment of drinking water, creation of cooperatives to strengthen internal production activities, construction of a health structure, construction of a flour mill, construction of housing structures and construction of roads.

Strengthening Health: Strengthening traditional medicine, collecting garbage and waste that remains in the territory and generates disease - prevention, planting native species that are used in traditional medicine.

The expectation is that the implementation of the REDD+ project in the Guajajara communities will bring improvements in livelihoods, education, health and environmental preservation efforts, ultimately contributing to sustainable development. The table below describes how the project's activities hope to contribute to the Sustainable Development Goals.

Table 24. National targets related to Sustainable development goals and REDD+ project

SDG	Brazilian targets
6	6.3 By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing the release of chemicals and hazardous materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally.
	6.b Support and strengthen the participation of local communities to improve water and sanitation management.
10	10.2 By 2030, empower and promote the social, economic and political inclusion of all, regardless of age, gender, disability, race, ethnicity, origin, religion, economic or other condition.
13	13.b Promote mechanisms for building capacity for climate change-related planning and effective management in least developed countries, including with a focus on women, youth, local and marginalized communities.
14	14.4 By 2020, effectively regulate harvesting, and end illegal, unreported and unregulated overfishing and destructive fishing practices, and implement science-based management plans, to restore fish populations in the shortest possible time, at least at levels that can produce maximum sustainable yield, as determined by its biological characteristics.
15	15.a Mobilize and significantly increase, from all sources, financial resources for the conservation and sustainable use of biodiversity and ecosystems.

Source: (ONU BR – NAÇÕES UNIDAS NO BRASIL – ONU BR., 2015).

1.18.2 Sustainable Development Contributions Activity Monitoring

Table 25. Sustainable Development Contributions during the monitoring period

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	3.8	Achieve universal health coverage, including financial risk protection, access to quality essential health-care services and access to safe, effective, quality and affordable essential medicines and vaccines for all.	Implemented activities to increase	The Council implemented activities to seek support for basic health care for the Guajajara, through strong internal governance that allows the improvement of the quality of life of people with access to health services.	The Council sent documents to the competent bodies and maintains governance focused on improving the quality of life, through access to basic rights, being a constant struggle on behalf of the populations (Annex 016).
2)	10.3	Ensure equal opportunity and reduce inequalities of outcome, including by eliminating discriminatory laws, policies and practices and promoting appropriate legislation, policies and action in this regard	Implemented activities to increase	The association sought, through representative offices under the territories, access to the issuance of identification documents such as birth certificates and work cards to give dignity to the Guajajara people.	Through an active search of official bodies with the support of partners from the Guajajara people, the council aims to ensure that people have access to basic documents to access services (Annex 15).
3)	13.2.	Total greenhouse gas emissions per year	Implemented activities to decrease	During the monitoring period, the activities of the project have contributed with the reduction of emissions of CO ₂ .	778.241 tCO ₂ e were reduced during the monitoring period (01/05/2021–31/12/2023).

4)	15.7	Take urgent action to end poaching and trafficking of protected species of flora and fauna and address both demand and supply of illegal wildlife products.	Implemented activities to decrease	Over the past few years, illegal invasion of territories for illegal logging and illegal hunting have been denounced by the Guajajara for the protection of biodiversity and people's safety.	The Guajajara officially demonstrated through complaints that are cataloged in the Violence Against Peoples Report Indigenous people in Brazil and also in the media and platforms that can collaborate with actions against illegal invasions (Annex 019).
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Source: KA'A PYAHU REDD+ Project

1.19 Additional Information Relevant to the Project

Leakage Management

The measures for leakage prevention within the project is addressed through the implementation of action lines, aiming to enable stakeholders associated with deforestation to improve their productive practices without the need to deforest or degrade new forest areas. To achieve this, capacity building in sustainable production systems is intended, which generate economic benefits for communities while also benefiting biodiversity, such as habitat connectivity, for example. Planned activities for this purpose include tree planting, the establishment of diverse production systems with species of different characteristics, and the use of organic and/or natural fertilizers, among others.

The proponents of the KA'A PYAHU REDD+ project control the leakages with management activities on the demand of products and services of the forest. Similarly, one of the proposed activities for the following years is to increase technification and agricultural productivity, which minimizes market leakages, all of this through the activities described in section 1.12.

Another advantage is that the implementation of multiple productive components increases profitability, employment and, in general, communal support.

Despite the lack of data, the model scenarios, study cases and general observations can provide information to prevent negative leakages and foment positive leakages, one advantage of multifaceted or integrated projects such as KA'A PYAHU REDD+ is that, utilizing reforestation, plantation and forest conservation programs will pursue to provide diverse and sustained economies which don't cause great interruptions in the market or unwanted migration or movement.

Between 2021 and 2023, an analysis was made on the coverage change of forests to other uses and their percentage compared with the change in the project area. From here on out the respective leakages were identified and were then discounted from the calculated emission reduction.

Verification period, through remote sensors to identify changes in forest cover. The monitoring protocol begins with the digital pre-processing of satellite images, where image acquisition is conducted for the evaluation year. Geometric corrections are applied to these images, and processes for cloud and shadow masking are implemented. Once the correction procedure is established, a mosaic of satellite images over the areas of interest is created, followed by a visual interpretation analysis of changes due to deforestation during the analysis period.

This analysis enables the establishment and determination of the Leakage Belt Forest Cover Monitoring Map, based on obtaining Forest/Non-Forest data provided by Mapbiomas for the crediting years for leakage.

Commercially Sensitive Information

According to the definition of *Commercially Sensitive Information* by Verra, it refers to data or details that, if made public, could harm the competitive advantage or proprietary interests of a business. This type of information, if disclosed publicly, could compromise the competitive position of the project proponents or reveal proprietary methods and strategies. The excluded information is not available in the public domain and is protected under confidentiality agreements with stakeholders. For this project, this includes sensitive items, such as:

- Financial Data: Detailed financial projections, including cash flow, revenue models, estimative prices and cost structures. Detailed cash flow statements have been excluded to protect proprietary financial strategies. The excluded financial data and projections are integral to maintaining a competitive edge in project financing and implementation.

- Legal Agreements: Specific terms of agreements between the project developer and indigenous communities have been redacted to honor confidentiality clauses and prevent competitive harm. Legal agreements have specific terms and conditions of contracts, especially those that involve indigenous communities, land use, and efforts to build strong governance around and inside the project management and permanence. The information is proprietary and not otherwise publicly available. The legal agreements contain sensitive terms that, if disclosed, could undermine negotiations or community relations.
- Personal Data: In accordance with LGPD (General Data Protection Law), we have not disclosed any personal information that is not authorized by the individual or stakeholder.
- Non-permanence Risk Tool: there are financial, legal agreements and other sensitive information as support to responses of the tool. So, if disclosed publicly, could compromise the competitive position of the project and its strategies

Further Information

- There is no further information about this project. All relevant information are included on each Section.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Table 26. Stakeholder identification of the KAA PYAHUU Project

Stakeholder Identification	The proponent of the KA'A PYAHU PROJECT is the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, which legally represents the Guajajara indigenous people of the territories covered by the project. There were initial rapprochement meetings between the council and Biofix, and thereafter face-to-face and virtual meetings for greater depth between the entities and knowledge of the internal governance of the Guajajara indigenous people, Free, Prior and Informed Consent and decision-making structures. In this process there was also the participation of external agents, in the case of Nature Carbon, which is a partner of the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara accompanies them throughout the process. Internal Stakeholders: Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara e Biofix Brasil LTDA. External Stakeholders: Nature Carbon.
Legal or customary tenure/access rights	The territories legally recognized and demarcated by Brazilian law that are part of the project are: Cana Brava Indigenous Land: Approved by Decree 246 - 10/30/1991, located in the jurisdiction of the Legal Amazon in the state of Maranhão, there is no presence of isolated indigenous peoples, it is not in the border strip and there are no overlaps. (Annex 001) Urucu-Juruá Indigenous Land: Approved by Decree 382 - 12/26/1991, located in the jurisdiction of the Legal Amazon in the state of Maranhão, there is no presence of isolated indigenous peoples, it is not in the border strip and there are no overlaps. (Annex 002 e 003) Lagoa Comprida Indigenous Land: Approved by Decree 313 - 10/30/1991, located in the jurisdiction of the Legal Amazon in the state of Maranhão, there is no presence of isolated

	indigenous peoples, it is not in the border strip and there are no overlaps. (Annex 004) Surrounding Indigenous Dominial: Indigenous Dominial by Decree 88,813 - 10/06/1983, located in the jurisdiction of the Legal Amazon in the state of Maranhão, there is no presence of isolated indigenous peoples, it is not in the border strip and there are no overlaps. (Annex 005) There are currently no land conflicts of any nature related to the territories that are part of the project.
Stakeholder diversity and changes over time	The Guajajara indigenous peoples, proponents of the KA'A PYAHU Project, call themselves Tenetehara and belong to the Tupi Guarani linguistic family. The three territories that are part of the project belong to this indigenous population, which is extensive and populates a large part of the state of Maranhão, located in the northeast of Brazil. The Guajajara are one of the most numerous indigenous peoples in Brazil. They inhabit more than 10 Indigenous Lands on the eastern edge of the Amazon, all located in Maranhão. Economy The main subsistence activities are farming, making handicrafts and fishing. In the dry season, from May to November, drilling, felling, burning, mowing and clearing are carried out, while from November to February planting and clearing are done. Some villages have large communal gardens prepared for community projects, to grow rice and fruit for sale. In the communities that take part in the projects, native seeds are planted so that the women can make handicrafts to generate income. The men fish to support their families and also to sell. Gathering is still practiced by almost all the Guajajara. Gathering activities are increasingly being replaced by fruit-growing in the villages and fields. The Guajajara currently plant a large number of fruit and palm trees. The forest product collected in greatest quantities for commercial purposes is honey. Culture Guajajara cosmology is typical of the Tupi-Guarani peoples, distinguishing four categories of supernatural beings, who are generically called karowara: (1) the creators or cultural heroes, responsible for the creation and transformation of the world, with Maíra and the twins Maíraira and Mucuraira being the most important and Zurupari, the creator of plagues and insects, venomous snakes and spiders, a much feared cultural hero; (2) the "owners" of the forests (Ka'a'zar), waters (Y'zar), game (Miar'i'zar) and trees (Wira'zar), who are hostile and much feared for their evil power; (3) the azang, wandering spirits of the dead, also much feared; and (4) the piwara, animal spirits. The life cycle is usually accompanied by a series of rituals. Among these, the initiation rituals, particularly those

	for girls, are the showiest and rich in meaning. There are also a series of rituals to ask Maíra for permission to plant, Miar'i'zar for permission to hunt and Y'zar for permission to fish. The Guajajara people adapted to new internal processes in the face of changes in interethnic relations with other indigenous peoples and also in relations with the state. The traditional way of life undergoes changes, without losing its essence, but which are fundamental for its cultural and social reproduction. Joining projects like this is part of the fundraising agenda for the preservation of forests and also for improving the lives of communities and even supporting cultural appreciation.
Expected changes in well-being	The KA'A PYAHU Project raises expectations of change and significant improvements for the Guajajara without altering their cultures, respecting the cultural diversity present in the territories. Through the meetings, gatherings and workshops, expectations of change through the project were identified for the Guajajara peoples: Support in the fight against illegal invasions of their territories by loggers who illegally remove timber, putting the local population in a dangerous situation; Support for productive activities such as beekeeping and honey production, planting native species for sustenance and handicrafts and other activities of this nature that are part of local subsistence and income generation in a sustainable way; Improving local governance and specialization through courses and workshops to strengthen the autonomy of the Guajajara people. Improvements in the areas of education and health by building support structures and strengthening indigenous school education and traditional Guajajara medicine. In summary, faced with the current scenario of abandonment of public policies, the Guajajara people of the four territories that are proponents of the project aim to strengthen their social, cultural and environmental lives so that their ways of life are properly valued through the benefits brought by REDD+
Location of stakeholders	The KA'A PYAHU Project is located in the state of Maranhão, in the northeast of Brazil, under the jurisdiction of the Legal Amazon. The project's proponents are the Guajajara peoples, inhabitants and holders of three indigenous lands (TI) and one indigenous domain area (DI): Terra Indígena Cana Brava, Terra Indígena Urucu-Juruá, Área Dominial Indígena Rodeador and Terra Indígena Lagoa Comprida. These are the territories directly impacted by the project, in addition to the other indigenous lands and communities also represented by the Conselho Supremo de Caciques e Lideranças Terra Indígena

	Cana Brava Guajajara, all belonging to the Guajajara indigenous people.
Location of resources	The total area of the indigenous land (declared area) is 165,544.30 hectares, (166,228.99 area calculated using GIS software based on the coordinate system WGS_1984_UTM Zone 22S). The resources that communities have access to are water, forest area, raw materials that come from the region, which are necessary resources for the lives of the Guajajara people.

Source: KA'A PYAHU REDD+ Project.

2.1.2 Stakeholder Consultation and Ongoing Communication

Table 27. Stakeholder consultation KAA PYAHUU project

Date of stakeholder consultation		16-November-2023
Stakeholder process	engagement	The process of approaching and consulting with the Guajajara indigenous people respected a series of stages, building a timeline since the first rapprochement meeting between Biofix and the people represented by the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara. - 04/25/2023, in the month that celebrates the day of indigenous peoples in Brazil, the first presentation meeting took place between the Guajajara people and the Biofix team, in this meeting, after the presentation on the carbon market in indigenous lands and the Biofix's work methodology for the leaders, the interest of community representatives in continuing the dialogue with Biofix was demonstrated (Annex 020). After this period, international meetings took place in the communities about the importance of the project and the internal consultation of the Guajajara people in relation to the KA'A PYAHU Project. - 10/16/2023, a meeting took place in the capital of the state of Maranhão, Imperatriz, with members of the Biofix team, Nature Carbon and representatives of the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara. At this meeting, Biofix made a detailed formal presentation of its working methodology with indigenous populations, presented the general concepts of the market and the stages of carbon projects with traditional populations, the roles and responsibilities within the project. At this meeting, the Letter of Intent of the Guajajara people represented by the association was signed to participate in the project with Biofix. (Annexes 010, 021, 029 and 029_A).

	- 10/20/2023, in Aldeia Coquinho, Guajajara village, an important meeting took place with Guajajara leaders and representatives organized by the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara. This internal meeting between indigenous peoples aimed to internally discuss more issues related to the KA'A PYAHU project and begin preparations for the general Free, Prior and Informed Consent assembly. (Annex 022). - 11/16/2023, the general Free, Prior and Informed Consent meeting took place, which will be detailed below in item 2.1.3 (Annex 023). 06/12/2023, on this day the official signing of the minutes of the general assembly by indigenous leaders took place, a committee was set up to collect the signatures of the four indigenous territories and domains. The meeting was also important to begin the survey of the retroactive projects of the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara (Annex 024). - 01/20/2024, since the general assembly, Biofix has met periodically with the leaders of the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara to align on the project, writing the document referring to the Ka'a Pyahu project, with the objective of keeping populations and communities updated about each stage of the project, fulfilling the continuity and partnership agreement between Biofix and the indigenous territories that participate in the project (Annex 025).
Consultation outcome	Since the first contacts with the Guajajara people represented by the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara Biofix has kept leaders informed about the processes and stages of carbon projects, from consent to the design and implementation stages. People are informed about the importance of active participation in all processes through continuous meetings. The choice of retroactive projects for the preparation of the project, the mapping of local actors, all of this respects the communication and protagonism of the people involved in the KA'A PYAHU project. The KA'A PYAHU Project is developed in compliance with local and national laws, statutes and regulations relating to environmental projects, REDD+ and climate change. Importantly, the formulation of the project is aligned with compliance with national and international legislation. This approach ensures strict adherence to regulations, promoting the effectiveness and legitimacy of the project in the legal and regulatory context.

Legislation:

International

UN Declaration on the Rights of Indigenous Peoples and ILO Convention 169 (Annex 038)

Federal Constitution 1988

Articles 231 and 232 of the Federal Constitution provide for the recognition of the social organization, customs, languages, beliefs and traditions of indigenous communities, as well as the right to the lands they traditionally occupy and the usufruct of their wealth. (Annex 039)

Federal legislation

The National Policy for Territorial and Environmental Management of Indigenous Lands - PNGATI, established by Decree 7.747/2012, has the following objectives: "to promote the participation of indigenous peoples, communities and organizations in environmental and territorial protection actions on indigenous lands", as well as "to promote actions aimed at recovering and restoring degraded areas on indigenous lands". (Annex 040)

LAW No. 6.001/73 - Indian Statute (Annex 041)

LAW No. 5.371/67 - establishing FUNAI (Annex 042)

State of Maranhão (Annex 043)

LAW No. 9.279 OF OCTOBER 20, 2010 (Establishes the State Environmental Education Policy and the State Environmental Education System of Maranhão)

LAW No. 9.370, OF MAY 13, 2011 (Adds item IV to art. 1 of Law No. 4.734/86, which prohibits the felling of babassu and makes other provisions)

LAW No. 9.413, OF JULY 13, 2011 (Regulates art. 241 of the Constitution of the State of Maranhão, Chapter III, Section VII of State Law No. 5405, of April 8, 1992, Chapter II, Section VIII of State Decree No. 13494, of November 12, 1993, and establishes the State System of Nature Conservation Units of Maranhão and makes other provisions)

LAW No. 10.169 OF DECEMBER 5, 2014 (Provides for the protection of all animals at the state level)

LAW No. 10.595, OF MAY 24, 2017 (Establishes the Green Maranhão Program, aimed at fostering and developing projects to support environmental conservation and recovery)

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Safeguards

Articulation with the Cancún Social and Environmental Safeguards. Social and environmental safeguards originate from the mitigation actions of the United Nations Framework Convention on Climate Change (UNFCCC), defining measures to reduce emissions associated with deforestation and forest degradation, conservation and increasing forest carbon stocks (REDD+). The Cancun Safeguards were established to guide countries implementing REDD+.

	The UNFCCC REDD+ Safeguards outline a global framework of social, environmental and governance principles. By complying with this global framework, countries can minimize risks and increase benefits, both carbon and non-carbon. They apply to all REDD+ initiatives, adapted to local, regional and national scales. (Annex 045) The UNFCCC establishes that the implementation of measures to reduce emissions from deforestation and forest degradation must follow the "Cancun Safeguards". These safeguards include complementary actions with national forest program objectives, transparency in forest governance structures, respect for the rights of indigenous peoples, effective stakeholder participation, and consistency with the conservation of natural forests and biological diversity, actions to avoid risks of reversals of REDD+ results, and actions to reduce the displacement of carbon emissions. The KAA PYAHU Project went through a process of socialization and prior, free and informed consultation (CPLI) with the indigenous communities, ensuring that their rights were respected. The consultation was carried out in accordance with international standards and Brazilian legislation. The communities' acceptance was formalized in General Assemblies. The KA'A PYAHU Project complies with the Cancun Safeguards established by the UNFCCC. The effective management of legal requirements and the implementation of socio-environmental safeguards guarantee the mitigation of adverse impacts, promoting environmental, social and economic benefits for the communities involved and contributing positively to the fight against climate change.
Ongoing communication	Continuous communication with the Guajajara people of the communities involved in the KA'A PYAHU project is essential for the smooth running of the project. From the first rapprochement meetings, the joint preparation and information meetings to generate Free, Prior and Informed Consent, until then, periodic meetings are held in virtual format fortnightly to monitor each stage of the formulation, current stage of the project, up to the most advanced stages. (Annexes 023,027,028 and 029). When necessary, face-to-face meetings will take place. Minutes are generated for all meetings and sufficient time is guaranteed to bring together leaders so that all meetings are representative of all territories and regions involved in the project. The meetings take place with the leaders of the four territories and the domain in Aldeia Coquinho and Biofix virtually.

	The procedure for documenting the outcomes of meetings, as informative assemblies or even meetings with Legal Representative or his team, is through minutes or reports, that are made for each meeting. These documents have the topics addressed in the meeting, the commitments for each stakeholder that participate, and the conclusions. With these, the assistance lists are attached. These minutes or reports are made and approved between the participants of meetings, after that these are delivered to the participants. Furthermore, these documents are included in project file management.
Stakeholder input	In all processes that culminated in the Free, Prior and Informed Consultation and on the day of the general consultation assembly, construction and study meetings were held with the people proposing the KA'A PYAHU project, in which indigenous protagonism was taken into account in all the steps. The construction of meeting formats, translators in the mother tongue, definition of retroactive projects, all stages of study and project formulation are carried out in partnership and co-participation with the indigenous communities involved. It is important to highlight that the contract was also made available in audio format, using terms in Tupi Guarani, the mother tongue of the Guajajara people. The form of partnership in which the KA'A PYAHU project is built is fundamental for the results to comply with the due social parameters and the safeguards described previously. All communications comply with the rules established by the internal governance of the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara.

Source: KA'A PYAHU REDD+ Project.

2.1.3 Free Prior and Informed Consent

Table 28. Free Prior and Informed Consent for the KA'A PYAHU Project

Obtaining consent	Free, prior and Informed Consent occurred during several preparatory stages culminating in the general meeting. As described in previous item 2.1.2, there were several meetings so that the communities represented by the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara were informed and guided decision-making. Before starting the process, local information must be gathered for the Free, Prior and Informed Consent undertaking, understanding the local context, carrying out prior research on elements of the local context that are important for carrying out the FPIC process together with local
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leaders. After this, it is essential to understand the legal and customary rights of people, identify and respect traditional decision-making structures, undertake an approach that takes cultural aspects into account and collaborate in the formulation and implementation: cultural norms must be integrated into the process of FPIC. Ensure the exchange of information through translators in the mother tongue, information must be presented in a language understandable to the community, in a timely manner and adapted to their culture. Ensure accountability: An ombudsman mechanism that can address FPIC violations must be established and the community must agree on how the project will be monitored to determine when the FPIC process will need to be renegotiated. Knowing this and with a commitment to transparency, all the steps mentioned were carried out from the first meetings to the assembly with the Guajajara people.

After the rapprochement, clarification and project presentation meetings (Annexes 020, 021, 022, 023, 024 and 025), the general meeting took place on 11/16/2023 (Annex 023). The assembly took place in the state of Maranhão, in the Coquinho village, respecting the internal governance that chose a location that was suitable for bringing together the leaders, was strategic and guaranteed representation. The agenda was the presentation of the four territories and the leaders involved in the project, checking the representation list to ensure that all interested parties, representatives of their communities were present, presentation of the KA'A PYAHU project in partnership with Biofix Brasil, internal deliberations and deliberation on the partnership contract signed after the consent of indigenous leaders.

The Biofix team explained, together with an interpreter in their mother tongue, the processes of a REDD+ project in the Brazilian voluntary market, the challenges and benefits that the project generates for the proposing communities. It was explained that the KA'A PYAHU Project does not negatively affect the means of subsistence, the relationship of communities with their territory, nor their social fabric. The activities to be carried out in the execution of the project's resources will be exposed and prioritized by the community itself, where Biofix is just a guiding and facilitating entity so that the REDD + project is maintained over time.

After the explanations, the leaders present at the meeting asked questions that were answered by Biofix regarding the formulation, implementation and benefit generation phases. Questions were also answered about what the project would affect the way of life of the communities, it was clarified that there will be no change in the Guajajara way of life with the KA'A PYAHU Project.

The assembly was the result of a process of prior and informative meetings about the KA'A PYAHU Project that

	culminated in a meeting with all the leaders of all the communities represented by the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara. The attendance lists with the signatures of all leaders are in annexes 27-37. After all processes carried out in accordance with national standards that govern projects developed in Indigenous Lands, which must respect the specific laws guaranteed by the 1988 Constitution, the result of the previous demands of the Indian Statute and the entire set of nationally recognized laws that protect the rights of indigenous peoples in Brazil, including the National Policy for the Sustainable Development of Traditional Peoples and Communities established by Decree no. 6,040/2007. (Annex 045)
Outcome of FPIC	After the entire process of information, doubts and consultation in the assembly, there was a moment of voting on whether or not the Guajajara communities would accept the KA'A PYAHU Project. The project was unanimously voted to approve. These were the decisions taken at the meeting: it is ratified at the meeting that Biofix Consultoria will be the developer of the REDD + KA'A PYAHU Project in the Cana Brava TI, Rodeador TI, Lagoa Comprida and Urucu Juruá. Due to the vote of the Guajajara representatives based on the experience that the company has in formulating projects and marketing Carbon Credits. The REDD + KA'A PYAHU project does not negatively affect livelihoods or the relationship between communities and their territory. Indigenous associations and leaders allow Biofix to carry out the studies necessary to structure the project, such as environmental and social studies of the territories participating in the project to calculate and obtain carbon bonds. The assembly decided to give its Free, Prior and Informed Consent for the implementation of the REDD + project in the Cana Brava TI, Rodeador TI, Lagoa Comprida and Urucu Juruá. Accepts the commitments and terms of the agreement, deciding to invest the project resources in the lines of action prioritized through the needs identification workshop. About the responsibilities of indigenous communities: collaborating with Biofix and the partner to execute and develop the project. Provide the documents and information that Biofix requests, including that necessary for the registration, issuance and sale of Carbon Credits. Allow the use of its legally demarcated territory for the development and execution of the Project, which will generate Carbon Credits. Notify Biofix about situations that may cause problems in the project, this includes informing about changes in management and direction that occur in the Association, in its legal representation or in decision-making bodies. Ensure execution of the project document. Collaborate with Biofix logistics and access in your territory when necessary to develop the necessary steps for the development of the project, informing about situations that impact the Project.

Furthermore, it is agreed that communities must fully participate in project socialization and approval activities, strive to conserve the forests in their territory and restore their degraded areas when there is a threat or damage to their integrity, with the support of the authorities. Do not hire other individuals or legal entities for the Commercialization or Issuance of Carbon Credits in your territory until the end of the contract with Biofix, guaranteeing your exclusivity.

The Nature Carbon partner's responsibilities are as follows: collaborate with the association and Biofix to execute and develop the Project, collect the requested documentation, dialogue, interact and maintain a good relationship with the ethnic authorities and administrative communities in the region, witness and monitor socialization activities, socio-environmental workshops and project actions in Indigenous communities, inform about situations that impact the Project. When necessary, send the Minutes of recognition or possession of traditional authorities.

Biofix's responsibilities: develop and execute the project, respecting all stages and changes that arise. Comply with all Brazilian legal regulations regarding indigenous communities. This includes socio-environmental safeguards, which are rules to strengthen the positive impacts and reduce the negative impacts of project actions. Carry out the sale and issuance of carbon credits in the Association's Indigenous communities. Execute the Issuance and Commercialization operations of carbon credits through a project, ensure that the communities represented by the association participate in the project, access information and exercise their autonomy, free determination and consent, among other responsibilities recorded in the minutes (Annex 023).

Approach made legally with the Guajajara people and committed to the rights of indigenous peoples, identification of internal governance, documentation of all stages of the project development processes, knowing the importance that all stages must be recorded and documented and these documents must be shared among the proposing indigenous peoples in language accessible to all interested parties. Provision of information and transparent access to the Guajajara people. Monitoring and communication at all stages: the stages must be recorded, monitored and shared between the parties, as well as active maintenance of communication. Respect for autonomy and self-determination: identification of cultural specificities and respect for the forms of decision-making and political and social structures of each people. Mapping and attention to indigenous representation and their rights: full attention to national and internal rights determined by each people during all stages and processes of project execution. Maintain an informative and transparent relationship with public bodies linked to the indigenous agenda: contact FUNAI, SESAI and CASAI in advance about the activities to be developed in the

	territory. FUNAI is Brazil's official indigenous body and has among its responsibilities to delimit, monitor and manage consultations with Indigenous Peoples and Indigenous Lands without this meaning the responsibility to approve or not the implementation of projects in the territories. (Annexes 027 and 028). For a full video description, see: https://www.youtube.com/watch?v=GF0zGsjZsWA .
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Source: KA'A PYAHU REDD+ Project.

2.1.4 Grievance Redress Procedure

Table 29. Grievance redress procedure

Development process	The KA'A PYAHU Project guarantees to respect and adopt all necessary measures to ensure the right to free, prior and informed consent (FPIC), in accordance with Convention No. 169 of the International Labor Organization (ILO), aggregate relevant information to REDD+ in the SIS, coordination or support from the CGU for conflict resolution. It was established that divergences, conflicts or controversies regarding the formulation, celebration, validity, execution, interpretation, termination and/or settlement of the contract will be resolved by mutual agreement between the parties, paying attention to the resolution mechanisms before the Ministry of the Environment and the National Indian Foundation. Bodies that in 2012 held a seminar to debate the indigenous issue within the scope of the implementation of REDD+ in Brazil. This resulted in the preparation of the document "Assumptions for the implementation of REDD+ in Indigenous Lands" (Annexes 025 and 023). Differences that cannot be resolved through the mediations set out above will be appealed to the Federal Attorney General's Office (PGU), which is a higher management body of the Federal Attorney General's Office (AGU), as provided for in the document, "Set of premises for implementing REDD+ in Indigenous Lands". (Annex 046)
Grievance redress procedure	The Project will use four main means of communication (verbal, written, in-person and virtual) with stakeholders in order to promote opportunities for discussion and participation throughout the development of the project, as well as ensuring that its development and implementation, including the Monitoring results and the validation and verification processes of the VCS Program are communicated to all interested parties, including access to all documents and information related to the project. With the aim of achieving a good quality of communication and exchange process between interested parties, a plan of periodic meetings with leaders will be implemented and consolidated in order to keep them informed about the project stages.

	- Receiving email for requests Communities will have a permanent consultation channel to make their requests and receive a response within the appropriate deadline. Through email: contatobrasil@biofix.com.br you can have direct contact with the company, clarify your doubts and/or requests
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Source: KA'A PYAHU REDD+ Project.

Table 30. Grievances received

Grievances received	Resolution and outcome
No grievances were received during the monitoring period.	No grievances were received during the monitoring period..

Source: KA'A PYAHU REDD+ Project.

2.1.5 Public Comments

Table 31. Public comments of the KAA PYAHU project

Comments received	Actions taken
The project has not yet gone through public comment. All comments received will be duly answered and evaluated by OVV.	The project has not yet gone through public comment. All comments received will be duly answered and evaluated by OVV..

Source: KA'A PYAHU REDD+ Project.

2.2 Risks to Stakeholders and the Environment

The possible risks previously identified in the KA'A PYAHU Project pay close attention to the current production of the National Commission for REDD+, established by Decree No. 11,548, of June 5, 2023, which is responsible for coordinating, monitoring, and reviewing the coordinate the development of requirements for access to payments for results of REDD+ policies and actions in Brazil, recognized by the United Nations Framework Convention on Climate Change (UNFCCC). The risks are also observed in accordance with Brazil's National Strategy for REDD+.

Additionally, we are guided by the REDD+ Safeguards mandated by the United Nations Framework Convention on Climate Change (UNFCCC), which describe a global framework of social, environmental and governance principles under which activities and actions should be implemented. By complying with the content of the aforementioned global framework, countries can minimize the risks of REDD+ measures and increase the possibility of achieving REDD+ results.

These safeguards seek, together, to reduce possible risks associated with the implementation of REDD+ projects and programs. At the same time, they maximize the potential for additional benefits from carbon sequestration. Likewise, these safeguards seek to guarantee the protection of rights and fulfillment of obligations of actors surrounding a REDD+ project. Among its benefits are protection against biodiversity loss, security of land ownership, training of stakeholders and strengthening forest governance, among other related aspects.

REDD+ safeguards apply to all REDD+ initiatives, including results-based payments, project design, and other actions aimed at reducing emissions associated with deforestation. Furthermore, they are adapted to local, regional and national scales and must be addressed and respected by all types of actors (communities, beneficiaries, implementers, financiers, donors, government, among others).

The UNFCCC establishes that the implementation of measures aimed at reducing emissions caused by deforestation and forest degradation must be carried out in accordance with the “Cancún Safeguards”, which appear in Annex X - Decision 1CP/16. Guided by these regulations and the work focused on the cultural specificities of the Guajajara people, we identified the following risks:

Table 32. Risk identified to the project

	Risks identified	Mitigation or preventative measure taken
Risks to stakeholder participation	Change in the productive activities of communities.	During the process of formulating the REDD+ project, the lines of action were defined together with certain communities with the support of the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara. Through understanding the needs of the Guajajara people, we understand that the Ka’a Pyahu Project, through previously mapped actions, does not harm the traditionality of the Guajajara people. This approach took into account not only the current priorities of communities, but also the priorities that have been identified over the years and reflected in these instruments. Therefore, to ensure that communities can continue with the productive activities in which they are involved, and even improve them, there is a process of technical and financial evaluation of the activities carried out within the scope of the project.
Working conditions	Through respect for the internal governance of the Guajajara indigenous people and guided by safeguards regarding autonomy and respect for internal structures, undue forms and conditions of work will be avoided.	The project has mechanisms to deal with internal governance, through periodic meetings, constant dialogues with leaders and chiefs and following and respecting the following safeguards: B. Transparent and effective national forest governance structures, taking into account national sovereignty and national legislation. Ensure that the government and relevant social actors, in particular indigenous peoples, traditional peoples and communities and traditional and family farmers, actively participate in the

		planning, implementation, monitoring and evaluation processes of policies related to the conservation, recovery and sustainable use of forests and natural ecosystems, especially in decision-making, with transparency in these decisions being ensured. These structures must guarantee the broad participation of society, compliance with current legislation, the guarantee of rights and sustainability, in accordance with the foundations and objectives of the Federal Constitution.
Safety of women and girls	No risk identified	The KA'A PYAHU Project does not identify any risk to the safety of women and girls in its implementation stages.
Safety of minority and marginalized groups, including children	No risk identified	The KA'A PYAHU Project does not pose a risk to the safety of marginalized minority groups, including children.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	Emission reversals and displacement	The KA'A PYAHU Project aims to preserve forest areas, rivers and also protect against risks to the environment such as fire, illegal hunting and illegal activities such as the removal of wood from the project's indigenous territories. GHG emitting sources will be monitored regularly, and must serve to guide command and control actions. The prior implementation of satellite monitoring already aims to overcome the risks of reversals and displacement of emissions, in addition to contributing to the conservation of fauna and flora in the territories proposing the project.

Source: KA'A PYAHU REDD+ Project.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Table 33. Rights related to work and labor in the project

Discrimination and sexual harassment	In the development of the project, there are two main actors responsible for ensuring the good implementation of the project: the development company BIOFIX CONSULTORIA LTDA. and the project proponent, the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara. Each of them has mechanisms to prevent discrimination and sexual harassment: BIOFIX CONSULTORIA LTDA. has mechanisms that include background checks on several national databases for individuals
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	entering into project implementation. Furthermore, within the company's internal regulations and policies, values associated with respect for differences, collaborative work with communities and any conduct that could lead to discrimination and sexual harassment are strictly prohibited. In turn, the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara also promotes respect in its internal regulations for all forms of life and the territories they inhabit, with special recognition of their history of constant struggle for the recognition of their rights.
Management experience	From the formulation to the implementation of the project, a team with varied technical capabilities is essential to guide the community and for the proper functioning of the project and capabilities that correspond to the demands of preparing and implementing the project. The following formulations are required: Quantifying emissions and understanding forestry, spatial and temporal analysis of ecosystems, approaching the community and understanding their needs, knowledge of biodiversity and its use, productive activities and their impacts on ecosystems. To meet these demands, the project has a team specialized in the environmental area and projects focused on nature-based solutions, an experienced legal team and anthropological knowledge for respectful and technical relationships with communities. Knowledge and previous experience in implementing land and carbon management projects with community involvement at project scale can be found on the company's website at: https://biofix.co/
Gender equity in labor and work	Throughout the approach and assembly with the Guajajara people, Biofix provided a joint work process with the communities that make up the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara and the interested parties that represent them, to fully guarantee the rights of participation of ethnic communities, which implies decision-making within the scope of project design by them as holders of the conservation initiative. In the consultation process, an intergenerational and gender approach was considered, with special attention to vulnerable and/or marginalized people.
Human trafficking, forced labor, and child labor	The proponents of the KA'A PYAHU Project comply with legal requirements related to human rights and abhor any form of forced labor, work carried out by children, as well as any activities that put people's lives at risk.

Source: KA'A PYAHU REDD+ Project.

2.3.2 Human Rights

Coordination with the Socio-Environmental Safeguards of Cancún and context of compliance with social and environmental safeguards for the KA'A PYAHU project

Social and environmental safeguards arise from the mitigation actions of the United Nations Framework Convention on Climate Change (UNFCCC), which defines in decision 1CP/16 the measures to reduce emissions associated with deforestation and forest degradation, conservation and increase of stocks of

forest carbon, better known as REDD+. Within these, the Cancún safeguards were established (where the conference took place), for which guidelines were agreed for REDD+ implementing countries.

The UNFCCC REDD+ Safeguards describe a global framework of social, environmental and governance principles under which activities and actions should be implemented. By complying with the content of the aforementioned global framework, countries can minimize the risks of REDD+ measures and increase the possibility of obtaining REDD+ benefits, both with and without carbon.

These safeguards seek, together, to reduce possible risks associated with the implementation of REDD+ projects and programs. At the same time, they maximize the potential for additional benefits from carbon sequestration. Likewise, these safeguards seek to guarantee the protection of rights and compliance with obligations of actors surrounding a REDD+ project. Among its benefits are protection against biodiversity loss, security of land ownership, empowerment of stakeholders and strengthening forest governance, among other related aspects.

REDD+ safeguards apply to all REDD+ initiatives, including results-based payments, project design and other actions aimed at reducing emissions associated with deforestation. Furthermore, they are adapted to local, regional and national scales and must be addressed and respected by all types of actors (communities, beneficiaries, implementers, financiers, donors, government, among others).

The UNFCCC establishes that the implementation of measures aimed at reducing emissions caused by deforestation and forest degradation must be carried out in accordance with the “Cancún Safeguards”, which are contained in Annex X – Decision 1CP/16, namely:

- 1) Complementary or consistent actions with the objectives of national forest programs and other relevant international conventions and agreements;
- 2) Transparent and effective national forest governance structures, taking into account national sovereignty and national legislation;
- 3) Respect for the knowledge and rights of indigenous peoples and members of local communities, taking into account relevant international obligations, national laws, and the United Nations Declaration on the Rights of Indigenous Peoples;
- 4) Full and effective participation of stakeholders, particularly indigenous peoples and local communities;
- 5) Actions consistent with the conservation of natural forests and biological diversity, ensuring that the actions referred to in paragraph 70 of Decision 1/CP 16 are not used for the conversion of natural forests but rather to encourage their protection and conservation and their ecosystem services, as well as to contribute to other social and environmental benefits;
- 6) Actions to avoid the risks of REDD+ results reversals; and
- 7) Actions to reduce the displacement of carbon emissions to other areas.

Indigenous Land Demarcation Process in Brazil and Indigenous Peoples' Rights to Territories

In Brazil, the demarcation of an Indigenous Land aims to guarantee indigenous people's right to land. Its duty is to establish the actual extent of indigenous possession, ensuring the protection of the demarcated boundaries and preventing occupation by third parties.

Since the approval of the Indian Statute in 1973, this formal recognition has been subject to an administrative procedure, provided for in Article 19 of Law 6001. Such procedure, which establishes the stages of the lengthy demarcation process, is regulated by an Executive decree and, over the years, has undergone several modifications. The last important modification occurred with Decree 1,775 of January 1996.

Demarcation is an exclusive competence of the Executive Power, according to the Federal Constitution, as it is merely an administrative process: the indigenous people's right to the land is original, that is, they were on these lands before the formation of the National State.

Under the terms of Decree 1775, the indigenous group involved, represented according to its own forms of organization, will participate in the procedure in all stages of land regularization, which has seven stages that will be described below:

Before the identification studies, there is a demand verification. The indigenous people make the territorial claim, and FUNAI sends an anthropology professional for Demand Qualification, that is, the execution of a preliminary characterization report of the claimed territory, the social, political, and economic context that drive the demand, and general information about the indigenous people who presented the request.

1. Identification and delimitation studies, under the responsibility of FUNAI. A Technical Group is formed here, coordinated by an anthropologist appointed by FUNAI, to conduct research and prepare the Circumstantial Report of Identification and Delimitation of the Indigenous Land (RCID) in accordance with the rules established by the Portaria MJ n 14/99. The traditionally occupied lands are supported by anthropological studies, supported by interdisciplinary research in environmental, historical, legal, agrarian, cartographic, and other related areas. The report also includes a land survey to identify and census any non-indigenous occupants in the demarcated area, a qualitative characterization of such occupation, a survey of data on any possession and/or ownership titles of lands incident to the territory claimed by the indigenous people and identified by the Technical Group. The Decree provides that public agencies must provide information on the area being identified. After the completion and delivery of the RCID to FUNAI with the characterization of the IL to be demarcated, the Report is submitted for approval by the federal indigenous assistance agency. After the summary is approved, it is published in the Official Gazette of the Union and in the Official Gazette of the federal unit where the demarcated area is located, accompanied by a descriptive memorial and map of the area. Such publication must be posted at the headquarters of the Municipal Prefecture where the property is located.

2. Administrative Contradiction and Contests. The contradiction implies the right of the States and municipalities where the area under demarcation is located, and other interested parties to express themselves, presenting to FUNAI reasons supported by evidence that they consider relevant, such as dominial titles, expert reports, opinions, witness statements, photographs, and maps, to claim compensation or demonstrate total or partial defects in the report according to their interests in the territory claimed by the indigenous people. The Decree establishes a deadline of up to ninety days after the publication of the summary of the RCDI for the beginning of the demarcation procedure, and the judgment of such procedure is the responsibility of the Minister of Justice.

3. Declaration of the limits of the Indigenous Land, under the responsibility of the Minister of Justice. This refers to the declaration, by means of an administrative order, of the limits of the indigenous land and determining its demarcation. Usually known as a declaratory order, this phase occurs after a deadline for administrative contradiction as mentioned above, and in case of existing contests, the declaration of the limits is decided by the Minister after judgment of the procedures of the Contradiction.

4. Physical Demarcation, under the responsibility of FUNAI. The setting of the landmarks that establish the physical limits of the TI is homologated by means of a decree and executed by technicians designated by FUNAI. After this stage, a land survey is carried out to assess the improvements implemented by non-indigenous occupants, under the responsibility of FUNAI, carried out jointly with the registration of non-indigenous occupants, under the responsibility of Incra. Based on the land survey of the RCDI, a technical team designated by FUNAI assesses the improvements, establishing financial values for compensation purposes in the case of occupations considered in good faith according to the legislation.

5. Homologation of demarcation, by the Presidency of the Republic. This act is carried out by presidential decree. At this stage of the process, non-Indigenous occupants are removed, with payment for

improvements considered in good faith, by FUNAI, and resettlement of non-Indigenous occupants who meet the profile of agrarian reform, by Incra;

6. Registration of the Indigenous Lands in the Union Heritage Secretariat, by FUNAI. According to the law, registration must occur within thirty days after the publication of the homologation decree.

Indigenous Rights over Demarcated Lands

The lands traditionally occupied by indigenous peoples are assets of the Union as provided for in article 20, item XI. In light of the classification of public assets proposed by national doctrine and article 231, paragraph 4, of the CF/88, which establishes that "the lands referred to in this article are inalienable and unavailable, and the rights over them are imprescriptible", it can be observed that indigenous lands are public assets of special use. The inalienability of common use and special use goods is expressly provided for in Article 100 of the Civil Code, which states that these goods are inalienable as long as they maintain their qualification. The unseizability is a consequence of the provision set forth in Article 100 of the CF/88, which stipulates that there is a specific way to satisfy claims against the defaulting Public Power. Public goods cannot be seized to satisfy creditors. Therefore, these goods cannot be encumbered with real security rights, as any encumbrance in this sense would be inconsequential. Imprescriptibility means that public goods, regardless of their category, are not subject to acquisitive prescription.

Therefore, the lands traditionally occupied by indigenous communities are characterized by their inalienability, unavailability, imprescriptibility, and unseizability. Also, indigenous lands are immune to expropriation by the Public Power, according to Federal Law No. 6.001/73, Article 38. The lands traditionally occupied by indigenous peoples constitute *Res Extra commercium*, since they cannot be leased or sold, acts or legal transactions that would restrict the full exercise of direct possession by the indigenous community or the tribal peoples. Article 231, § 2 of the Constitution determines that the lands traditionally occupied by indigenous peoples are intended for their permanent possession, with exclusive use of the wealth of the soil, rivers, and lakes therein.

Indigenous peoples are ensured permanent possession of the lands they occupy without any temporal limit being set. For this reason, their loss to third parties, even if in good faith, is inadmissible, and any title of ownership that infringes on the Union's domain or the possession of the indigenous peoples is of no value. Article 231, § 6 of the Federal Constitution states that "acts aimed at the occupation, ownership, and possession of the lands referred to in this article are null and void, with no legal effect [...]". The Indian Statute (Law no. 6.001/73) provides a definition of indigenous usufruct in its art. 24:

"Article 24. The usufruct guaranteed to Indians or natives includes the right to possess, use and enjoy natural resources and all the utilities existing on the occupied lands, as well as the product of the economic exploitation of such natural resources and utilities

§ 1. The use of springs and water from the stretches of rivers within the occupied lands is also included in the usufruct, which extends to its accessories and their accruals.

§ 2. The Indian is guaranteed the exclusive exercise of hunting and fishing in the areas occupied by him, and persuasive policing measures that may have to be applied to him shall be executed."

The constitutional right of exclusive usufruct means that indigenous people can take from their land's natural resources all fruits, utilities, and revenues, as long as they do not compromise their environmental sustainability. It is noteworthy that the right of usufruct cannot be alienated by indigenous people, which does not mean that they are compelled to use their property directly. They may partner with third parties and be advised in projects aimed at exploiting their natural resources. Therefore, indigenous lands are inalienable; for this reason, indigenous people cannot engage in projects that result in the loss of their land's possession or compromise the sustainability of their resources, as the basis of indigenous land ownership is the need for utilization. In addition to the Constitution prohibiting the alienation of indigenous lands, the Indian Statute prohibits the transfer of possession of indigenous land to third parties, as provided for in its art. 18, caput: "Indigenous lands may not be the object of leasehold

or of any legal act or transaction that restricts the full exercise of direct possession by the indigenous community or natives."

Within the definition of exclusive indigenous usufruct, a distinction is made between the use of natural resources to meet the internal needs of indigenous ethnicities in conformity with their uses, customs, and traditions and the production of surpluses for commercialization, even if it aims at their subsistence. In the first case, the resources and goods of the area can be used without any restriction. In the second case, the extraction of these surpluses must follow the legal standards of environmental protection, i.e., it must be bounded by applicable legislation, especially environmental legislation.

Legislation Competence and Carbon Credits in Indigenous Areas - Territorial Issues of Indigenous Law in Brazil

As mentioned earlier in this document, in relation to indigenous rights and usufruct under Indigenous Lands it is important to pay attention to the following specific issue, which refers to the Brazilian legislation regarding carbon credits in indigenous areas.

There is broad recognition and respect for constitutional, legal, and customary rights associated with land tenure and usufruct, the formal destination of occupied lands, and the use of natural resources of indigenous peoples, including full respect for the UN Declaration on the Rights of Indigenous Peoples and ILO Convention 169.

The Federal Constitution of Brazil expresses their self-determination and establishes competencies to legislate on indigenous matters.

Art. 4 The Federative Republic of Brazil is governed in its international relations by the following principles

III - self-determination of peoples;

Articles 231 and 232 of the Federal Constitution provide for the recognition of the social organization, customs, languages, beliefs, and traditions of indigenous communities, as well as the right to the lands they traditionally occupy and the usufruct of their wealth.

The legal competence and the fruition of the use and generation of emissions reductions on Indigenous Lands are of Federal responsibility, since the "exclusive" usufructuaries are the indigenous peoples and communities, since they, despite not holding the title (which belongs to the Union), have "their permanent possession, and the exclusive usufruct of the riches of the soil, rivers, and lakes existing therein is their responsibility;

The goods and usufruct of indigenous patrimony can be declared as indigenous patrimonial income for the purposes of contracting and commercializing carbon credits, in the terms provided by arts. 39 and 40 of Law N° 6.001/73 c/c art. 231, §2 of the Federal Constitution of 1988:

Art. 231. Their social organization, customs, languages, beliefs and traditions are recognized to the Indians, as well as their original rights over the lands they traditionally occupy, the Union being responsible for demarcating them and protecting and enforcing respect for all their assets.

§ The lands traditionally occupied by the Indians are those inhabited by them on a permanent basis, those used for their productive activities, those indispensable for the preservation of the environmental resources necessary for their well-being and those necessary for their physical and cultural reproduction, according to their uses, customs and traditions.

§ The lands traditionally occupied by Indians are destined for their permanent possession, and the exclusive use of the riches of the soil, rivers and lakes existing therein is their responsibility (emphasis added)

§ 3 The exploitation of water resources, including energy potentials, the research and mining of mineral wealth on indigenous land may only be carried out with the authorization of the National Congress, after

hearing the affected communities, and they shall be guaranteed a share in the results of the mining, in accordance with the law.

§ 4. The lands referred to in this article are inalienable and unavailable, and the rights thereto are infeasible.

Art 39. The following constitute assets of the Indigenous Heritage

- I - the lands belonging to the domain of tribal groups or indigenous communities;
- II - the exclusive usufruct of the natural wealth and all utilities existing in the lands occupied by tribal groups or indigenous communities and in the areas reserved for them; (emphasis added)

Art. 40 - The owners of the Indigenous Patrimony are

- I - the indigenous population of the Country, with respect to goods or income belonging to or destined for forest dwellers, without discrimination of persons or tribal groups;
- II - the tribal group or determined indigenous community, regarding the possession and use of the lands exclusively occupied by them or reserved to them;
- III - the indigenous community or tribal group named in the property title deed, in relation to the respective real estate or movable property.

In this way, the legal framework should be respected by the States, since the indigenous peoples and populations are the holders of the exclusive use of the natural resources derived from their areas, and it is not possible for the State to impose any rules or obligations in this regard, but only to create conditions of cooperation that allow indigenous people to achieve their own objectives.

That is, it is generally understood that it is impossible for the State to legislate any issue involving indigenous law (art. 22, XIV) and also agrarian law (art. 22, I), with the aggravating factor that these are Union lands (art. 20, XI). All articles of the Brazilian Federal Constitution.

KA'A PYAHU Project is aware of Law No. 11578 OF 11/01/2021, published in the DOE - MA on Nov 3, 2021, which establishes the Policy for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation, of Conservation of Forest Carbon Stocks, Sustainable Forest Management and Increase in Forest Carbon Stocks (REDD+), Management of Environmental Assets and Payment for Environmental Services (PSA) of the State of Maranhão, known as the REDD+ and PSA Jurisdictional System, and amends State Law No. 11,000, of April 2, 2019, to expand the scope of activity of Maranhão Parcerias - MAPA.

The project is also aware of Decree No. 37969 OF 10/26/2022, published in the DOE - MA on Oct 26, 2022, which regulates State Law No. 11,578, of November 1, 2021, which established the Policy for Reducing Gas Emissions Greenhouse Effect from Deforestation and Forest Degradation, the Conservation of Forest Carbon Stocks, the Sustainable Management of Forests and the Increase of Forest Carbon Stocks (REDD+), the Management of Environmental Assets and Payment for Environmental Services (PSA) of State of Maranhão, called the REDD+ and PSA Jurisdictional System. Especially:

Art. 35. REDD+ and PSA Programs and projects, public or private, to be evaluated within the scope of the State Jurisdictional System must be based on systems, reports, information and documents that ensure the capacity for Measurement, Verification and Registration (MRV), as well as being certified by an independent technical-scientific entity, according to recognized certification standards.

Single paragraph. Preparation and support actions for REDD+ and PSA projects may be promoted, such as feasibility studies, socio-environmental analyses, preparation of emissions scenarios and business plans, as required by regulatory act, endorsed by the Deliberative Council

The KA'A PYAHU Project remains attentive to all implementations, institutions and regulations that may affect its conduct.

2.3.3 Indigenous Peoples and Cultural Heritage

The KA'A PYAHU Project recognizes the cultural riches and vast material and immaterial heritage of the Guajajara people, proponents of the project. Based on the study and careful listening of the people who are part of the project, all actions that will be carried out from the moment of certification and sale of carbon credits, will comply with the autonomy and alignment of Guajajara cultural precepts and way of life with the lines of action established in accordance with socio-environmental safeguards.

The construction of the project permeates the understanding that the preservation of the fauna and flora of the Indigenous Lands Lagoa Comprida, Urucu Juruá, Cana Brava and Rodeador, dialogue directly with the forms of life, combining nature and culture according to the indigenous perspective. In this way, the objective of promoting preservation and conservation of natural assets and appreciation and generation of social benefits demonstrates the respect and importance of the KA'A PYAHU project with the protection of the cultural heritage of the Guajajara people.

Through strict compliance with Cancun's socio-environmental safeguards, compliance with the guidelines previously established by CONAREDD+ (Annexes) and respect for internal governance, the project is capable of being a partner for the cultural and natural assets of the Guajajara people, protagonists of the KA'A PYAHU project. The following safeguards are highlighted by this project and demonstrate the commitment to the Guajajara people:

The project follows the principles or premises defined jointly by FUNAI and the MMA, in a seminar held in March 2012, whose main objective was to meet the interests of indigenous peoples, respond to the objectives of the PNMC and guide the design and implementation of concrete applications of REDD+ in indigenous territories. We highlight some of the Principles/Assumptions that are part of the document and govern the KA'A PYAHU project:

- The historical contribution of indigenous peoples to the maintenance of forest stocks is recognized, through traditional resource management as well as territorial management strategies;
- It is recognized that the historical deforestation accumulated in indigenous lands is not very significant and is associated with the specific forms of indigenous occupation of the territory, without this meaning that traditional ways of life are considered "vectors of deforestation or degradation";
- It is stated that the main focus of REDD+ initiatives on indigenous lands must be to prevent historical patterns of deforestation observed in other areas of use and occupation from entering indigenous lands;
- It is agreed, therefore, that the priority role of indigenous peoples for the purpose of combating climate change is the long-term conservation of carbon stocks stored in their forested areas;
- It is considered that the national REDD+ strategy, through an indigenous component, must fulfill the role and seek synergies between the implementation of the National Policy on Climate Change (PNMC) and the National Policy on Environmental Management in Indigenous Lands (PNGATI);
- It is reiterated that the definition of an indigenous component in the national REDD+ strategy will have as a principle respect for national and international legislation in force, including prior, free and informed consultation;
- Ecosystem services/REDD+ initiatives are considered to be carried out preferably by providers of ecosystem services, that is, by indigenous peoples themselves, with support and recognition by the State, in order to strengthen the principle of self-determination provided for in ILO Convention 169;

- It is understood that the definition of ecosystem services/REDD+ initiatives requires concrete consideration of technical and scientific aspects, such as ensuring the permanence of achieved results and avoiding displacement (environmental integrity);
- It is reiterated that the indigenous component of the national strategy will be implemented in clear coordination with national, state and local policies and programs;
- It is stated that a system of monitoring and transparency in the execution of indigenous ecosystem services/REDD+ initiatives will be established through the national REDD+ strategy, in full articulation with existing information systems, such as the MMA's REDD+ Brazil website;
- The need to provide a record of complaints caused by non-compliance or disrespect for socio-environmental safeguards, including indigenous rights, is stated, as well as a clear conflict resolution mechanism within the scope of the transparency and monitoring system of the REDD+ strategy, supported by the actions of the competent bodies (AGU, PGU, when applicable);

By following these and other premises established in the document Set of premises for implementing REDD+ in Indigenous Lands, the Ka'a Pyahu project is committed to valuing, preserving and supporting Guajajara cultural, material and immaterial heritage.

2.3.4 Property Rights

Table 34. Property rights in the project

Rights to territories and resources	Throughout this document it was demonstrated that the ownership of the territory where the REDD+ project is being implemented belongs to the Guajajara people, respecting the legal determination of usufruct over the indigenous territories proposing the project. Based on this principle, the association contract is signed between BIOFIX and the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara, in which the council's ownership of the territory is explicitly declared. The usufruct of the territories where the KA'A PYAHU Project takes place belongs to the communities belonging to the Guajajara ethnic group, which are legally represented by the indigenous association Conselho Supremo dos Caciques e Lideranças da Terra Indígena Cana Brava (Annex 011). The Cana Brava Indigenous Land, approved by DECREE No. 246, OF OCTOBER 29, 1991 (Annex 001); Urucu-Juruá Indigenous Land, approved by DECREE No. 382, OF DECEMBER 24, 1991 (Annexes 002), there was a rectification of the geographic coordinates of Decree No. 382, of December 24, 1991, which approved the administrative demarcation of the Urucu Indigenous Land /Juruá on FEBRUARY 10, 2003 (Annex 003); Lagoa Comprida Indigenous Land, DECREE OF FEBRUARY 10, 2003 (Annex 004) and Rodeador Indigenous Domain Area approved by DECREE No. 88,813, OF OCTOBER 4, 1983 (Annex 005), are under the possession of the people proposing the project, therefore owners carbon credits and tradable carbon credits. The Guajajara indigenous people who are part of the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara , making use not only of their rights of autonomy and
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	self-determination, but also of the legal capacity that has been legally granted to them, which allows them to enter into contracts under the terms of the civil and commercial law, through the signing of the Partnership and Cooperation Agreement (Annex 012), granted to Biofix Brasil LTDA, as the project's structuring entity, the express power to generate the issuance and commercialization of carbon credits (certified reductions in carbon emissions). GHG) originating from the territories in which the Community Council exercises its jurisdiction. The above, subject to the exercise of autonomy and self-determination of the communities, who freely and voluntarily decided to enter into this contract and jointly execute the project. Furthermore, BIOFIX is committed to complying with the socio-environmental safeguards contemplated in the regulations and policies of the environment and sustainable development sector for REDD+ projects. Based on these considerations, for the formulation and implementation of the REDD+ project, several spaces for socialization and signing of free, prior and informed consent were developed, processes described in section 2.1.2.
Respect for property rights	The KA'A PYAHU Project in all its stages of implementation to guarantee the protection and preservation of the Guajajara indigenous people and their rights, will comply with the related legislation already mentioned in this document of rights over the territories, as well as the premises and safeguards that protect and provide the implementation of the project in accordance with Guajajara cultural ways and traditions, see section 2.3.3.

Source: KA'A PYAHU REDD+ Project.

2.3.5 Benefit Sharing

Not applicable. The beneficiary communities of the KA'A PYAHU Project have the right to access and remain on the strips of land determined by law, protected by the Union. The 1988 Constitution stipulated four conditions for the notion of traditionally occupied land to qualify: a) be permanently inhabited; b) be used for productive activities; c) be essential to the preservation of environmental resources necessary for the well-being of the indigenous people; d) be necessary for physical and cultural reproduction, according to their uses, customs and traditions. Article 231 of the Federal Constitution recognizes the right to exclusive enjoyment of the riches of the soil, rivers and lakes existing on the lands they traditionally occupy. Such usufruct is non-transferable (Annexes 001, 002,003, 004 and 005).

2.4 Ecosystem Health

Table 35. Risk identified in ecosystem health

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	No risk identified	The project's objectives include forest conservation through the lines of action

		described in section 1.12, including the monitoring and deforestation control line of action. Therefore, no negative impacts on biodiversity and ecosystems are expected. On the contrary, the maintenance of forests and ecosystems guarantees biodiversity.
Soil degradation and soil erosion	No risk identified	At the ecological level, there are different relationships between soils and forests, which are necessary and have important repercussions on each other and on the environment in general. Soil is an important component of forests and forest ecosystems as it helps regulate important ecosystem processes such as nutrient uptake, decomposition and water availability. Soils, among other functions, provide anchorage, water and nutrients to trees. In turn, trees and other plants and vegetation types are an important factor in the creation of new soil when leaves and vegetation decay and decompose ¹⁰ . Therefore, conservation of the forests in the project area is not expected to have negative effects on soils. Otherwise, the relationships and dynamics between soils and forests are guaranteed.
Water consumption and stress	No risk identified	Project activities correspond to unplanned avoided deforestation. It does not correspond with an ARR project. Therefore there are no associated risks with water consumption.
Usage of fertilizers	No risk identified	Similar to the previous case, it is not an ARR or AFM project. Therefore no fertilizers are used in the project activities.

Source: KA'A PYAHU REDD+ Project.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes ☐ No

As showed in 1.14.2.5. section, there are 5 species classified as Critically Endangered **CR** and 4 species classified as Endangered.

Table 36. Threatened species in the project's area of influence

Species and habitat	
<i>Nyctibius leucopterus</i>	<i>Since this is a REDD+ project, in which forest conservation is carried out, it is expected that the species that are threatened and all other species will not be affected. On the contrary, by conserving forests, their habitats are conserved and their habitat fragmentation is avoided.</i>
<i>Psophia obscura</i>	
<i>Crax fasciolata pinima</i>	

¹⁰ <https://www.fao.org/soils-2015/news/news-detail/es/c/285875>

<i>Limnodromus griseus</i>	
<i>Lepidothrix iris</i>	
<i>Thalasseus maximus</i>	
<i>Chiropotes satanas</i>	
<i>Cebus kaapori</i>	
<i>Leopardus tigrinus</i>	

2.4.2 Introduction of species

During this monitoring period no activities of planting or species were introduced.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A	N/A	N/A	N/A

Existing invasive species	Mitigation measures to prevent spread or continued existence of invasive species
N/A	N/A

2.4.3 Ecosystem conversion

Not applicable because it is a REDD+ project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Table 37. Methodologies, Modules and tools for the KA'A PYAHU REDD+ project

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0007	REDD+ Methodology Framework (REDD+MF).	7.0
Module	VMD0001	Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools (CP-AB)	1.2
Module	VMD0007	Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation (BL-UP)	3.3
Module	VMD0010	Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation	1.2
Module	VMD0015	Methods for monitoring of GHG emissions and removals in REDD and CIW projects (M-REDD)	2.2
Module	VMD0016	Methods for stratification of the project area (X-STR)	1.3
Module	VMD0017	Estimation of uncertainty for REDD project activities (X-UNC)	2.2
Tool	VT0001	Tool for the demonstration and Assessment of additionality in vcs Agriculture, Forestry and Other land Use (AFOLU) project activities	3.0

3.2 Applicability of Methodology

Table 38. Specific applicability of methodologies, modules, and tools in REDD activities

Methodology ID	Applicability condition	Justification of compliance
VM0007	-Land in the project area has qualified as forest (following the definition used by VCS; in addition, see Section 5.1.2) for at least the 10 years prior to the project start date. Mangrove forests are excluded from any tree height requirement in a forest definition, as the consist of (close to) 100% mangrove	The analysis of land use categories in the project is based mainly on Mapbiomas which utilizes a satellite imagery dataset used composed by Landsat 5 (Thematic Mapper), Landsat 7 (Enhanced Thematic Mapper Plus) and Landsat 8 (Operational Land Imager), Landsat

	species, which often do not reach the same height as other tree species and occupy contiguous areas and their functioning as a forest is independent of tree height. -Baseline deforestation and forest degradation in the project area fall within one or more of the following categories: • Unplanned deforestation (VCS category AUDD) • Planned deforestation/degradation (VCS category APD) -Baseline agents of deforestation must: a. clear the land for tree harvesting, settlements, crop production (agriculturalist) or ranching or aquaculture, where such clearing for crop production or ranching or aquaculture does not amount to large scale industrial agriculture or aquaculture activities b. have no documented and uncontested legal right to deforest the and for these purposes; and c. be either residents in the reference region for deforestation or immigrants.	sensors. The MapBiomass classification scheme is a hierarchical system with a combination of LULC classes compatible with the Food and Agriculture Organization (FAO) and IBGE classification systems. -The procedures finding into the boundaries of the project are in the category of Unplanned deforestation (AUDD). - The agents of deforestation that threaten the indigenous territories and the project proponents, are the result of human actions Illegal logging: One of the agents of deforestation is the illegal exploitation of timber resources. The invasion of territories in search of wood for illegal trade, threatens natural resources and also the lives of people in the territory. Through episodes of conflict, threats and even death, making it one of the reports of violence against indigenous peoples in Brazil produced by the Indigenous Missionary Council, an important document and instrument of national denunciation (Annex 019) The Guajajara people report continued invasions by loggers and farmers who produce soy, causing socio-environmental damage. There are also reports of wild animal hunters, who threaten communities. The illegal removal of wood from the Urucu Juruá Indigenous Land has been taking place since September 2019 and according to estimates by the Guajajara, around 70% of the forest in the Indigenous Land has already been destroyed.
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		Illegal Pastures: There are reports of illegal land leasing and illegal pastures in the territories of the project proponents. Around 75% of the forest cover in the state of Maranhão was deforested to provide space for agriculture and livestock, points out the National Institute for Space Research, INPE. Approximately 70% of the remaining native forest is found in the 16 indigenous lands and conservation units of Maranhão. What makes areas attractive for extractivism.
VMD0001	This module is applicable to all forest types and age classes.	For the analysis of the project, the Mapbiomas ¹¹ data are used. In the case of forest cover, this classification covers the different forest phytophysionomies for each biome. These phytophysionomies correspond to the ecosystems presented in the NREF ¹² Brazil, and their equivalence with FAO.
VMD0007	The module must be applied to all project activities where the baseline agents of deforestation: (i) clear the land for tree harvesting, settlements, crop production (agriculturalist) or ranching or aquaculture, where such clearing for crop production or ranching or aquaculture does not amount to large scale industrial agriculture or aquaculture activities. (ii) have no documented and uncontested legal right to deforest the and for these purposes; (iii) and be either residents in the reference region for deforestation or immigrants.	The project complies with the applicability conditions expressed for the VM0007 methodology.
VMD0010	Applicable to estimate changes in carbon stocks and emissions due to displacement of activities that cause deforestation of land.	The module VMD007 (BL-UP) is mandatory for the project activity (unplanned deforestation). Therefore, it is mandatory to apply module VMD0010 to determine emissions in the leakage belt.

¹¹ <https://brasil.mapbiomas.org/codigos-de-legenda/>

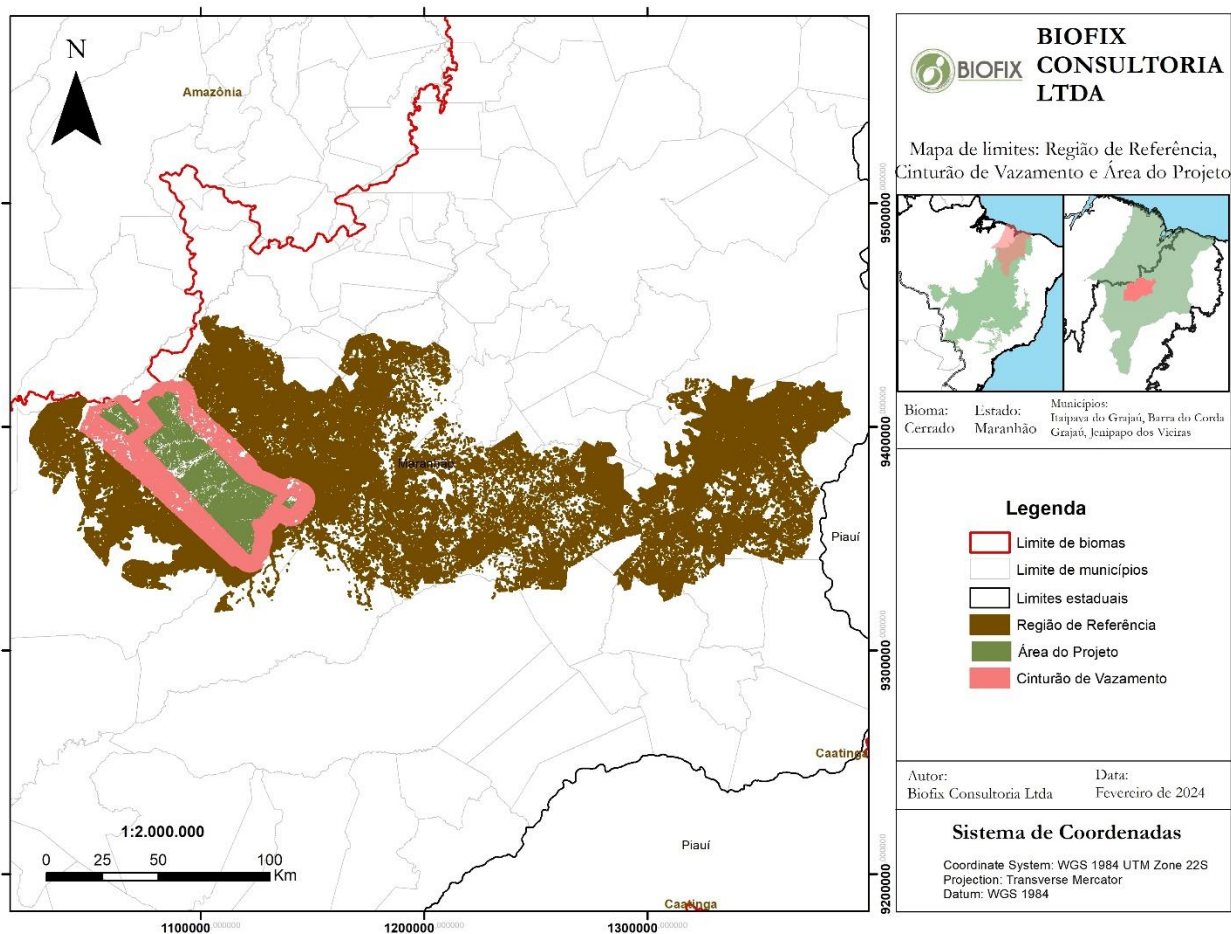
¹² <https://redd.unfccc.int/submissions.html?country=bra>

	Activities subject to displacement are the conversion of forest land to other land uses. Mandatory module if BL-UP was used.	In addition, the conditions for feasible scenarios of displacement of emissions to the leakage belt are fulfilled.
VMD0015	Strata as defined in the relevant baseline modules are fixed and may not be changed without baseline revision. The module is mandatory for REDD, CIW-REDD, RWE-REDD and stand-alone CIW project activities.	This module is mandatory because it is a REDD+ project.
VMD0016	The project area may be stratified ex ante, and this stratification may be revised ex post for monitoring purposes. Established strata may be merged if reasons for their establishment have disappeared or have proven irrelevant to key variables for estimating net GHG emissions or removals. A map displaying the final delineation of strata must be included in the project description. Areas of individual strata naturally sum to the total project area; any discrepancies must be reconciled.	The forest strata are defined according to the NREF Brazil forest types. In the case of the project, the forest strata correspond to the transitional forest phytophysiognomies between the Cerrado and Amazon biomes.
VMD0017	This module is mandatory when using methodology REDD+ MF. It is applicable for estimating the uncertainty of estimates of emissions and removals of CO₂-e generated from REDD and WRC project activities.	The analysis of the project data uses Mapbiomas as a source, which works with a confidence interval of 95%, and a maximum standard error of 5% (Souza, et al., 2020).

3.3 Project Boundary

The figure below shows the following boundaries: Project Area (PA), Reference region for project rate of deforestation (RRD) and Leakage Belt (LB).

Figure 25. Project boundaries: Project Area, Reference Region and Leakage Belt

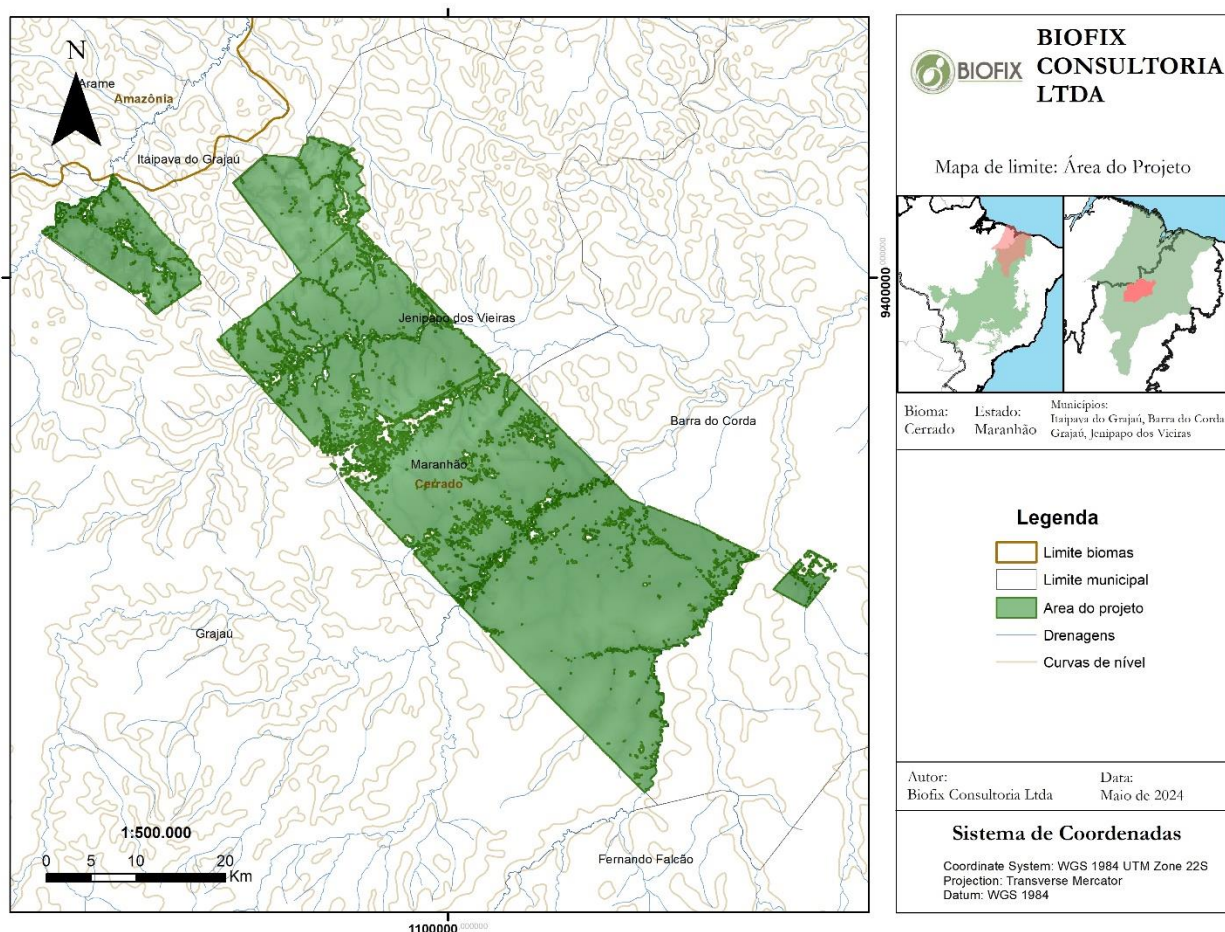


Source: KA'A PYAHU REDD+ Project.

3.3.1 Project Area

The KA'A PYAHU REDD+ project is located in the state of Maranhão, in the northeast region of Brazil, under the jurisdiction of the Legal Amazon. Specifically, the project is located on indigenous lands of the Guajajara peoples: Cana Brava Indigenous Land, Urucu-Juruá Indigenous Land, Rodeador Indigenous Dominion Area and Lagoa Comprida Indigenous Land.

From this area, the area that remained as stable forest at the beginning of the project (2021) was established. The result of the analysis yielded the Project Area of 150,800 ha of forest. This area is dominated by the Submontane Semi Deciduous Seasonal Forest (Fs) Phytophysiology (MMA, 2017). According with the 4th GHG inventory, this phytophysiology is one of the types of forest formations, which meet the definition of forest: “Minimun area of 0.5 hectares with trees of minimum height of 5 meters and minimum canopy coverage of 10 percent, or trees capable of reaching these limits in situ. Does not include area predominantly used for agricultural or urban purpose” (Brasil, 2020).



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3.3.2.1 Similarity analysis between Reference Region and Project Area

In accordance with VM0007 methodology, module VMD0007-BL-UP_v3.3, the following criteria were considered in the similarity analysis between the Reference Region (RRD) and the Project Area. First, the main agents of deforestation were addressed, then landscape factors, followed by transportation networks and human infrastructure, social factors, policies, and regulations and finally with the exclusion of planned deforestation.

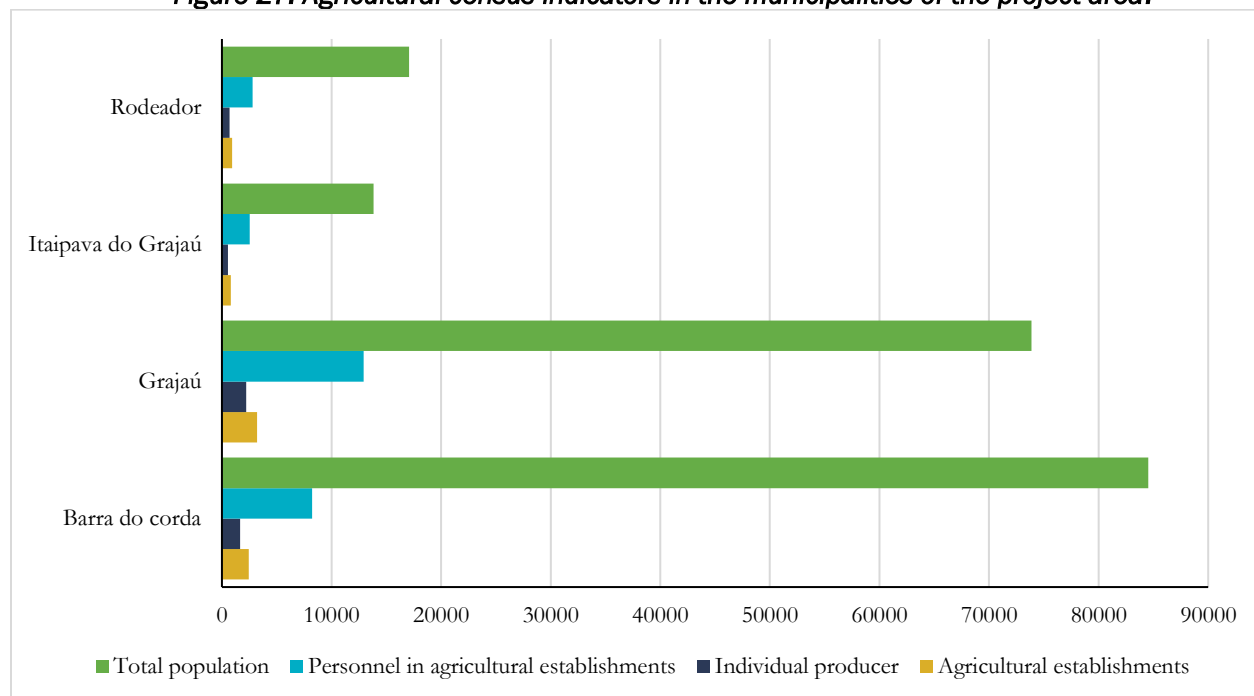
Main agents of deforestation

It was found that the main drivers of deforestation in the RRD at the beginning of the historical reference period were the same as those expected to cause deforestation in the project area during the project period. As noted in PDD Sections 3.4.1 and 5.1.4, at the end of the historical reference period (2010-2020), the main post-deforestation land use in the entire RRD was grassland and agriculture. That is, of the 128,656 ha deforested, 53% were converted to pasture and 26% to agriculture, so it is assumed that the main drivers of deforestation in the PA are also related to agriculture and grasslands, which gives a selection of two main drivers of deforestation: agriculture and grasslands.

As shown in in section 1.14.1.6 and section 3.5. (Step 1a), agriculture is an important productive activity for the Guajajara people and also in the municipalities surrounding the Indigenous Lands. Some traditional crops are cassava, corn, rice, pumpkin, watermelon, beans, broad beans, yams, yams, sesame, and peanuts.

A search was carried out on the proportion of farmers versus ranchers in the municipalities both in the reference region and in the project area. The data were taken from the 2017 agricultural census, taking into account the variables: Individual producer, Agricultural establishments and Personnel employed in agricultural establishments¹³. These population data of producers were compared with the total population of the municipalities.

Figure 27. Agricultural census indicators in the municipalities of the project area.



Source: Agricultural Census 2017.

¹³ <https://censoagro2017.ibge.gov.br/>

In a similar way, data were obtained for the municipalities of the reference region, which partially covers 25 municipalities. From the sum of the data of the variables mentioned above from all the municipalities, both in the project area and in the reference region, the following results are obtained:

Table 39. Agricultural census indicators in the RRD and PA.

PA				RRD			Similarity (±20%)
Variable ¹⁴	Population (#)	Total population (#)	Proportion (%)	Population (#)	Total population (#)	Proportion (%)	
Individual producer	5,103	189,308	2.70	27,121	920,251	2.95	Comply
Agricultural establishments	7,406	189,308	3.91	36,695	920,251	3.99	Comply
Personnel in agricultural establishments	26,502	189,308	14.00	111,252	920,251	12.09	Comply

Source: Agricultural Census 2017.

The table indicates that the proportion of individual producers with respect to the total population was 2.70% in the municipalities of the project area and 2.95% in the municipalities of the reference region, which means compliance with the proportion between these two areas. Similarly, the population in agricultural establishments is 14% in the project area, versus 12.09% in the reference region, complying with the range of similarity established by the methodology.

Migration

According to the VMD0007 module, a data search was carried out to establish the proportion of migrants with respect to the total population. The 2022 census does not have this type of information at the municipal level, therefore the data from the 2010 population census¹⁵ were used. The emigration rate was calculated as the relationship between the migrant population and the resident population in the municipalities of the project area, resulting in:

Table 40. Migration rate in the project area municipalities

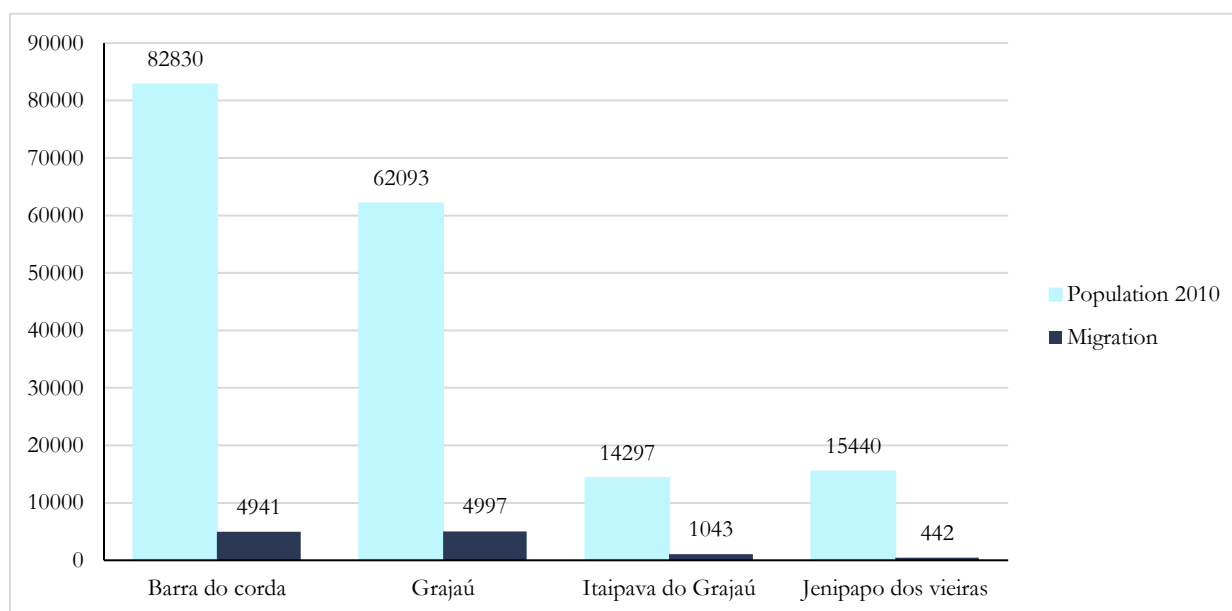
PA			
Variable	Total Population (#)	Migration (#)	Proportion (%)
Barra do Corda	82,830	4,941	5.97
Grajaú	62,093	4,997	8.05
Itaipava do Grajaú	14,297	1,043	7.30
Jenipapo dos Vieiras	15,440	442	2.86
Total	174,660	11,423	6.04¹⁶

¹⁴ The translation "agricultural" comes from "agropecuário" This means that it also includes livestock and not only that related to crops.

¹⁵ <https://censo2010.ibge.gov.br/resultados.html>

¹⁶ Average migration rate in municipalities. If the direct proportion is taken, it has a value of 6.54

Figure 28. Total population and migration in the project area municipalities



Source: IBGE 2010 population census

This same data search was carried out for the municipalities of the reference region, in order to compare the proportion of migrants between the project area and the reference region, as shown below:

Table 41. Proportion of migrants in the municipalities of the project area and the reference region

PA			RRD			Similarity (±20%)
Total population (#)	Migration (#)	Proportion (%)	Total population (#)	Migration (#)	Proportion (%)	
174,660	11,423	6.54 ¹⁷	893,354	46,934	5.25	Comply

Source: IBGE 2010 population census

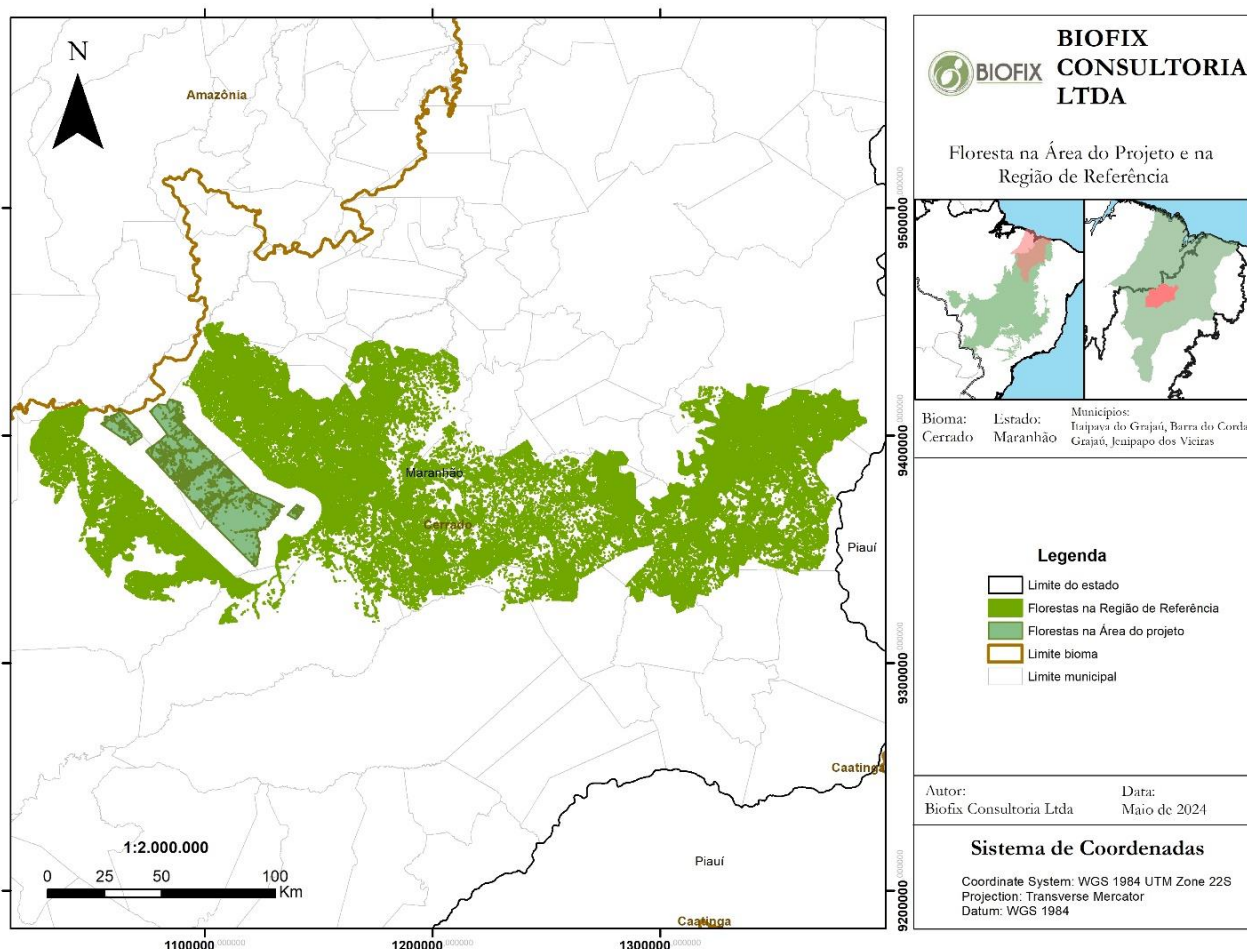
The values obtained indicate that the similarity of the proportion of migrants is met both in the project area (6.54%) and in the reference region (5.25%). It is worth clarifying that the proportion was obtained from the sum of the migrant and total population of the municipalities. However, if the value is obtained as the average of the migration rate of each municipality, the similarity condition is also met, with a value of 6.04% for the project area and 5.17% for the reference region.

¹⁷ Is the value obtained by dividing the sum of the migrant population of all the municipalities by the sum of the population of all the municipalities.

Landscape factors

Forest classes

Figure 29. Forest class into the PA and the RRD



Source: KA'A PYAHU REDD+ Project.

According to the forest classification used by Brazil, which considers lands of “minimum area of 0.5 hectares with trees of minimum height of 5 meters and minimum canopy coverage of 10 percent, or trees capable of reaching these limits in situ. Does not include area predominantly used for agricultural or urban purpose”, both the Project Area and the RRD are entirely covered by forest. This forest definition comprise various types of various phytophysionomies in the different Brazilian biomes (Brasil, 2020).

Table 42. Forest classes into the PA and the RRD

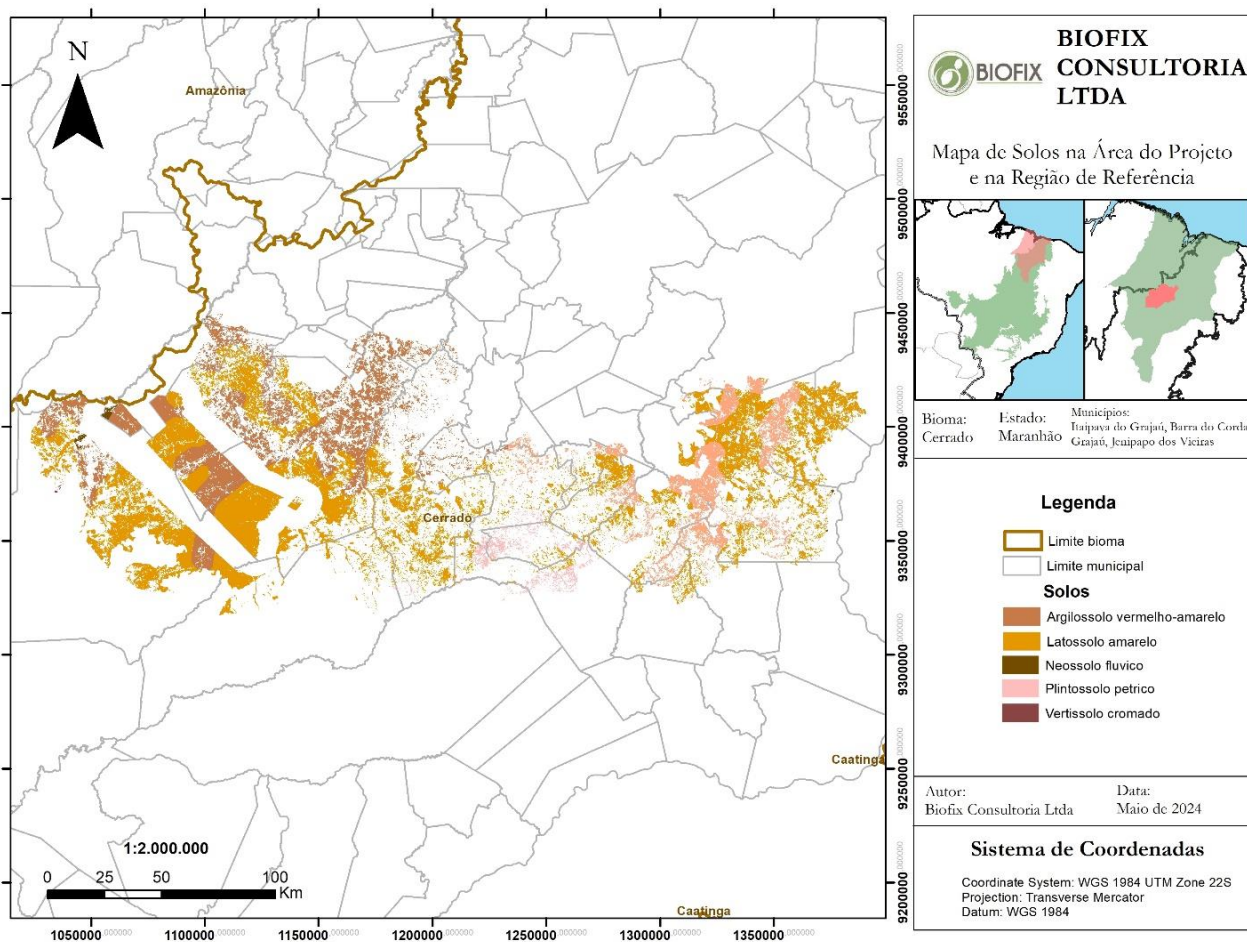
Forest type	PA (ha)	Proportion (%)	RRD (ha)	Proportion (%)	Similarity (±20%)
Forest	150,800	100	843,490	100	Comply
Count	150,800	100	843,490	100	

Source: KA'A PYAHU REDD+ Project.

Soil

According to the 2001 Brazilian Soil Map (EMBRAPA, 2001), the sub-order soils of the Reference region are: yellow- latosol, Red-yellow Argisols, Petric plinthosol and Fluvic Neosols sub-order. The main soil suborder both in the project area and in the reference region corresponds to Yellow-Latosol, with a proportion of approximately 60% in both cases. The second sub-order of soil corresponds to Red-yellow Argisols, which in the case of the project area occupies 40% and in the reference region 22%.

Figure 30. Soil type map in the Project Area and Reference Region RRD.



Source: KA'A PYAHU REDD+ Project.

Table 43. Soil type distribution in the RRD and PA.

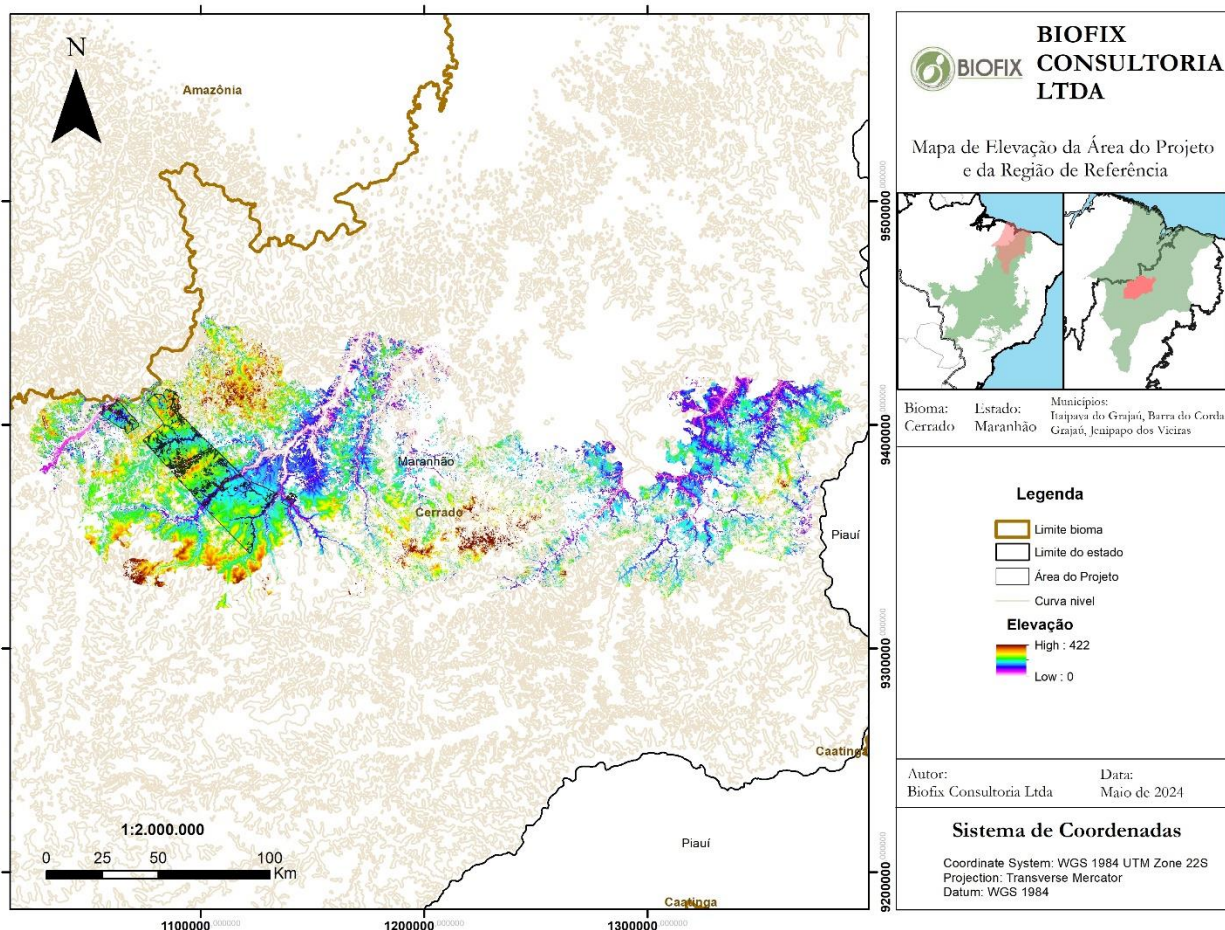
Soil type	PA (ha)	Proportion (%)	RRD (ha)	Proportion (%)	Similarity (±20%)
Yellow-Latosol	89,524.31	59.37	509,027.52	60.35	Comply
Red-yellow Argisols	59,698.62	39.59	188,960.53	22.41	Comply (±42)
Fluvic Neosols	1,577.38	1.05	1,680.28	0.20	No comply
Petric plinthosol	0.00	0.00	143,618.06	17.02	N/A
Vertisol	0.00	0.00	204.32	0.02	N/A
Count	150,800	100	843,490	100	

Source: KA'A PYAHU REDD+ Project.

Elevation

In terms of elevation, both the project area and the reference region are located in an altitudinal range between 0 and 500 meters above sea level.

Figure 31. Elevation range map in the Project Area and Reference Region RRD



Source: KA'A PYAHU REDD+ Project.

Table 44. Elevation classes in the RRD and PA

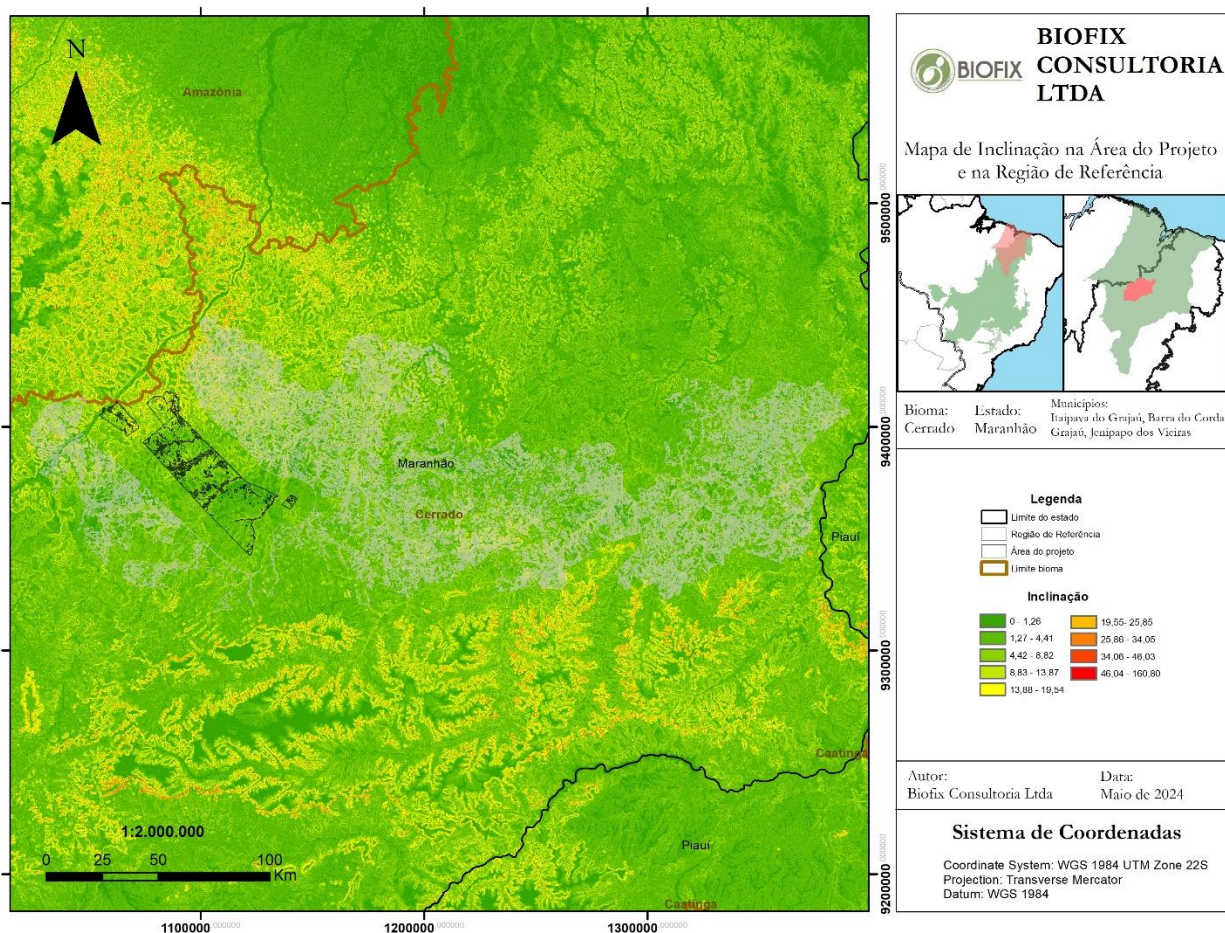
Elevation (m)	PA (ha)	Proportion (%)	RRD (ha)	Proportion (%)	Similarity (±20%)
0-500	150,800	100	843,490	100	Comply
Count	150,800	100	843,490	100	

Source: KA'A PYAHU REDD+ Project.

Slope

In terms of slope, both the project area and the reference region have a gentle slope (slope < 15%).

Figure 32. Slope class map in the Project Area and Reference Region RRD



Source: KA'A PYAHU REDD+ Project.

Table 45. Slope class distribution in the RRD and PA

Slope (%)	8PA (ha)	Proportion (%)	RRD (ha)	Proportion (%)	Similarity (±20%)
0-15	150,800	100	843,490	100	Comply
Count	150,800	100	843,490	100	

Source: KA'A PYAHU REDD+ Project.

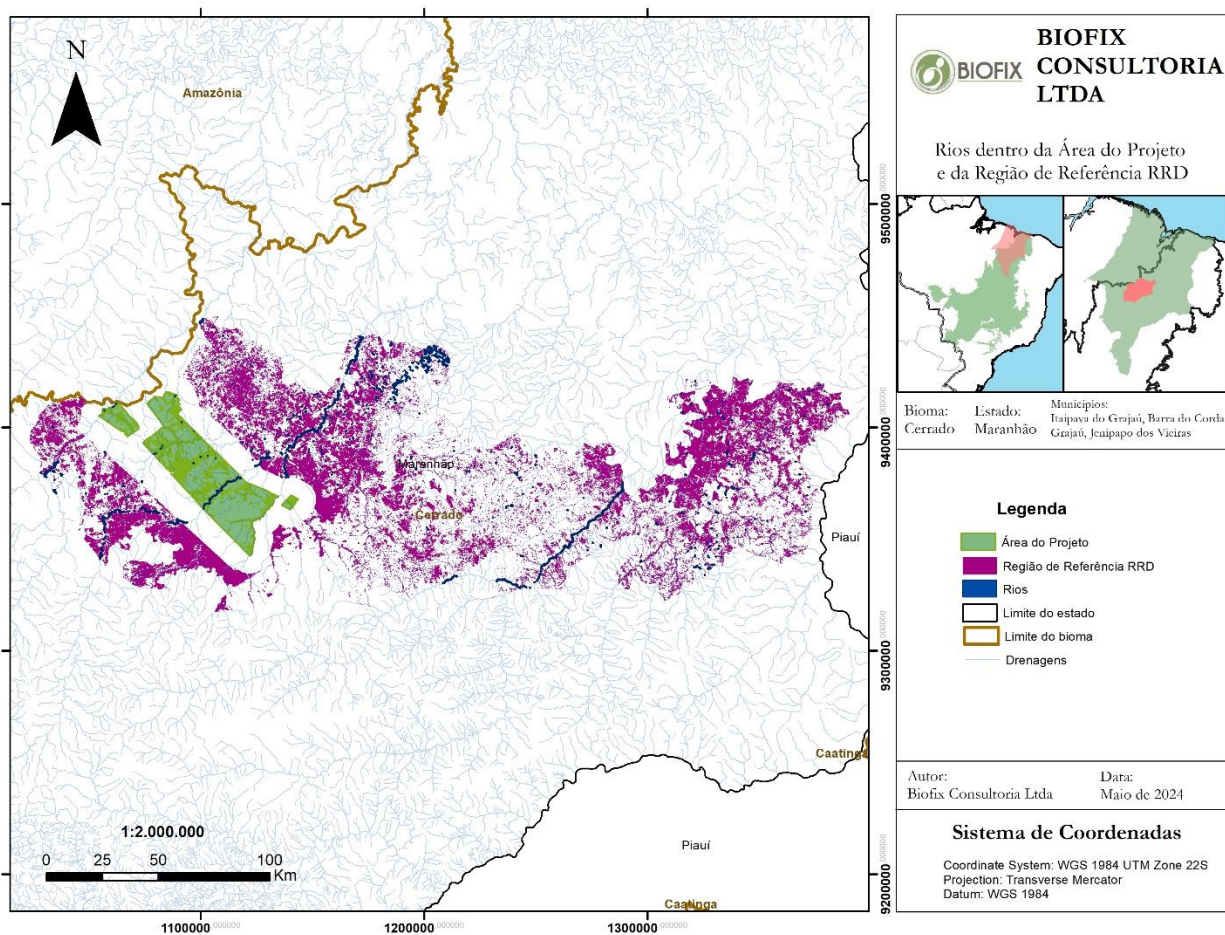
Transportation networks and human infrastructure

The VMD0007 BL-UP v3.3 module states that infrastructure such as roads, navigable rivers, and settlements, that increase the likelihood of deforestation and that exist historically in the RRD must be directly comparable to those that are expected to exist within the project area during the project term.

River density

The river density in the RRD is about 141.3 m/km² and the density in the Project Area is 154.1 m/km², which means the river density of the RRD is less than the river density from the PA, getting in accordance with VMD0007 criteria: “Where navigable rivers are present in the project area, navigable river/stream density (m/km²) for the RRD is the same, less than, or does not exceed by more than 20% of the Project Area”.

Figure 33. Rivers inside the Project Area and Reference Region RRD



Source: KA'A PYAHU REDD+ Project.

Table 46. Rivers density in the Project Area and Reference Region RRD

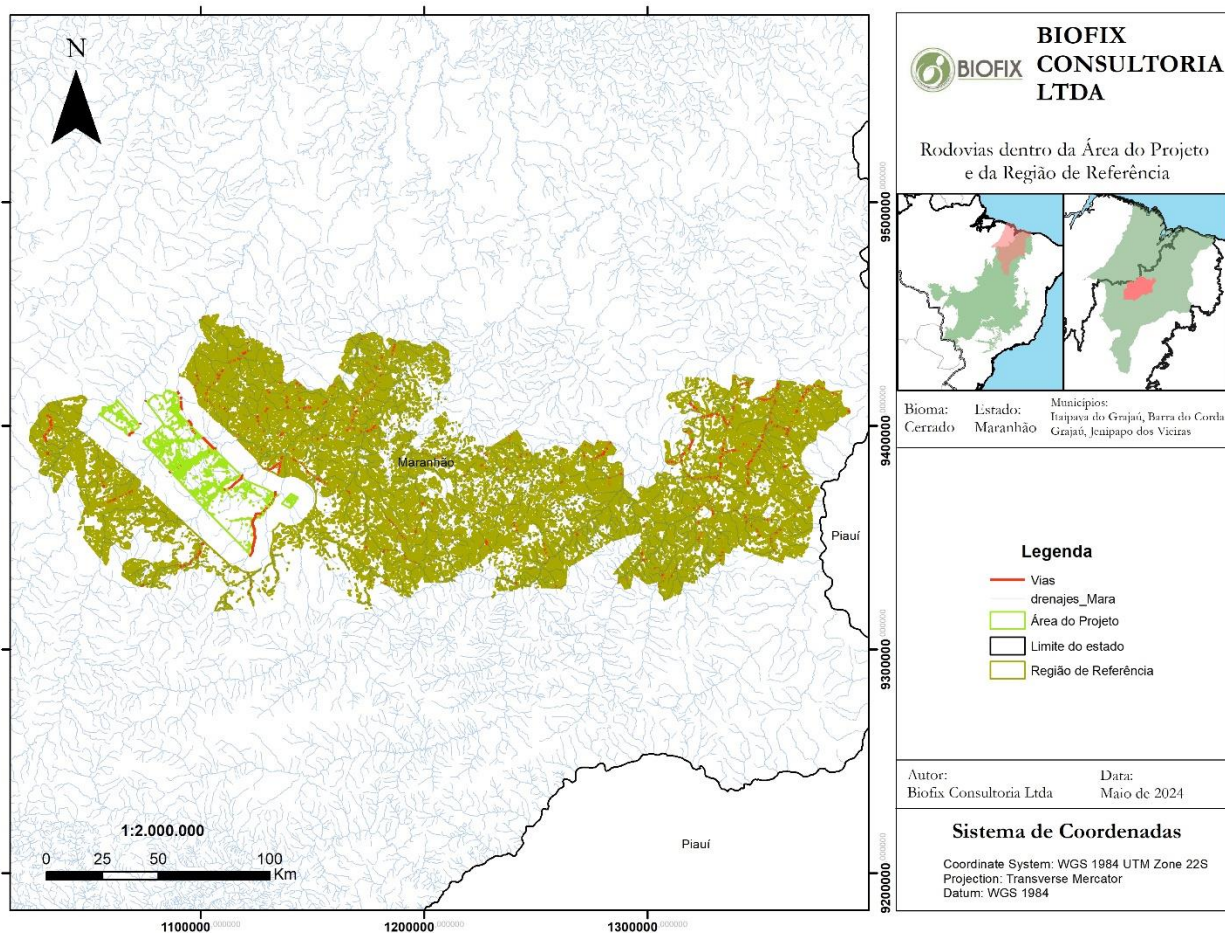
Long rivers PA (m)	Area PA (km ²)	River density PA (m/km ²)	Long rivers RRD (m)	Area RRD (km ²)	River density RRD (m/km ²)	Similarity (±20%)
232,35	1508	154.1	1,191,906,7	8,434.9	141.3	Comply

Source: KA'A PYAHU REDD+ Project.

Road density

To obtain the road density for the PA, a 1.0 Km buffer was established over this area, and subsequently, the existing roads within this area were determined. Finally, the density was calculated, using the area with the generated buffer, and dividing it by the total length of the roads. The following map shows the roads inside the PA and the buffer generated, to meet the criteria for similarity between PA and RRD.

Figure 34. Roads inside the PA including a buffer of 1.0 km and in the RRD



Source: KA'A PYAHU REDD+ Project.

The analysis of the difference between the PA and RRD was conducted in terms of road density (m/km^2) resulting in a value of $44.8 m/km^2$ in the Project Area and $32.56 m/km^2$ for the RRD, giving a road density result in the RRD of less than in the PA, complying with the criteria established by the methodology.

Table 47. Road density analysis between the RRD and PA

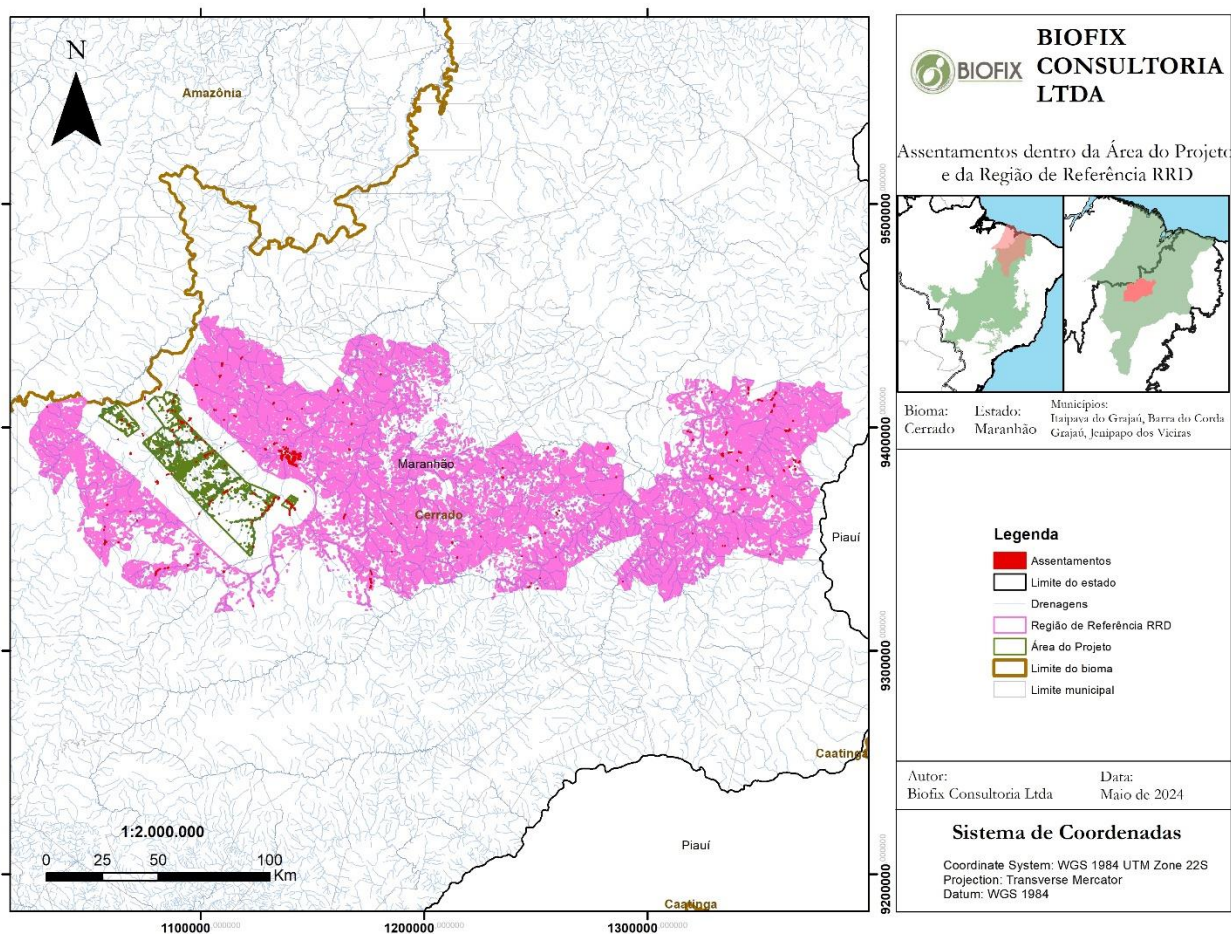
Long roads PA (m)	Area PA (km^2)	River density PA (m/km^2)	Long roads RRD (m)	Area RRD (km^2)	River density RRD (m/km^2)	Similarity ($\pm 20\%$)
67,625.08	1508	44.84	274,624.63	8,434.9	32.56	Comply

Source: KA'A PYAHU REDD+ Project.

Settlement density

The settlement density assessment was carried out using the VMD0007 BL-UP module. Thus, a spatial analysis was made in the Project Area and in the Reference Region (RRD), making a buffer of 1km around the areas, and identifying the settlements inside this area, according to official information on urbanized areas in Brazil (IBGE, 2022). Once they were identified, the density was calculated as the table below shows. Finally, it is observed that the settlement density of the Reference Region (RRD) does not exceed more than 20% of the Project Area settlement density.

Figure 35. Settlement density in the Project Area and Reference Region RRD



Source: KA'A PYAHU REDD+ Project.

Table 48. Settlement density in the Project Area and Reference Region RRD

Settlements PA (#)	Area PA (km ²)	Settlement density PA (#/km ²)	Settlements RRD (#)	Area RRD (km ²)	Settlement density RRD (#/km ²)	Similarity (±20%)
65	1,508	0,04	166	8,434.9	0,02	Comply

Source: KA'A PYAHU REDD+ Project.

3.3.3 Leakage belt (LB)

The delimitation of the Leakage Belt (LB) was carried out following the methodology described by the VMD0007 Module BL-UP v3.3 making an analysis of similarity over landscape factors (forest types, soil types, slope ratio, elevation classes), transportation factors (river density, road density and settlement density), policies and regulations and social factors.

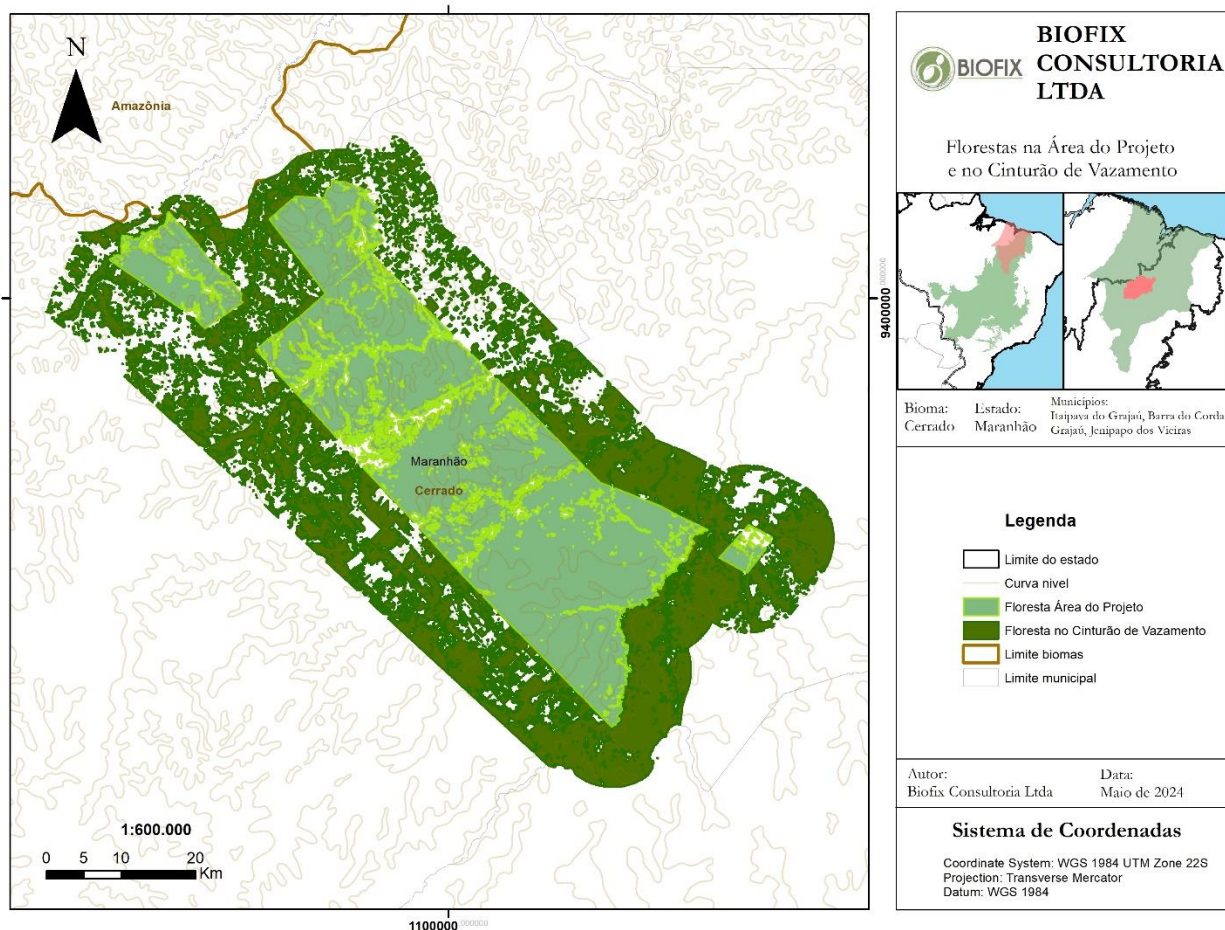
According with the VMD0007 Module, the minimum leakage belt area must be equal to at least 90% of the project area. The project area is 150.800 has, and the leakage belt is 151.444 has. This means that the minimum size of the Leakage belt meets the condition ($L.B. \geq 135.720$ hectares).

3.3.3.1 Landscape factors

Forest types

According to the forest classification used by Brazil, which considers lands of “*minimum area of 0.5 hectares with trees of minimum height of 5 meters and minimum canopy coverage of 10 percent, or trees capable of reaching these limits in situ. Does not include area predominantly used for agricultural or urban purpose*”, both the Project Area and the Leakage Belt are entirely covered by forest. This forest definition comprise various types of various phytophysiognomies in the different Brazilian biomes (Brasil, 2020).

Figure 36. Forest in the Project Area and the Leakage Belt



Source: KA'A PYAHU REDD+ Project.

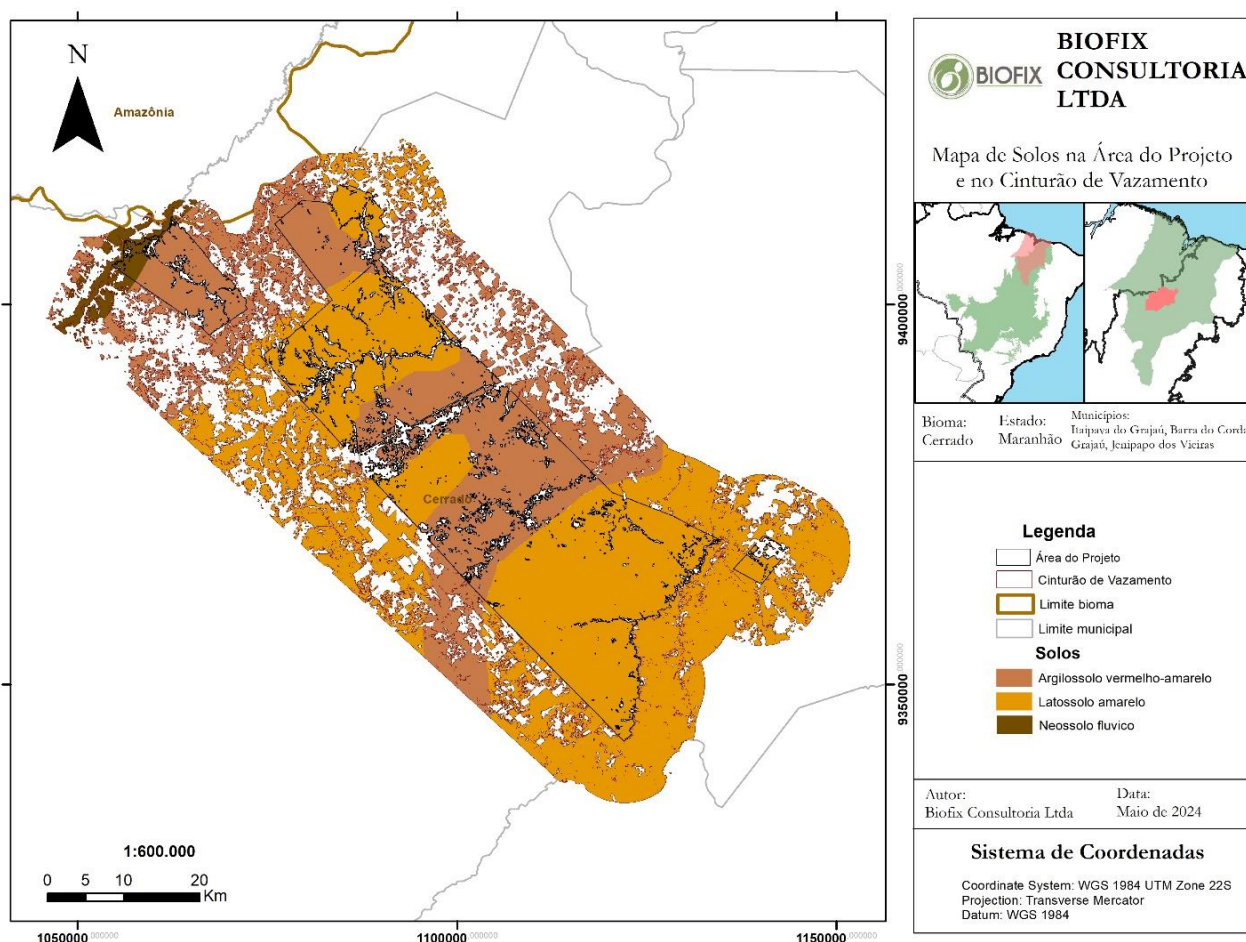
Table 49. Forest in the Project Area and the Leakage Belt

Forest type	PA (ha)	Proportion (%)	RRD (ha)	Proportion (%)	Similarity (±20%)
Forest	150.800	100	151.444	100	Comply
Count	150.800	100	151.444	100	

Source: KA'A PYAHU REDD+ Project.

Soil

According to the 2001 Brazilian Soil Map (EMBRAPA, 2001), the sub-order soils of the Leakage Belt are the same of the project area: yellow- latosol, Red-yellow Argisols, and Fluvic Neosols sub-order. The main soil suborder both in the project area and in the leakage belt corresponds to Yellow-Latosol, with a proportion of approximately 60% in both cases. The second sub-order of soil corresponds to Red-yellow Argisols, which in the case of the project area occupies 40% and in the leakage belt 30.4%.

Figure 37. Soil type distribution in the Project Area and the Leakage Belt


Source: KA'A PYAHU REDD+ Project.

Table 50. Soil type distribution in the Project Area and the Leakage Belt

Soil type	PA (ha)	Proportion (%)	LB (ha)	Proportion (%)	Similarity (±20%)
Yellow-Latosol	89,524.31	59.37	100,791.9	66.55	Comply
Red-yellow Argisols	59,698.62	39.59	46,022.59	30.39	Comply (±25)
Fluvic Neosols	1,577.38	1.05	4,629.86	3.06	No comply
Count	150,800	100	843,490	100	

Source: KA'A PYAHU REDD+ Project.

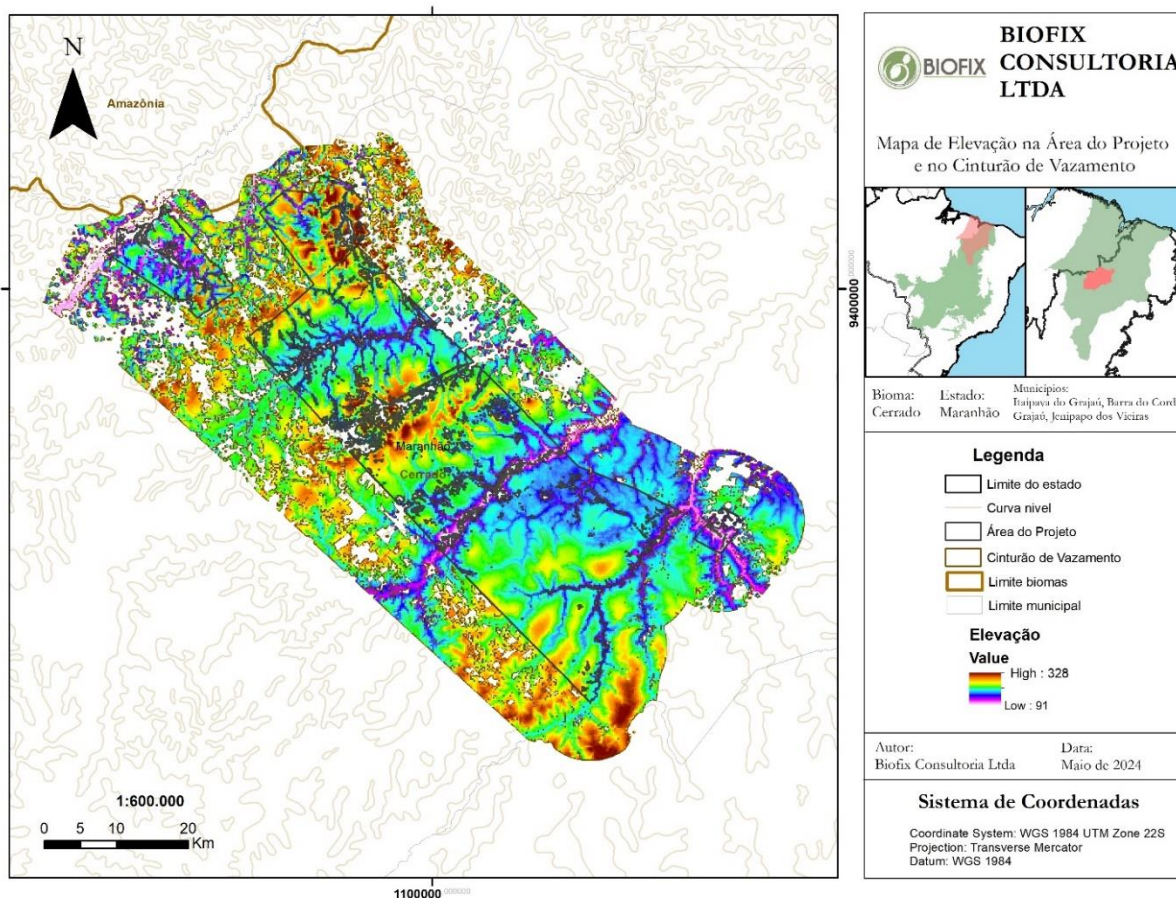
Elevation

In terms of elevation, both the project area and the leakage belt are located in an altitudinal range between 0 and 500 meters above sea level.

Table 51. Elevation class between the Project Area and the Leakage Belt

Elevation (m)	PA (ha)	Proportion (%)	RRD (ha)	Proportion (%)	Similarity (±20%)
0-500	150.800	100	151.444	100	Comply
Count	150.800	100	151.444	100	

Source: KA'A PYAHU REDD+ Project.

Figure 38. Elevation range map in the Project Area and the Leakage Belt


Source: KA'A PYAHU REDD+ Project.

Slope

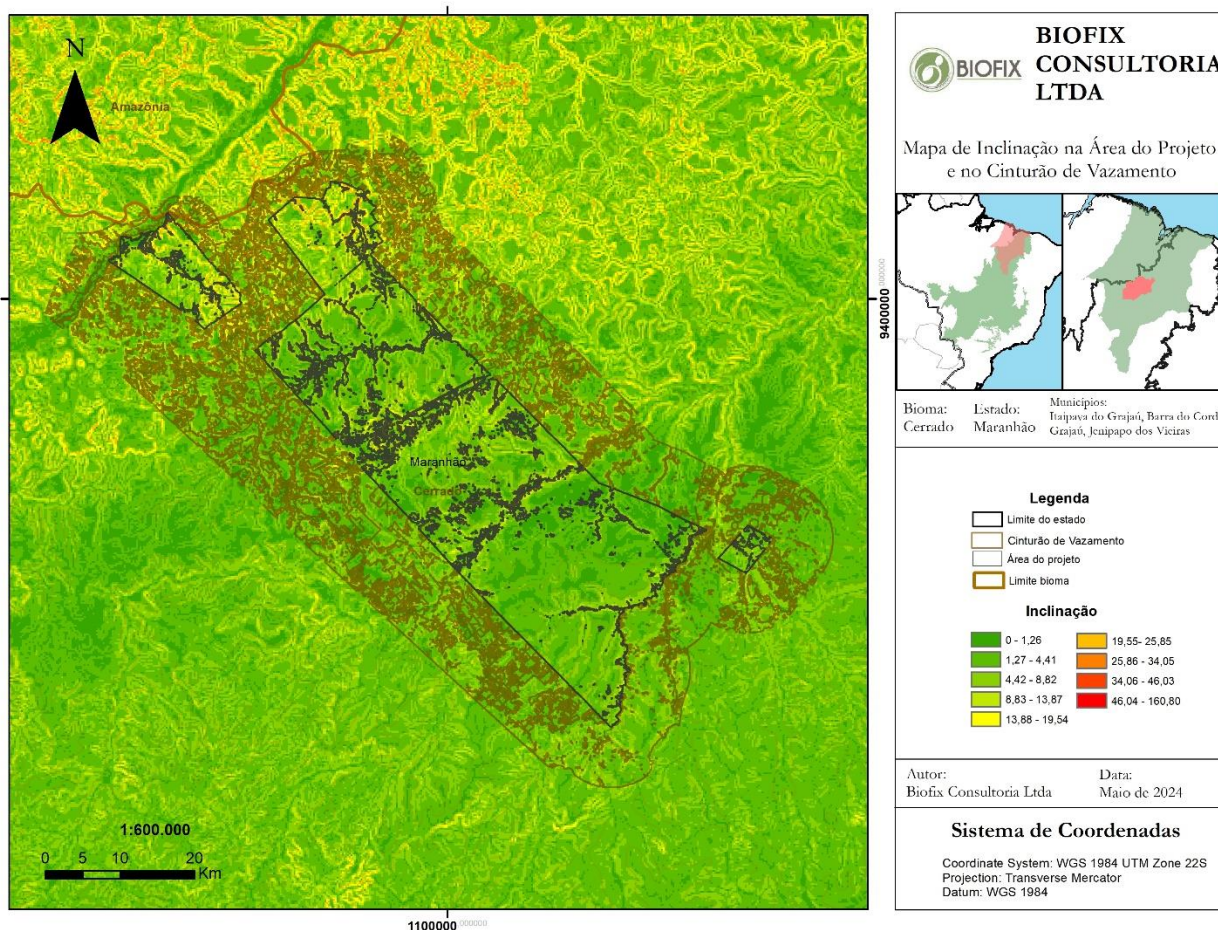
In terms of slope, both the project area and the Leakage belt have a gentle slope (slope < 15%).

Table 52. Slope class distribution in the Project Area and the Leakage Belt

Slope (%)	PA (ha)	Proportion (%)	RRD (ha)	Proportion (%)	Similarity (±20%)
0-500	150.800	100	151.444	100	Comply
Count	150.800	100	151.444	100	

Source: KA'A PYAHU REDD+ Project.

Figure 39. Map of slope class distribution in the Project Area and the Leakage Belt



Source: KA'A PYAHU REDD+ Project.

3.3.3.2 Transportation factors

River density

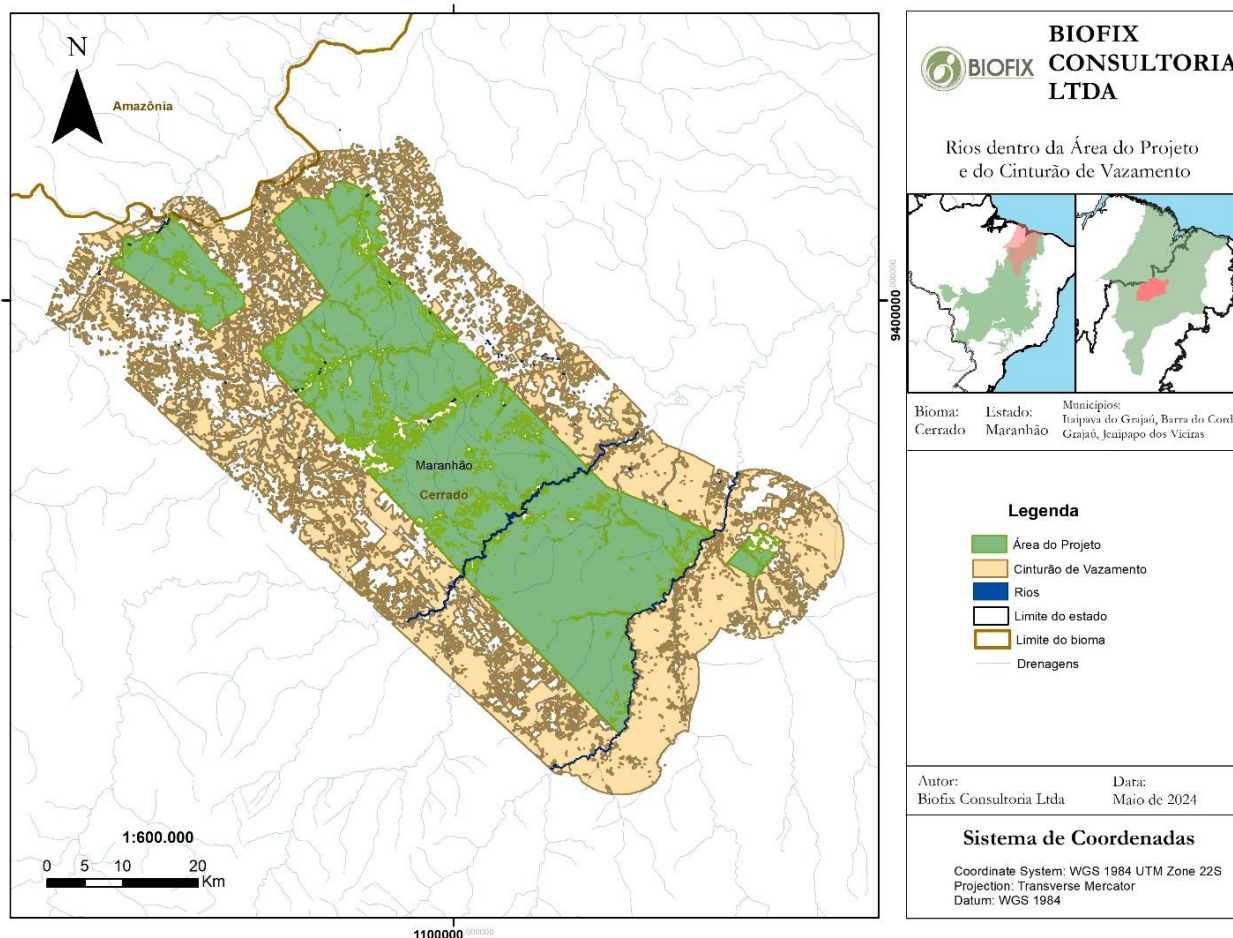
The calculation of the river density for the LB is 163.2 m/km² and in the PA is 154.1 m/km². This means that the river density of the LB does not exceed 20% of the PA river density, in compliance with the VMD0007 module.

Table 53. Rivers density in the Project Area and the Leakage belt.

Long rivers PA (m)	Area PA (km ²)	River density PA (m/km ²)	Long rivers RRD (m)	Leakage Belt (km ²)	River density LB (m/km ²)	Similarity (±20%)
232,357	1,508	154.1	247,117.3	1,514.4	163.2	Comply

Source: KA'A PYAHU REDD+ Project

Figure 40. River density map in the Project Area and the Leakage Belt



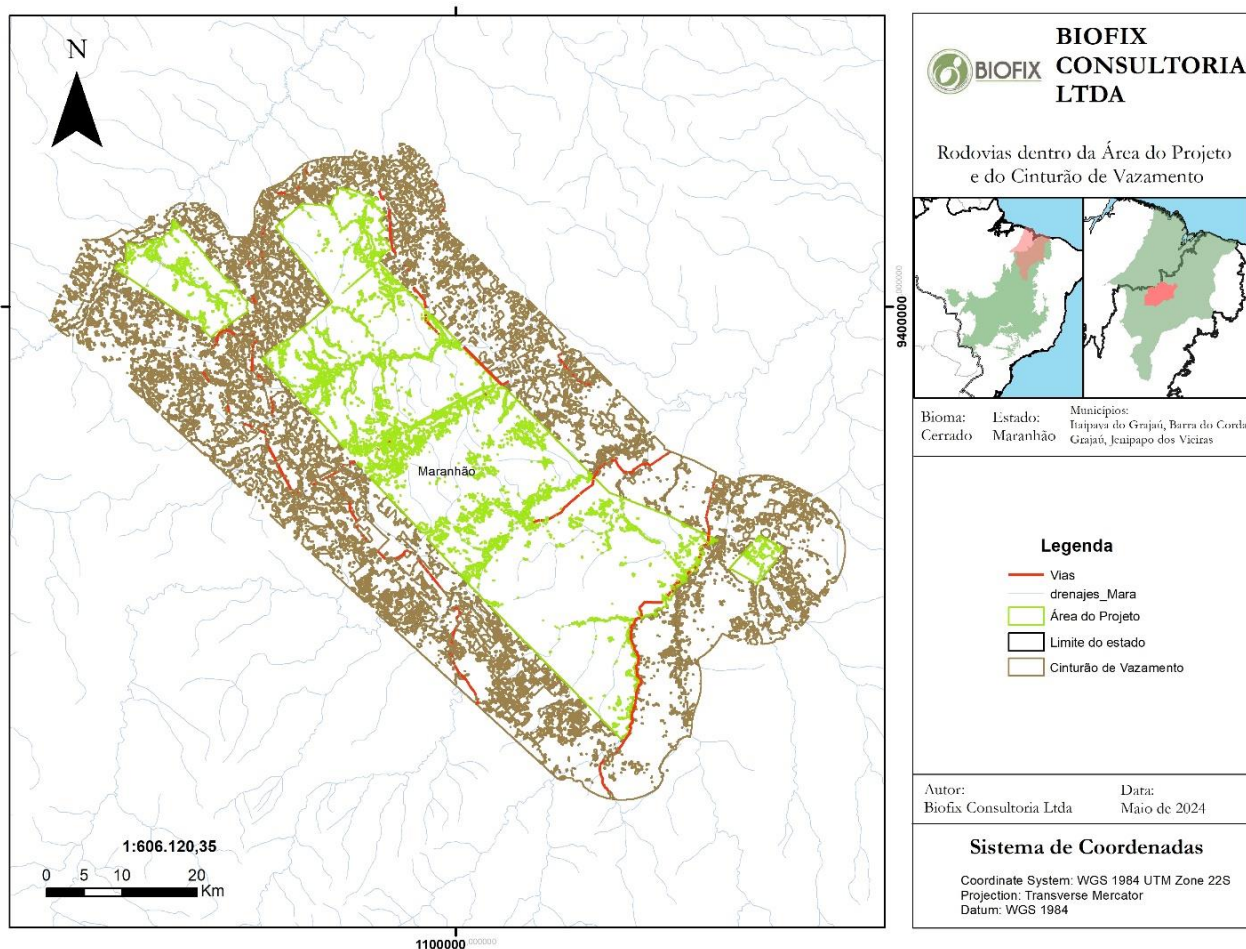
Source: KA'A PYAHU REDD+ Project.

Road density

To obtain the road density for the PA, a 1.0 Km buffer was established over this area, and subsequently, the existing roads within this area were determined. Finally, the density was calculated, using the area with the generated buffer, and dividing it by the total length of the roads.

The analysis of the difference between the PA and LB was conducted in terms of road density (m/km²) resulting in a value of 44.8 m/km² in the Project Area and 53.7 m/km² for the RRD, giving a road density result in the LB of (±20%) than in the PA, complying with the criteria established by the methodology.

Figure 41. Roads inside the PA including a buffer of 1.0 km and in the LB.



Source: KA'A PYAHU REDD+ Project.

Table 54. Road density analysis between the LB and PA

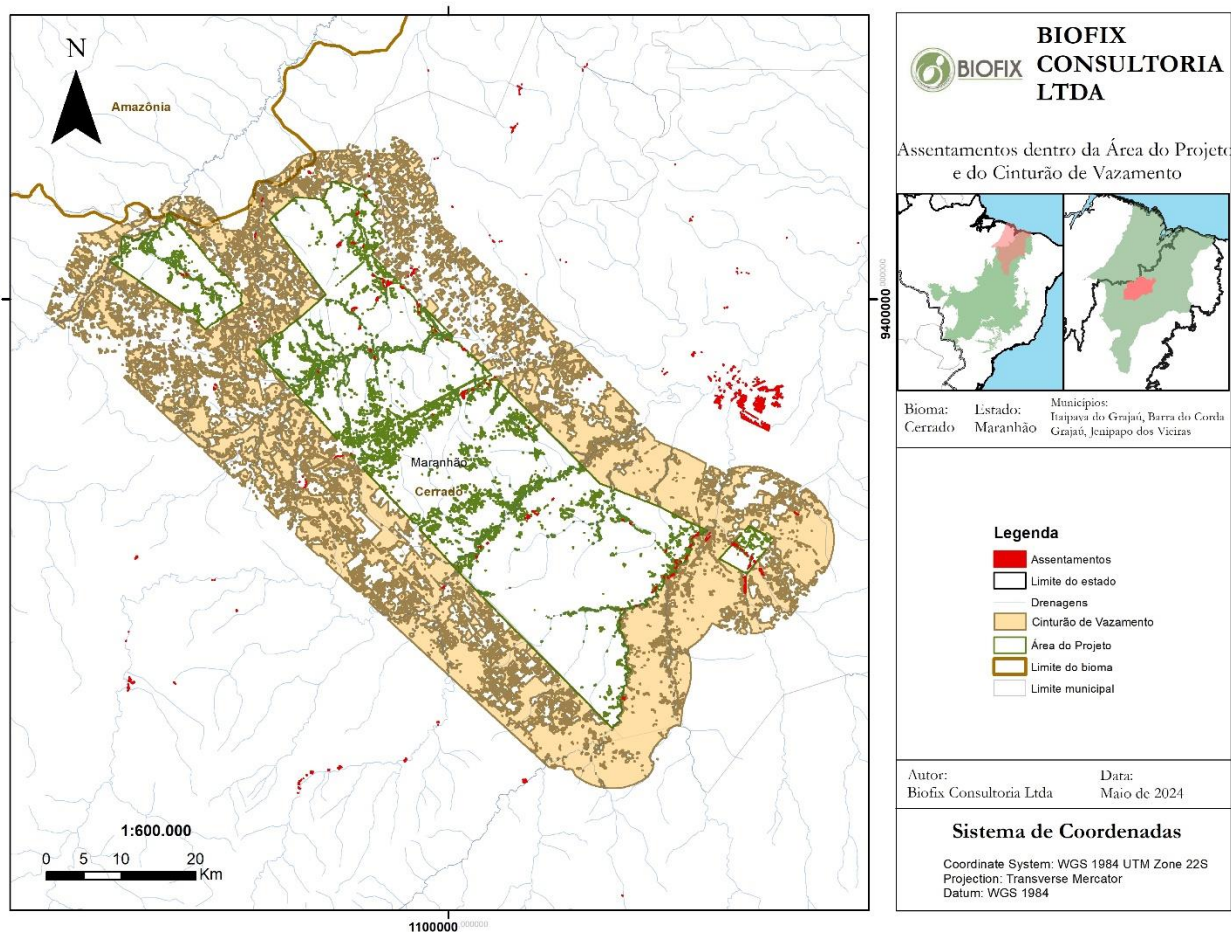
Long roads PA (m)	Area PA (km ²)	River density PA (m/km ²)	Long roads RRD (m)	Area LB (km ²)	River density RRD (m/km ²)	Similarity (±20%)
67,625.01	1508	44.84	81,361.6	1,514.4	53.7	Comply

Source: KA'A PYAHU REDD+ Project.

Settlement density

The settlement density assessment was carried out using the VMD0007 BL-UP module. Thus, a spatial analysis was made in the Project Area and in the Leakage Belt LB, making a buffer of 1km around the areas, and identifying the settlements inside this area, according to official information on urbanized areas in Brazil (IBGE, 2022). Once they were identified, the density was calculated as the table below shows. Finally, it is observed that the settlement density of the Leakage Belt does not exceed more than 20% of the Project Area settlement density.

Figure 42. Settlement density map in the Project Area and the Leakage Belt



Source: KA'A PYAHU REDD+ Project.

Table 55. Settlement density analysis between the Project Area and the Leakage Belt

Settlements PA (#)	Area PA (km ²)	Settlement density PA (#/km ²)	Settlements LB (#)	Area RRD (km ²)	Settlement density LB(#/km ²)	Similarity (±20%)
65	1,508	0,04	29	1,514.4	0,02	Comply

Source: KA'A PYAHU REDD+ Project.

3.3.4 Temporal boundaries

The KA'A PYAHU REDD+ project has the following time limits: the reference period is 11 years, from January 01, 2010, to December 31, 2020.

While the project's crediting period is from May 01, 2021, to April 31, 2041, and the period of the first verification is from May 01, 2021, to December 31, 2023.

3.3.5 Carbon Pools

Table 56. Carbon pools included in the estimate of net emission reductions from REDD activity

Carbon Pool	Included	Rationale/ Explanation of the choice
Aboveground	Yes	A large carbon pool is affected by the project activities
Belowground	Yes	Expected to increase due to project activities
Deadwood	No	Excluded due to lack of information.
Litter	No	Excluded per VCS AFOLU requirements.
Soil organic carbon	No	Excluded
Harvested wood products	No	Excluded due to lack of information.

Source: KA'A PYAHU REDD+ Project.

Table 57. GHG emission sources included or excluded from project boundaries for REDD activity

	Source	Gas	¿Included?	Rationale/ Explanation
Base Line	Biomass burning	CO ₂	Yes	Emissions are included in carbon pool changes.
		CH ₄	No	Not required for REDD projects as per VCS AFOLU requirements, it is conservatively omitted, except when implementing CFE activities.
		N ₂ O	No	Not required for REDD projects as per VCS AFOLU requirements, emissions from woody biomass burning are assumed to be negligible and conservatively excluded except when CFE activities are implemented.
		Other	No	No other GHG gases were considered in this project activity.
	Combustion of fossil fuels	CO ₂	No	They are not significant in the project area.
		CH ₄	No	Not required for REDD projects as per VCS AFOLU requirements, therefore conservatively omitted, except when implementing CFE activities.
		N ₂ O	No	Not required for REDD projects as per VCS AFOLU requirements, emissions from woody biomass burning are assumed to be negligible and conservatively excluded except when CFE activities are implemented.
		Other	No	No other GHG gases were considered in this project activity.
	Use of fertilizers	CO ₂	No	According to the methodology, potential emissions are negligible
		CH ₄	No	Omitted to hold the conservation principle
		N ₂ O	No	N/A
		Other	No	N/A
Project	Biomass burning	CO ₂	Yes	The emissions are included in the changes in the carbon stock.
		CH ₄	No	Not required for REDD projects as per VCS AFOLU requirements, conservatively omitted, except when implementing CFE activities.
		N ₂ O	No	Not required for REDD projects as per VCS AFOLU requirements, emissions from woody biomass burning are assumed to be negligible and conservatively excluded except when CFE activities are implemented.
		Other	No	No other GHG gases were considered in this project activity.
		CO ₂	No	They are not significant in the project area

	Combustion of fossil fuels	CH ₄	No	Not required for REDD projects as per VCS AFOLU requirements, conservatively omitted except when implementing CFE activities
		N ₂ O	No	Not required for REDD projects as per VCS AFOLU requirements, emissions from woody biomass burning are assumed to be negligible and conservatively excluded except when CFE activities are implemented.
		Other	No	No other GHG gases were considered in this project activity.
	Use of fertilizers	CO ₂	No	According to the methodology, potential emissions are negligible
		CH ₄	No	Omitted to hold the conservation principle
		N ₂ O	No	Omitted to hold the conservation principle
		Other	No	No other GHG gases were considered in this project activity.

Source: KA'A PYAHU REDD+ Project.

3.4 Baseline Scenario

According to section 6.1 (determination of the most plausible baseline scenario) of the VM0007 v1.7 methodology, the stepwise approach below must be followed in addition to the tool VT0001 (Tool for the demonstration and assessment of additionality in VCS Agriculture, Forestry and other land use (AFOLU) project activities" V.3) to determine the most plausible baseline scenario. It is worth noting that this tool was used to demonstrate the additionality of the project.

Step 1. Reuse the plausible alternative land use scenarios to the REDD project activity that have been listed as an outcome of sub-step 1b of the additionality tool VT0001

In section 3.5 of this document related to the additionality of the project, as a product of steps 1a and 1b, the following credible and consistent alternative uses were identified, following the conditions (Continuation of the pre-project land-use; Project activity performed on the land within the project boundary without being registered as a VCS REDD project; and activities similar to the proposed project on at least part of the land within the project boundary of the proposed REDD project):

- *Land Use Scenario 1: Agricultural crops - buriti (Mauritia Flexuosa), cassava (Manihot esculenta Crantz), soybean (Glycine max (L) Merrill) and maize (Zea mays).*

Based on what is supported in sections 1.14.1.6 and 3.5 (sub-step 1.a) of this document regarding the type of agriculture practiced by communities, low-scale commercial crops such as cassava, corn, rice, squash, watermelon, beans, yam, taro, sesame, and peanuts are combined. This deforestation driver is also due to limited technical capacity to establish ecologically sound productive alternatives. This implies that producers use unsustainable ancestral activities, such as burning cover crops, which also necessitates expanding intervention areas to maintain basic production for self-sufficiency. This deforestation factor is identified in the reference area, the project area, and the leakage area as typical practices in the Cerrado biome.

As shown in section 3.5 (sub-step 1.a) the area of agriculture within indigenous territories has increased in the historic reference period. Between 2010 and 2017 there was an increase in agricultural areas, of which more than 95% of the area was dedicated soybean cultivation. However, since 2018, there has been a decrease in the area of agriculture, especially due to a decrease in soybean cultivation, which now occupies less than 5% of the proposed area.

- *Land Use Scenario Scenario 2: Conversion of forest lands into livestock farming due to unplanned deforestation*

In Guajajara territories, this deforestation driver is considered. There are a few families primarily raising livestock for subsistence, with commercial aspects limited to family economies. Besides cattle, there are also pig, goat, sheep, and fish farms primarily for family subsistence. Local records indicate instances of land invasion by ranchers for cattle grazing, leading to local conflicts.

As shown in section 3.5 (sub-step 1.a) the area of pastures within indigenous territories has increased in recent years by approximately 300% between 2010 and 2020. This situation is a reflection of what is happening at the municipal level. In all municipalities in the project area, between 2010 and 2020, there was an average increase of 125% in the area dedicated to pasture. In the municipality of Grajaú there was an increase of 60,270 hectares (139.6%), being the municipality with the greatest increase in pasture land.

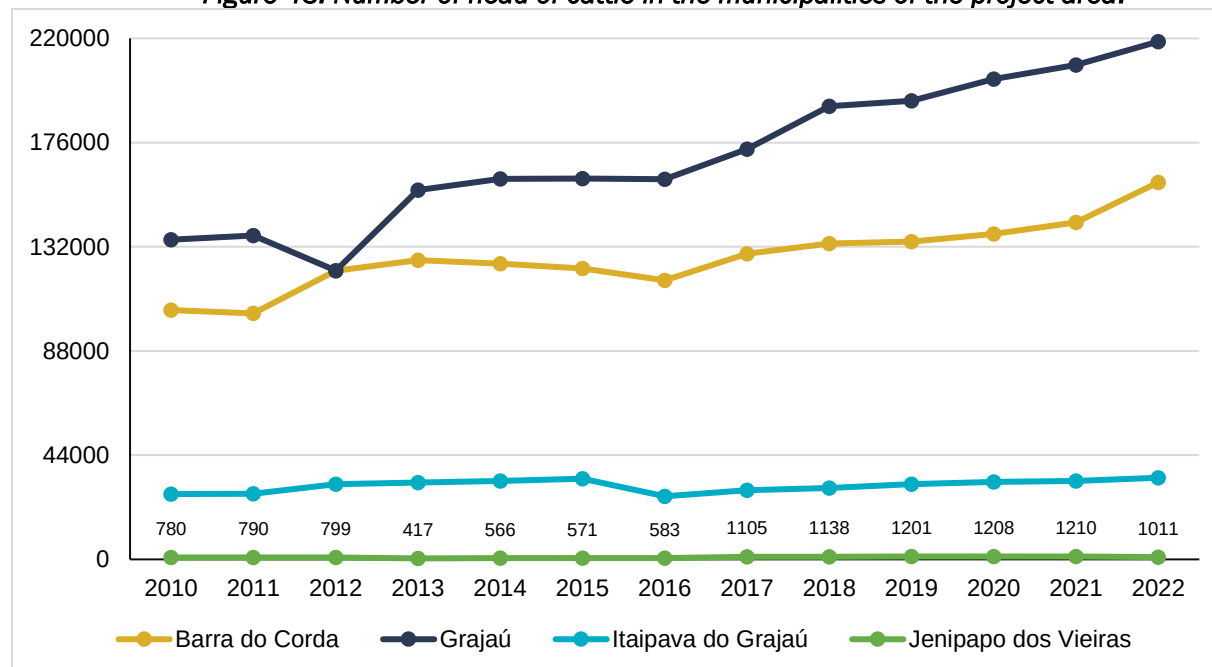
Table 58. Pasture area in the indigenous territories and municipalities of the project area in the historical reference period

Indigenous land/Municipality	2010 (ha)	2020 (ha)	Increase 2010-2020 (ha)	Increase (%)
Urucu Juruá	1,360.58	4,387.88	3,027.31	322.50
Cana Brava Guajajara	446.16	816.25	370.09	182.95
Lagoa Comprida	12.66	47.41	34.75	374.46
Rodeador	319.29	776.87	457.58	243.31
Barra do corda	90,473.46	115,576.03	25,102.57	127.75
Grajaú	152,227.84	212,497.83	60,269.99	139.59
Itaipava do Grajaú	45,023.49	52,167.63	7,144.14	115.87
Jenipapo dos Vieiras	52,220.01	62,812.55	10,592.54	120.28

Source. Adapted from Mapbiomas data.

Similarly, according to IBGE data, the number of cattle in the municipalities has increased significantly in recent years. This shows a correlation between the increase in pasture area and the effective increase in the number of cattle. Thus, the municipality of Grajaú, which presented the greatest increase in pasture area, was the municipality that had the greatest increase in the number of head of cattle, going from 135015 (2010) to 202849 (2020), for an increase of 67,834 head of cattle.

Figure 43. Number of head of cattle in the municipalities of the project area.



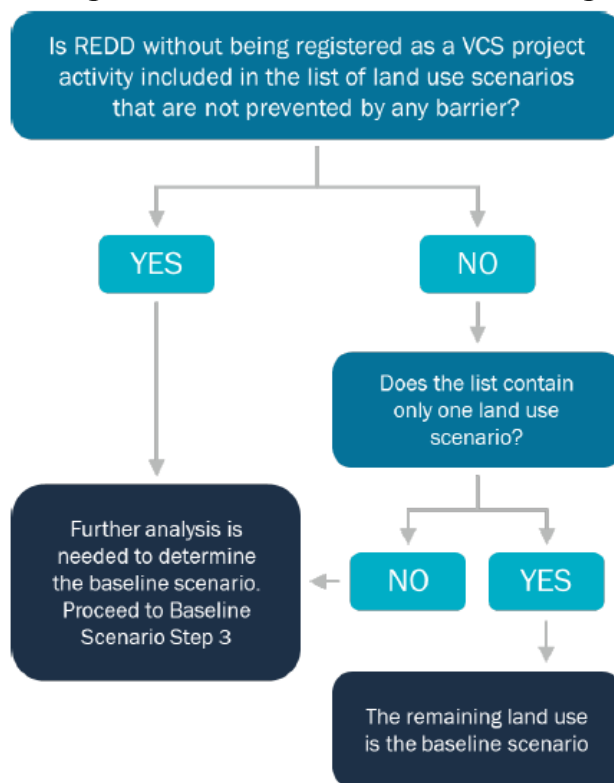
Source: Adapted from <https://cidades.ibge.gov.br/pesquisas>

- *Land use scenario 3: Forest conservation without registration of the VCS AFOLU project.*

The third scenario corresponds to conservation activities that were carried out with the KA'A PYAHU project without being registered as a VCS AFOLU project. Such activities are presented in detail in the section 1.12.

Step 2. Where the VT0001 barrier analysis is used to demonstrate additionality, apply the decision tree figure to list of all alternative land use scenarios from step 1, that are not prevented by any barrier:

Figure 44. Tree figure decision baseline scenario according to the VM 0007



Source: VM0007 REDD+ Methodology framework V.1.7.

As explained in step 3 of additionality section 3.5, two types of barriers were identified:

- ✓ Investment barriers: Lack of access to credit - Debt finance is not available for land use scenarios.
- ✓ Technological barriers: Lack access to necessary materials Technical capabilities.

As explained in step 3b, both alternative land use scenarios 1(*Agricultural crops*) and 3 (*Forest conservation without registration of the VCS AFOLU project*) present these financial and technical bars. This is basically due to different economic and social factors of the indigenous territories: 80% of the indigenous population does not have a monthly income, the literacy rate is 50%, lack of coverage in services such as electricity and water, coming from the general sewage or rainwater system. Also, social and economic factors of the municipalities where the project is developed: incidence of poverty, GDP per capita, position of the municipalities with respect to GDP, human development index and position of agriculture as an economic activity (see in detail step 3b, section 3.5).

Therefore, the land use scenario 2 (*Conversion of forest land into cattle*) is not prevented by any barrier, then, according with the VM 0007, this remaining land use scenario is the baseline scenario.

3.5 Additionality

To identify the baseline scenario and demonstrate the additionality of the KA'A PYAHU Project, the "Tool for the demonstration and assessment of additionality in VCS Agriculture, Forestry and other land use (AFOLU) project activities" V.3 was used. The additionality analysis is detailed below, following the tool's steps:

Step 1: Identification of alternative land use scenarios for the proposed activity of the VCS AFOLU project.

Sub-step 1a: Identify credible alternative land use scenarios for the proposed VCS AFOLU project

Three alternative land use scenarios were identified in the indigenous land, where the project is located. To demonstrate that the scenarios are credible and realistic, an analysis of historical land use and previous activities in the project area was carried out. Similarly, to demonstrate the coherence and consistency of the scenarios with the normative framework, bibliographic consultation was carried out on territorial planning documents at municipal and local level, development plans at regional level and norms at national level.

Land Use Scenario 1: Agricultural crops - buriti (*Mauritia Flexuosa*), cassava (*Manihot esculenta Crantz*), soybean (*Glycine max (L) Merrill*) and maize (*Zea mays*).

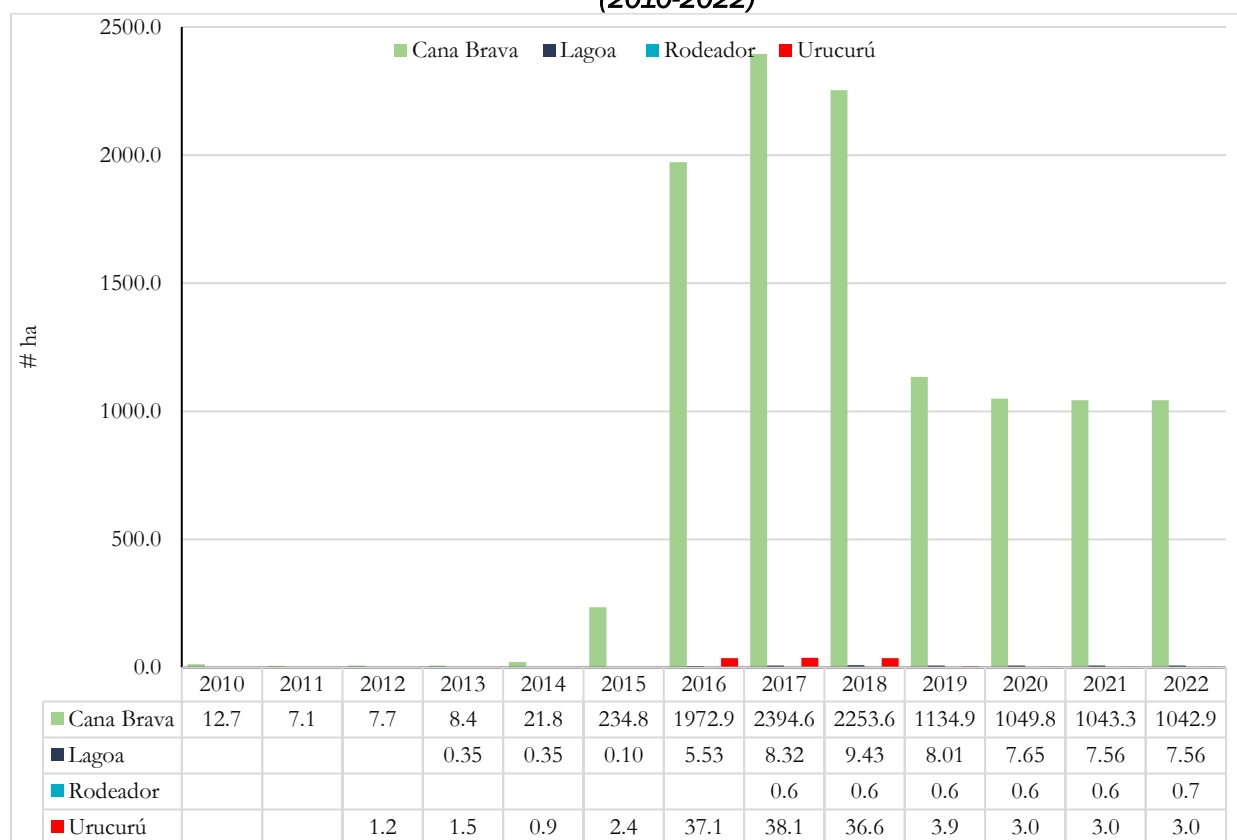
The first alternative land use scenario involves agricultural cultivation of native species essential for family nutrition and part of the ritual festival agenda of the Guajajara people.

As described in section 1.18.1, agriculture is an important productive activity for the Guajajara people and also in the municipalities surrounding Indigenous Lands. According to data from the State Secretariat of Agriculture, Livestock, and Fisheries - SAGRIMA, through the Superintendence of Research and Geoprocessing - SPG, which produced the "Profile of Maranhão Agriculture Bulletin based on Municipal Agricultural Production - PAM 2020" (Annex 046)¹⁸ from the Brazilian Institute of Geography and Statistics (IBGE, 2021), data supporting this alternative land use scenario were collected. According to 2022 data from MapBiomias, in the municipality of Barra do Corda there is a total of 39,373,495 hectares destined for soybean planting, the municipality of Grajaú, for the same purpose, allocates 39,392,965 hectares, and Jenipapo dos Vieiras, in turn, 39,393,124 hectares.

More specifically in the indigenous territories that are part of the project, according to Mapbiomas data, in recent years the area dedicated to agriculture has increased, showing a peak of growth in 2016. For example, in the Cana Brava indigenous territory, the number of hectares under agriculture increased from 235 hectares in 2015 to 1972 hectares in 2016. That is, it increased in a proportion of approximately 8 times the area in one year. Similarly, in the Urucu/Juruá indigenous territory, the area went from 2.4 hectares in 2015 to 37 hectares in 2016. In the Cana Brava indigenous territory, the coverage of agricultural land in 2010 was 12.7 hectares and in 2020, the year before the start of the project, it was 1,050 hectares. That is an increase of 1,037 hectares, and a cumulative difference of 9,098 hectares in the historical reference period in the project area

¹⁸ THE STATE SECRETARIAT OF AGRICULTURE, LIVESTOCK AND FISHING - SAGRIMA, through the Research and Geoprocessing - SPG, periodically presents the Bulletin: Profile of Maranhão Agriculture based on Production data Municipal Agricultural - PAM 2020 from the Brazilian Institute of Geography and Statistics (IBGE, 2021). The objective of this Bulletin is to outline a brief profile of agriculture in the state of Maranhão. The bulletin highlights: the role of agriculture in the economy and job market in Maranhão; to the main crops produced; the main municipalities producers; regional vocations and the relevance of production of Maranhão in the regional and national context.

Figure 45. Area under agricultural land cover in the indigenous territories of the project area (2010-2022)



Source. Adapted from Mapbiomas data.

A large part of the agricultural areas in the indigenous territories of the project area are related to temporary soybean crops, especially in the years prior to 2018. For example, between 2010 and 2017, of the annual area dedicated to agriculture, soybeans accounted for an average of 97% of these agricultural areas. Since 2018, the percentage has progressively reversed, with soybean crops occupying less than 3% of agricultural areas in recent years, and other temporary crops, occupying the largest percentage of areas.

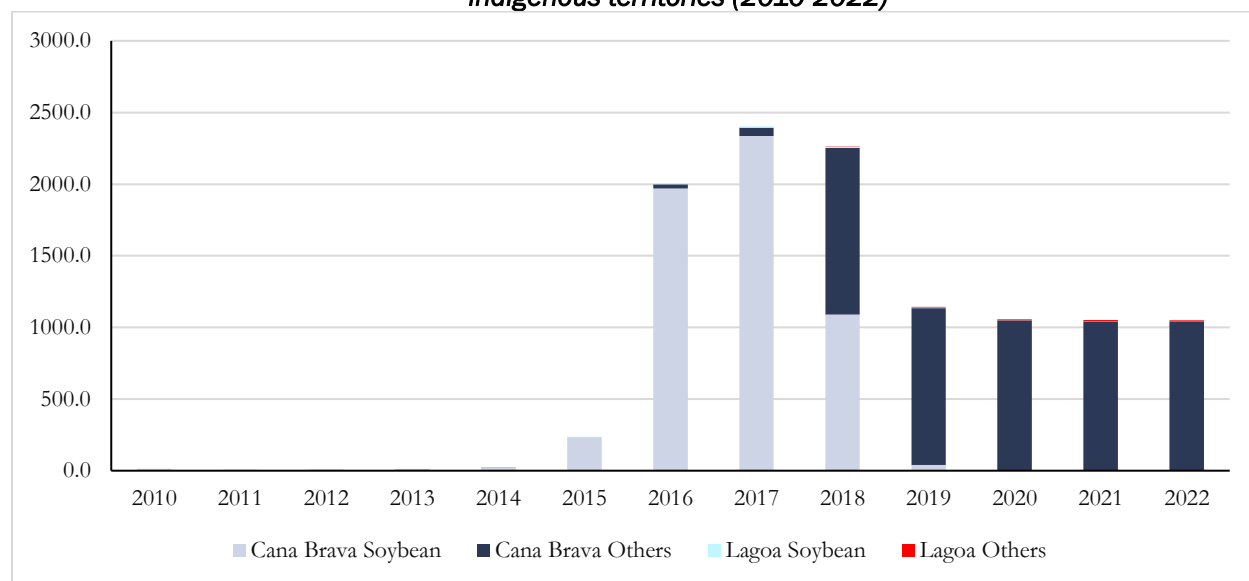
Other crops include the planting and native culture of buriti, corn and cassava, which are Guajajara cultural activities; and were presented during the safeguards workshop. Furthermore, these crops are in the technical literature on the Guajajara people. (Frasao, 2019) , in " The indigenous food of Maranhão: A Theoretical Focus on the food culture of the Tenetehara-Guajajara Indians" demonstrates the importance of these species in Guajajara food and intangible culture. Soy, according to the latest census mapping of the municipalities of Jenipapo dos Vieiras, Grajaú and Barra do Corda, drives local economies and presents a large flow of crops in the regions covered by the project¹⁹.

According to the Agricultural Census 2017, developed by the Brazilian Institute of Geography and Statistics, agricultural activities carried out in Indigenous Lands where the project is being developed include the following types of family farming, according to the FAO/Inra study (1996), attempting to draw a distinction between patronal and family holdings, which considers: family establishments that simultaneously meet the following conditions: a) work direction is carried out by the producer; b) no expenses with contract labor; c) number of permanent and temporary employees is less than or equal to

¹⁹ more information about soybean planting in Barra do Corda: <https://www.agrolink.com.br/regional/ma/barra-do-corda/estatistica>; Grajaú: <https://www.agrolink.com.br/regional/ma/grajau/estatistica> y Jenipapo dos Vieiras: <https://www.agrolink.com.br/regional/ma/jenipapo-dos-vieiras> and MapBiomas.

four temporary workers (without permanent employees) or three temporary workers (in case of at most one permanent employee), calculated on an annual average; d) total area is less than or equal to 500 hectares for Southeast and South regions and 1000 hectares for other regions (Schmitz & da Mota, 2006)

Figure 46. Area under soybean and other temporary crops in the Cana Brava and Lagoa Comprida indigenous territories (2010-2022)



Source. Adapted from Mapbiomas data

Table 59. Agricultural establishments, by FAO/INCRA family farming classification, according to Declared Indigenous Lands – 2017.

Indigenous land	Total	Unfamiliar	AF type A	AF type B	AF type C	AF type D
Urucu Juruá	93		1	33	44	15
Cana Brava Guajajara	731	4	56	230	197	244
Lagoa Comprida	144	1	7	37	58	41
Rodeador	41		2	16	18	5

Source: Brazilian Institute of Geography and Statistics, adapted.

Land Use Scenario Scenario 2: Conversion of forest lands into livestock farming due to unplanned deforestation

The second alternative land use scenario involves the conversion of forest lands into livestock farming due to unplanned deforestation. This scenario arises from the continuation of previous land use activities and the absence of the KA'A PYAHU project.

As mentioned in section 3.4.1.2, unplanned deforestation is one of the agents of deforestation in the project area and its surroundings, where the Guajajara people engage in agricultural, livestock, and extractive activities on a small scale.

The Guajajara people from the project's proposing territories report that families raise pigs, cattle, goats, sheep, poultry, and chickens, and there are also fish farms. However, none of these activities are intended

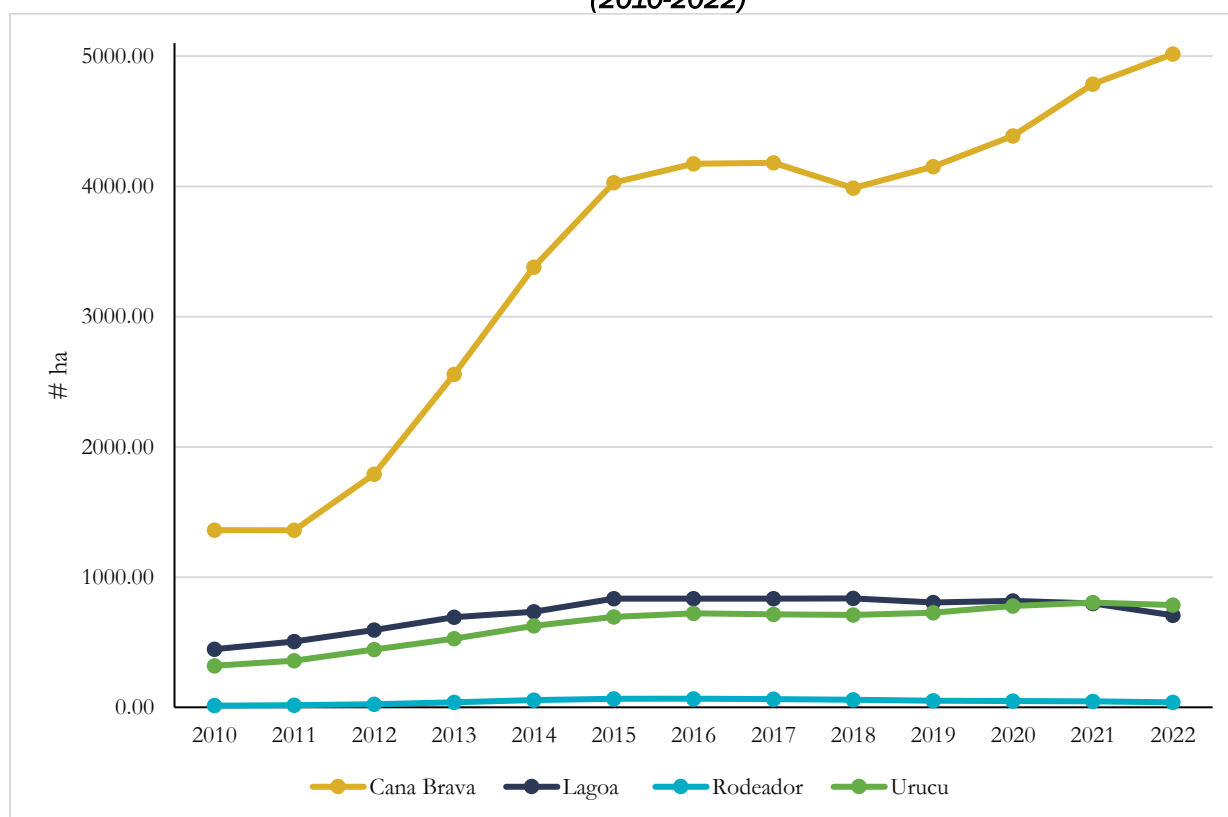
for large-scale commercial use; they are restricted to family subsistence, and occasional sales contribute minimally to their livelihoods. Community leaders stated that there is currently no documentation regarding the number of livestock in each territory, no official physical records of these activities, and that they are not significant in terms of quantity.

According to land use information from the Agricultural Census provided by the Brazilian Institute of Geography and Statistics (2022) in the project area, cattle ranching is the main economic activity in the agricultural sector of Maranhão state. The cattle herd, numbering 3,902.6 thousand head, is primarily used for meat production.

Commercial cattle ranching is prohibited on indigenous lands in Brazil, but according to reports from the Guajajara people, there is illegal invasion by local ranchers who use parts of the territories as pasture.

According to data obtained from Mapbiomas, pasture cover in the indigenous territories that are part of the project has increased significantly in recent years, especially between 2010 and 2017 in which the growth occurred in a logarithmic manner. In all indigenous territories, the pasture area between 2010 and 2020 (historical reference period), doubled and in some cases even tripled: In the Cana Brava territory, it went from 1,360 hectares (2010) to 4,387 hectares (2020), with an increase of 323% of the area. The Lagoa Comprida territory increased from 446 hectares to 816 hectares between 2010 and 2020, an increase of 182%. In the Rodeador territory, for the same years, pasture cover increased from 12 hectares to 47 hectares, an increase of 374%. Finally, in the Urucu territory, there was an increase of 243%, from 319 hectares to 776 hectares of pasture between 2010 and 2020.

Figure 47. Area in pasture land use category in the indigenous territories of the project area (2010-2022)



Source. Adapted from Mapbiomas data

Land use scenario 3: Forest conservation without registration of the VCS AFOLU project.

The third scenario corresponds to conservation activities that were carried out with the KA'A PYAHU project without being registered as a VCS AFOLU project. Such activities are presented in detail in the section 1.12.

It is important to highlight that these activities do not involve economic activities, therefore, they do not generate financial or economic benefits.

Outcome of Substep 1a: List of credible alternative land use scenarios that would have occurred within the project area boundaries:

- Land use scenario 1: Agricultural crops.
- Land use scenario 2: Conversion of forest land into cattle.
- Land use scenario 3: Forest conservation without registration of the VCS AFOLU project.

Substep 1b: Consistency of reliable land use scenarios with applicable laws and regulations

To demonstrate the coherence and consistency of scenarios with the regulatory framework, a bibliographic consultation was conducted on municipal and local territorial planning documents, regional development plans, and national-level norms.

Land Use Scenario 1: Agricultural crops - buriti (*Mauritia Flexuosa*), cassava (*Manihot esculenta Crantz*), soybean (*Glycine max (L) Merrill*) and maize (*Zea mays*).

According to the Classification and Quantification of Land in the State of Maranhão regarding Use, Agricultural Suitability, and Legal Protection Conditions (Annex 049) indigenous lands in Maranhão total 19,065.49 km². Protected integral and sustainable use conservation units occupy 14,399.39 km² and 45,965.74 km² respectively. The total area protected by conservation units and indigenous lands is 79,169.33 km², which is 23.85% of the state's total area. The calculated areas do not include maritime areas within some conservation units.

Of the total land in the Amazon Biome, 35.3% was occupied by natural vegetation, 31.7% by pastures, and 27.9% by agriculture. In the Cerrado Biome, 61.0% of the land was occupied by pastures, 29.5% by natural vegetation, and 8.4% by agriculture. Of the land occupied by agriculture, 64.3% was in the Amazon Biome; of the land occupied by pastures, 77.0% was in the Cerrado Biome. Most of the natural vegetation (59.8%) was in the Cerrado Biome. Agriculture occupied 6.2% of protected areas, 19.3% of areas near drainage networks, and 18.8% of unprotected areas far from drainage networks. Pastures occupied 40.1% of protected areas, 54.7% of areas near drainage networks, and 55.2% of unprotected areas. Natural vegetation remained on 47.1% of protected areas, 24% of areas near drainage networks, and 25.3% of unprotected areas.

According to IBGE CITIES²⁰ and IBGE Agro 2017²¹, in the municipality of Barra do Corda, agriculture corresponds to 53% of the GDP, in Jenipapo dos Vieiras, 49,973 hectares are dedicated to agricultural activities, and Grajaú has 477,768 hectares dedicated to agricultural crops.

According to the "Bulletin: Profile of Maranhão Agriculture based on Municipal Agricultural Production Data - PAM 2018 from the Brazilian Institute of Geography and Statistics (IBGE, 2018) (Annex 047) prepared with the support of the State Secretariat of Agriculture, Livestock and Fisheries - SAGRIMA. In maize cultivation, Maranhão produced 1,230,971 tons in the 2017/2018 harvest, with the southern

²⁰ <https://www.ibge.gov.br/en/cities-and-states.html>

²¹ <https://censoagro2017.ibge.gov.br/>

region of the state leading in grain production. In cassava cultivation, Maranhão ranks 1st in the northeastern region and 7th in quantity produced in Brazil. In the 2017/2018 harvest, it produced 681,018 tons. Buriti also has significant importance in the land use of the Guajajara and indigenous peoples of the region.

Another important source of data is the Agropecuarian Zoning - ZAMA²², which is a guiding instrument for policies in the productive sector, contributing significantly to minimizing losses in agriculture, thus constituting an essential piece in the execution of public policies and agricultural security, and constituting an important tool for socioeconomic integration and technological aggregation. In addition, the Macro-Ecological-Economic Zoning (MacroZEE)²³ of the State of Maranhão, established by article 242 of the Constitution of the State of Maranhão, is a guiding document for land use and occupation and rational use of natural resources, whose guidelines now guide public policies aimed at sustainable development and promoting the well-being of the population of the State of Maranhão.

In other words, in addition to being part of land use planning, the scenario of agricultural crops is coherent within the regulatory framework and is also part of the regional vision.

Land Use Scenario Scenario 2: Conversion of forest lands into livestock farming due to unplanned deforestation

According to the State Foundation for Statistical Analysis System (SEADE)²⁴, in the state of Maranhão, beef cattle farming is the main economic activity in the agricultural sector of the state. Its herd, totaling 3,902.6 thousand head, is almost entirely destined for meat production. Maranhão has the second-largest bovine herd in the Northeast region, behind only Bahia. Beef and dairy activities, along with associated pasture reform, account for more than 90% of the labor force in livestock farming.

The latest survey conducted in 2022 by the Brazilian Institute of Geography and Statistics reveals that the municipalities of Barra do Corda, Jenipapo dos Vieiras, and Grajaú have livestock activities, especially cattle herding, as one of their main land use activities.

The Guajajara people from the indigenous territories proposing the KA'A PYAHU project have cattle, poultry, and pig pastures for family consumption, with minimal commercialization for subsistence.

Land use scenario 3: Forest conservation without registration of the VCS AFOLU project.

In section 1.15 of the PDD, the laws and regulatory framework under which the KA'A PYAHU project is framed were listed.

Outcome of step 1b:

According to the information presented, the three alternative land use scenarios are in compliance with mandatory legislation:

1. Agricultural crops
2. Conversion of forest land into cattle

²² ZAMA is a technical instrument that seeks to support actions aimed at state Agriculture, aiming to reduce losses and create perspectives that increase production and productivity in this sector, resulting in economic improvements and people's quality of life. In this way, Agricultural Zoning represents an essential instrument in the execution of public policies and agricultural security, constituting an important support tool for the socioeconomic and environmental development of Maranhão.

²³ Ecological-Economic Zoning (ZEE) is an instrument for planning and ordering the Brazilian territory, harmonizing economic, social and environmental relations. It demands an effective institutional sharing effort, aimed at the integration of territorial public actions and policies, as well as articulation with civil society, integrating their interests around a pact for territorial management. DECREE No. 35,177, OF SEPTEMBER 12, 2019 Law No. 11269 OF May 28, 2020.

²⁴ The State Statistical Data Analysis System Foundation, better known as the Seade Foundation, is an entity of the Planning and Management Secretariat of the Government of the state of São Paulo. It is a national reference center in the production and dissemination of socioeconomic and demographic analyzes and statistics

3. Forest conservation without registration of the VCS AFOLU project

Sub-step 1c. Selection of the baseline scenario:

The analysis presented in the following steps (3a and 3b) allows the baseline scenario to be determined. As a result of barrier analysis (step 3b), the only scenario that is not prevented by any barriers is scenario 2: Conversion of forest land into cattle ranching due to unplanned deforestation. While scenario 1: Agricultural crops and scenario 3: Forest conservation without registration of the VCS AFOLU project are prevented by the barriers analyzed.

Step 3: Barrier analysis

According to the "Tool for the demonstration and assessment of additionality in VCS Agriculture, Forestry and other land use (AFOLU) project activities V3.0", barrier analysis can be performed instead of investment analysis (step 2), therefore, this step is used to determine the additionality of the project.

Sub-step 3a. Identify barriers that would prevent the implementation of the type of 2.3.1 proposed project activity

According to the T-ADD tool, the barriers that the previously identified land use scenarios could potentially face are as follows:

- Investment barriers
 - Lack of access to credit

Regarding access to financial services in the country, especially concerning credit, there is low access for several reasons:

- People often do not know what products they need or do not understand them.
- They know what products they need but do not have access to them.
- Financial institutions are not aware of the needs of those who are excluded.
- When institutions are aware of the needs, they may not have the necessary products to include them.
- They have the products or can offer them, but do not do so due to risks, costs, and/or regulatory restrictions.

The balance of the 2021 General Budget of the Union carried out by INESC (Institute of Socioeconomic Studies) (Annex 048) showed that, in 2021, R\$ 746.34 million was authorized for the budget of FUNAI (National Indian Foundation), but only R\$ 139.80 million was allocated to the entity's final actions, those directly impacting indigenous communities, such as the protection and promotion of indigenous peoples' rights, environmental and territorial management, incentive for sustainability in line with their lifestyles, actions of an assistive nature, such as distribution of basic food baskets, incentive for participation in public policies, among others. The agenda of the 2024 Balance of the Union 2023, points out that it is necessary to develop financial mechanisms for the acquisition of funds for the protection and management of indigenous lands and indigenous lands.

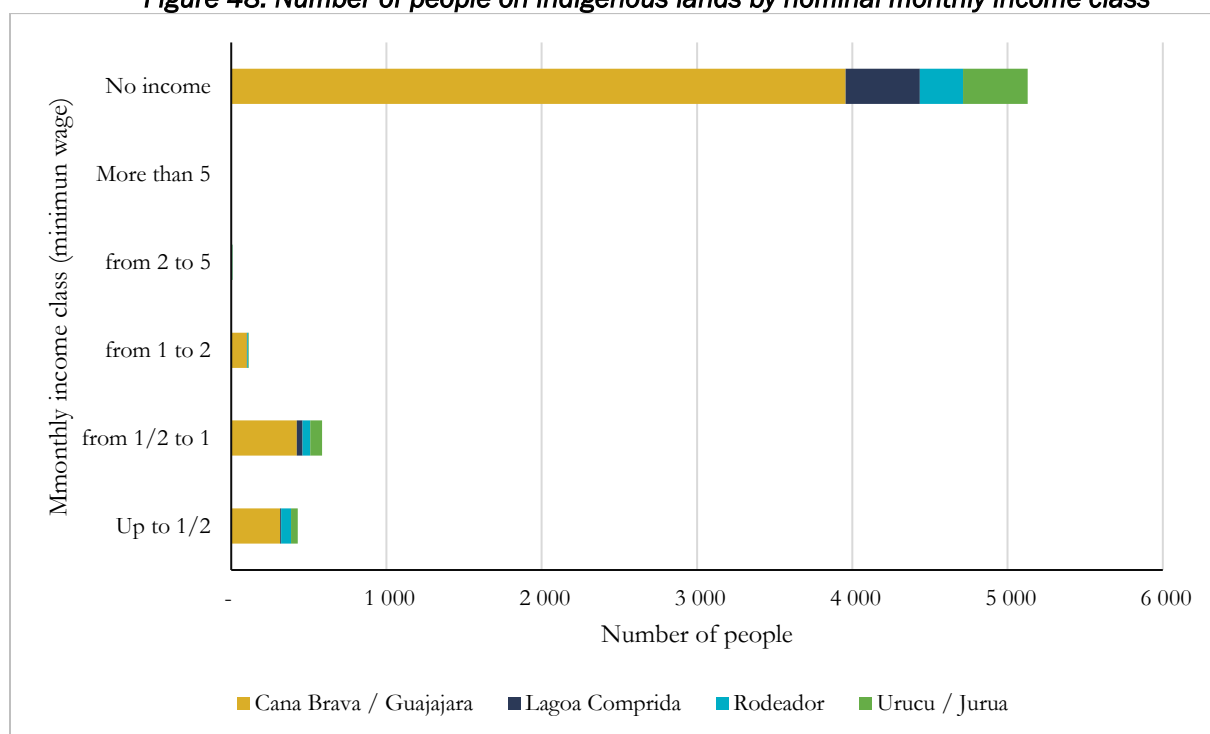
Global Environment Fund (GEF), in 2023, concluded that less than 1% of the international aid to address climate change goes directly to traditional communities and indigenous peoples - or to projects that ensure the possession of their lands and forest management. Despite this, studies reinforce the important role of these groups for effective and fair environmental protection.

In Brazil, indigenous and traditional communities still face many bureaucratic barriers in their own countries, as legal requirements in many cases are complex and restrict the release of international funding. In several cases, communities do not even have legal status to receive and manage large funds, and many live in remote areas without access to banks.

Indigenous populations in Brazil, like the Guajajara, still face very basic needs such as creating associations, regularization, and accessing credit lines offered in the country, which demands a complex bureaucratic structure that often prevents traditional populations from accessing public policies for the conservation and preservation of their territories.

In addition, the economic context of the inhabitants of the indigenous territories is very limited. According to data from the 2010 Demographic Census conducted by the IBGE²⁵, in the chapter related to indigenous territories, On average in the 4 territories, 80% of the inhabitants had no income: for example, of the 4,803 inhabitants over 10 years of age counted in the Cana Brava territory, 82% (equivalent to 3,957 people) had no income and only 2.2% had an income of 1 to more than one minimum wage²⁶. In the indigenous territory of Lagoa Comprida, of the 524 people counted, 92% (478 inhabitants) had no income, and less than 1% had incomes equivalent to 1 or more minimum wages. This situation demonstrates a limitation of the financial capacity of the inhabitants to autonomously develop conservation activities and monitor deforestation control.

Figure 48. Number of people on indigenous lands by nominal monthly income class



Source. Adapted from IBGE Census 2010.

➤ Technological barriers

- Lack of access to necessary materials, for example, planting materials.

Access to materials, inputs, and equipment necessary for the development of alternative land use projects is related, among other factors, to public service coverage, connectivity, and ease of transportation within municipalities.

²⁵ <https://censo2010.ibge.gov.br/>

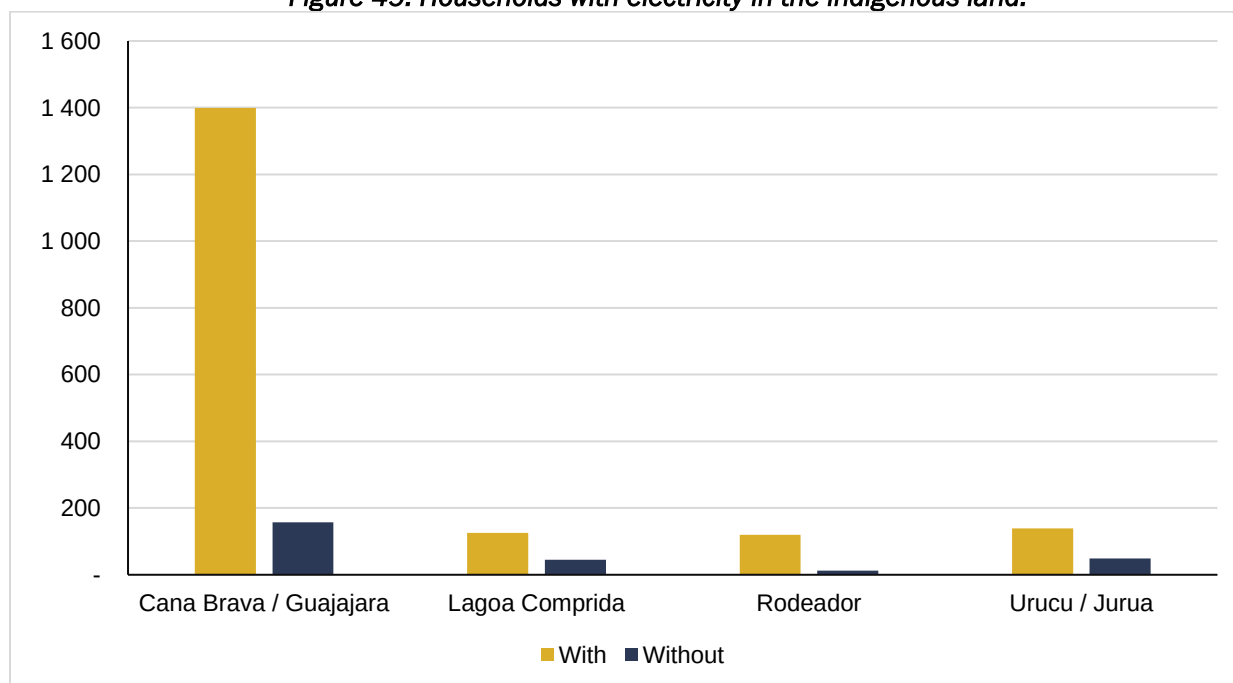
²⁶ Minimum wage used: R\$ 510.00

In Brazil, the main actions executed by the Government aimed at promoting access to information by family farmers primarily involve the Ministry of Agrarian Development (MDA) and the Ministry of Communications (MC); however, these actions can be coordinated with other ministries.

The combination of technology, knowledge, and innovation capacity has been essential for the development of Brazilian agriculture. However, small-scale farms, for the most part, have not kept pace with the development observed on large rural properties in recent decades. Therefore, to continue promoting rural development, it is necessary to provide technological and innovative alternatives for family farmers, especially smaller ones, to enhance productive inclusion and consequent income generation, improve quality of life, and above all, facilitate the family succession process. Moreover, innovation on small-scale farms will also contribute to adopting practices that optimize natural resource use, thus making family production increasingly environmentally sustainable.

To enable productive inclusion of family farmers in the different emerging opportunities, it is necessary to increasingly stimulate their professionalization and entrepreneurial capacity. Entrepreneurship, for example, will favor technological development and sector innovation. However, both are equally necessary for the modernization of family agriculture. Modernization involves using inputs, processes, machines, and equipment appropriate to the segment and the conditions of family farmers, enabling significant gains in productivity and sustainability. However, this comes at a cost: it depends on more investments and the development of innovative strategies in knowledge and technology creation and transfer. For family farmers to feel confident and secure in adopting new technologies or even more sophisticated property management processes, it is fundamental that the process of appropriating these technologies be accompanied by a support and encouragement network. This is undoubtedly an effective way to help as many producers as possible, especially the most vulnerable, encouraging them to participate in the growth flow. With such actions, there is also an opportunity to promote more comprehensive productive inclusion (de Carvalho Bittencourt, 2020).

Figure 49. Households with electricity in the indigenous land.



Source. Adapted from IBGE Census 2010.

In Brazilian indigenous territories, the difficulty in accessing technology is even greater and directly impacts territorial monitoring and family security. Power lines and internet access in these communities are still not a reality, leaving many technologically isolated and unable to access higher technological capacities to support land-related activities. According to IBGE 2010 census data, approximately 18% of

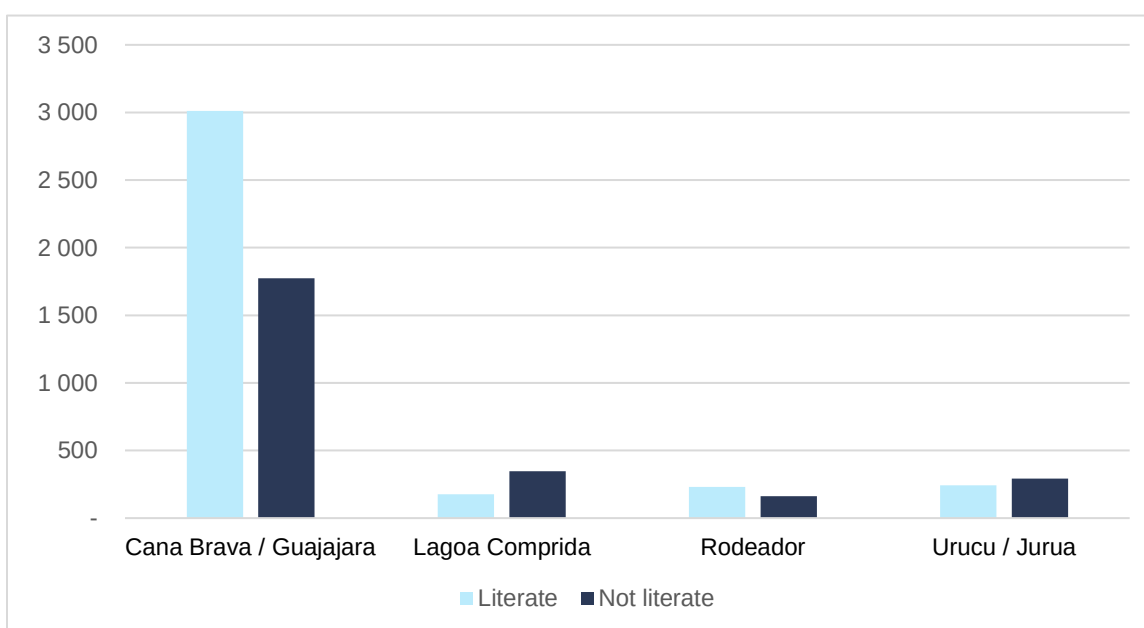
the homes in the indigenous territories in the project area do not have electricity. In the Urucu and Lagoa Comprida territories, this percentage is as high as 26%, i.e., one in four households does not have electricity.

- Technical capabilities

In general terms, technical capacity is understood as the knowledge of technical competencies or specific skills related to the correct performance of a specific professional field. The improvement and acquisition of technical capabilities are related to the acquisition and application of practical knowledge acquired in specialized academic contexts, such as specific advanced courses and the development of professional practices (Álvarez-Santullano, de Prada-Creo, & Pino-Juste, 2018).

In section 1.14 the literacy rates and technical capacities of the municipalities involved in the project are detailed. More specifically in the indigenous territories of the project area, according to IBGE data, approximately 50% of the people are not literate. Particularly noteworthy are the Lagoa Comprida and Urucu territories, where the illiteracy rate is 66% and 55%, respectively, i.e., more than half of the people are illiterate.

Figure 50. Literacy conditions in the indigenous territories of the project area



Source. Adapted from IBGE Census 2010.

This situation is also reflected in access to other basic services. According to the same IBGE source, approximately 91% of the indigenous households in the project area do not have a general sewage or rainwater drainage system or septic tank. Likewise, the communities' water supply is very varied and the conditions are not potable. Approximately 48% of the households in Cana Brava and 68% of the households in Lagoa Comprida, the main form of water supply comes from the general distribution network. The second form of water supply in the Cana Brava territory is through rivers and lakes (27% of the households). In the case of the Uruju territory, the main form of water supply (61%) is from wells and springs in the village, as well as from rivers and lakes (31%). In the case of the Rodeador territory, 100% of the households are supplied by rivers and lakes.

The high percentage of illiteracy, low employment of the population and low wage income, together with the low coverage of basic services described above, such as water, sewerage, sanitation and electricity, are factors that converge in the low overall technical capacity of the municipalities in the project area. In

other words, the supply of people with technical and higher education is low, which is a limitation for the execution of different types of projects, due to the lack of trained professionals, among other factors.

Outcome of sub-step 3a: List of barriers that may prevent one or more land use scenarios identified:

- ✓ Investment barriers: Lack of access to credit - Debt finance is not available for land use scenarios.
- ✓ Technological barriers: Lack access to necessary materials, for example planting materials – Technical capabilities.

Sub-step 3b: Elimination of land use scenarios that are prevented by the identified barriers

Land use scenarios 1 and 3 present investment barriers. According to the 2021 IBGE data²⁷, Maranhão is the state with the lowest level of the human development index (IDH) in Brazil with a value of 0.676 and is the state with the lowest Nominal monthly household income per capita, with a value of 945 R\$. The Human Development Index (HDI) is the geometric mean of the normalized indices of each of the three dimensions: a long and healthy life, knowledge and a decent standard of living. That is, life expectancy at birth, average years of adult schooling and gross income per capita²⁸.

More specifically, the municipalities covered by the project area have poverty percentages above 55%. Compared to the 5,570 municipalities in Brazil, the GDP per capita of the municipalities in the project area have very low values, placing them in the last positions, as in the case of Jenipapo dos Veiras and Itaipava do Grajaú, which are above the 5,500th position. Similarly, in comparison with the 217 municipalities in the state of Maranhão, the same municipalities are almost above the 200th position. The municipality of Grajaú, with a value of 16,265 R, ranks 31st in the state of Maranhão, and 3,580 out of the 5570 in Brazil.

Table 60. Economic and social indicators of the municipalities in the project area

Indicator	Municipality			
	Barra do Corda	Grajaú	Itaipava do Grajaú	Jenipapo dos Veiras
Adequate sanitation (%)	13.50	4.80	2.80	2.60
Poverty incidence (%)	60.04	56.37	57.60	56.08
GDP per capita (2021) R\$	10,004.24	16,265.45	6,641.29	7,344.62
GDP Position in relation to other municipalities in Brazil ²⁹	4,889	3,580	5,546	5,500
GDP Position in relation to other municipalities in the state of Maranhão ³⁰	83	31	196	174
Human Development Index (HDI)	0.606	0.609	0.518	0.490
Employed population [2022] (%)	11.00	12.01	4.27	4.32
Economic activity Agriculture (x1,000 R\$) [2021]	72,029.12	187,242.72	16,933.80	33,490.63
Economic activity Agriculture Position in relation to other municipalities in the state of Maranhão	35	15	187	88

Source. Adapted from IBGE Cidades.

²⁷<https://cidades.ibge.gov.br/brasil/ma/pesquisa/37/30255?tipo=ranking>

²⁸ <https://hdr.undp.org/data-center/human-development-index#/indicies/HDI>

²⁹ Brazil has 5,570 municipalities

³⁰ The state of Maranhão has 270 municipalities

On the other hand, in terms of work and employment rates, the project area municipalities have very low values, which do not exceed 12% of the population. Likewise, in terms of the economic activity of agriculture, it is not the main activity of these municipalities in comparison with other municipalities in the state.

The high poverty incidence and the low economic development of the municipalities in the project area show the economic limitations to promote agriculture and conservation projects, since the municipalities present priorities such as exceeding levels of poverty and lower the proportion of people with unsatisfied basic needs, as well as access to vital public services.

Similarly, scenario 1 and scenario 3 present technological barriers. In the sub-step 3a, the table shows the high rates of illiteracy of the population of the indigenous land of the project area. This, added to the investment barriers, shows a technical limitation to promote and develop activities based on agriculture and forest conservation. For instance, concerning coverage monitoring and ensuring the long-term sustainability of agricultural activities, there are already limitations in terms of process modernization and input acquisition.

Outcome of sub-step 3b: List of land use scenarios that are not prevented by any barrier:

- Land use scenario 2: Conversion of forest land into cattle

Step 4. Common practice analysis

Protected Areas

According to the Brazilian Ministry of the Environment, protected areas encompass Conservation Units (CUs), mosaics, and ecological corridors, which are considered essential from an economic standpoint for conserving socio-biodiversity. They also provide environmental services and generate business opportunities.

Spaces like parks, national forests, forest mosaics, and CUs maintained by the Amazon Protected Areas Program (ARPA)³¹ are responsible for directly producing part of the water destined for human consumption and preventing the emission of billions of tons of carbon into the atmosphere. Ecological corridors, which connect fragments of natural areas, are increasingly important. Defined in the National System of Conservation Units (SNUC) as portions of natural or semi-natural ecosystems connecting CUs, they facilitate gene flow and the movement of biota - the set of living organisms within an ecosystem, including flora, fauna, fungi, and other groups of organisms. They also facilitate species dispersal and recolonization of degraded areas and ensure the survival of populations that require larger areas than individual units.

To enable the maintenance and conservation of all protected areas, the government employs various policy strategies contained in different instruments such as the National UCs Registry, the National Strategic Plan for Protected Areas (PNAP), and nationally-reaching programs and projects. (Annex 050)

In Brazil, there is Law 14,119/21 (Annex 051), which regulates payment for environmental services. It is a form of incentive for conservation and sustainable development through remuneration in exchange for preserved goods. The text creates a national policy, but the creation of a collegiate body to define resource allocation and a national registry on payments was vetoed.

³¹ The Amazon Protected Areas Program (Arpa) is considered one of the largest in the world in the sphere of environmental conservation. The first phase of the Arpa Program, carried out between 2002 and 2009, supported the creation and consolidation of 46 UCs, totaling 23.4 million hectares, in addition to supporting the consolidation of another 8.5 million hectares of previously created UCs. The creation and implementation of SisARPA - a computerized system for coordinating and managing the Arpa Program - was also supported.

The text provides for the creation of a federal program for payment for environmental services (PEPSA), focusing on actions to maintain, restore, or improve vegetative cover in priority conservation areas, combat habitat fragmentation, and establish biodiversity corridors and water resource conservation.

Payments for environmental services can take various forms: direct (monetary or non-monetary), provision of social improvements to rural and urban communities, compensation linked to certificates for reducing emissions from deforestation and degradation, loan for use (comodato), green bonds (green bonds), and Environmental Reserve Quota instituted by the Forest Code (Law 12,651 of 2012, Annex 051). Revenues obtained from charging for the use of water resources, as provided by Law 9,433 of 1997, may be used for payment for environmental services, but this depends on the decision of the watershed committee. Other payment modalities may be established by regulatory acts of the Brazilian Institute of the Environment and Renewable Natural Resources (IBAMA), which will be the managing body of the national policy.

The text prohibits the use of public funds for payment for environmental services to individuals and legal entities in default regarding a conduct adjustment agreement or commitment signed with environmental agencies and also on embargoed areas.

Mechanism of Payment for Environmental Services – PSA

PSA (Payment for Environmental Services) is a financial mechanism to compensate rural producers, family farmers, settlers, as well as traditional communities and indigenous peoples, for the environmental services provided on their properties that generate benefits for society as a whole (Annex 052).

Environmental Services according to Federal Law No. 14,119 dated January 13, 2021:

Provisioning services: those that provide environmental goods or products used by humans for consumption or commercialization, such as water, food, wood, fibers, and extracts, among others.

Supporting services: those that maintain the continuity of life on Earth, such as nutrient cycling, waste decomposition, soil fertility production, maintenance or renewal, pollination, seed dispersal, control of potential pest populations and potential disease vectors, protection against ultraviolet solar radiation, and maintenance of biodiversity and genetic heritage.

Regulating services: those that contribute to maintaining the stability of ecosystem processes, such as carbon sequestration, air purification, moderation of extreme climatic events, maintenance of hydrological cycle balance, flood and drought minimization, and control of critical erosion and landslide processes.

Cultural services: non-material benefits provided by ecosystems, including recreation, tourism, cultural identity, spiritual and aesthetic experiences, and intellectual development, among others.

Barriers related to protected areas

Within the scope of the collective territories adjudication procedure, responsibilities are attributed by Law No. 14,701 of October 20, 2023, (Annex 053) which regulates Article 231 of the Federal Constitution, to regulate the recognition, demarcation, use, and management of indigenous lands. Regarding the use and management of indigenous lands, Article 19 states that indigenous communities, through their own decision-making processes and conflict resolution methods, choose the form of use and occupation of their lands. Article 20 specifies that indigenous usufruct does not override the national defense and sovereignty policy interests.

Barriers related to PSA

In Brazil, to advance the implementation of federal policy and provide more legal certainty to existing PSA programs, it is necessary to proceed with the regulation process by the Executive Branch.

The law has not yet been regulated. There is great anticipation among institutions involved in PSA, as well as among local government entities that may lack guidelines for their own regulatory frameworks.

Civil society has been discussing important points to be considered in this process, such as monitoring and transparency mechanisms to attract investments to this agenda, and a governance structure that includes equal representation from the government, productive sector, and civil society.

The new law introduces a mechanism to recognize efforts and good practices adopted by producers - payment for environmental services - which encourages positive transformations in forest conservation, soil preservation, biodiversity protection, water security, and climate change mitigation efforts in Brazil.

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☒ No

Although Brazil is classified as a Non-Annex 1 country under the UNFCCC, there are no mandatory requirements for the project's activities. All initiatives originating in Brazil still operate within a voluntary context. However, there is a framework of constitution, laws, and regulations that guide how these activities, when chosen as a land-use option, can be implemented. Below, we provide a detailed outline of the relevant rules that enable this voluntary initiative, which, in this case, is also considered private and individual under the concept of the right to self-determination of indigenous peoples as recognized by the Brazilian Constitution. The Legal Aspects of Additionality and the Project Design Document (PDD) of the Kaa Pyahu Project can be found above: FOREST CODE - Law 12.651/2012:

- Art. 41. The federal Executive Branch is authorized to establish, without prejudice to compliance with environmental legislation, a program to support and encourage environmental conservation, as well as the adoption of technologies and good practices that reconcile agricultural and forestry productivity with the reduction of environmental impacts, as a way of promoting ecologically sustainable development, always observing the criteria of progressiveness, covering the following categories and lines of action: (Amended by Law 12.727, of 2012).

- § 4º Activities of maintenance of Permanent Preservation Areas, Legal Reserves, and restricted use areas are eligible for any payments or incentives for environmental services, constituting additionality for the purposes of national and international markets for certified emission reductions of greenhouse gases. (Emphasis added)

Indigenous Heritage Assets and Rights:

The goods and usufruct of indigenous heritage can be declared as a right to indigenous heritage income for the purposes of contracting and trading carbon credits, as provided by Articles 39 and 40 of Law No. 6.001/73 in conjunction with Article 231, §2º of the Federal Constitution of 1988:

- Art. 231. The social organization, customs, languages, beliefs, and traditions of the indigenous peoples, as well as their original rights to the lands they traditionally occupy, are recognized, with the Union being responsible for demarcating them, protecting and ensuring respect for all their assets.

- § 1.º The lands traditionally occupied by indigenous peoples are those they inhabit on a permanent basis, use for their productive activities, are essential for the preservation of the environmental resources necessary for their well-being, and are necessary for their physical and cultural reproduction, according to their uses, customs, and traditions.

- § 2.º The lands traditionally occupied by indigenous peoples are intended for their permanent possession, with the exclusive usufruct of the riches of the soil, rivers, and lakes therein belonging to them. (Emphasis added)

- § 3.º The use of water resources, including their energy potentials, the research and extraction of mineral riches on indigenous lands can only be carried out with the authorization of the National Congress, after consulting the affected communities, ensuring their participation in the mining results as provided by law.

- § 4º The lands referred to in this article are inalienable and indisposable, and the rights over them are imprescriptible.

- Art. 39. The assets of the Indigenous Heritage include:

- I . The lands belonging to the domain of tribal groups or indigenous communities;
- II . The exclusive usufruct of the natural resources and all utilities existing in the lands occupied by tribal groups or indigenous communities and in the areas reserved for them; (Emphasis added)

- Art. 40. The holders of Indigenous Heritage are:

- I . The indigenous population of the country, regarding assets or income belonging to or intended for indigenous peoples, without discrimination of persons or tribal groups;
- II . The specific tribal group or indigenous community, concerning the possession and usufruct of the lands exclusively occupied by them, or reserved for them;
- III . The indigenous community or tribal group named in the acquisition title of the property, concerning the respective real estate or movable assets.

The legal system must respect the provisions of the legal framework, with indigenous populations and peoples being the exclusive holders of the use of natural resources derived from their areas, and it is not possible for the State to impose any rules or obligations in this regard, but only to create cooperation conditions that allow indigenous people to achieve their own goals.

Given the nature of the project, as well as the contracting occurring in the free exercise of the right to self-determination and self-government of indigenous peoples, prior and express approval by FUNAI or the Federal Public Ministry is not required. However, by convention between the Parties, it was agreed that the Project would be informed to the referred bodies. It is opportune to mention Instruction No. 01/2021 of FUNAI, which endorsed the environmental licensing based on the legitimacy of commercialization by the indigenous themselves of their environmental riches, inaugurating the figure of the INDIGENOUS ENTREPRENEUR, which can be realized through their respective pure associations - only indigenous - and mixed organizations - indigenous and non-indigenous - as is the case.

LEGITIMACY OF CARBON CREDITS TRADING ON INDIGENOUS LANDS :

The Brazilian legal system, in accordance with international law, has endorsed the legitimacy of trading carbon credits as an effective method and technology to contribute to the reduction of greenhouse gas (GHG) emissions, including by indigenous communities and tribal peoples, with no specific prohibition for indigenous people, as entrepreneurs, to promote the exploitation and trading of carbon credits existing in their territories (ITs);

- International Framework :

- The Kyoto Protocol, the Warsaw Framework for REDD+, and the Paris Agreement on Climate Change all aim to endorse the legitimacy of the participation and inclusion of indigenous peoples and tribal communities in the global policy of reducing GHG emissions;

- The Kyoto Protocol defined emission reduction targets for developed countries and those with economies in transition to capitalism at the time, considered historically responsible for the current climate change, promulgated in Brazil by Decree No. 5.445, of May 12, 2005, covering Emissions Trading, Joint Implementation, and Clean Development Mechanism (CDM);

- The Warsaw Framework for REDD+ adopted during the 19th Conference of the Parties (COP-19) of the United Nations Framework Convention on Climate Change (UNFCCC) in Warsaw, Poland, in November 2013, created an international REDD architecture intended to provide financial incentives to developing countries with policies aimed at reducing GHG emissions from deforestation and forest degradation, sustainable forest management, and increasing carbon stocks. In this International Diploma of the Warsaw Framework, REDD+ Safeguards were adopted, establishing general guidelines to be observed by signatory countries, also known as Cancun Safeguards, aiming to ensure that REDD+ initiatives adequately address sensitive issues such as the rights of indigenous peoples and traditional communities, social participation, preservation of natural ecosystems, permanence of REDD+ results achieved, and the risk of shifting deforestation and forest degradation pressures to other areas;

- The Cancun Safeguards also ensure that the application of resources transferred as payment for REDD+ results does not cause negative impacts, resources that can be applied and used in indigenous lands (ITs);

- The Paris Agreement, conventioned in December 2015 by the signatory countries of the UNFCCC during the 21st Conference of the Parties (COP 21), expressly contemplated the rights of indigenous peoples and tribal communities, aiming at reducing carbon dioxide emissions from 2020 onwards, reaffirming the commitment to keep the global average temperature increase below 2°C above pre-industrial levels, as well as limiting the temperature increase to 1.5°C above pre-industrial levels;

- Convention No. 169 of the International Labor Organization (ILO) on Indigenous and Tribal Peoples, concluded in Geneva and promulgated by Executive Decree No. 5.051/2004, expressly ensured the right to participation, consultation, and choice in public policies, works, services, and activities of any kind, to be executed in their respective territories - lands traditionally occupied by these communities -, as a way to maintain and protect their material and immaterial culture intact;

- The first cases involving the trading of carbon credits in the voluntary market on indigenous lands to be submitted for analysis and manifestation by FUNAI was the "Suruí Carbon Project," subject of Process No. 08620.000721/2011-95, whose Technical Opinion No. 22/CGMT/DPT/2011 and Note No. 039//PGF/PFE/FUNAI/2011-CAF expressly recognized that indigenous peoples hold the right to the ownership of carbon credits and that the realization of these projects can be a viable alternative for generating income sustainably;

- Technical Information No. 21 (2848944-FUNAI), considering the existence of various contracts signed between certain indigenous communities and companies willing to trade carbon credits, issued guidelines to endorse the legitimacy of this contractual modality on the following themes: I - Indigenous

Peoples and REDD+ in Brazil: General Considerations and Recommendations, drafted by FUNAI (2849006) and II - Premises agreed upon between the National Indigenous Foundation and the Ministry of the Environment for the development of the Indigenous Component of the National REDD+ Strategy (2849015);

- Indigenous peoples and communities can be institutionally classified as INDIGENOUS ENTREPRENEURS, revealing that environmental licensing is not mandatory for executing the eventual contracts signed, as provided by Articles 2, 4, item III of Law No. 6.001/73 in conjunction with Articles 1 and 2 of Joint Instruction No. 1, of February 22, 2021, formulated by the Ministry of Justice and Public Security/National Indigenous Foundation, in accordance with Instruction No. 15, of May 18, 2018, of IBAMA;

- The goods and usufruct of the indigenous heritage can be declared as a right to indigenous heritage income for the purposes of contracting and trading carbon credits, as provided by Articles 39, 40, 43 to 45 of Law No. 6.001/73 in conjunction with Article 231, §2º of the Federal Constitution of 1988;

- There is no INDIGENOUS GUARDIANSHIP REGIME applicable to 'integrated' and 'in the process of integration' indigenous peoples, a limitation that applies only

The uploaded file contains a single-page image with text regarding the legal aspects of additionality and Project Design Document (PDD) for the Kaa Pyahu Project. It specifically references the Brazilian Forest Code (Law 12.651/2012) and various rights and regulations surrounding indigenous heritage assets and the legitimacy of trading carbon credits on indigenous lands.

3.5.2 Additionality Methods

To demonstrate the additionality of the project (section 3.5), the steps of the "Tool for the demonstration and assessment of additionality in VCS Agriculture, Forestry and other land use (AFOLU) project activities" V.3 were followed.

3.6 Methodology Deviations

Above biomass and Below-ground biomass

It is important to clarify that the values of the carbon reservoirs are taken from the Brazil's National Forest Reference Emission Level for Results-based Payments for REDD+ under the United Nations Framework Convention on Climate Change (Brasil, 2024), which is compatible with the confidence levels and sampling errors of the VCS standard 4.7. The NREF values were taken having in mind the application associated with modules VMD001, VMD0004, VMD0015, VMD0016 and VMD0017.

The UNFCCC regarding the Reference Emission Levels of Forest Emissions by parties has established different guidelines set forth in COP15, COP 16, COP 17, and COP 19. In these decisions, the Convention invites countries to transparently establish Reference Emission Levels for Forests, considering historical data, national circumstances, and providing estimates that reduce uncertainties (paragraphs 1 and 7 of decision 4/CP. 15, paragraph 71b of decision 1/CP. 16). Additionally, the Convention emphasizes that for the establishment of these levels, it is advisable to adopt a phased approach, incorporating better data and methodologies (Chapter 12, decision 12/CP. 17), in an integrated, transparent, and information-based manner that is clear, comprehensive, coherent, and accurate (Annex decision 12/CP. 17, decision 13/CP. 19).

In compliance with these decisions and reaffirming its commitment to Climate Change, especially in terms of results-based payments, Brazil developed the first Forest Emissions Reference Level (NREF) for the Amazon biome (Brasil, 2014), meeting the complete, transparent, consistent and accurate principles. Subsequently, in 2017, Brazil presented the second NREF especially for the Cerrado biome (FREL Cerrado). This document, modified in 2018 ensures transparency, completeness, consistency, and precision, managing a global accuracy of 94.5%.

In 2018, Brazil submit the Brazil's submission of a Forest Reference Emission Level (FREL) for reducing emissions from deforestation in the Amazonia biome for REDD+ results-based payments under the UNFCCC from 2016 to 2020 (Brasil, 2018). In 2022, Brazil submit the fourth FREL, this time including all biomes, and finally in march 2024, present a national forest reference emission level (FREL) for a technical assessment under the United Nations Framework Convention on Climate Change (Brasil, 2024). As with other previous FREL documents, this latest document complies with the principles of transparent, complete, consistent and accurate information, meeting the requirements of methodology VM0007 v.1.7. Therefore, the values for carbon reserves in forests for each reservoir, to estimate emissions reductions for the KA'A PYAHU REDD+ project, were taken from the FREL 2024.

VMD0010 LK-ASU

According with 3.15.14 section of the VCS Standard V4.7., "*Where the applied methodology requires the quantification of activity-shifting leakage, projects may apply the optional default activity-shifting leakage deduction of 15 percent to the gross GHG emission reductions and/or removals*". Therefore, the KA'A PYAHU REDD+ Project apply the 15% by default for quantification of estimated GHG emission reductions and removals.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

During the monitoring period (01/05/2021–31/12/2023), a series of activities aligned with the action lines of the REDD+ project was executed, as outlined below:

Active pursuit of the rights of the Guajajara population: During the monitoring period, the association Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara strengthened as the representative of the peoples from the four project proposing territories and actively fought for the recognition of their rights guaranteed by the Brazilian state, such as access to basic documents.

Participation and collective struggle to mitigate actions against local rights and the environment: The project proponent association fought against the illegal use of their territories for exploitation without consulting the peoples and conducting an environmental impact assessment, through mitigation and compensatory actions.

Taking action against wildlife trafficking and illegal logging: The peoples reported, monitored, and safeguarded their territories against illegal exploitation of native resources.

Attention to and care for local health and education: Steps were taken to strengthen access to specific indigenous health care through competent state bodies and also to enhance indigenous school education.

These actions were proposed by the Conselho Supremo de Caciques e Lideranças Terra Indígena Cana Brava Guajajara according to the realities and demands of the Guajajara peoples in order to promote well-being, improve quality of life, and care for the diversity and environmental preservation of the four territories. Urgent activities for service and quality of life improvement were prioritized through claims for the Guajajara peoples from strong and organized governance. The implementation of the REDD+ project aims to promote activities that contribute to long-term reduction of greenhouse gas emissions through the execution of action lines and social and environmental safeguards. Activities initiated before the monitoring period are part of the current validation and verification process, and it is expected that over time, recently implemented activities will result in environmental and social benefits.

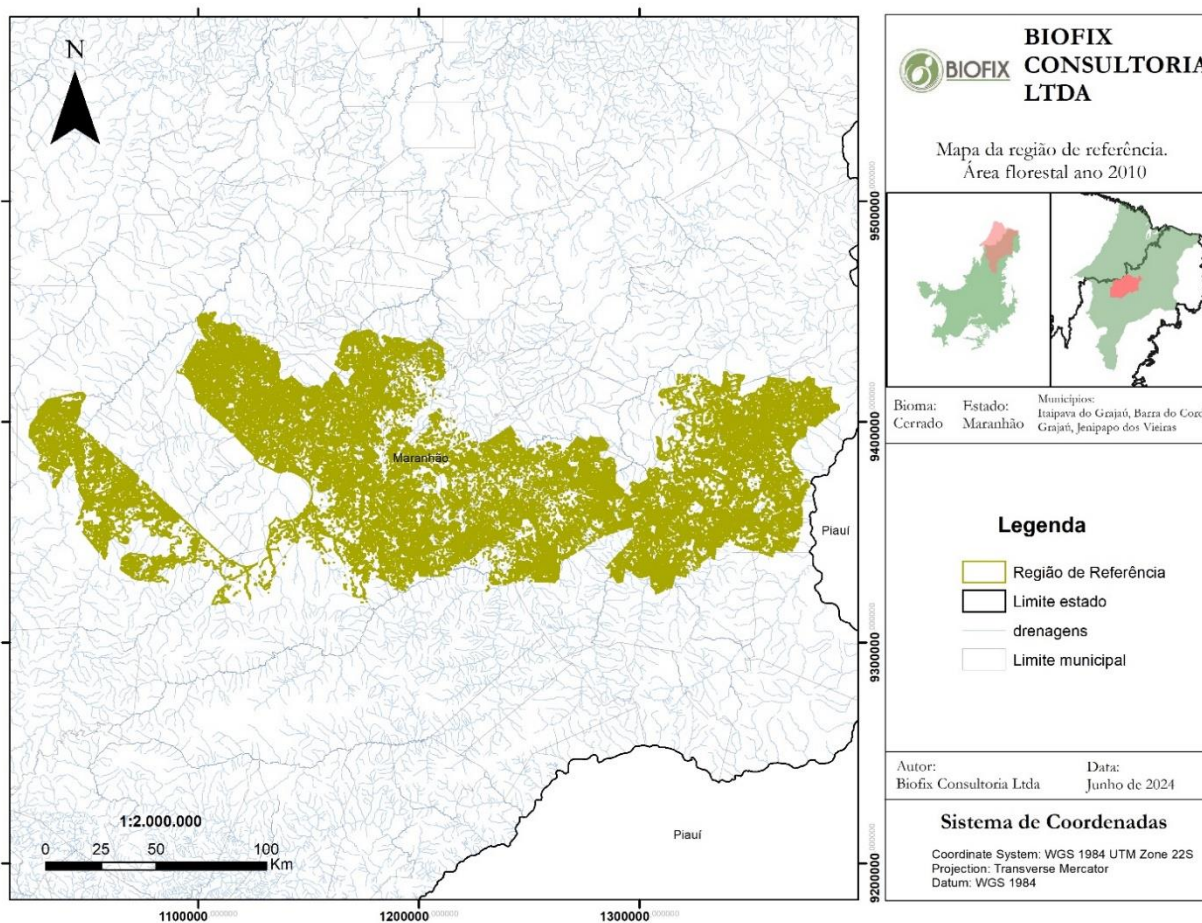
5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

To carry out the process and quantification of the Baseline Emissions, the indications of the methodology modules VM0007 BL-UP. The reference deforestation rate was calculated from the Reference Region (RRD). The deforestation rate is based on the simple historical average approach given the amount of data and the deforestation trend in the reference area; the population driver approach was not used. The following sections summarize the analyzes and equations to determine the baseline, the detailed calculations are found in Annex 55.

5.1.1 Reference Region to project the Deforestation rate (RRD)

Figure 51. Map of the reference region. Forest area year 2010 in the RRD.



Source: KA'A PYAHU REDD+ Project.

This section corresponds to part 1 (Definition of boundaries) of the module BL-UP (VMD 0007, V 3.3). The delimitation of the RRD Reference Region was carried out following the module described in section 3.3.2. of this document, through the geospatial analysis of landscape and transport factors.

As required by the BL-UP module, the RRD must respond to the following conditions:

$$MREF = RAF * PA$$

$$RAF = 7.500 * PA^{-0,7}$$

The project area is 150,800 has. According with the module therefore the RAF (Reference Area Factor) is 1.78, and the MREF is 268,275 has. This means that the minimum size of the RRD must be 268,275 ha. The resulting area of the RRD was 843,490 ha, in compliance with the minimum area given of the MREF. This area does not include the project area, or the leakage belt.

As also stated in 3.3.2 PDD section, it was sought that the conditions of the RRD and the Project Area (PA) were similar in the biophysical conditions, conditions of land tenure, transport and infrastructure and the presence of agents and drivers of deforestation.

5.1.2 Calculation of the deforestation rate

This section corresponds to part 2 (Estimation of annual areas of unplanned deforestation) of the module BL-UP (VMD 0007, V 3.3). According to the module, the first step (2.1) is the analysis of historical deforestation, to quantify the historical deforestation rate during the historical reference period within the RRD. For this purpose, data from 4 years of deforestation 2010, 2015, 2016 and 2020, fulfilling the 11 years of the reference period were obtained. The module establishes three approaches to calculate the deforestation rate: average simple historical, linear regressions and nonlinear regressions.

Table 61. Forest and Non-forest areas in the RRD (2010-2020).

CLASS	2010	2015	2016	2020
NO FOREST (ha)	0	73,671	78,495	128,657
FOREST (ha)	843,491	769,814	764,996	714,834
TOTAL (ha)	843,491	843,491	843,491	843,491

Source: KA'A PYAHU REDD+ Project.

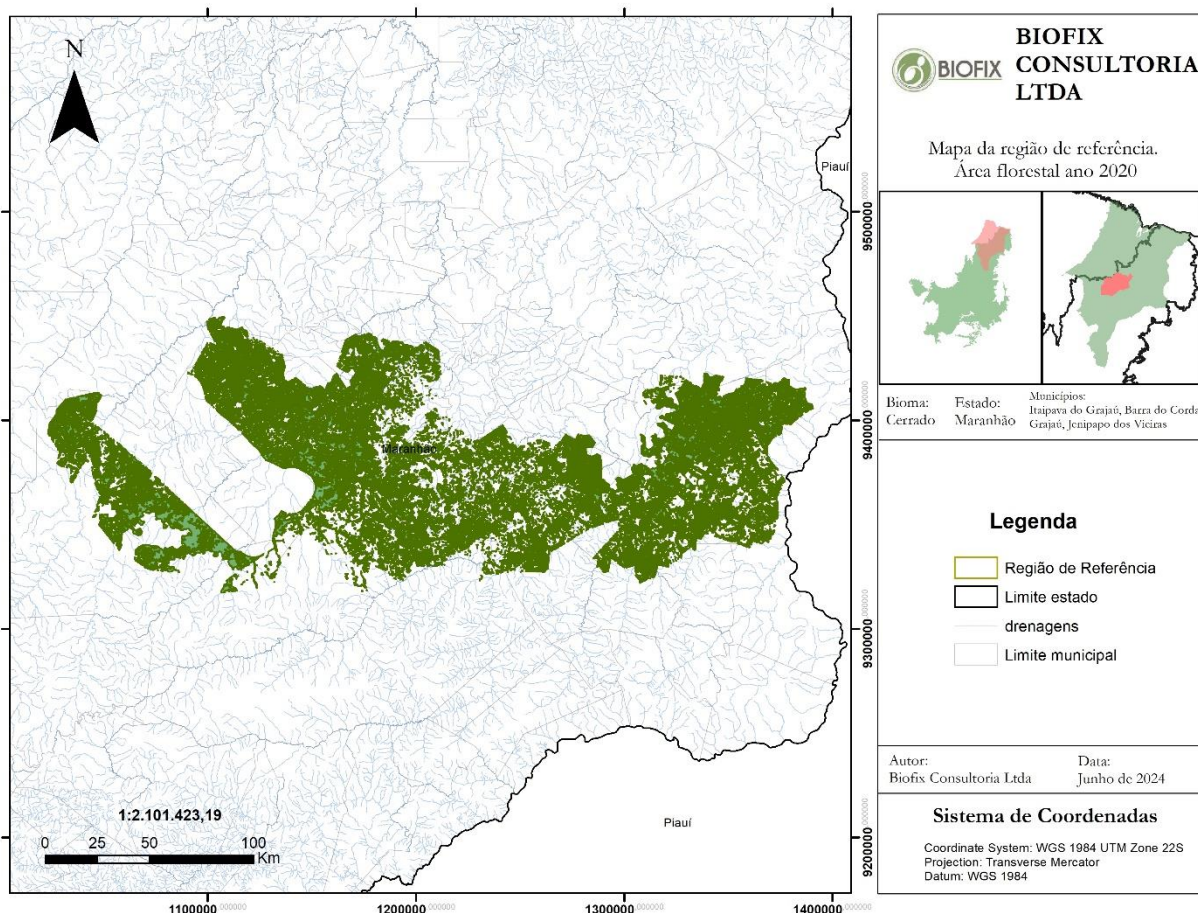
From the above data, the following deforestation data for the RRD area can be determined.

Table 62. Historic deforestation in RRD

Time	Year	Forest area	Deforestation (ha)
1	2010	843,491	12,279
2	2011	No data	-
3	2012	No data	-
4	2013	No data	-
5	2014	No data	-
6	2015	769,814	12,279
7	2016	764,996	4,818
8	2017	No data	-
9	2018	No data	-
10	2019	No data	-
11	2020	714,831	10,032

Source: KA'A PYAHU REDD+ Project.

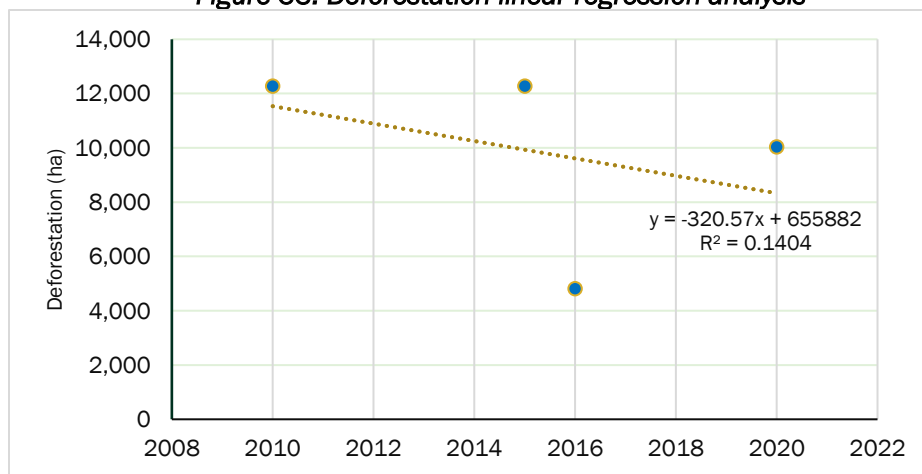
Figure 52. Regional Forest Cover / Non-Forest Cover Benchmark Map 2020.



Source. KA'A PYAHU REDD+ Project.

The module states (step 2.2, Estimation of the annual areas of unplanned baseline deforestation in the RRD) that if there are less than five points in time are used in the analysis, a linear regression must be used. Therefore, a linear regression was performed with a result of $r^2:0.14$. That is, the regression does not comply with the requirement that any regression must be significant, and must have an $r^2 \geq 0.75$.

Figure 53. Deforestation linear regression analysis



Source. KA'A PYAHU REDD+ Project.

Similarly, a non-linear regression cannot be applied since there are fewer than 5 points in time are used in the analysis. Therefore, the method for determining the historical deforestation rate in the RRD area is the simple historical average (equation 3 of the BL - UP).

$$A_{BSL,RRD,unplanned,t} = A_{RRD,unplanned,hrp} * T_{hrp}$$

Where,

$A_{BSL,RRD,unplanned,t}$ = Projected area of unplanned baseline deforestation in the RRD in year t; ha.

$A_{RRD,unplanned,hrp}$ = Total area of deforested during the historical reference period in the RRD; ha

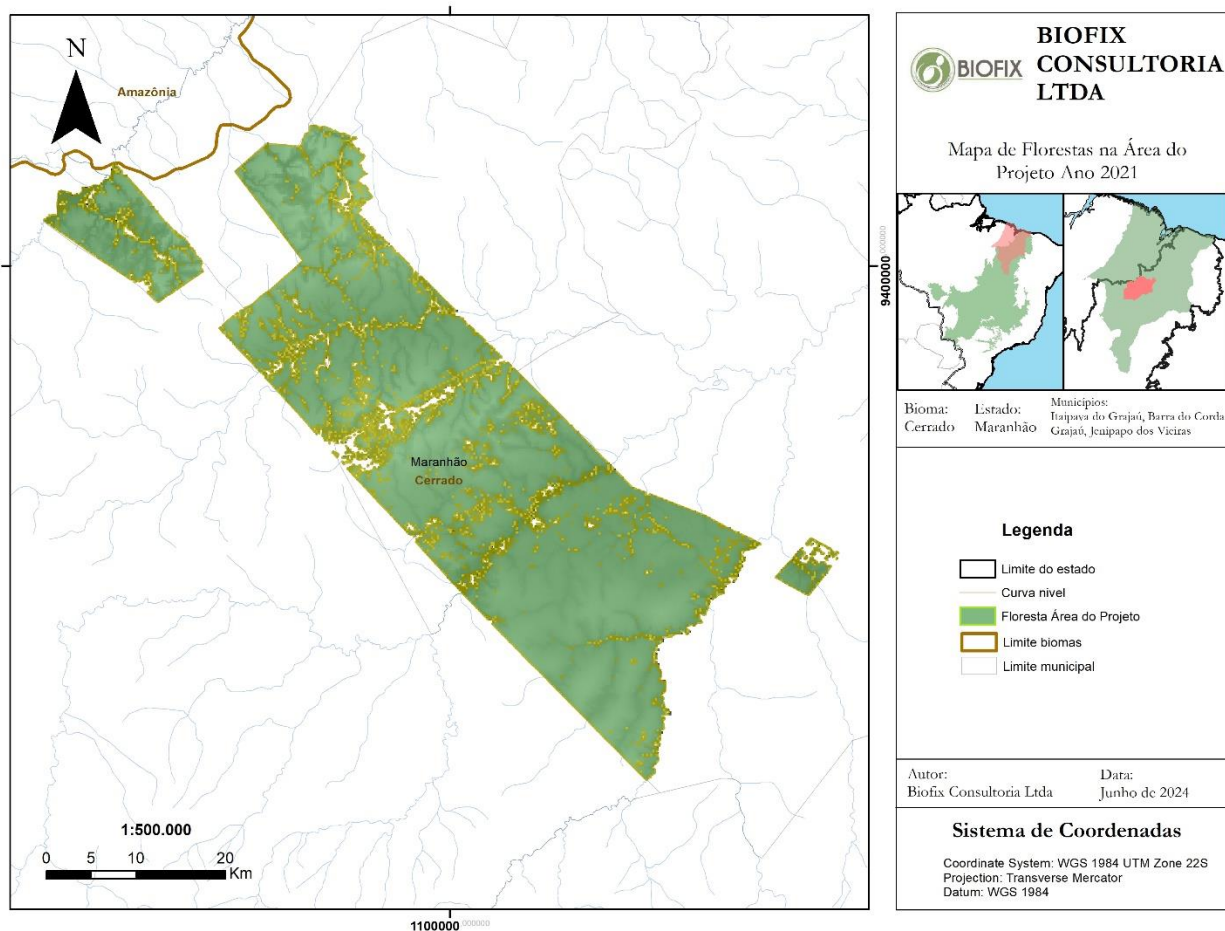
T_{hrp} = Duration of the historical reference period in years; yr.

Obtaining as a result of a deforestation of 128,657 ha (between 2010 and 2020), in the RRD, to obtain a value of $A_{BSL,RRD,unplanned,t} = 11,696$ ha/year.

5.1.3 Estimated annual areas of unplanned baseline deforestation in the project area

This section corresponds to step 2.3 of the module BL-UP (VMD007 V3.3). The projected unplanned baseline deforestation in the project area is estimated as follows, with the baseline information presented in the following table.

Figure 54. Project Forest Cover Benchmark Map 2021



Source. KA'A PYAHU REDD+ Project.

Table 63. Forests in the project area.

CLASS/Year	2021	2023
NO FOREST (ha)	0	555
FOREST (ha)	150,800	150,245
TOTAL (ha)	150,800	150,800

Source. KA'A PYAHU REDD+ Project.

$$A_{BSL,PA,unplanned,t} = A_{BSL,RRD,unplanned,hrp} * P_{PA}$$

$A_{BSL,PA,unplanned,t}$ = Projected area of unplanned baseline deforestation in the project area in year t; ha.

$A_{BSL,RRD,unplanned,t}$ = Projected area of unplanned baseline deforestation in the RRD in year t; ha.

P_{PA} = Ratio of the project area to the total area of RRD; dimensionless.

The P_{PA} is 0.21. To determine the area of unplanned deforestation of the baseline in the project area, it is important to clarify that the official information used as a source for determining deforested areas from Mapbiomas presented for each year. For the year 2021 in the KAA PYAHU REDD+ project, deforestation was estimated with the calendar year value. Then, the deforestation was prorated in the number of effective days of quantification. That is, the deforestation value was divided into 365 days and it was multiplied by 245 (as the start date is 01/05/2021). Similarly, for the last year it was divided into 365 and multiplied by the remaining days, as shown below:

Table 64. Unplanned deforestation in the project area in the baseline

Year	$A_{BSL,PA,unplanned,t}$ (ha)
2021	1,656
2022	2,467
2023	2,467
2024	2,467
2025	2,467
2026	2,467
2027	2,467
2028	2,467
2029	2,467
2030	2,467
2031	2,467
2032	2,467
2033	2,467
2034	2,467
2035	2,467
2036	2,467
2037	2,467
2038	2,467
2039	2,467
2040	2,467
2041	811

Source. KA'A PYAHU REDD+ Project.

5.1.4 Carbon Reservoirs

This section corresponds to step 4 of the modulo BL-UP (VMD 007 V3.3). To estimate the carbon stock changes and GreenHouse Gas Emissions, the post-deforestation carbon stocks were calculated considering the option 2 (historical area-weighted average) of the section 4.2.2 of the module. It means

Figure 55. Land use change post-deforestation in the Reference Region (2010-2020)

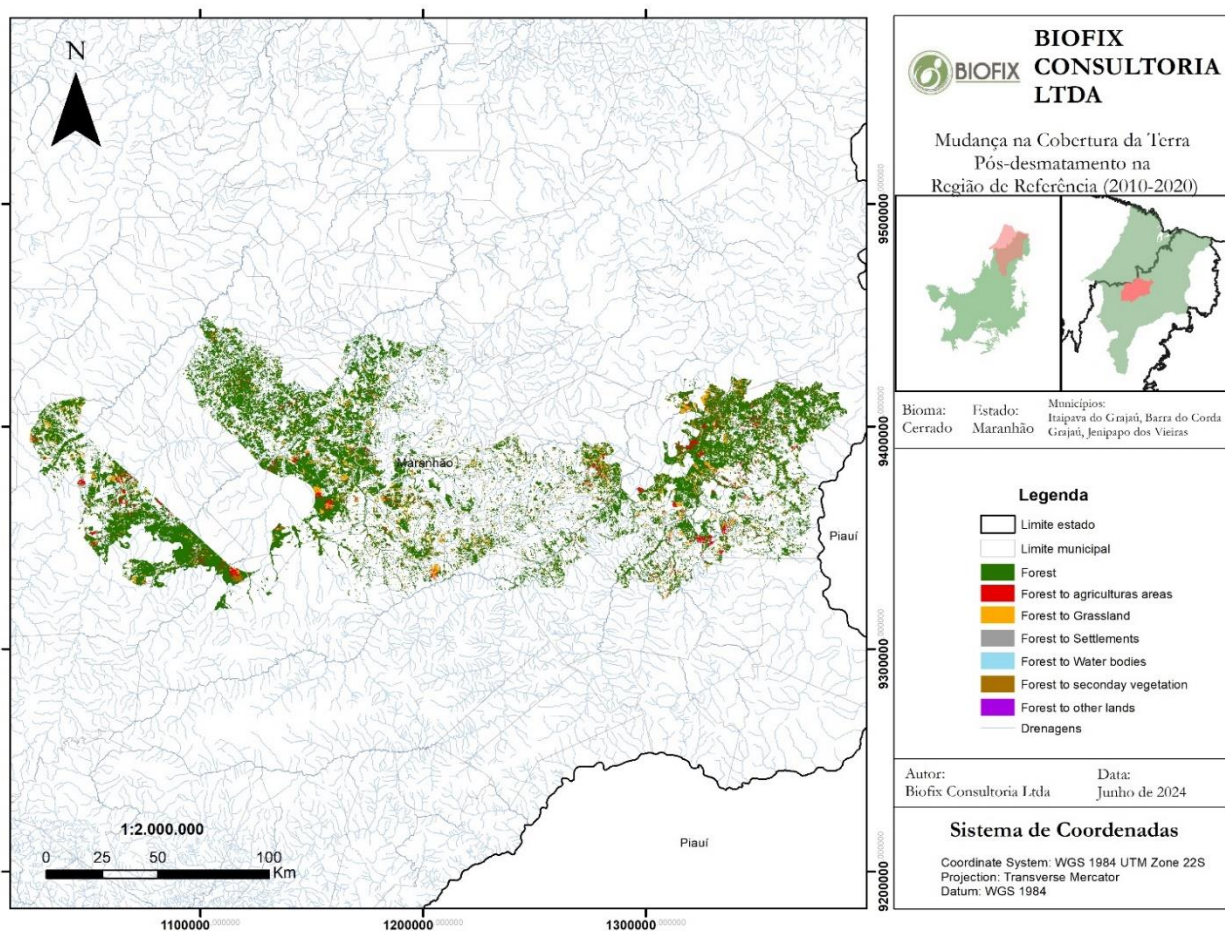


Table 65. Matrix Land use change post-deforestation in the RRD (2010 to 2020)

Land use	2010		2020			
	Area (ha)	Area (%)	Land use	Area (ha)	Area (%)	
Forest	843,490.71	100	forest		714,834.45	
			Non forest	Agriculture	33,376.05	25.94
				Grassland	68,118.12	52.95
				settlements	414.27	0.32
				Water bodies	76.23	0.06
				Other lands	1,330.92	1.03
				Secondary vegetation	25,340.67	19.70
Non forest total			128,656.26	100.00		

According with the section 4.2.2 of the module, a historical mix of post-deforestation land uses (initiated during the historical reference period) is assumed to be representative of future changes. From the historical weighted land use change matrix, the area-weighted average of carbon stocks for each land use is calculated, as shown below:

Table 66. Carbon reservoir stock by land-use post deforestation in the reference region

Deforestation 2010-2020		Area		Non-tree average carbon stock (t.C/ha)		Post-def carbon stock by land-use (t.C/ha)	
Land use post-deforestation		ha	(%)	Aboveground biomass	Belowground biomass	Aboveground biomass	Belowground biomass
Non forest	Agriculture	33,376.05	25.94	5.38	0.04	1.40	0.01
	Grassland	68,118.12	52.95	7.57	1.10	4.01	0.58
	Settlements	414.27	0.32	0.00	0.00	0.00	0.00
	Water bodies	76.23	0.06	0.00	0.00	0.00	0.00
	Other lands	1,330.92	1.03	0.00	0.00	0.00	0.00
	Secondary vegetation	25,340.67	19.70	17.58	3.52	3.46	0.69
Non-forest total		128,656.26	100.00	30.53	4.66	8.87	1.29

Source: (Brasil, 2020), (Miteva, Kennedy, & Baumgarten, 2014), and KA'A PYAHU REDD+ Project.

Then, stock changes in each pool were calculated by subtracting post-deforestation carbon stocks from forest carbon stocks

Table 67. Carbon reservoir stock values in the baseline scenario

Reservoir	Carbon stock (t C/ha)		Carbon stock (tCO ₂ e/ha)		
	Pre-def	Non-tree post def	Pre-def	Non-tree post def	Baseline carbon stock change
Aboveground biomass	39.96	8.87	146.52	32.51	114.01
Belowground biomass	7.99	1.29	29.30	4.71	24.58

Source. KA'A PYAHU REDD+ Project.

5.2 Project Emissions

To develop this item the next three step were followed, according to section 5 VMD0015.

STEP 1. Selection and analyses of land-use and land-cover (LU/LC) change data

Processing LU/LC Change Data:

The data to establish changes in land use and land cover stem from the input generated by Mapbiomas Forest, non-natural forest formation, farming, non-vegetated areas, layers and land cover data for the baseline period, corresponding to the years 2010 and 2020. For the project area, the data correspond to the years 2021 and 2023. Once these inputs were obtained, the quantification of deforestation was carried out by subtracting the forest layer of the initial year from the forest area of the final evaluation year. The resulting area in this case represents the deforested area during this period, and an analysis of land cover change was performed on this area.

Considering that the Forest/ Non natural forest formation and cover data layers generated by Mapbiomas use Landsat satellite images (Landsat 5, Landsat 7 and Landsat) and Google Earth Engine (GEE) platform

with machine learning algorithms (Souza, et al., 2020), the quantification and identification of land cover change were based on the same input. Based on this, it is established that these images have a spatial resolution of 30 meters, complying with the specifications in module VMD0015 for the quantification procedure.

The satellite images obtained for the analyses were sourced from Google Earth Engine, a platform that enables the search and download of satellite images from Landsat satellites. These images were obtained for the entire extent of the reference region and the project area, covering the years of analysis and monitoring. Image selection aimed to obtain scenes with cloud cover of less than 30%.

After obtaining satellite images for the reference region and the project area, quantification of deforested areas and change in land cover was conducted through visual interpretation processes. Results of change areas for the project area were stored in the layer named "LandCover_Change_Project_Area_2021_2023", and for the reference region, it was labeled as "LandCover_Change_Reference_Region_2010_2020". These results were provided in Annex 40. GEODATABASE_PROYECTO_KAA_PYAHUU_v1".

Post-processing and accuracy assessment

1. Map area change detected in the imagery. See section 5.1.3 (Stratification of the project area).
2. Calculate the area of each category of change within the project area and, where required, the leakage belt. See section 7.3 (Matrix Land use change post-deforestation in the Project Area (2021 - 2023))
 - a. At the end of each monitoring period:
 - Calculate the area of each category of change within the project area and, where required, the leakage belt. See section 7.3 (Matrix Land use change post-deforestation in the Project Area (2021 - 2023))
 - Update the Forest Cover Benchmark Maps for the project area (Section 7.3) and leakage belt (section 7.4).

The section 2b (step 1 VMD0015) does not apply, since it is the first validation and verification of the project, therefore, the baseline is not revalidated.

Step 2. Interpretation and analyses

Monitoring deforestation:

This step produces an estimate of the emissions resulting from any deforestation that occurs within the project area and leakage belt. See sections 7.3 and 7.4.

Step 3. Documentation

- a. Data sources and pre-processing

Satellite images were used to complement the information generated by Mapbiomas. As mentioned previously, Mapbiomas works with satellite images, the cloud computing Google Earth Engine and experts in remote sensing and computing, data science, and biomes. Additionally, the satellite imagery dataset used in the MapBiomas is composed by the Thematic Mapper (TM), Enhanced Thematic Mapper Plus (ETM+) and the Operational Land Imager (OLI), Landsat sensors, on board Landsat 5, Landsat 7 and Landsat 8. These images have a spatial resolution of 30 meters, adhering to the specifications of module VMD0015.

As mentioned before, these satellite images were chosen for land use and land cover change analysis due to their ease of acquisition. They are the same data source used to produce the Forest/Non-Forest input annually.

Regarding the spectral bands used, two main band compositions were employed: true color composition (bands 3, 2, 1 for Landsat 7 and bands 4, 3, 2 for Landsat 8) and false color composition (bands 4, 3, 2 for Landsat 7 and bands 5, 4, 3 for Landsat 8). These images were downloaded and subsequently visualized using ArcGIS software version 10.8, particularly through the module called ArcMap.

b. Data classification

The classification was followed according to the information generated by Mapbiomas, which is a hierarchical system with a combination of LULC classes compatible with the Food and Agriculture Organization (FAO) and IBGE classification systems. This classification has three levels: at the first level there are six classes: forest (1), non-forest formation (2), farming (3), non-vegetated area (4), water (5) and not observed (6). The level 2 has 12 classes that also have a combination of LULC classes. Finally The MapBiomas LULC Classification System can be linked to other international and national classification systems (Souza, et al., 2020). Therefore, the final identification of land use and changes within the deforested area between the years 2010 and 2020 for the reference region and between the years 2021 and 2023 for the project area.

Following the methodological protocol, when satellite images exhibited cloud cover over the area of interest, this data was supplemented using optical images from the Sentinel-2 satellite. If such data was unavailable, radar images from the Sentinel-1 satellite were utilized. These auxiliary images were also procured through the Google Earth Engine platform.

For the interpretation of land use and land cover changes, ArcGIS 10.8 software was employed, specifically using the ArcMap module. This enabled the creation of a geographic database and its storage in a file, which was subsequently provided through Annex 55.

c. Classification accuracy assessment

Following the information from Mapbiomas, the LULC mapping product accuracy at level 1 present an 89.13% of global accuracy. According to the level 1 categories, the forest formations present an accuracy of 92.41%. Similarly, at level 2 categories, the natural forest formation present an accuracy of 92.80%

d. Changes in Data sources and pre-processing / Data classification

For the development of the project document and the monitoring report, the same data sources were used, and as a result, no modifications have been made to them currently.

5.2.1 Emissions from deforestation

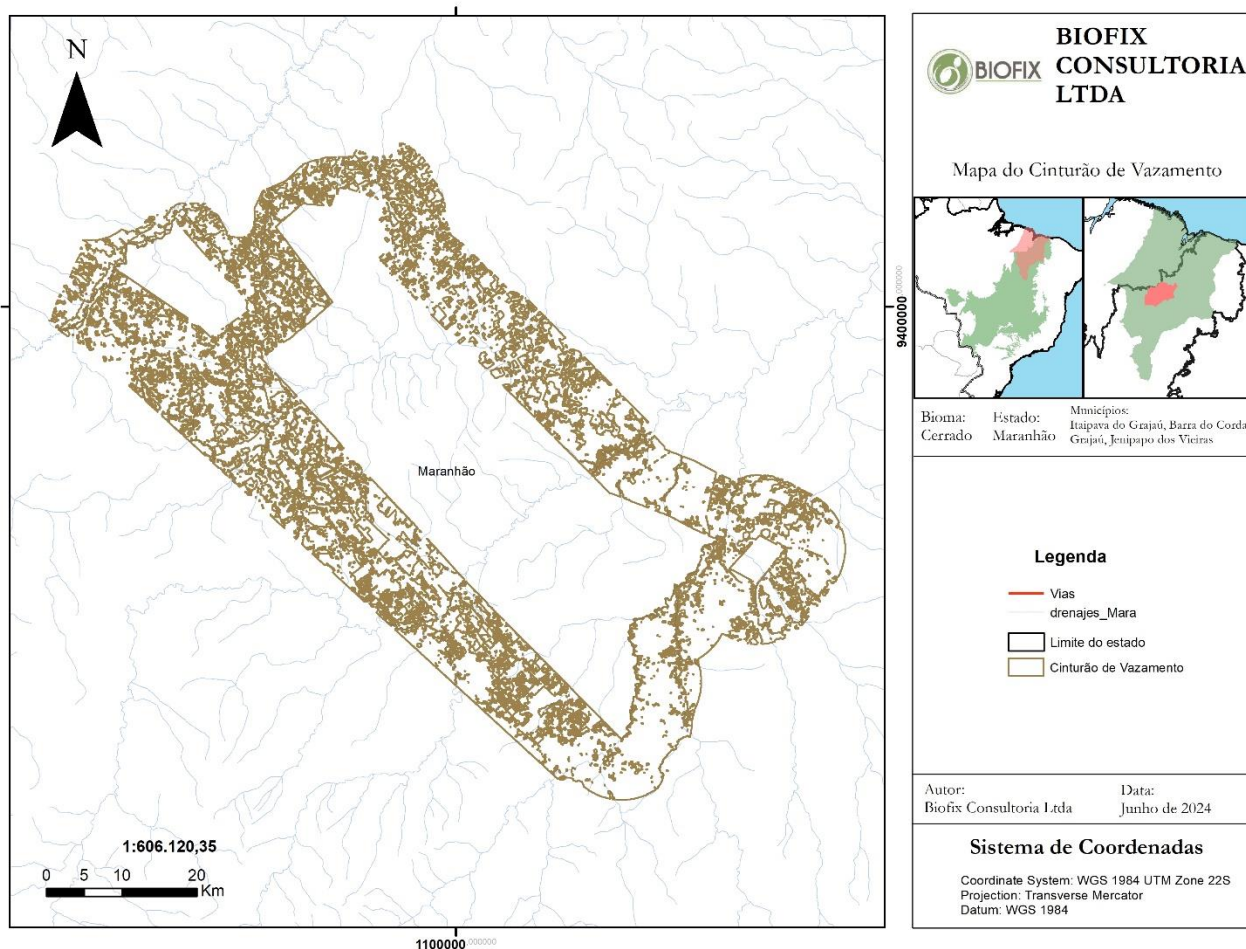
The project does not estimate emissions from planned deforestation. Therefore, in the baseline scenario, it is assumed that no emissions will be generated from deforestation.

In a different way, in the project scenario, the emissions corresponding to the deforestation that occurred in the years of the monitoring period. To quantify these emissions the equation 6, of the section 5.2 of the module VMD0015 V2.3, will be used.

5.3 Leakage Emissions

The delimitation of the Leakage Belt was carried out following the methodology described in section 3.3.3. of this document; through the geospatial analysis of landscape and transport factors. By the year 2021, the area of the leakage belt, that is, the forest area is 151,444 hectares.

Figure 56. Leakage Belt Forest Cover Benchmark Map 2021



Source. KA'A PYAHU REDD+ Project.

Table 68. Forest area in the leakage belt

CLASS/Year	2021	2023
NO FOREST (ha)	0	1,864
FOREST (ha)	151,444	149,580
TOTAL (ha)	151,444	151,444

Source. KA'A PYAHU REDD+ Project.

According to the step 2.3 of the module BL-UP (VMD007 V3.3), the projected unplanned baseline deforestation in the leakage belt is estimated as follows:

$$A_{BSL, LB, unplanned, t} = A_{BSL, RRD, unplanned, hrp} * P_{LK}$$

$A_{BSL, LB, unplanned, t}$ = Projected area of unplanned baseline deforestation in the leakage belt in year t; ha.

$A_{BSL, RRD, unplanned, t}$ = Projected area of unplanned baseline deforestation in the RRD in year t; ha.

P_{LK} = Ratio of the leakage belt to the total area of RRD; dimensionless.

The P_{LK} is 0.18, resulting in:

Table 69. Projected area of unplanned deforestation from the baseline in the leakage belt

Year	<i>ABSL, LB, unplanned, t (ha)</i>
2021	1,431
2022	2,132
2023	2,132
2024	2,132
2025	2,132
2026	2,132
2027	2,132
2028	2,132
2029	2,132
2030	2,132
2031	2,132
2032	2,132
2033	2,132
2034	2,132
2035	2,132
2036	2,132
2037	2,132
2038	2,132
2039	2,132
2040	2,132
2041	701

Source. KA'A PYAHU REDD+ Project.

According with the section 3.15.14 of the VCS Standard V.4.7, projects may apply the optional default activity-shifting leakage deduction of 15% to the gross GHG emission reductions. Therefore, the project assumes a default value of 15% of leakage. Using the carbon stocks from section 5.1.4, the baseline scenario emissions in the leakage belt are:

Table 70. Estimation of GHG Emissions in the Leakage Belt

Year	<i>ABSL, LB, unplanned, t (ha)</i>	<i>ΔCBSL, LK, unplanned</i>
2021	215	29,749
2022	320	44,320
2023	320	44,320
2024	320	44,320
2025	320	44,320
2026	320	44,320
2027	320	44,320
2028	320	44,320
2029	320	44,320
2030	320	44,320
2031	320	44,320
2032	320	44,320
2033	320	44,320
2034	320	44,320
2035	320	44,320
2036	320	44,320
2037	320	44,320
2038	320	44,320
2039	320	44,320

2040	320	44,320
2041	105	14,571

Source. KA'A PYAHU REDD+ Project.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

This section corresponds to equation 22 of the BL-UP module (VMD0007 V3.3) were used to determine the potential for reductions due to unplanned deforestation.

$$\Delta C_{BSL,i,t} = A_{unplanned,i,t} * (\Delta C_{AB_{tree,i}} - \Delta C_{WP,i} + \Delta C_{AB_{non-tree,i}} + \Delta C_{LI,i})$$

$$+ \left(\sum_{t=10}^t A_{unplanned,i,t} \right) * (\Delta C_{BB_{tree,i}} + \Delta C_{BB_{non-tree,i}} + \Delta C_{DW,i}) * \left(\frac{1}{10} \right)$$

$$+ \left(\sum_{t=20}^t A_{unplanned,i,t} \right) * (C_{WP100,i} + \Delta C_{SOC,i}) * \left(\frac{1}{20} \right)$$

Where:

$\Delta C_{BSL,i,t}$	Sum of the baseline carbon stock change in all terrestrial pools in stratum i in year t , t CO ₂ e (calculated separately for the project area [PA] and the leakage belt [LB])
$A_{unplanned,i,t}$	Area of unplanned deforestation in forest stratum i in year t , ha
$C_{WP,i}$	Carbon stock entering the wood products pool from stratum i , t CO ₂ e ha ⁻¹
$C_{WP100,i}$	Carbon stock entering the wood products pool at the time of deforestation that is expected to be emitted over 100-years from stratum i , t CO ₂ e ha ⁻¹
$\Delta C_{AB_{tree,i}}$	Baseline carbon stock change in aboveground tree biomass in stratum i , t CO ₂ e ha ⁻¹
$\Delta C_{BB_{tree,i}}$	Baseline carbon stock change in belowground tree biomass in stratum i , t CO ₂ e ha ⁻¹
$\Delta C_{AB_{non-tree,i}}$	Baseline carbon stock change in aboveground non-tree biomass in stratum i , t CO ₂ e ha ⁻¹
$\Delta C_{BB_{non-tree,i}}$	Baseline carbon stock change in belowground non-tree biomass in stratum i , t CO ₂ e ha ⁻¹
$\Delta C_{DW,i}$	Baseline carbon stock change in dead wood in stratum i , t CO ₂ e ha ⁻¹
$\Delta C_{LI,i}$	Baseline carbon stock change in litter in stratum i , t CO ₂ e ha ⁻¹
$\Delta C_{SOC,i}$	Baseline carbon stock change in terrestrial soil organic carbon in stratum i , t CO ₂ e ha ⁻¹
i	1, 2, 3, ... M strata

Table 71. Estimated baseline GHG emissions in the project area

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
01-May-2021 to 31-Dec-2021	229,535	0	29,749	199,786	0	199,786
01-Jan-2022 to 31-Dec-2022	341,961	0	44,320	297,640	0	297,640

01-Jan-2023 to 31-Dec-2023	341,961	0	44,320	297,640	0	297,640
01-Jan-2024 to 31-Dec-2024	341,961	0	44,320	297,640	0	297,640
01-Jan-2025 to 31-Dec-2025	341,961	0	44,320	297,640	0	297,640
01-Jan-2026 to 31-Dec-2026	341,961	0	44,320	297,640	0	297,640
01-Jan-2027 to 31-Dec-2027	341,961	0	44,320	297,640	0	297,640
01-Jan-2028 to 31-Dec-2028	341,961	0	44,320	297,640	0	297,640
01-Jan-2029 to 31-Dec-2029	341,961	0	44,320	297,640	0	297,640
01-Jan-2030 to 31-Dec-2030	341,961	0	44,320	297,640	0	297,640
01-Jan-2031 to 31-Dec-2031	341,961	0	44,320	297,640	0	297,640
01-Jan-2032 to 31-Dec-2032	341,961	0	44,320	297,640	0	297,640
01-Jan-2033 to 31-Dec-2033	341,961	0	44,320	297,640	0	297,640
01-Jan-2034 to 31-Dec-2034	341,961	0	44,320	297,640	0	297,640
01-Jan-2035 to 31-Dec-2035	341,961	0	44,320	297,640	0	297,640
01-Jan-2036 to 31-Dec-2036	341,961	0	44,320	297,640	0	297,640
01-Jan-2037 to 31-Dec-2037	341,961	0	44,320	297,640	0	297,640
01-Jan-2038 to 31-Dec-2038	341,961	0	44,320	297,640	0	297,640
01-Jan-2039 to 31-Dec-2039	341,961	0	44,320	297,640	0	297,640
01-Jan-2040 to 31-Dec-2040	341,961	0	44,320	297,640	0	297,640
01-Jan-2041 to 31-April-2041	112,425	0	14,571	97,854	0	97,854
Total	6,839,213	0	886,407	5,952,805	0	5,952,805

Source. KA'A PYAHU REDD+ Project.

Applying the digital non-permanence risk tool (T-BAR) v.4.2, which yielded 20%, the following Buffer and its verified carbon units are obtained:

Table 72. Estimated reductions VCUs in the project area

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
01-May-2021 to 31-Dec-2021	229,535	0	29,749	39,957	159,289		159,289
01-Jan-2022 to 31-Dec-2022	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2023 to 31-Dec-2023	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2024 to 31-Dec-2024	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2025 to 31-Dec-2025	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2026 to 31-Dec-2026	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2027 to 31-Dec-2027	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2028 to 31-Dec-2028	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2029 to 31-Dec-2029	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2030 to 31-Dec-2030	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2031 to 31-Dec-2031	341,961	0	44,320	59,528	232,159		238,112
01-Jan-2032 to 31-Dec-2032	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2033 to 31-Dec-2033	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2034 to 31-Dec-2034	341,961	0	44,320	59,528	238,112		238,112

01-Jan-2035 to 31-Dec-2035	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2036 to 31-Dec-2036	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2037 to 31-Dec-2037	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2038 to 31-Dec-2038	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2039 to 31-Dec-2039	341,961	0	44,320	59,528	238,112		238,112
01-Jan-2040 to 31-Dec-2040	112,425	0	44,320	59,528	238,112		238,112
01-Jan-2041 to 31-April-2041	229,535	0	29,749	19,571	78,283		78,283
Total	6,839,213	0	886,407	1,190,591	4,762,244		4,762,244

Source. KA'A PYAHU REDD+ Project.

6 MONITORING

6.1 Data and Parameters Available at Validation

The data and parameters used in estimating GHG emission reductions for each of the modules for the validation period are presented below:

Table 73. Data and parameters VM0007: METHODOLOGICAL FRAMEWORK (REDD-MF) v 1.7

Data / Parameter:	ΔC BSL,unplanned
Data unit:	t CO2e
Description:	Net greenhouse gas emissions in the baseline from unplanned deforestation
Source of data	Module BL-UP.
Value applied	5,952,805
Justification of choice of data or description of measurement methods and procedures applied	See Module BL-UP
Purpose of Data	Calculation of baseline emissions
Comments	N/A

Table 74. Data and parameters VMD0007: Estimation of changes in baseline carbon stocks and greenhouse gas emissions from unplanned deforestation (BL-UP) v 3.3

Data / Parameter:	Baseline Deforestation Maps
Data unit:	N/A
Description:	Maps showing the location of deforested hectares in each year of the baseline period
Source of data	Section 5.1.1
Value applied	See maps section 5.1.1
Justification of choice of data or description of measurement methods and procedures applied	Information on land cover change and deforestation from Mapbiomas brasil to estimate and to analysis ex ante and ex post deforestation and loss of coverage.
Purpose of Data	Calculation of baseline
Comments	Must be updated each time the baseline is revisited (at least every 10 years). Accuracy 92%

Data / Parameter:	LB
Data unit:	ha
Description:	Leakage belt area
Source of data	Remote sensing data Annex 55. Tab DATOS_SIG
Value applied	151,444
Justification of choice of data or description of measurement methods and procedures applied	Area where there may be a displacement of emissions that the project must partially assume and discount from the baseline
Purpose of Data	Calculation of baseline leakage
Comments	Must be updated each time the baseline is revisited (at least every 10 years)

Data / Parameter:	PA
Data unit:	ha
Description:	Unplanned deforestation project area
Source of data	Remote sensing data. Annex 55. Tab DATOS_SIG
Value applied	150,800
Justification of choice of data or description of measurement methods and procedures applied	Estimation of the project area that can be deforested by deforestation agents and drivers and that is within the area of the project proponent
Purpose of Data	Calculation of baseline deforestation
Comments	Must be estimated at time zero, this estimate must be used for ex-ante purposes

Data / Parameter:	PLK
Data unit:	Dimensionless
Description:	Ratio of the area of the leakage belt to the total area of RRD
Source of data	Remote sensing data and Baseline Maps.
Value applied	0.18
Justification of choice of data or description of measurement methods and procedures applied	Used for calculating project area of unplanned baseline deforestation in the leakage belt area. Calculated from the result of remotely sensed data analysis
Purpose of Data	Calculation of baseline emissions and leakage.

Comments	Monitored at least once every 10 years (when the baseline is revisited). Must be estimated at time zero, this estimate must be used for ex-ante purposes.
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Data / Parameter:	PPA
Data unit:	Dimensionless
Description:	Ratio of the project area to the total area of RRD
Source of data	Remote sensing data and Baseline Maps.
Value applied	0.21
Justification of choice of data or description of measurement methods and procedures applied	Calculated as the ratio of the forest data of the project area and the reference region.
Purpose of Data	Calculate projected area of unplanned baseline deforestation in the project area
Comments	Monitored at least once every 10 years (when the baseline is revisited). Must be estimated at time zero, this estimate must be used for ex-ante purposes

Data / Parameter:	RRD
Data unit:	ha
Description:	Geographic boundaries of the reference area for projection of rate of deforestation
Source of data	GPS coordinates and/or remote sensing data
Value applied	843,491
Justification of choice of data or description of measurement methods and procedures applied	Established in accordance with the requirements established in the BL-UP module. For the factors of transport, landscape and deforestation rates, different official sources from Brazil cited in section 3.3.1 were used.
Purpose of Data	Projection of rate of deforestation
Comments	Must be updated each time the baseline is revisited (at least every 10 years)

Data / Parameter:	Thrp
Data unit:	yr
Description:	Duration of the historical reference period in years
Source of data	Module BL-UP requirement

Value applied	11
Justification of choice of data or description of measurement methods and procedures applied	The start date of the historical reference period must be between 9 and 12 years prior to the project start date and the end date must be within two years prior to the project start date.
Purpose of Data	Projection of rate of deforestation
Comments	Must be between 9 and 12 years

Data / Parameter:	ARRD,unplanned,hrp
Data unit:	Ha
Description:	Total area deforested during the historical reference period in RRD
Source of data	Remote sensing data. Annex 55. Tab DATOS_SIG
Value applied	128,657
Justification of choice of data or description of measurement methods and procedures applied	See M-REDD and BL-UP modules
Purpose of Data	Calculation of baseline emissions
Comments	Monitored at least once every 10 years (when the baseline is revisited)

Data / Parameter:	ABSL,PA,unplanned,t
Data unit:	Ha
Description:	Project area of unplanned baseline deforestation in the project area in year t
Source of data	Remote sensing data. Annex 55. Tab BL – UP
Value applied	2,467 ha/year
Justification of choice of data or description of measurement methods and procedures applied	See M-REDD and BL-UP
Purpose of Data	Calculation of baseline emissions
Comments	Monitored at least once every 10 years (when the baseline is revisited)

Data / Parameter:	ABSL,LB,unplanned,t
Data unit:	Ha

Description:	Project area of unplanned baseline deforestation in the leakage area in year t
Source of data	Remote sensing data. Annex 55. Tab BL_LK
Value applied	2,132 ha/year
Justification of choice of data or description of measurement methods and procedures applied	See M-REDD and LK-ASU
Purpose of Data	Estimation of emission from unplanned deforestation in the leakage belt
Comments	Monitored at least once every 10 years (when the baseline is revisited)

Data / Parameter:	CAB_tree,i
Data unit:	t CO2e ha-1
Description:	Carbon stock in aboveground biomass in trees in stratum
Source of data	Brazil's National Forest Reference Emission Level (2022) and Fourth greenhouse gas inventory, AFOLU sector.
Value applied	NREF 146.52 Baseline 114.01
Justification of choice of data or description of measurement methods and procedures applied	Brazil's National Forest Reference Emission Level for Results-based Payments for REDD+ under the United Nations Framework Convention on Climate Change
Purpose of Data	Calculate the total carbon stock in the project area
Comments	

Data / Parameter:	CAB_nontree,i
Data unit:	t CO2e ha-1
Description:	Carbon stock in aboveground non tree vegetation
Source of data	Different sources used on carbon stock for different land uses such as pasture, agriculture, secondary vegetation, in the Cerrado biome. See section 5.1.4
Value applied	32.51
Justification of choice of data or description of measurement methods and procedures applied	Official sources (fourth greenhouse gas inventory, AFOLU sector) for weighting the post-deforestation carbon stock

Purpose of Data	Estimation of baseline emissions
Comments	

Data / Parameter:	CBB_tree,i
Data unit:	t CO2e ha-1
Description:	Carbon stock in belowground biomass in trees in stratum i
Source of data	Brazil's National Forest Reference Emission Level (2022) and fourth greenhouse gas inventory, AFOLU sector.
Value applied	NREF 29.30 Baseline 24.58
Justification of choice of data or description of measurement methods and procedures applied	The values of the NREF 2019 are used to comply with article 41 of Resolution 1447 of 2018.
Purpose of Data	Calculation of baseline emissions
Comments	

Data / Parameter:	CBB_nontree,i
Data unit:	t CO2e ha-1
Description:	Carbon stock in belowground non tree vegetation
Source of data	Different sources used on carbon stock for different land uses such as pasture, agriculture, secondary vegetation, in the Cerrado biome. See section 5.1.4
Value applied	LB 4.71
Justification of choice of data or description of measurement methods and procedures applied	Official sources (fourth greenhouse gas inventory, AFOLU sector) for weighting the post-deforestation carbon stock
Purpose of Data	Estimation of baseline emissions
Comments	

Data / Parameter:	Regional Forest Cover / Non-Forest Cover Benchmark Map
Data unit:	N/A
Description:	Map showing the location of forest land within the reference region at the beginning of the crediting period
Source of data	Deforestation information available from Mapbiomas. See section 5.1.2.
Value applied	N/A

Justification of choice of data or description of measurement methods and procedures applied	Module BL-UP.
Purpose of Data	Calculate the change in land use after deforestation in the reference period. This allows calculating the change matrix to estimate total carbon by reservoirs and net emissions.
Comments	

Data / Parameter:	Project Forest Cover Benchmark Map
Data unit:	N/A
Description:	Map showing the location of forest land within the project area at the beginning of each monitoring period.
Source of data	
Value applied	N/A
Justification of choice of data or description of measurement methods and procedures applied	Module BL-UP
Purpose of Data	Calculation of baseline emissions and project emissions
Comments	Deforestation information available from Mapbiomas. See section 5.1.3.

Data / Parameter:	Leakage Belt Forest Cover Benchmark Map
Data unit:	N/A
Description:	Map showing the location of forest land within the leakage belt area at the beginning of each monitoring period.
Source of data	Deforestation information available from Mapbiomas. See section 5.3.
Value applied	N/A
Justification of choice of data or description of measurement methods and procedures applied	Module BL-UP
Purpose of Data	Calculate the deforestation in the leakage belt. This allows calculating the net emissions.
Comments	

Table 75. Data and parameters VMD0001: Estimation of carbon stocks in the above and belowground biomass in live tree and non-tree pools (CP-AB) v.1.2.

Data / Parameter:	CF
Data unit:	t C t d.m. ⁻¹
Description:	Carbon fraction of dry matter in t C t-1 d.m.
Source of data	Brazil's National Forest Reference Emission Level (2022) and fourth greenhouse gas inventory, AFOLU sector.
Value applied	0,47
Justification of choice of data or description of measurement methods and procedures applied	Value officially used in Brazil's national reference forest emissions levels
Purpose of Data	calculation of baseline and project emissions
Comments	

6.2 Data and Parameters Monitored

The data and parameters to be used in estimating GHG emission reductions are presented below, for each of the modules for the monitoring period, following the modules:

Table 76. Data and parameters VM0007: METHODOLOGICAL FRAMEWORK (REDD-MF) v 1.7.

Data / Parameter:	ΔCWPS-REDD
Data unit:	t CO ₂ e
Description:	Net GHG emissions in the REDD project crediting period up to year t*
Source of data	Information generated by Mapbiomas and processed by the project developer to apply the BL-UP Module and R-MEDD module
Description of measurement methods and procedures to be applied	From the quantification of deforestation with the maps of forest cover changes that Mapbiomas draws annually, emissions from deforestation are quantified
Value applied	0
Monitoring equipment	Remote data sensing
QA/QC procedures to be applied	Information from Mapbiomas, processing data images/information. The Mapbiomas works with a confidence interval of 95%, a maximum standard error of 5%, and an accuracy for natural forest formation of 92% (Souza, et al., 2020)
Purpose of Data	Calculation of project emissions
Calculation method	M-REDD module

Comments	It is expected that no emissions will be generated due to unplanned deforestation, since there is an indigenous guard of the project, as well as the lines of action that monitor and control deforestation are established. However, in the monitored period, deforestation was evident, as shown in section 7.2 for this same parameter.
Data / Parameter:	CLK-AS,unnplanned
Data unit:	t CO ₂ e
Description	Net GHG emissions due to activity shifting for projects preventing unplanned deforestation
Source of data	Information generated by Mapiomas, processed by the project developer applying the LK-ASU module and M-REDD module.
Description of measurement methods and procedures to be applied	M-REDD module and LK-ASU module
Frequency of monitoring/recording	Every monitoring period
Value applied	886,407
Monitoring equipment	Remote data sensing
QA/QC procedures to be applied	GIS analysis of coverage land use in the leakage belt.
Purpose of data	Calculation of leakage emissions
Calculation method	M-REDD module and LK-ASU module
Comments	Estimate from the baseline for the years of the crediting period

Table 77. Data and parameters VMD0015: Methods for monitoring of GHG emissions and removals in REDD and CIW projects (M-REDD) v 2.3

Data / Parameter:	Project Forest Cover Monitoring Map
Data unit:	N/A
Description	Map showing the location of forest land within the project area at the beginning of each monitoring period.
Source of data	Update of digital maps Mapbiomas
Description of measurement methods and procedures to be applied	See section 5.2 of this document, where the 3 steps of section 5 of the M-REDD module are developed
Frequency of monitoring/recording	Annual
Value applied	Will be generated at the beginning and end of each monitoring period.

Monitoring equipment	Remote data sensing
QA/QC procedures to be applied	Quality review of satellite images and official maps to complete information with other geographic information systems
Purpose of data	Calculation of project emissions
Calculation method	GIS. Module M-REDD
Comments	Accuracy for natural forest formation of 92% (Souza, et al., 2020)

Data / Parameter:	Leakage Belt Forest Cover Monitoring Map
Data unit:	N/A
Description	Map showing the location of forest land within the leakage belt area at the beginning of each monitoring period.
Source of data	Update of digital maps Mapbiomas
Description of measurement methods and procedures to be applied	See section 5.2 of this document, where the 3 steps of section 5 of the M-REDD module are developed
Frequency of monitoring/recording	Annual
Value applied	Will be generated at the beginning and end of each monitoring period.
Monitoring equipment	Remote data sensing
QA/QC procedures to be applied	Quality review of satellite images and official maps to complete information with other geographic information systems
Purpose of data	Calculation of leakage
Calculation method	GIS. Module M-REDD
Comments	Accuracy for natural forest formation of 92% (Souza, et al., 2020)

Data / Parameter:	ADefPA,i,u,t
Data unit:	ha
Description	Area of recorded deforestation in the project area in stratum I converted to land use u in year t
Source of data	Analysis date remote.

Description of measurement methods and procedures to be applied	M-REDD module
Frequency of monitoring/recording	Every monitoring period
Value applied	0
Monitoring equipment	Remote data sensing
QA/QC procedures to be applied	GIS analysis of coverage land forest changes. Quality review of satellite images and official maps to complete information
Purpose of data	Calculation of project emissions
Calculation method	Step 1 Module M-REDD
Comments	It is expected that no emissions will be generated due to unplanned deforestation, since there is an indigenous guard of the project, as well as the lines of action that monitor and control deforestation are established. However, in the monitored period, deforestation was evident, as shown in section 7.2 for this same parameter.

Data / Parameter:	ADefLB,i,u,t
Data unit:	ha
Description	Area of recorded deforestation in the leakage belt in stratum i converted to land use u in year t
Source of data	Analysis date remote.
Description of measurement methods and procedures to be applied	M-REDD module
Frequency of monitoring/recording	Every monitoring period
Value applied	6,396
Monitoring equipment	Remote data sensing
QA/QC procedures to be applied	GIS analysis of coverage land forest changes. Quality review of satellite images and official maps to complete information
Purpose of data	Calculation of leakage
Calculation method	Step 1 Module M-REDD
Comments	Estimate from the baseline for the years of the crediting period

Data / Parameter:	CAB_tree,i
Data unit:	t CO2e ha-1
Description	Carbon stock in aboveground biomass in trees in stratum I for the project crediting period
Source of data	Brazil's National Forest Reference Emission Level (2022) and fourth greenhouse gas inventory, AFOLU sector.
Description of measurement methods and procedures to be applied	The most updated NREF values for Brazil will be used and the amount of carbon stock to discounting the lost changes.
Frequency of monitoring/recording	Annual
Value applied	NREF 146.52 C.P. 114.01
Monitoring equipment QA/QC procedures to be applied	GIS analysis of coverage changes for the discount on the baseline values.
Purpose of data	Calculation of project emissions
Calculation method	Step 2 Module M-REDD
Comments	Estimation from the baseline

Data / Parameter:	CAB_nontree,i
Data unit:	t CO2e ha-1
Description	Carbon stock in aboveground biomass in non trees in stratum I for the project crediting period
Source of data	Different sources used on carbon stock for different land uses such as pasture, agriculture, secondary vegetation, in the Cerrado biome. See section 5.1.4
Description of measurement methods and procedures to be applied	Different values for brazil land uses carbon stock will be used and the amount of carbon stock to discounting the lost changes
Frequency of monitoring/recording	Annual
Value applied	32.51
Monitoring equipment QA/QC procedures to be applied	GIS analysis of coverage changes for the discount on the baseline values.
Purpose of data	Calculation of project emissions
Calculation method	Step 2 Module M-REDD
Comments	Estimation from the baseline

Data / Parameter:	CBB_tree,i
Data unit:	t CO2e ha-1
Description	Carbon stock in belowground biomass in trees in stratum In the project crediting period
Source of data	Brazil's National Forest Reference Emission Level (2022) and fourth greenhouse gas inventory, AFOLU sector.
Description of measurement methods and procedures to be applied	The most updated NREF values for Brazil will be used and the amount of carbon stock to discounting the lost changes.
Frequency of monitoring/recording	Annual
Value applied	NREF 29.30 CP 24.58
Monitoring equipment QA/QC procedures to be applied	GIS analysis of coverage changes for the discount on the baseline values.
Purpose of data	Calculation of project emissions
Calculation method	Step 2 Module M-REDD
Comments	Estimation from the baseline

Data / Parameter:	CBB_nontree,i
Data unit:	t CO2e ha-1
Description	Carbon stock in belowground biomass in non trees in stratum In the project crediting period
Source of data	Different sources used on carbon stock for different land uses such as pasture, agriculture, secondary vegetation, in the Cerrado biome. See section 5.1.4
Description of measurement methods and procedures to be applied	Different values for brazil land uses carbon stock will be used and the amount of carbon stock to discounting the lost changes
Frequency of monitoring/recording	Annual
Value applied	4.71
Monitoring equipment QA/QC procedures to be applied	GIS analysis of coverage changes for the discount on the baseline values.
Purpose of data	Calculation of project emissions
Calculation method	Step 2 Module M-REDD
Comments	Estimation from the baseline

6.3 Monitoring Plan

In accordance with the provisions of methodology VM0007 (REDD+ MF), the development of the monitoring plan should include the following monitoring tasks: Monitoring of project implementation; monitoring the actual carbon stock changes and greenhouse gas emissions; monitoring the leakage carbon stock changes and greenhouse gas emissions, and the estimation of ex-post net carbon stock changes and greenhouse gas emissions.

Each of the tasks mentioned above must comply with the information from the technical description of the monitoring task, the data, and parameters to be collected, an overview of the data collection procedure, a quality control and assurance procedure, data archiving, and the organization and responsibilities of the parties involved.

To comply with these aspects, the following table will show the corresponding information to the technical description of the monitoring task, data to be collected, and the overview of data collection.

Table 78. Monitoring task

Task	Technical description of the monitoring task	Data to be collected	Overview of data collection procedures
Monitoring of project implementation	The monitoring of project implementation starts from the establishment of techniques through remote sensors to identify changes in forest cover, both for the project area and for the leakage belt. The monitoring protocol starts from the digital pre-processing of satellite images, in which image acquisition is carried out for the evaluation year. Corrections for geometry are made on these images, and cloud and shadow masking processes are applied. Once the correction procedure is established, a mosaic of satellite images over the areas of interest is created, and subsequently a	Project Forest Cover Monitoring Map Leakage Belt Forest Cover Monitoring Map	The establishment of the Project Forest Cover Monitoring Map and the Leakage Belt Forest Cover Monitoring Map is based on obtaining the Forest/Non-Forest formation input data provided by Mapbiomas for the years of the crediting period, inside the project area. Once these inputs are obtained, the loss of forest cover between the crediting periods is determined by using remote sensing, primarily from the Landsat optical satellite imagery. These satellite images have a spatial resolution of 30 meters and are used to identify changes in existing land cover within the areas of forest loss. The accuracy of the map must

	visual interpretation analysis of changes due to deforestation is performed during the analysis period. In addition to this, an early warning analysis is carried out in areas with high deforestation potential.		exceed 90%, meeting the requirements set by module VMD0015. The generation of new maps will occur during the assessment of the next crediting period and will be carried out yearly.
Monitoring of actual carbon stock changes and greenhouse gas emissions	The current monitoring calculation for the change in carbon reserves and greenhouse gas emissions begins with the selection and analysis of the change in forest coverage measured within the project area. For the change in coverage, remote sensors are used to highlight the changes. Once the areas of forest cover change have been analyzed, we proceed to calculate the change in the project area, calculating the total deforested zones. Finally, we proceed to evaluate the emissions resulting from deforestation generated within the project area.	AdefPA,l,u,t (Area of recorded deforestation in the project area in stratum l converted to land use u in year t) CAB_tree,l (Carbon stock in aboveground biomass in trees in stratum l in the monitoring period) CBB_tree,l (Carbon stock in belowground biomass in trees in stratum l in the monitoring period)	The establishment of the data collection and parameters set for this monitoring task stems from obtaining the Brazil's National Forest Reference Emission Level for Results-based Payments for REDD+, associated with the calculations of carbon reserves in aboveground biomass, carbon reserves in underground biomass. For the deforestation area in the project area, this will be obtained through the collection of information on forest cover loss, demonstrating land use change through satellite images, primarily from Landsat satellites. The monitoring of the data to be collected will be carried out yearly.
Monitoring of leakage carbon stock changes and greenhouse gas emissions	The monitoring of changes in carbon reserves from greenhouse gas leaks is based on module VMD0010 (LK-ASU). In this, six main steps must be performed, starting from the estimation of changes in the carbon reserves of the leakage belt. Subsequently, the estimation of the proportion of the area deforested by local and immigrant deforestation	AdefLB,l,u,t (Area of recorded deforestation in the leakage belt in stratum l converted to land use u in year t)	The area of deforestation stored in the leakage belt, converted to current use, stems from the use of remote sensors. These sensors determine the change in forest area due to the activities of deforestation agents. The change in the area of the leakage belt is quantified in association with changes in coverage, and this must be monitored annually. The acquisition of coverage changes

	agents in the baseline is performed. After this, the estimation of displaced deforestation that occurred in the project area to the leakage belt must be carried out. Once the previous step has been completed, the estimation of deforestation displaced from the project area to the leakage belt area must be carried out. Then, the emissions from prevention activities in the leaks must be calculated, and finally, the estimation of the total leaks associated with unplanned deforestation is made.		in the leakage belt begins with obtaining data from Mapbiomas, and subsequently, an evaluation of coverage changes is carried out using optical images from Landsat satellites. This is done to identify land use and to quantify greenhouse gas emissions.
Estimation of ex post net carbon stock changes and greenhouse gas emissions	The process of estimating changes in carbon reserves in the ex-post of greenhouse gas emissions is based on module VMD0015 (M-REDD). For the calculation, the process of identifying land cover changes is performed, which is obtained through remote sensors, primarily from Landsat and Sentinel satellites. Subsequently, the quantification of the lost forest area is carried out in the reference region, project area, and leakage belt. Afterward, the calculation of emissions resulting from deforestation is performed.	CAB_tree,i (Carbon stock in aboveground biomass in trees in stratum i in the monitoring period) CBB_tree,i (Carbon stock in belowground biomass in trees in stratum i in the monitoring period)	The parameters to be monitored start from identifying the estimated cover changes within spatial boundaries. Once the area of forest change is obtained, estimates are made with the data established in the NREF, which are associated with the type of sink, and the change factors associated with conversions are also obtained through the Brazil's National Forest Reference Emission Level for Results-based Payments for REDD+. The monitoring frequency of the selected parameters will be carried out yearly

Source. KA'A PYAHU REDD+ project

Quality control and quality assurance procedure

Quality control, as well as the acquisition and centralization of the information required for project monitoring, will be carried out by BIOFIX CONSULTORIA LTDA., including the evaluation of the uncertainty of the data that are relevant, in the baseline scenario.

To comply with the reduction of uncertainty in the quantification of emissions and removals, local parameters were selected based on the sink values established by the Brazil's National Forest Reference Emission Level for Results-based Payments for REDD+ under the United Nations Framework Convention on Climate Change (Brasil, 2024) specially for obtaining values of above-ground biomass, below-ground biomass, and soil organic carbon.

Data archiving

The data obtained from the monitoring plan will be stored and managed through Google Drive, which is a data storage service where the data is stored in the cloud. This service allows copying files from the computer to be saved in the cloud, and also allows creating documents and folders directly from the platform, associating functions similar to the files presented in the Microsoft Office package.

The storage of information is carried out by BIOFIX CONSULTORIA LTDA, and consists of a folder-based structure, which is used to facilitate the search for specific files, depending on the stage of project development.

Regarding monitoring data, these are stored in an initial folder called "PROJETOS". Within this folder, one should access a new one called "5. CERTIFICADOS E EM IMPLEMENTAÇÃO ", once inside it, proceed to search for the folder with the project's name, which in this case is "2. KA'A PYAHU", and once in this folder, select the folder named "8. IMPLEMENTAÇÃO, RECURSOS". Inside the aforementioned folder, the data related to the monitoring process, as well as the relevant documents that will be stored there.

Organization and responsibilities of the parties involved in all of the above

The organization and responsibilities of the parties involved in the monitoring plan will be handled by BIOFIX CONSULTORIA LTDA and the project proponent.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

The monitoring period includes the years 01/05/2021 to 31/12/2023, in which the project proponents of the project has been carrying out activities in favor of the conservation and improvement of the environmental conditions of its territory, which lead to a quantification of reduction of GHG emissions.

7.1 Data and Parameters Monitored

Table 79. Data and parameters VM0007: METHODOLOGICAL FRAMEWORK (REDD-MF) v 1.7

Data / Parameter:	ΔCWPS-REDD
Data unit:	t CO ₂ e
Description:	Net GHG emissions in the REDD project monitoring period
Value applied	72,685
Comments	Monitoring period: 01/05/2021-31/12/2023

Data / Parameter:	CLK-AS,unplanned
Data unit:	t CO ₂ e
Description	Net greenhouse gas emissions due to activity shifting for projects preventing unplanned deforestation preventing unplanned deforestation.
Value applied	118,498
Comments	Monitoring period: 01/05/2021-31/12/2023

Table 80. Data and parameters VMD0015: Methods for monitoring of GHG emissions and removals in REDD and CIW projects (M-REDD) v 2.3

Data / Parameter:	Project Forest Cover Monitoring Map
Data unit:	N/A
Description	Map showing the location of forest land within the project area at the beginning of each monitoring period.
Value applied	Section 5.13 and section 7.3.
Comments	Accuracy 92%

Data / Parameter:	Leakage Belt Forest Cover Monitoring Map
Data unit:	N/A
Description	Map showing the location of forest land within the leakage belt area at the beginning of each monitoring period.
Value applied	See section 5.3 and section 7.4
Comments	Accuracy 92%

Data / Parameter:	ADefPA,i,u,t
Data unit:	ha
Description	Area of recorded deforestation in the project area in stratum I converted to land use u in year t
Value applied	494.2
Comments	Monitoring period: 01/05/2021-31/12/2023

Data / Parameter:	ADefLB,i,u,t
Data unit:	ha
Description	Area of recorded deforestation in the leakage belt in stratum I converted to land use u in year t
Value applied	1,864
Comments	Monitoring period: 01/05/2021-31/12/2023

Data / Parameter:	CAB_tree,i
Data unit:	t CO2e ha-1
Description	Carbon stock in aboveground biomass in trees in stratum I in the monitoring period
Value applied	NREF : 146.52 MR 120.60
Comments	Monitoring period: 01/05/2021-31/12/2023

Data / Parameter:	CAB_nontree,i
Data unit:	t CO2e ha-1

Description	Carbon stock in aboveground biomass in non trees in stratum I in the monitoring period
Value applied	MR 25.92
Comments	Monitoring period: 01/05/2021-31/12/2023

Data / Parameter:	CBB_tree,i
Data unit:	t CO2e ha-1
Description	Carbon stock in belowground biomass in trees in stratum I in the monitoring period
Value applied	NREF : 29.30 MR: 26.48
Comments	Monitoring period: 01/05/2021-31/12/2023

Data / Parameter:	CBB_nontree,i
Data unit:	t CO2e ha-1
Description	Carbon stock in belowground biomass in non trees in stratum I in the monitoring period
Value applied	MR 2.82
Comments	Monitoring period: 01/05/2021-31/12/2023

7.2 Baseline Emissions

The baseline emissions reduction was made as described in section 5.1, that is, for the monitoring period only deforestation activity is monitored, as shown in the following table.

Table 81. Estimated emission reduction for the baseline of the monitoring period

Year	<i>ABSL,PA,unplanned,t (ha)</i>	Estimated baseline emissions (tCO _{2e}) ³²
2021	1,656	243,599
2022	2,467	362,913
2023	2,467	362,913
Total	6,591	969,425

Source. KA'A PYAHU REDD+ Project.

³² Updated baseline with stock changes from post-deforestation during the monitoring period (2021-2023).

7.3 Project Emissions

The reduction of emissions of the project is determined by discounting the deforestation that occurred in the project area. To determine the reduction of real emissions in the project area, the deforestation between 01/05/2015 and 31/12/2023 was calculated, resulting in 494.2 ha deforested, distributed in 124.2 ha in the 245 days corresponding for the 2021, and 185 ha annual average in the following 2.

Figure 57. Map of land use change in the monitoring period 2021-2023 in the project area

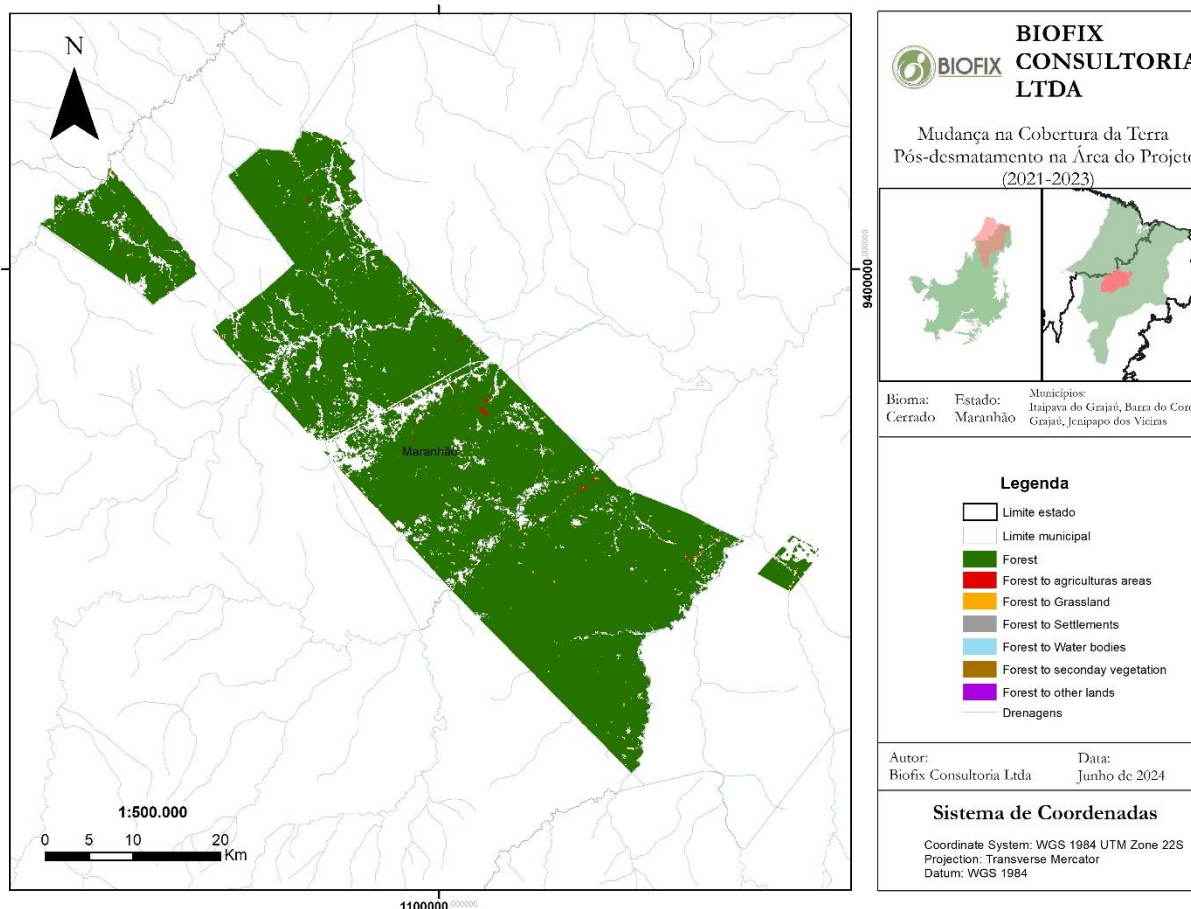


Table 82. Matrix Land use change post-deforestation in the Project Area (2021 - 2023)

Land use	2021		2023			
	Area (ha)	Area (%)	Land use	Area (ha)	Area (%)	
Forest	150,800	100	Forest	150,245.19		
			Non forest	Agriculture	226.71	40.84
				Grassland	298.35	53.75
				Settlements	0.00	0.00
				Water bodies	1.53	0.28
				Other lands	3.15	0.57
				Secondary vegetation	25.38	4.57
Non forest total			555.12	100.00		

Source. KA'A PYAHU REDD+ Project.

Once the matrix land use change post-deforestation in the project area (2021 – 2023) was determined, the post-deforestation carbon stocks were calculated, having accounted the long-term average stocks on the land-use during the monitoring period (2021-2023). As explained in section 5.1.4 of this document, the post-deforestation carbon stocks were calculated considering the option 2 (historical area-weighted average) of the section 4.2.2 of the module BL-UP (VMD0007 v3.3), resulting in:

Table 83. Carbon reservoir stock by land-use post deforestation in the Project Area

Deforestation 2021-2023		Area		Non-tree average carbon stock (t.C/ha)		Post-def carbon stock by land-use (t.C/ha)	
Land use post-deforestation		ha	(%)	Aboveground biomass	Belowground biomass	Aboveground biomass	Belowground biomass
Non forest	Agriculture	226.71	40.84	5.38	0.04	2.20	0.02
	Grassland	298.35	53.75	7.57	1.10	4.07	0.59
	Settlements	0.00	0.00	0.00	0.00	0.00	0.00
	Water bodies	1.53	0.28	0.00	0.00	0.00	0.00
	Other lands	3.15	0.57	0.00	0.00	0.00	0.00
	Secondary vegetation	25.38	4.57	17.58	3.52	0.80	0.16
Non-forest total		555.12	100.00	30.53	4.66	7.07	0.77

Source. KA'A PYAHU REDD+ Project.

Then, stock changes in each pool were calculated by subtracting post-deforestation carbon stocks from forest carbon stocks.

Table 84. Carbon reservoir stock values in the monitoring period (2021-2023)

Reservoir	Carbon stock (t C/ha)		Carbon stock (tCO ₂ e/ha)		
	Pre-def	Non-tree post def	Pre-def	Non-tree post def	Baseline carbon stock change
Aboveground biomass	39.96	7.07	146.52	25.92	120.60
Belowground biomass	7.99	0.77	29.30	2.82	26.48

Source. KA'A PYAHU REDD+ Project.

According with the equation 3, step 2, section 5.2.1. (monitoring deforestation) of the module M-REDD (VMD0015, V2.3), the net carbon stock change as a result of deforestation is equal to the area deforested multiplied by the emission per unit area:

$$\Delta C_{P,defPA,i,t} = \sum_{u=1}^U (A_{DefPA,u,i,t} *)$$

Where,

$\Delta C_{P,defPA,i,t}$ = Net carbon stock change as a result of deforestation in the project case in the project area in stratum i in year.

$A_{DefPA,u,i,t}$ = Area of recorded deforestation in the project area stratum i converted to land use u in year t; ha

$\Delta C_{\text{pools},P,\text{Def},u,i,t}$ = Net carbon stock changes in all pools in the project case in land use u in stratum i in year t ; t CO₂-e ha⁻¹

Resulting in 72,686 tCO₂ emissions during the monitoring period (2021-2023) in the Project Area.

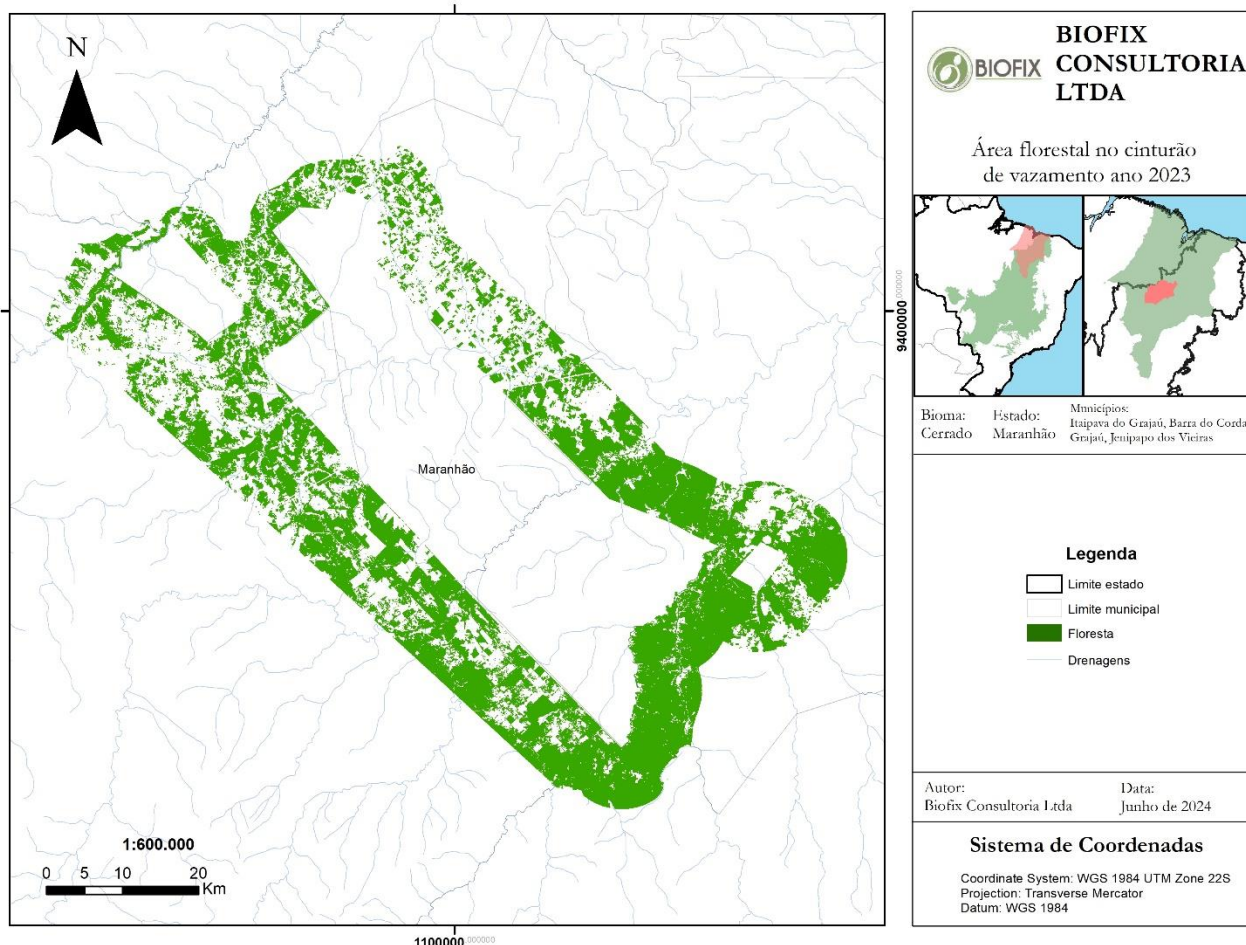
Table 85. Project Emissions during the monitoring period (2021-2023).

Year	$A_{\text{Def},PA}$ (ha)	$\Delta C_{P\text{def}PA}$ (tCO ₂ e)
2021	124.2	18,265.65
2022	185.0	27,210.60
2023	185.0	27,210.60
Total	494.2	72,686.84

Source. KA'A PYAHU REDD+ Project.

7.4 Leakage Emissions

Figure 58. Map of forests in the leakage belt year 2023



Source. KA'A PYAHU REDD+ Project.

During the monitoring period, 1,864 hectares were deforested. As mentioned in section 3.6 of methodological deviations, according to the VCS standard section 3.15.14, an optional 15% is applied by

default for the project. Following step 3 of the module VMD0010 (LK-ASU) V1.2, emissions in the leakage belt is estimated as follows:

$$\Delta C_{LK-ASU-LB} = \Delta C_{P, LB} - \Delta C_{BSL, LK, unplanned}$$

Where,

$\Delta C_{LK-ASU-LB}$ = Net CO₂ emissions due to unplanned deforestation displaced from the project area to the leakage belt up to year t* (t CO₂e).

$\Delta C_{BSL, LK, unplanned}$ = Net CO₂ emissions in the baseline from unplanned deforestation in the leakage belt up to year t* (t CO₂e).

Table 86. Emissions within the leakage belt in the KA'A PYAHUD project during the monitoring period

year	$A_{BSL, LB, unplanned, t}$ (ha)	$\Delta C_{BSL, LK, unplanned}$ (tCO ₂)	A_{DefLB} (ha)	$C_{P, LB, unplanned}$ (t CO ₂ e)	$\Delta C_{LK-ASU-LB}$
2021	215	29,749	417	61,348	29,776
2022	320	44,320	621	91,397	44,361
2023	320	44,320	621	91,397	44,361
Total	854	212,572	1,864	244,142	118,498

Source. KA'A PYAHU REDD+ Project.

7.5 GHG Emission Reductions and Carbon Dioxide Removals

According with 8.4.2 section of the methodology VM0007, the total net emission reductions of the KA'A PYAHU REDD+ project was calculated from the following equation:

$$NER_{REDD} = \Delta C_{BSL-REDD} - \Delta C_{WPS-REDD} - \Delta C_{LK-REDD}$$

NER_{REDD} = Total net GHG emission reductions of the REDD project activity up to year t* (t CO₂e).

$\Delta C_{BSL-REDD}$ = Net GHG emissions in the REDD baseline scenario up to year t* (t CO₂e).

$\Delta C_{WPS-REDD}$ = Net GHG emissions in the REDD project scenario up to year t* - from Module M-REDD (t CO₂e).

$\Delta C_{LK-REDD}$ = Net GHG emissions due to leakage from the REDD project activity up to year t* (t CO₂e).

Table 87. Net emission reduction of the KA'A PYAHU REDD+ project

Vintage period	Baseline emissions (tCO ₂ e) ³³	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
01-May-2021 to 31-Dec-2021	243,599	18,264	29,776	195,558	0	195,558
01-Jan-2022 to 31-Dec-2022	362,913	27,210	44,361	291,341	0	291,341
01-Jan-2023 to 31-Dec-2023	362,913	27,210	44,361	291,341	0	291,341
Total	969,425	72,685	118,498	778,241	0	778,241

³³ Adjusted baseline values based on stock changes in carbon pools during the monitored period in the project area.

Source. KA'A PYAHU REDD+ Project.

Table 88. Non-permanence risk rating in the project

State the non-permanence risk rating (%)	20
Has the non-permanence risk report been attached as either an appendix or a separate document?	☒ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e the Long-term Average (LTA).	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No If no, provide justification.
State, in tCO ₂ e, the expected total GHG benefit to date	778,241
If a loss occurred (including a loss event or reversal), state the amount of tCO ₂ e lost:	

Source. KA'A PYAHU REDD+ Project.

Table 89. Total VCU Issuance for the KA'A PYAHU REDD+ Project during the monitoring period

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Buffer pool allocation (tCO ₂ e)	Reductions VCUs (tCO ₂ e)	Removals VCUs (tCO ₂ e)	Total VCU issuance (tCO ₂ e)
01-May-2021 to 31-Dec-2021	243,599	18,264	29,776	39,112	156,446	0	156,446
01-Jan-2022 to 31-Dec-2022	362,913	27,210	44,361	58,268	233,073	0	233,073
01-Jan-2023 to 31-Dec-2023	362,913	27,210	44,361	58,268	233,073	0	233,073
Total	969,425	72,685	118,498	155,648	622,592	0	622,592

Source. KA'A PYAHU REDD+ Project.

Table 90. Estimated ex-ante GHG emission reductions and the achieved emission reductions for this monitoring period of the KA'A PYAHU REDD+ project.

Vintage period	Ex-ante estimated reductions	Achieved reductions	Percent difference	Explanation for the difference
01-May-2021 to 31-Dec-2021	199,786	195,558	2.17	The differences presented between the removal values calculated ex-ante and the monitoring period do not exceed 2.2%. The values used as emission factors between the baseline and the monitoring period are different, since in the first case, it is done from the calculation of stock changes due to deforestation in the reference region in the historical period reference (2010-2020). For the second, the factors were obtained from
01-Jan-2022 to 31-Dec-2022	297,640	291,341	2.17	
01-Jan-2023 to 31-Dec-2023	297,640	291,341	2.17	

				stock changes ex-post deforestation during the monitoring period (2021-2023) in the project area. Likewise, the difference also arises because in the baseline scenario, emissions were not assumed in the project area. That is, deforestation or efficiency percentage was not assumed based on an estimated efficiency percentage of the project.
Total	795,066	778,241	2.17	

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APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification
<u>Non-permanence Risk Tool</u>	Cash flow, Revenue models, estimative prices and cost structures	Financial Data has been excluded to protect proprietary financial strategies. The excluded financial data and projections are integral to maintaining a competitive edge in project financing and implementation.
<u>2.1</u> <u>2.1.2</u> <u>2.1.3</u> <u>2.2</u>	Legal Agreements cited on these Section	Specific terms of agreements between the project developer and indigenous communities have been redacted to honor confidentiality clauses and prevent competitive harm. Legal agreements have specific terms and conditions of contracts, especially those that involve indigenous communities, land use, and efforts to build strong governance around and inside the project management and permanence. The information is proprietary and not otherwise publicly available. The legal agreements contain sensitive terms that, if disclosed, could undermine negotiations or community relations.
5.4.	Supports uploaded on Digital Non-permanence Risk Tool which based the % cited on this Section	There is data (ex. cash flow), sensitive information, as support to responses of the tool. So, if disclosed publicly, could compromise the competitive position of the project and its strategies
ANNEX: 26, 27, 28, 30, 31, 32, 33, 34, 35, 36 e 37.	Personal data on LPIC documents (ex. Minutes, Attendance list, village Representative Subscription list with names, ID number, phone)	In accordance with LGPD (General Data Protection Law), we have not disclosed any personal information that is not authorized by the individual or stakeholder.

APPENDIX

Annex 001.	DECREE No. 246, OF OCTOBER 29, 1991 CANA BRAVA
Annex 002.	DECREE No. 382 OF DECEMBER 24, 1991 URUCU JURUÁ
Annex 003.	DECREE OF FEBRUARY 10, 2003 URUCU JURUÁ
Annex 004.	DECREE No. 313 OF October 29, 1991 Lagoa Comprida
Annex 005.	DECREE No. 88,813, OF OCTOBER 4, 1983 RODEADOR
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Annex 006b.	VILLAGES TI CANABRAVA - REGIÃO ILHA SÃO PEDRO E CACIMBA VELHO
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